

Cooper GM. The Cell: A Molecular Approach. 2nd Ed. Washington, DC: ASM Press, Sinauer Assoc., 2000.  
Lodish H, Berk A, Zipursky SL, Matsudaira P, Baltimore D, Darnell J. Molecular Cell Biology. 4th Ed. New York: W.H. Freeman, 2000. The section on cell structure is particularly good.  
Alberts B, Johnson A, Lewis J, Raff M, Roberts K, Walter P. Molecular Biology of the Cell. 4th Ed. New York: Garland Science, 2002.



## REVIEW QUESTIONS—CHAPTER 10

1. Which of the following is a characteristic of the plasma membrane?
  - (A) It is composed principally of triacylglycerols and cholesterol.
  - (B) It contains principally nonpolar lipids.
  - (C) It contains phospholipids with their acyl groups extending into the cytosol.
  - (D) It contains more phosphatidylserine in the inner than the outer leaflet.
  - (E) It contains oligosaccharides sandwiched between the inner and outer leaflets.
2. Transmembrane proteins
  - (A) can usually be dissociated from membranes without disrupting the lipid bilayer.
  - (B) are classified as peripheral membrane proteins.
  - (C) contain hydrophobic amino acid residues at their carboxy terminus.
  - (D) contain hydrophilic amino acid residues extending into the lipid bilayer.
  - (E) contain membrane-spanning regions that are  $\alpha$ -helices.

Use the following case history for questions 3 and 4.

A patient had a sudden heart attack caused by inadequate blood flow through the vessels of the heart. As a consequence, there was an inadequate supply of oxygen to generate ATP in his cardiomyocytes.

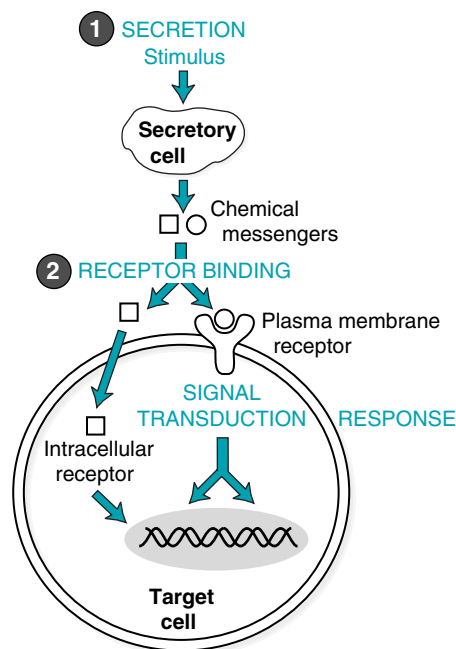
3. The compartment of the cardiomyocyte most directly involved in ATP generation is the
  - (A) mitochondrion.
  - (B) peroxisome.
  - (C) lysosome.
  - (D) nucleus.
  - (E) Golgi.
4. The transport process most directly affected by his decreased rate of ATP generation is
  - (A) active transport of  $\text{Na}^+$  out of the cell.
  - (B) facilitative transport of glucose into the cell.
  - (C) passive diffusion of  $\text{H}_2\text{O}$  into the cell.
  - (D) ion flux through ligand-gated channels.
  - (E) receptor-mediated endocytosis.

Use the following case history for questions 5 and 6.

A patient in a nursing home who developed severe diarrhea was diagnosed with a *Clostridium difficile* infection. The severe diarrhea associated with *C. difficile* is caused principally by two toxins that are UDP-glucosyltransferases. These toxins modify a monomeric G protein, thereby disrupting cellular attachments associated with the actin skeleton.

5. On the basis of its effect, which of the following G protein families is the most likely target of this modification?
  - (A) Ras
  - (B) Ran
  - (C) Rho
  - (D) Rab
  - (E) Arf

# 11 Cell Signaling by Chemical Messengers



**Fig. 11.1.** General features of chemical messengers.

Within a complex organism such as the human, different organs, tissues, and individual cell types have developed specialized functions. Yet each cell must contribute in an integrated way as the body grows, differentiates, and adapts to changing conditions. Such integration requires communication that is carried out by chemical messengers traveling from one cell to another or by direct contact of cells with the extracellular matrix or with each other. The eventual goal of such signals is to change actions carried out in target cells by intracellular proteins (metabolic enzymes, gene regulatory proteins, ion channels, or cytoskeletal proteins). In this chapter, we present an overview of signaling by chemical messengers.

**Chemical messengers.** Chemical messengers (also called signaling molecules) transmit messages between cells. They are **secreted** from one cell in **response to a specific stimulus** and **travel to a target cell**, where they **bind to a specific receptor** and **elicit a response** (Fig. 11.1). In the nervous system, these chemical messengers are called **neurotransmitters**; in the **endocrine system**, they are **hormones**, and in the **immune system**, they are called **cytokines**. Additional chemical messengers include **retinoids**, **eicosanoids**, and **growth factors**. Depending on the distance between the secreting and target cells, chemical messengers can be classified as **endocrine** (travel in the blood), **paracrine** (travel between nearby cells), or **autocrine** (act on the same cell or on nearby cells of the same type).

**Receptors and Signal Transduction.** Receptors are proteins containing a binding site specific for a single chemical messenger and another binding site involved in transmitting the message (see Fig. 11.1). The second binding site may interact with another protein or with DNA. They may be either **plasma membrane receptors** (which span the plasma membrane and contain an extracellular binding domain for the messenger) or **intracellular binding proteins** (for messengers able to diffuse into the cell) (see Fig. 11.1). Most plasma membrane receptors fall into the categories of **ion channel receptors**, **tyrosine kinase receptors**, **tyrosine-kinase associated receptors** (**JAK-STAT receptors**), **serine-threonine kinase receptors**, or **heptahelical receptors** (proteins with seven  $\alpha$ -helices spanning the membrane). When a chemical messenger binds to a receptor, the signal it is carrying must be converted into an intracellular response. This conversion is called **signal transduction**.

**Signal Transduction for Intracellular Receptors.** Most intracellular receptors are **gene-specific transcription factors**, proteins that bind to DNA and regulate the transcription of certain genes (Gene transcription is the process of copying the genetic code from DNA to RNA.).

**Signal Transduction for Plasma Membrane Receptors.** Mechanisms of signal transduction that follow the binding of signaling molecules to plasma membrane receptors include phosphorylation of receptors at tyrosine residues (**receptor tyrosine kinase activity**), conformational changes in **signal transducer proteins** (e.g., proteins with **SH2 domains**, the **monomeric G protein Ras**, **heterotrimeric G proteins**) or increases in the levels of **intracellular second messengers**. Second messengers are nonprotein molecules generated inside the cell in response to

hormone binding that continue transmission of the message. Examples include 3',5'-cyclic AMP (cAMP), inositol trisphosphate ( $IP_3$ ), and diacylglycerol (DAG).

Signaling often requires a rapid response and rapid termination of the message, which may be achieved by degradation of the messenger or second messenger, the automatic G protein clock, deactivation of signal transduction kinases by phosphatases, or other means.



## THE WAITING ROOM



**Mya Sthenia** is a 37-year-old woman who complains of increasing muscle fatigue in her lower extremities with walking. If she rests for 5 to 10 minutes, her leg strength returns to normal. She also notes that if she talks on the phone, her ability to form words gradually decreases. By evening, her upper eyelids droop to the point that she has to pull her upper lids back in order to see normally. These symptoms are becoming increasingly severe. When Mya is asked to sustain an upward gaze, her upper eyelids eventually drift downward involuntarily. When she is asked to hold both arms straight out in front of her for as long as she is able, both arms begin to drift downward within minutes. Her physician suspects that **Mya Sthenia** has myasthenia gravis and orders a test to determine whether she has antibodies in her blood directed against the acetylcholine receptor.



**Ann O'Rexia**, who suffers from anorexia nervosa, has increased her weight to 102 lb from a low of 85 lb (see Chapter 9). On the advice of her physician, she has been eating more to prevent fatigue during her daily jogging regimen. She runs about 10 miles before breakfast every second day and forces herself to drink a high-energy supplement immediately afterward.



**Dennis Veere** was hospitalized for dehydration resulting from cholera toxin (see Chapter 10). In his intestinal mucosal cells, cholera A toxin indirectly activated the CFTR channel, resulting in secretion of chloride ion and  $Na^+$  ion into the intestinal lumen. Ion secretion was followed by loss of water, resulting in vomiting and watery diarrhea. Dennis is being treated for hypovolemic shock.



Acetylcholine is released by neurons and acts on acetylcholine receptors at neuromuscular junctions to stimulate muscular contraction. Myasthenia gravis is an acquired autoimmune disease in which the patient has developed pathogenic antibodies against these receptors. **Mya Sthenia's** decreasing ability to form words and her other symptoms of muscle weakness are being caused by the inability of acetylcholine to stimulate repeated muscle contraction when the numbers of effective acetylcholine receptors at neuromuscular junctions are greatly reduced.



Endocrine hormones enable **Ann O'Rexia** to mobilize fuels from her adipose tissue during her periods of fasting and during jogging. While she fasts overnight,  $\alpha$ -cells of her pancreas increase secretion of the polypeptide hormone glucagon. The stress of prolonged fasting and chronic exercise stimulates release of cortisol, a steroid hormone, from her adrenal cortex. The exercise of jogging also increases secretion of the hormones epinephrine and norepinephrine from the adrenal medulla. Each of these hormones is being released in response to a specific signal and causes a characteristic response in a target tissue, enabling her to exercise. However, each of these hormones binds to a different type of receptor and works in a different way.



**Ann O'Rexia's** fasting is accompanied by high levels of the endocrine hormone glucagon, which is secreted in response to low blood glucose levels. It enters the blood and acts on the liver to stimulate a number of pathways, including the release of glucose from glycogen stores (glycogenolysis) (see Chapter 3). The specificity of its action is determined by the location of receptors. Although liver parenchymal cells have glucagon receptors, skeletal muscle and many other tissues do not. Therefore, glucagon cannot stimulate glycogenolysis in these tissues.

## I. GENERAL FEATURES OF CHEMICAL MESSENGERS

Certain universal characteristics to chemical messenger systems are illustrated in Figure 11.1. Signaling generally follows the sequence: (1) the chemical messenger is secreted from a specific cell in response to a stimulus; (2) the messenger diffuses or is transported through blood or other extracellular fluid to the target cell; (3) a receptor in the target cell (a plasma membrane receptor or intracellular receptor) specifically binds the messenger; (4) binding of the messenger to the receptor elicits a response; (5) the signal ceases and is terminated. Chemical messengers elicit their response in the target cell without being metabolized by the cell.

Another general feature of chemical messenger systems is that the specificity of the response is dictated by the type of receptor and its location. Generally, each receptor binds only one specific chemical messenger, and each receptor initiates a characteristic signal transduction pathway that will ultimately activate or inhibit certain processes in the cell. Only certain cells, the target cells, carry receptors for that messenger and are capable of responding to its message.

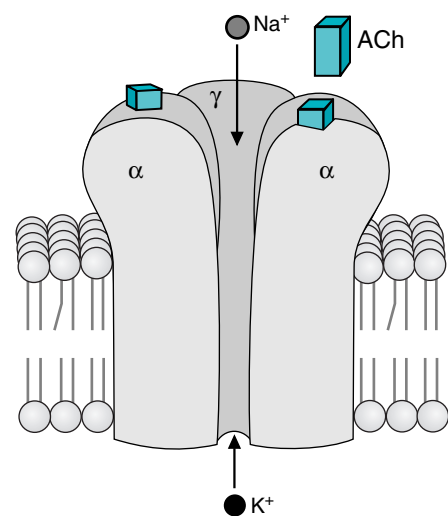
The means of signal termination is an exceedingly important aspect of cell signaling, and failure to terminate a message contributes to a number of diseases, such as cancer.



Most chemical messengers (including neurotransmitters, cytokines, and endocrine hormones) are contained in vesicles that fuse with a region of the cell membrane when the cell receives a stimulus to release the messenger. Most secretory cells use a similar set of proteins to enable vesicle fusion, and fusion is usually triggered by  $\text{Ca}^{2+}$  influx, as seen with the release of acetylcholine.



Myasthenia gravis is a disease of autoimmunity caused by the production of an antibody directed against the acetylcholine receptor in skeletal muscle. In this disease, B and T lymphocytes cooperate in producing a variety of antibodies against the nicotinic acetylcholine receptor. The antibodies then bind to various locations in the receptor and cross-link the receptors, forming a multi-receptor antibody complex. The complex is endocytosed and incorporated into lysosomes, where it is degraded. **Mya Sthenia**, therefore, has fewer functional receptors for acetylcholine to activate.



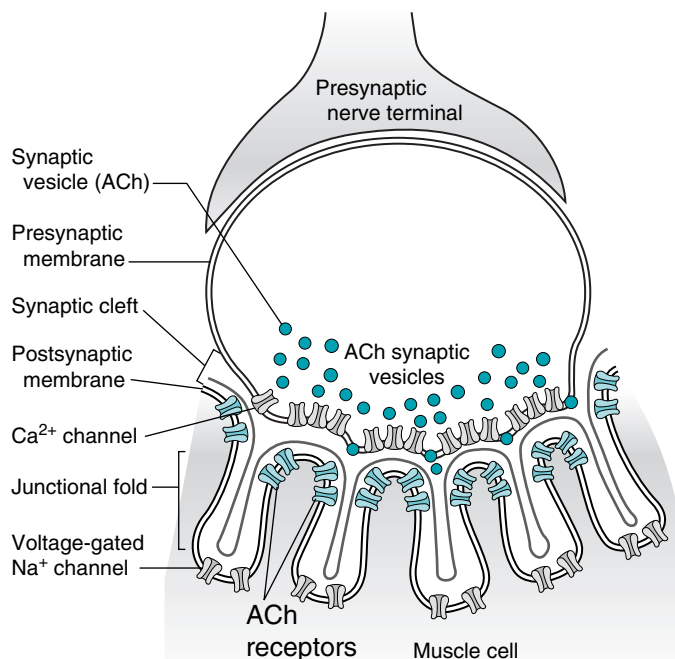
**Fig. 11.3.** The nicotinic acetylcholine receptor. Each receptor is composed of five subunits, and each subunit has four membrane-spanning helical regions. The two  $\alpha$  subunits are identical and contain binding sites for acetylcholine. When two acetylcholine molecules are bound, the subunits change their conformation so that the channel in the center of the receptor is open, allowing  $\text{K}^+$  ions to diffuse out and  $\text{Na}^+$  ions to diffuse in.

## A. General Features of Chemical Messenger Systems Applied to the Nicotinic Acetylcholine Receptor

The individual steps involved in cell signaling by chemical messengers are illustrated with acetylcholine, a neurotransmitter that acts on nicotinic acetylcholine receptors on the plasma membrane of certain muscle cells. This system exhibits the classic features of chemical messenger release and specificity of response.

Neurotransmitters are secreted from neurons in response to an electrical stimulus called the action potential (a voltage difference across the plasma membrane, caused by changes in  $\text{Na}^+$  and  $\text{K}^+$  gradients, that is propagated along a nerve). The neurotransmitters diffuse across a synapse to another excitable cell, where they elicit a response (Fig. 11.2). Acetylcholine is the neurotransmitter at neuromuscular junctions, where it transmits a signal from a motor nerve to a muscle fiber that elicits contraction of the fiber. Before release, acetylcholine is sequestered in vesicles clustered near an active zone in the presynaptic membrane. This membrane also has voltage-gated  $\text{Ca}^{2+}$  channels that open when the action potential reaches them, resulting in an influx of  $\text{Ca}^{2+}$ .  $\text{Ca}^{2+}$  triggers fusion of the vesicles with the plasma membrane, and acetylcholine is released into the synaptic cleft. Thus, the chemical messenger is released from a specific cell in response to a specific stimulus.

Acetylcholine diffuses across the synaptic cleft to bind to plasma membrane receptors on the muscle cells called nicotinic acetylcholine receptors (Fig. 11.3). The subunits are assembled around a channel, which has a funnel-shaped opening in the center. As acetylcholine binds to the receptor, a conformational change opens the narrow portion of the channel (the gate), allowing  $\text{Na}^+$  to diffuse in and  $\text{K}^+$  to diffuse out (A uniform property of all receptors is that signal transduction begins with conformational changes in the receptor.). The change in ion concentration



**Fig. 11.2.** Acetylcholine receptors at the neuromuscular junction. A motor nerve terminates in several branches; each branch terminates in a bulb-shaped structure called the presynaptic bouton. Each bouton synapses with a region of the muscle fiber containing junctional folds. At the crest of each fold, there is a high concentration of acetylcholine receptors, which are gated ion channels.



activates a sequence of events that eventually triggers the cellular response—contraction of the fiber.

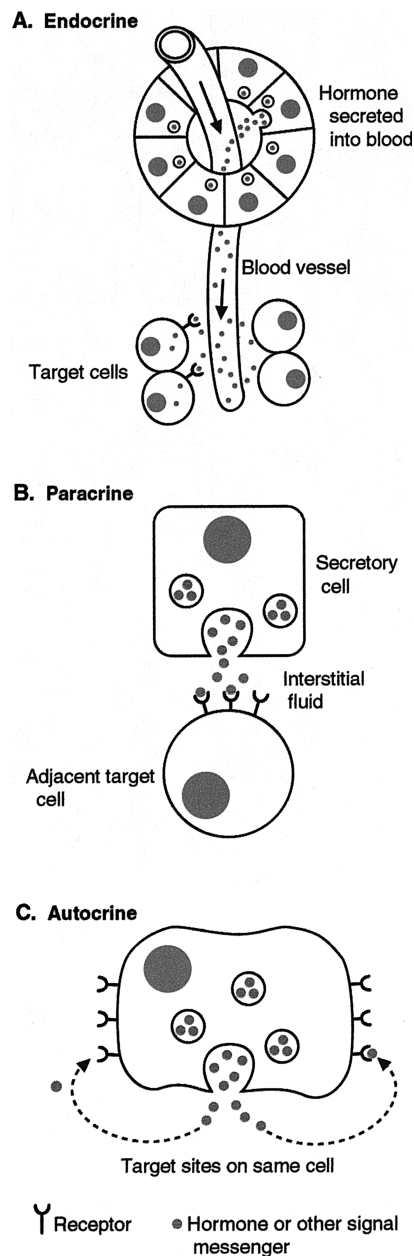
Once acetylcholine secretion stops, the message is rapidly terminated by acetylcholinesterase, an enzyme located on the postsynaptic membrane that cleaves acetylcholine. It is also terminated by diffusion of acetylcholine away from the synapse. Rapid termination of message is a characteristic of systems requiring a rapid response from the target cell.

## B. Endocrine, Paracrine, and Autocrine

The actions of chemical messengers are often classified as endocrine, paracrine, or autocrine (Fig. 11.4). Each endocrine hormone is secreted by a specific cell type



**Mya Sthenia** was tested with an inhibitor of acetylcholinesterase, edrophonium chloride, administered intravenously (see Chapter 8, Fig. 8.18). After this drug inactivates acetylcholinesterase, acetylcholine that is released from the nerve terminal accumulates in the synaptic cleft. Even though Mya expresses fewer acetylcholine receptors on her muscle cells (due to the auto-antibody-induced degradation of receptors), by increasing the local concentration of acetylcholine, these receptors have a higher probability of being occupied and activated. Therefore, acute intravenous administration of this short-acting drug briefly improves muscular weakness in patients with myasthenia gravis.



**Fig. 11.4.** Endocrine, autocrine, and paracrine actions of hormones and other chemical messengers.

(generally in an endocrine gland), enters the blood, and exerts its actions on specific target cells, which may be some distance away. In contrast to endocrine hormones, paracrine actions are those performed on nearby cells, and the location of the cells plays a role in specificity of the response. Synaptic transmission by acetylcholine and other neurotransmitters (sometimes called neurocrine signaling) is an example of paracrine signaling. Acetylcholine activates only those acetylcholine receptors located across the synaptic cleft from the signaling nerve and not all muscles with acetylcholine receptors. Paracrine actions are also very important in limiting the immune response to a specific location in the body, a feature that helps prevent the development of autoimmune disease. Autocrine actions involve a messenger acting on the cell from which it is secreted, or on nearby cells that are the same type as the secreting cells.

### C. Types of Chemical Messengers

Three major signaling systems in the body employ chemical messengers: the nervous system, the endocrine system, and the immune system. Some messengers are difficult to place in just one such category.

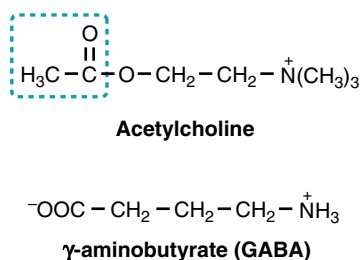
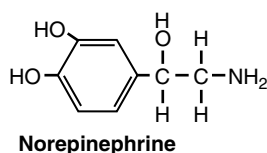
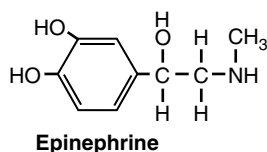


Fig. 11.5. Small molecule neurotransmitters.



The catecholamine hormone epinephrine (also called adrenaline) is the fight, fright, and flight hormone. Epinephrine and the structurally similar hormone norepinephrine are released from the adrenal medulla in response to a variety of immediate stresses, including pain, hemorrhage, exercise, hypoglycemia, and hypoxia. Thus, as **Ann O'Rexia** begins to jog, there is a rapid release of epinephrine and norepinephrine into the blood.



#### 1. THE NERVOUS SYSTEM

The nervous system secretes two types of messengers: small molecule neurotransmitters, often called biogenic amines, and neuropeptides. Small molecule neurotransmitters are nitrogen-containing molecules, which can be amino acids or are derivatives of amino acids (e.g., acetylcholine and  $\gamma$ -aminobutyrate, Fig. 11.5). Neuropeptides are usually small peptides (between 4 and 35 amino acids), secreted by neurons, that act as neurotransmitters at synaptic junctions or are secreted into the blood to act as neurohormones.

#### 2. THE ENDOCRINE SYSTEM

Endocrine hormones are defined as compounds, secreted from specific endocrine cells in endocrine glands, that reach their target cells by transport through the blood. Insulin, for example, is an endocrine hormone secreted from the  $\beta$  cells of the pancreas. Classic hormones are generally divided into the structural categories of polypeptide hormones (e.g., insulin—see Chapter 6, Fig. 6.15 for the structure of insulin), catecholamines such as epinephrine (which is also a neurotransmitter), steroid hormones (which are derived from cholesterol), and thyroid hormone (which is derived from tyrosine). Many of these endocrine hormones also exert paracrine or autocrine actions. The hormones that regulate metabolism are discussed throughout this chapter and in subsequent chapters of this text.

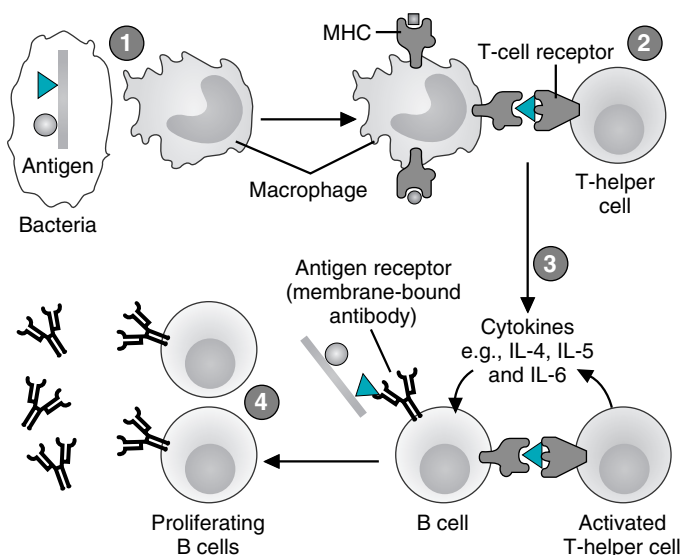
Some compounds normally considered hormones are more difficult to categorize. For example, retinoids, which are derivatives of vitamin A (also called retinol) and vitamin D (which is also derived from cholesterol) are usually classified as hormones, although they are not synthesized in endocrine cells.

#### 3. THE IMMUNE SYSTEM

The messengers of the immune system, called cytokines, are small proteins with a molecular weight of approximately 20,000 daltons. Cytokines regulate a network of responses designed to kill invading microorganisms. The different classes of cytokines (interleukins, tumor necrosis factors, interferons, and colony-stimulating factors) are secreted by cells of the immune system and usually alter the behavior of other cells in the immune system by activating the transcription of genes for proteins involved in the immune response.



**Selection and Proliferation of B Cells Producing the Desired Antibody.** Interleukins, a class of cytokine, illustrate some of the signaling involved in the immune response. Interleukins are polypeptide factors with molecular weights ranging from 15,000 to 25,000 Daltons. They participate in a part of the immune response called humoral immunity, which is carried out by a population of lymphoid B cells producing just one antibody against one particular antigen. The proliferation of cells producing that particular antibody is mediated by receptors and by certain interleukins.



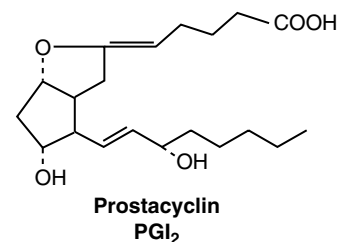
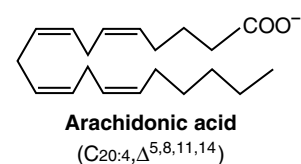
Bacteria phagocytized by macrophages are digested by lysosomes. (1) A partially digested fragment of the bacterial protein (the blue antigen) is presented on the extracellular surface of the macrophage by a membrane protein called an MHC (**m**ajor **h**istocompatibility **c**omplex). (2) Certain lymphoid cells called T-helper cells contain receptors that can bind to the displayed antigen–MHC complex, a process that activates the T-cell (direct cell-to-cell signaling, requiring recognition molecules). (3) The activated T-helper cell then finds and binds to a B cell whose antigen receptor binds a soluble fragment of that same bacterially derived antigen molecule (again, direct cell-to-cell signaling). The bound T cell secretes interleukins, which act on the B cell (a paracrine signal). The interleukins thus stimulate proliferation of only those B cells capable of synthesizing and secreting the desirable antibody. Furthermore, the interleukins determine which class of antibody is produced.

#### 4. THE EICOSANOIDS

The eicosanoids (including prostaglandins [PG], thromboxanes, and leukotrienes) control cellular function in response to injury (Fig.11.6) These compounds are all derived from arachidonic acid, a 20-carbon polyunsaturated fatty acid that is usually present in cells as part of the membrane lipid phosphatidylcholine (see Chapter 5, Fig. 5.21). Although almost every cell in the body produces an eicosanoid in response to tissue injury, different cells produce different eicosanoids. The eicosanoids act principally in paracrine and autocrine functions, affecting the cells that produce them or their neighboring cells. For example, vascular endothelial cells (cells lining the vessel wall) secrete the prostaglandin  $\text{PGI}_2$  (prostacyclin), which acts on nearby smooth muscle cells to cause vasodilation (expansion of the blood vessel).

#### 5. GROWTH FACTORS

Growth factors are polypeptides that function through stimulation of cellular proliferation. For example, platelets aggregating at the site of injury to a blood vessel secrete



**Fig. 11.6.** Eicosanoids are derived from arachidonic acid and retain its original 20 carbons (thus the name *eicosanoids*). All prostaglandins, such as prostacyclin, also have an internal ring.



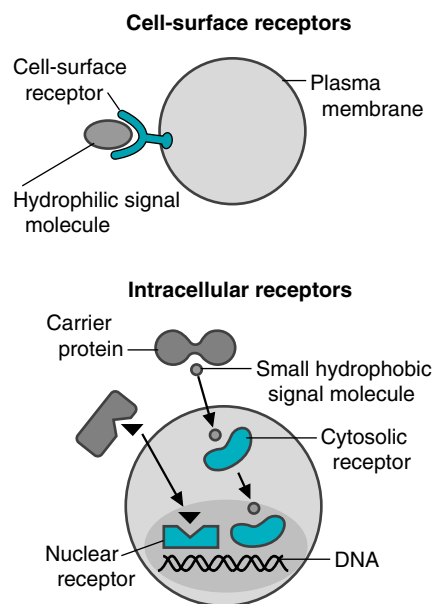
**Lotta Topaigne** suffered enormously painful gout attacks affecting her great toe (see Clinical Comments, Chapter 8). The extreme pain was caused by the release of a leukotriene that stimulated pain receptors. The precipitated urate crystals in her big toe stimulated recruited inflammatory cells to release the leukotriene.

PDGF (platelet-derived growth factor). PDGF stimulates the proliferation of nearby smooth muscle cells, which eventually form a plaque covering the injured site. Some growth factors are considered hormones, and some have been called cytokines.

*Each of the hundreds of chemical messengers has its own specific receptor, which will usually bind no other messenger.*

## II. INTRACELLULAR TRANSCRIPTION FACTOR RECEPTORS

### A. Intracellular Versus Plasma Membrane Receptors



**Fig. 11.7.** Intracellular vs. plasma membrane receptors. Plasma membrane receptors have extracellular binding domains. Intracellular receptors bind steroid hormones or other messengers able to diffuse through the plasma membrane. Their receptors may reside in the cytoplasm and translocate to the nucleus, reside in the nucleus bound to DNA, or reside in the nucleus bound to other proteins.

The structural properties of a messenger determine, to some extent, the type of receptor it binds. Most receptors fall into two broad categories: intracellular receptors or plasma membrane receptors (Fig. 11.7). Messengers using intracellular receptors must be hydrophobic molecules able to diffuse through the plasma membrane into cells. In contrast, polar molecules such as peptide hormones, cytokines, and catecholamines cannot rapidly cross the plasma membrane and must bind to a plasma membrane receptor.

Most of the intracellular receptors for lipophilic messengers are gene-specific transcription factors. A transcription factor is a protein that binds to a specific site on DNA and regulates the rate of transcription of a gene (i.e., synthesis of the mRNA). External signaling molecules bind to transcription factors that bind to a specific sequence on DNA and regulate the expression of only certain genes; they are called gene-specific or site-specific transcription factors.

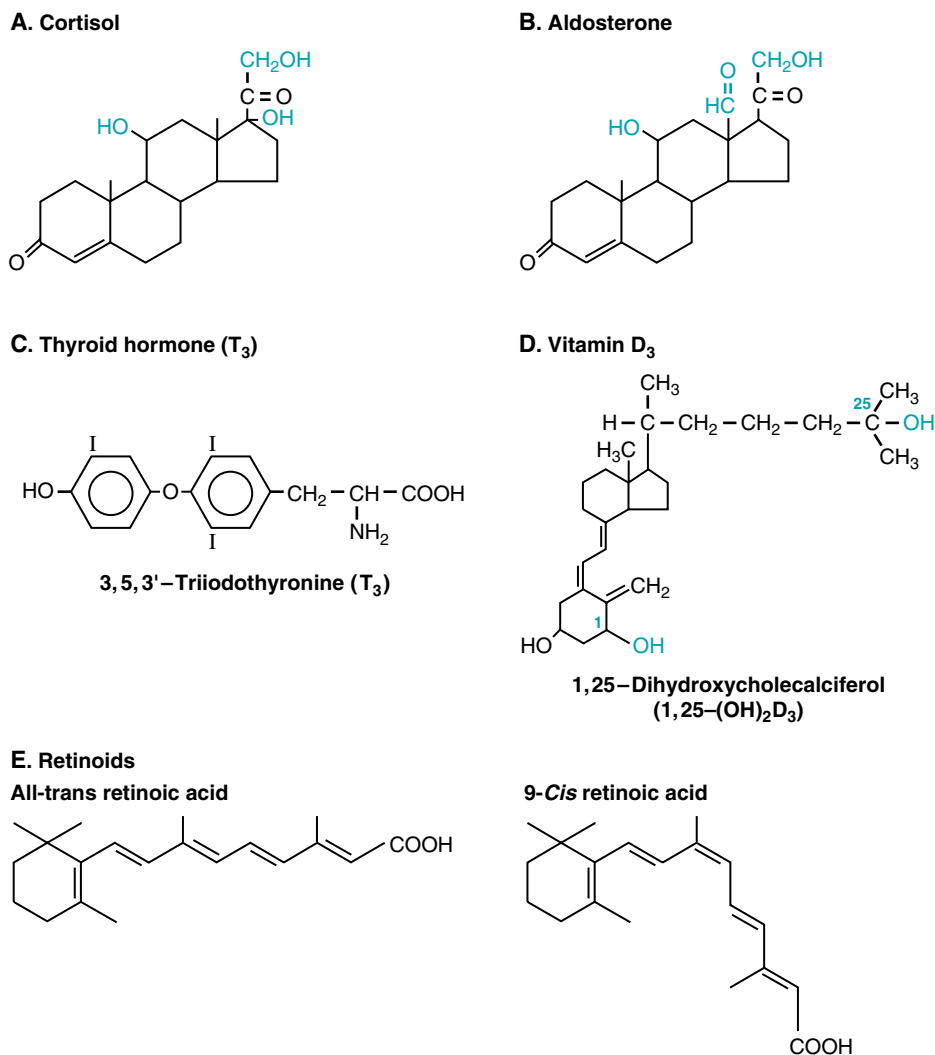
### B. The Steroid Hormone/Thyroid Hormone Superfamily of Receptors

Lipophilic hormones that use intracellular gene-specific transcription factors include the steroid hormones, thyroid hormone, retinoic acid (active form of vitamin A), and vitamin D (Fig. 11.8). Because these compounds are water-insoluble, they are transported in the blood bound to serum albumin, which has a hydrophobic binding pocket, or to a more specific transport protein, such as steroid hormone-binding globulin (SHBG) and thyroid hormone-binding globulin (TBG). The intracellular receptors for these hormones are structurally similar and are referred to as the steroid hormone/thyroid hormone superfamily of receptors.

The steroid hormone/thyroid hormone superfamily of receptors reside primarily in the nucleus, although some are found in the cytoplasm. The glucocorticoid receptor, for example, exists as cytoplasmic multimeric complexes associated with heat shock proteins. When the hormone cortisol (a glucocorticoid) binds, the receptor undergoes a conformational change and dissociates from the heat shock proteins, exposing a nuclear translocation signal (see Chapter 10, Section VI.) The receptors dimerize, and the complex (including bound hormone) translocates to the nucleus, where it binds to a portion of the DNA called the hormone response element (e.g.,



The steroid hormone cortisol is synthesized and released from the adrenal cortex in response to the polypeptide hormone ACTH (adrenal corticotrophic hormone). Chronic stress (pain, hypoglycemia, hemorrhage, and exercise) signals are passed from the brain cortex to the hypothalamus to the anterior pituitary, which releases ACTH. Cortisol acts on tissues to change enzyme levels and redistribute nutrients in preparation for acute stress. For example, it increases transcription of the genes for regulatory enzymes in the pathway of gluconeogenesis, thereby increasing the content of these enzymes (called gene-specific activation of transcription, or induction of protein synthesis). Induction of gluconeogenic enzymes prepares the liver to respond rapidly to hypoglycemia with increased synthesis of glucose. **Ann O'Rexia**, who has been frequently fasting and exercising, has an increased capacity for gluconeogenesis in her liver.



**Fig. 11.8.** Steroid hormone/thyroid hormone superfamily. A. Cortisol (a glucocorticoid). B. Aldosterone (an androgen). C. Thyroid hormone D. Vitamin D<sub>3</sub>. E. Retinoids.

the glucocorticoid receptor binds to the glucocorticoid response element, GRE). Most of the intracellular receptors reside principally in the nucleus, and some of these are constitutively bound, as dimers, to their response element in DNA (e.g., the thyroid hormone receptor). Binding of the hormone changes its activity and its ability to associate with, or disassociate from, DNA. Regulation of gene transcription by these receptors is described in Chapter 16.



Recently several nuclear receptors have been identified that play important roles in intermediary metabolism, and they have become the target of lipid-lowering drugs. These include the peroxisome proliferator activated receptors (PPAR  $\alpha$ ,  $\beta$  and  $\gamma$ ), the liver X-activated receptor (LXR), the farnesoid X-activated receptors (FXR), and the pregnane X receptor (PXR). These receptors form heterodimers with the 9-*cis* retinoic acid receptor (RXR) and bind to their appropriate response elements in DNA in an inactive state. When the activating ligand binds to the receptor (oxysterols for LXR, bile salts for FXR, secondary bile salts for PXR, and fatty acids and their derivatives for the PPARs), the complex is activated, and gene expression is altered. Unlike the cortisol receptor, these receptors reside in the nucleus and are activated once their ligands enter the nucleus and bind to them.





Signal transduction pathways, like a river, run in one direction. From a given point in a signal transduction pathway, events closer to the receptor are referred to as “upstream,” and events closer to the response are referred to as “downstream.”



Protein kinases transfer a phosphate group from ATP to the hydroxyl group of a specific amino acid residue in the protein. Tyrosine kinases transfer the phosphate group to the hydroxyl group of a specific tyrosine residue and serine/threonine protein kinases to the hydroxyl of a specific serine or threonine residue (serine is more often phosphorylated than threonine in target proteins). Different protein kinases have specificity for distinct amino acid sequences (containing a tyrosine, serine, or threonine). Thus, two different protein kinases target distinct sequences (and usually different proteins) for phosphorylation. A protein containing both target sequences could be a substrate for both protein kinases.

### III. PLASMA MEMBRANE RECEPTORS AND SIGNAL TRANSDUCTION

All plasma membrane receptors are proteins with certain features in common: an extracellular domain that binds the chemical messenger, one or more membrane-spanning domains that are  $\alpha$ -helices, and an intracellular domain that initiates signal transduction. As the ligand binds to the extracellular domain of its receptor, it causes a conformational change that is communicated to the intracellular domain through the rigid  $\alpha$ -helix of the transmembrane domain. The activated intracellular domain initiates a characteristic signal transduction pathway that usually involves the binding of a specific intracellular signal transduction protein.

The pathways of signal transduction for plasma membrane receptors have two major types of effects on the cell: (1) rapid and immediate effects on cellular ion levels or activation/inhibition of enzymes and/or (2) slower changes in the rate of gene expression for a specific set of proteins. Often, a signal transduction pathway will diverge to produce both kinds of effects.

#### A. Major Classes of Plasma Membrane Receptors

Individual plasma membrane receptors are grouped into the categories of ion channel receptors, receptors that are kinases or bind kinases, and receptors that work through second messengers. This classification is based on the receptor's general structure and means of signal transduction.

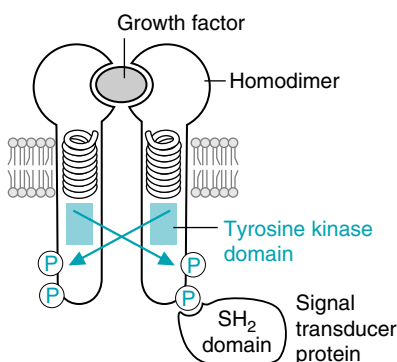
##### 1. ION CHANNEL RECEPTORS

The ion channel receptors are similar in structure to the nicotinic acetylcholine receptor (see Fig. 11.3). Signal transduction consists of the conformational change when ligand binds. Most small molecule neurotransmitters and some neuropeptides use ion channel receptors.

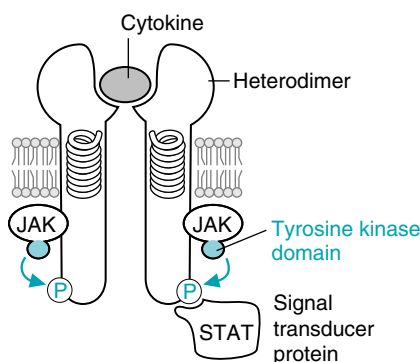
##### 2. RECEPTORS THAT ARE KINASES OR BIND KINASES

Several types of receptors that are kinases or bind kinases are illustrated in Figure 11.9. Their common feature is that the intracellular domain of the receptor (or an associated protein) is a kinase that is activated when the messenger binds to the extracellular domain. The receptor kinase phosphorylates an amino acid residue on the receptor (autophosphorylation) or an associated protein. The message is propagated through signal transducer proteins that bind to the activated messenger–receptor complex (e.g., Grb2, STAT, or Smad).

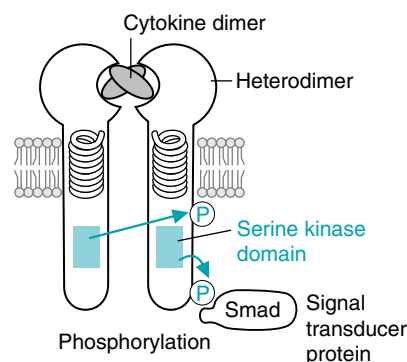
#### A. Tyrosine kinase receptor



#### B. Jak-Stat receptors



#### C. Serine/threonine kinase receptors



**Fig. 11.9.** Receptors that are kinases or bind kinases. The kinase domains are shown in blue, and the phosphorylation sites are indicated with blue arrows. A. Tyrosine kinase receptors. B. JAK-STAT receptors. C. Serine/threonine kinase receptors.

### 3. HEPTAHELICAL RECEPTORS

Heptahelical receptors (which contain 7-membrane spanning  $\alpha$ -helices) are the most common type of plasma membrane receptor. They work through second messengers, which are small nonprotein compounds, such as cAMP, generated inside the cell in response to messenger binding to the receptor (Fig. 11.10). They continue intracellular transmission of the message from the hormone/cytokine/neurotransmitter, which is the “first” messenger. Second messengers are present in low concentrations so that their concentration, and hence the message, can be rapidly initiated and terminated.

## B. Signal Transduction through Tyrosine Kinase Receptors

The tyrosine kinase receptors are summarized in Figure 11.9A. They generally exist in the membrane as monomers with a single membrane-spanning helix. One molecule of the growth factor generally binds two molecules of the receptor and promotes their dimerization (Fig. 11.11). Once the receptor dimer has formed, the intracellular tyrosine kinase domains of the receptor phosphorylate each other on certain tyrosine residues (autophosphorylation). The phosphotyrosine residues form specific binding sites for signal transducer proteins.

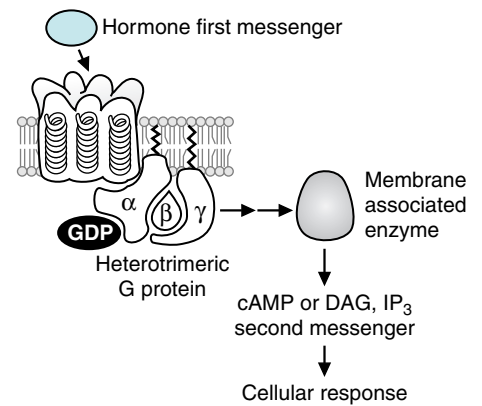
### 1. RAS AND THE MAP KINASE PATHWAY

One of the domains of the receptor containing a phosphotyrosine residue forms a binding site for intracellular proteins with a specific three-dimensional structure known as the SH2 domain (the Src homology 2 domain, named for the first protein in which it was found, the src protein of the Rous sarcoma virus). The adaptor



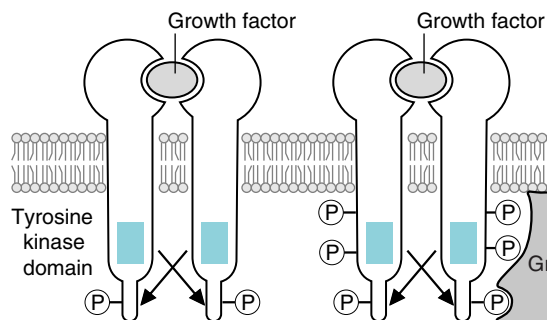
Although many different signal transducer proteins have SH2 domains, and many receptors have phosphotyrosine residues, each signal transducer protein is specific for one type of receptor. This specificity of binding results from the fact that each phosphotyrosine residue has a different amino acid sequence around it that forms the binding domain. Likewise, the SH2 domain of the transducer protein is only part of its binding domain.

### Heptahelical receptors



**Fig. 11.10.** Heptahelical Receptors and Second Messengers. The secreted chemical messenger (hormone, cytokine, or neurotransmitter) is the first messenger, which binds to a plasma membrane receptor such as the heptahelical receptors. The activated hormone–receptor complex activates a heterotrimeric G protein and via stimulation of membrane-bound enzymes, different G-proteins lead to generation of one or more intracellular second messengers, such as cAMP, diacylglycerol (DAG), or inositol trisphosphate ( $IP_3$ ).

#### 1. Growth factor binding and dimerization



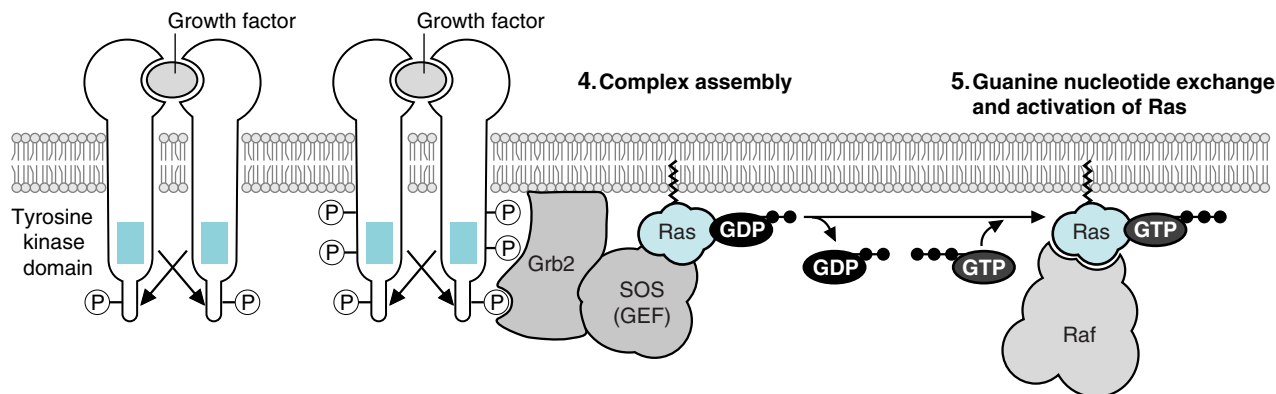
#### 2. Auto-crossphosphorylation

#### 3. Binding of adaptor proteins such as Grb2

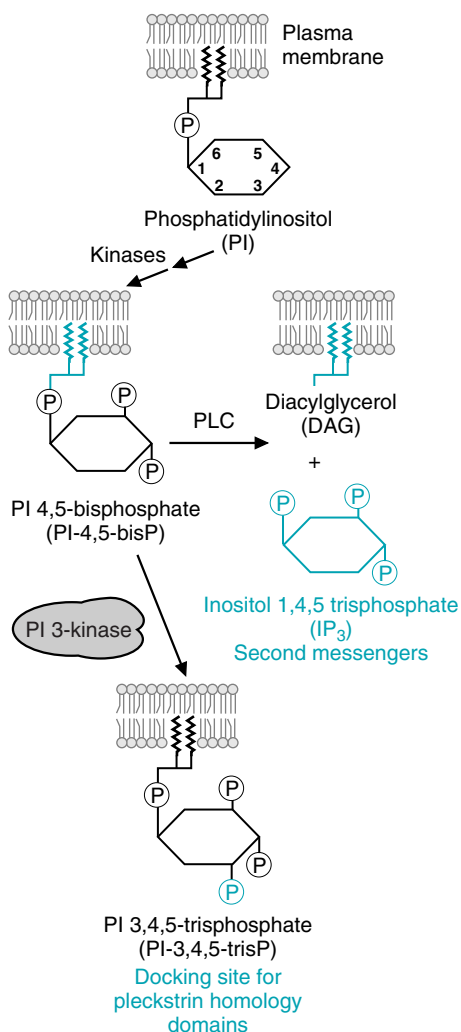
#### 4. Complex assembly

#### 5. Guanine nucleotide exchange and activation of Ras

#### 6. Ras binds raf and initiates MAP kinase pathway



**Fig. 11.11.** Signal transduction by tyrosine kinase receptors. (1) Binding and dimerization. (2) Autophosphorylation. (3) Binding of Grb2 and SOS. (4) SOS is a GEF (guanine nucleotide exchange protein) that binds Ras, a monomeric G protein anchored to the plasma membrane. (5) GEF activates the exchange of GTP for bound GDP on Ras. (6) Activated Ras containing GTP binds the target enzyme Raf, thereby activating it.



**Fig. 11.12.** Major route for generation of the phosphatidylinositol signal molecules, inositol 1,4,5-trisphosphate (IP<sub>3</sub>) and phosphatidylinositol 3,4,5 trisphosphate (PI-3,4,5-trisP). PI 3-kinase phosphorylates PI-4,5-bisP and PI-4P at the 3 position. Prime symbols are sometimes used in these names to denote the inositol ring. DAG is also a second messenger.

protein Grb2, which is bound to a membrane phosphoinositide, is one of the proteins with an SH2 domain that binds to phosphotyrosine residues on growth factor receptors. Binding to the receptor causes a conformational change in Grb2 that activates another binding site called an SH3 domain. These activated SH3 domains bind the protein SOS (SOS is an acronym for “son of sevenless,” a name unrelated to the function or structure of the compound). SOS is a guanine nucleotide exchange factor (GEF) for Ras, a monomeric G protein located in the plasma membrane (see Chapter 9, Section III.C.2.) SOS activates exchange of guanosine triphosphate (GTP) for guanosine diphosphate (GDP) on Ras, causing a conformational change in Ras that promotes binding of the protein Raf. Raf is a serine protein kinase that is also called MAPKKK (mitogen activated protein kinase kinase kinase.) Raf begins a sequence of successive phosphorylation steps called a phosphorylation cascade (When a kinase in a cascade is phosphorylated, it binds and phosphorylates the next enzyme in the cascade.). The MAP kinase cascade terminates at a gene transcription factor, thereby regulating transcription of certain genes involved in cell survival and proliferation.

Many tyrosine kinase receptors (as well as heptahelical receptors) also have additional signaling pathways involving phosphatidylinositol phosphates.

## 2. PHOSPHATIDYLINOSITOL PHOSPHATES IN SIGNAL TRANSDUCTION

Phosphatidylinositol phosphates serve two different functions in signal transduction: (1) Phosphatidylinositol 4',5' bisphosphate (PI-4,5-bisP) can be cleaved to generate the two intracellular second messengers, diacylglycerol (DAG) and inositol trisphosphate (IP<sub>3</sub>); and (2) Phosphatidylinositol 3',4',5' trisphosphate (PI-3,4,5-trisP) can serve as a plasma membrane docking site for signal transduction proteins.

Phosphatidyl inositol, which is present in the inner leaflet of the plasma membrane, is converted to PI-4,5-bisP by kinases that phosphorylate the inositol ring at the 4' and 5' positions (Fig. 11.12). PI-4,5-bisP, which has three phosphate groups, is cleaved by a phospholipase C-isozyme to generate IP<sub>3</sub> and DAG. The phospholipase isozyyme C<sub>γ</sub> (PLC<sub>γ</sub>) is activated by tyrosine kinase growth factor receptors, and phospholipase C<sub>β</sub> is activated by a heptahelical receptor-G protein signal transduction pathway.

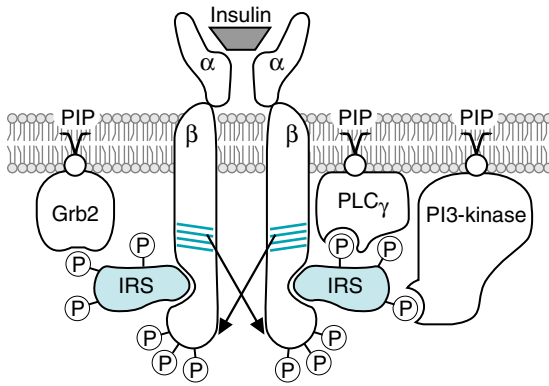
PI-4,5-bisP can also be phosphorylated at the 3' position of inositol by the enzyme phosphatidylinositol 3' kinase (PI 3-kinase) to form PI -3,4,5- trisP (see Fig. 11.12). PI-3,4,5- tris P (and PI -3,4 bis P) form membrane docking sites for proteins containing a certain sequence of amino acids called the pleckstrin homology (PH) domain. PI 3- kinase contains an SH2 domain and is activated by binding to a specific phosphotyrosine site on a tyrosine kinase receptor or receptor-associated protein.

## 3. THE INSULIN RECEPTOR

The insulin receptor, a member of the tyrosine kinase family of receptors, provides a good example of divergence in the pathway of signal transduction. Unlike other growth factor receptors, the insulin receptor exists in the membrane as a preformed dimer, with each half containing an α and a β subunit (Fig. 11.13). The β subunits



Insulin is a growth factor that is essential for cell viability and growth. It increases general protein synthesis, which strongly affects muscle mass. However, it also regulates immediate nutrient availability and storage, including glucose transport into skeletal muscle and glycogen synthesis. Thus, **Di Abietes** and other patients with type I diabetes mellitus who lack insulin rapidly develop hyperglycemia once insulin levels drop too low. They also exhibit muscle “wasting.” To mediate the diverse regulatory roles of insulin, the signal transduction pathway diverges after activation of the receptor and phosphorylation of IRS, which has multiple binding sites for different signal mediator proteins.



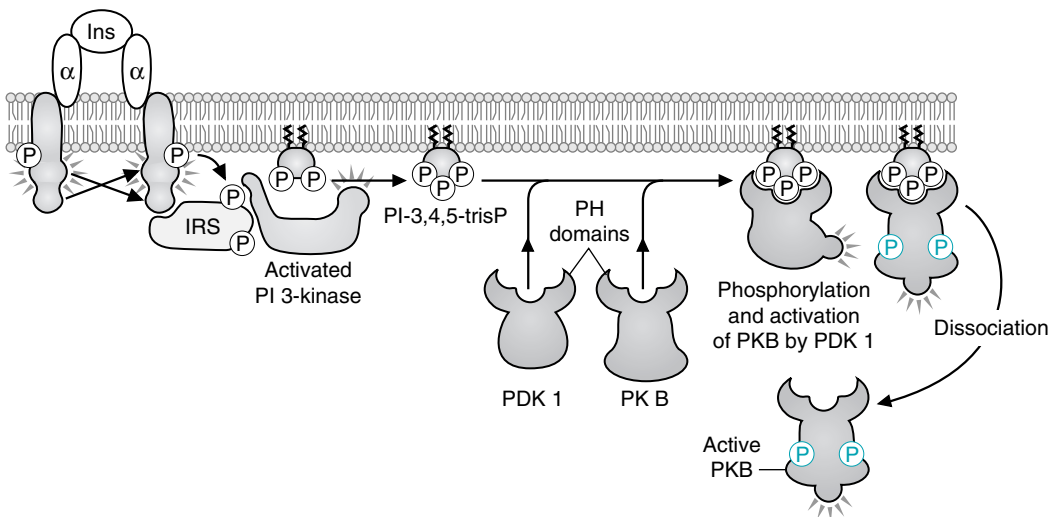
**Fig. 11.13.** Insulin receptor signaling. The insulin receptor is a dimer of two membrane-spanning  $\alpha$ - $\beta$  pairs. The tyrosine kinase domains are shown in blue, and arrows indicate auto-cross-phosphorylation. The activated receptor binds IRS molecules (insulin receptor substrates) and phosphorylates IRS at multiple sites, thereby forming binding sites for proteins with SH2 domains: Grb2, phospholipase C $\gamma$ (PLC $\gamma$ ), and PI 3-kinase. These proteins are associated with various phosphatidylinositol phosphates (all designated with PIP) in the plasma membrane.

autophosphorylate each other when insulin binds, thereby activating the receptor. The activated phosphorylated receptor binds a protein called IRS (insulin receptor substrate). The activated receptor kinase phosphorylates IRS at multiple sites, creating multiple binding sites for different proteins with SH2 domains. One of the sites binds Grb2, leading to activation of Ras and the MAP kinase pathway. Grb2 is anchored to PI-3,4,5-trisP in the plasma membrane through its PH (pleckstrin homology) domain. At another phosphotyrosine site, PI 3-kinase binds and is activated. At a third site, phospholipase C $\gamma$ (PLC $\gamma$ ) binds and is activated. The insulin receptor can also transmit signals through a direct docking with other signal transduction intermediates.

The signal pathway initiated by the insulin receptor complex involving PI 3-kinase leads to activation of protein kinase B, a serine-threonine kinase that mediates many of the downstream effects of insulin (Fig. 11.14). PI 3-kinase binds and phosphorylates PI-4,5-bis P in the membrane to form PI-3,4,5-trisP. Protein kinase



Protein kinase B is a serine-threonine kinase, also known as Akt. One of the signal transduction pathways from protein kinase B (Akt) leads to the effects of insulin on glucose metabolism. Other pathways, long associated with Akt, result in the phosphorylation of a host of other proteins that affect cell growth, cell cycle entry, and cell survival. In general, phosphorylation of these proteins by Akt inhibits their action and promotes cell survival.



**Fig. 11.14.** The insulin receptor–protein kinase B signaling pathway. Abbreviations: Ins, insulin; IRS, insulin receptor substrate; PH domains, pleckstrin homology domains; PDK1, phosphoinositide-dependent protein kinase 1; PKB, protein kinase B. The final phosphorylation step that activates PKB is shown in blue.

B and PDK1 (phosphoinositide-dependent kinase-1) are recruited to the membrane by their PH domains, where PDK1 phosphorylates and activates protein kinase B. Many other signal transducer proteins have PH domains and are docked at the membrane, where they can find and bind each other. Thus, the insulin signal diverges again and again. Insulin is covered in more detail in Chapters 26, 36 and 43.

### C. Signal Transduction by JAK-STAT Receptors

Tyrosine kinase-associated receptors called Jak-STAT receptors are often used by cytokines to regulate the proliferation of certain cells involved in the immune response (see Fig. 11.9B). The receptor itself has no intrinsic kinase activity but binds (associates with) the tyrosine kinase Jak (**janus kinase**). Their signal transducer proteins, called STATs (**signal transducer and activator of transcription**), are themselves gene-specific transcription factors. Thus, Jak-STAT receptors have a more direct route for propagation of the signal to the nucleus than tyrosine kinase receptors.

 Although Jak is an acronym for **janus kinase**, it has been suggested that it stands for “just another kinase”. It was named for Janus, a two-headed god of the Romans.

Each receptor monomer has an extracellular domain, a membrane-spanning region, and an intracellular domain. As the cytokine binds to these receptors, they form dimers (either homodimers or heterodimers, between two distinct receptor molecules) and may cluster (Fig. 11.15). The activated Jaks phosphorylate each other and intracellular tyrosine residues on the receptor, forming phosphotyrosine-binding sites for the SH2 domain of a STAT. STATs are inactive in the cytoplasm until they bind to the receptor complex, where they are also phosphorylated by the bound JAK. Phosphorylation changes the conformation of the STAT, causing it to dissociate from the receptor and dimerize with another phosphorylated STAT, thereby forming an activated transcription factor. The STAT dimer translocates to the nucleus and binds to a response element on DNA, thereby regulating gene transcription.

There are many different STAT proteins, each with a slightly different amino acid sequence. Receptors for different cytokines bind different STATs, which then form heterodimers in various combinations. This microheterogeneity allows different cytokines to target different genes.

### D. Receptor Serine/Threonine Kinases

Proteins in the transforming growth factor superfamily use receptors that have serine/threonine kinase activity and associate with proteins from the Smad family, which are gene-specific transcription factors (see Fig. 11.9C). This superfamily includes transforming growth factor  $\beta$  (TGF- $\beta$ ), a cytokine/hormone involved in tissue repair, immune regulation, and cell proliferation, and bone morphogenetic proteins (BMPs), which control proliferation, differentiation, and cell death during development.

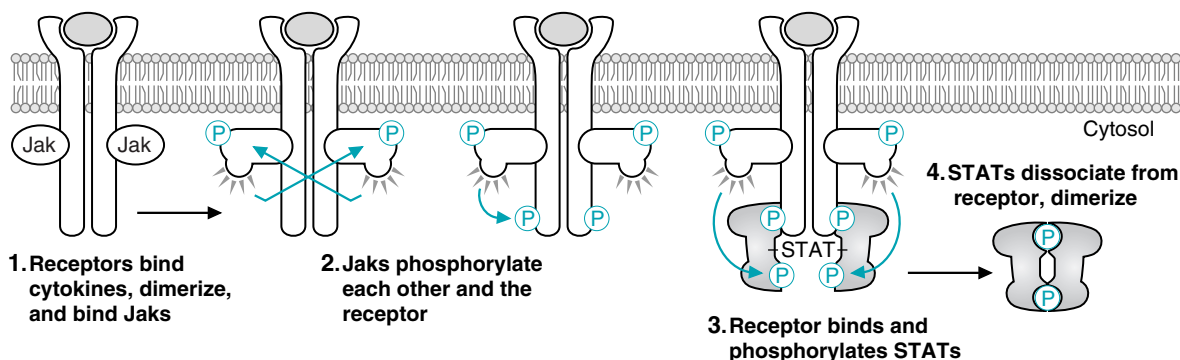
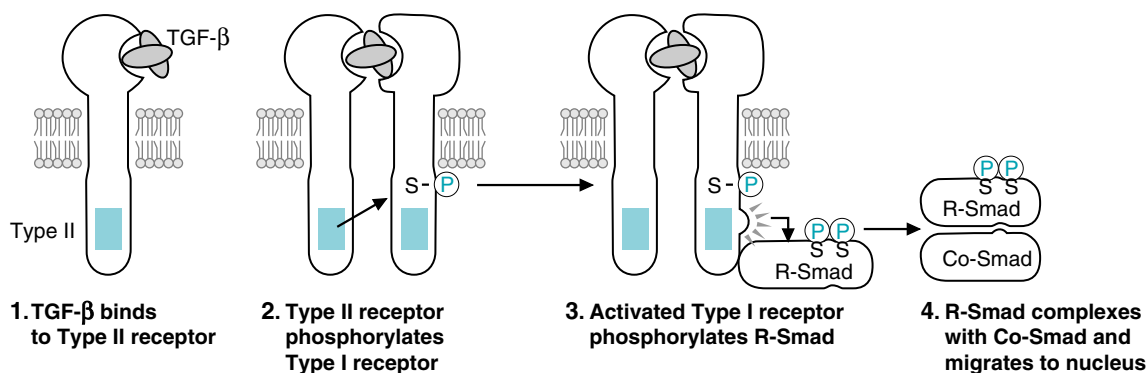


Fig. 11.15. Steps in Jak-STAT receptor signaling.





**Fig. 11.16.** Serine/threonine receptors and Smad proteins. TGF- $\beta$  (transforming growth factor  $\beta$ ), which is composed of two identical subunits, communicates through a receptor dimer of type I and type II subunits that have serine kinase domains. The type I receptor phosphorylates an R-Smad (receptor-specific Smad), which binds a Co-Smad (common Smad, also called Smad 4).

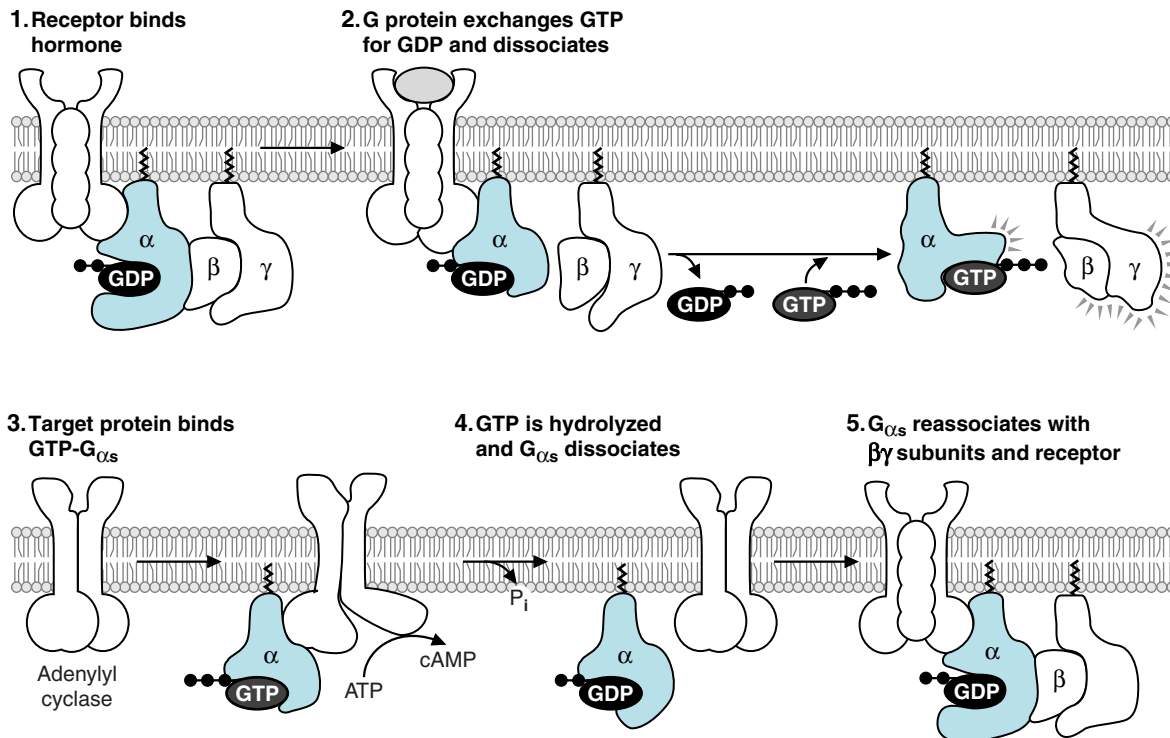
A simplified version of TGF- $\beta$ 1 binding to its receptor complex and activating Smads is illustrated in Fig. 11.16. The TGF- $\beta$  receptor complex is composed of two different single membrane-spanning receptor subunits (type I and type II), which have different functions even though they both have serine kinase domains. TGF- $\beta$  binds to a type II receptor. The activated type II receptor recruits a type I receptor, which it phosphorylates at a serine residue, forming an activated receptor complex. The type I receptor then binds a receptor-specific Smad protein (called R-Smads), which it phosphorylates at serine residues. The phosphorylated R-Smad undergoes a conformational change and dissociates from the receptor. It then forms a complex with another member of the Smad family, Smad 4 (Smad 4 is known as the common Smad, Co-Smad, and is not phosphorylated). The Smad complex, which may contain several Smads, translocates to the nucleus, where it activates or inhibits the transcription of target genes. Receptors for different ligands bind different Smads, which bind to different sites on DNA and regulate the transcription of different genes.

## E. Signal Transduction through Heptahelical Receptors

The heptahelical receptors are named for their 7-membrane spanning domains, which are  $\alpha$ -helices (see Fig. 11.10; see also Chapter 7, Fig. 7.10). Although hundreds of hormones and neurotransmitters work through heptahelical receptors, the extracellular binding domain of each receptor is specific for just one polypeptide hormone, catecholamine, or neurotransmitter (or its close structural analog). Heptahelical receptors have no intrinsic kinase activity but initiate signal transduction through heterotrimeric G proteins composed of  $\alpha$ ,  $\beta$  and  $\gamma$  subunits. However, different types of heptahelical receptors bind different G proteins, and different G proteins exert different effects on their target proteins.

### 1. HETEROTRIMERIC G PROTEINS

The function of heterotrimeric G proteins is illustrated in Figure 11.17 using a hormone that activates adenylyl cyclase (e.g., glucagon or epinephrine). While the  $\alpha$  subunit contains bound GDP, it remains associated with the  $\beta$  and  $\gamma$  subunits, either free in the membrane or bound to an unoccupied receptor (see Fig. 11.17, part 1). When the hormone binds, it causes a conformational change in the receptor that activates GDP dissociation and GTP binding. The exchange of GTP for bound GDP causes dissociation of the  $\alpha$  subunit from the receptor and from the  $\beta\gamma$  subunits (see Fig. 11.17, part 2). The  $\alpha$  and  $\beta$  subunits are tethered to the intracellular side of the



**Fig. 11.17.** Heptahelical receptors and heterotrimeric G proteins. (1) The intracellular domains of the receptor form a binding site for a G protein containing GDP bound to the  $\alpha$ -subunit. (2) Hormone binding to the receptor promotes the exchange of GTP for GDP. As a result, the complex disassembles, releasing the G protein  $\alpha$ -subunit from the  $\beta\gamma$  complex. (3) The  $G_{\alpha s}$  subunit binds to a target enzyme, thereby changing its activity. The  $\beta\gamma$  complex may simultaneously target another protein and change its activity. (4) Over time, bound GTP is hydrolysed to GDP, causing dissociation of the  $\alpha$ -subunit from Adenyl cyclase. The GDP- $\alpha$ -subunit reassociates with the  $\beta\gamma$  subunit and the receptor.



The importance of signal termination is illustrated by the “internal clock” of G proteins, which is the rate of spontaneous hydrolysis of GTP to GDP. Mutations in *ras* (the gene encoding Ras) that decrease the rate of GTP hydrolysis are found in about 20 to 30% of all human cancers, including approximately 25% of lung cancers, 50% of colon cancers, and more than 90% of pancreatic cancers. In these mutations of Ras, GTP hydrolysis is decreased and Ras remains locked in the active GTP-bound form, rather than alternating normally between inactive and active state in response to extracellular signals. Consequently, MAP kinase pathways are continuously stimulated and drive cell proliferation, even in the absence of growth factors that would be required for *ras* activation in normal cells.

plasma membrane through lipid anchors, but can still move around on the membrane surface. The GTP- $\alpha$  subunit binds its target enzyme in the membrane, thereby changing its activity. In this example, the  $\alpha$ -subunit binds and activates Adenyl cyclase, thereby increasing synthesis of cAMP (see Fig. 11.17, part 3).

With time, the  $G_{\alpha s}$  subunit inactivates itself by hydrolyzing its own bound GTP to GDP and  $P_i$ . This action is unrelated to the number of cAMP molecules formed. Like the monomeric G proteins, the GDP- $\alpha$  subunit then dissociates from its target protein, Adenyl cyclase (see Fig. 11.16, part 4). It reforms the trimeric G protein complex, which may return to bind the empty hormone receptor. As a result of this GTPase “internal clock,” sustained elevations of hormone levels are necessary for continued signal transduction and elevation of cAMP.

A large number of different heterotrimeric G protein complexes are generally categorized according to the activity of the  $\alpha$  subunit (Table 11.1). The 20 or more different isoforms of  $G_{\alpha}$  fall into four broad categories:  $G_{\alpha s}$ ,  $G_{\alpha i/o}$ ,  $G_{\alpha q/11}$ , and  $G_{\alpha 12/13}$ .  $G_{\alpha s}$  refers to  $\alpha$  subunits, which, like the one in Figure 11.17, stimulate Adenyl cyclase (hence the *s*).  $G_{\alpha}$  subunits that inhibit Adenyl cyclase are called  $G_{\alpha i}$ . The  $\beta\gamma$  subunits likewise exist as different isoforms, which also transmit messages.



Acetylcholine has two types of receptors: nicotinic ion channel receptors, the receptors inhibited by antibodies in myasthenia gravis, and muscarinic receptors, which exist as a variety of subtypes. The M2 muscarinic receptors activate a  $G_{\alpha i/o}$  heterotrimeric G protein in which release of the  $\beta\gamma$  subunit controls  $K^+$  channels and pacemaker activity in the heart. Epinephrine has several types and subtypes of heptahelical receptors:  $\beta$  receptors work through a  $G_{\alpha s}$  and stimulate Adenyl cyclase;  $\alpha_2$  receptors in other cells work through a  $G_{\alpha i}$  protein and inhibit Adenyl cyclase;  $\alpha_1$  receptors work through  $G_{\alpha q}$  subunits and activate phospholipase  $C\beta$ . This variety in receptor types allows a messenger to have different actions in different cells.

**Table 11.1 Subunits of Heterotrimeric G Proteins**

| G $\alpha$ subunit                                                                      | Action                                               | Some Physiologic Uses                                                                                                                                                                                                    |
|-----------------------------------------------------------------------------------------|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\alpha_s$ ; G $\alpha$ (s) *                                                           | stimulates adenylyl cyclase                          | Glucagon and epinephrine to regulate metabolic enzymes, regulatory polypeptide hormones to control steroid hormone and thyroid hormone synthesis, and by some neurotransmitters (e.g., dopamine) to control ion channels |
| $\alpha_{i/o}$ ; G $\alpha$ (i/o)<br>(signal also flows through $\beta\gamma$ subunits) | inhibits adenylyl cyclase                            | Epinephrine, many neurotransmitters including acetylcholine, dopamine, serotonin. Has a role in the transducin pathway, which mediates detection of light in the eye.                                                    |
| $\alpha_{q/11}$ ; G $\alpha$ (q/11)                                                     | activates phospholipase C $_{\beta}$                 | Epinephrine, acetylcholine, histamine, thyroid-stimulating hormone (TSH), interleukin 8, somatostatin, angiotensin                                                                                                       |
| $\alpha_{12/13}$ ; G $\alpha$ (12/13)                                                   | Physiologic connections are not yet well established | Thromboxane A <sub>2</sub> , lysophosphatidic acid                                                                                                                                                                       |

\*There is a growing tendency to designate the heterotrimeric G protein subunits without using subscripts so that they are actually visible to the naked eye.

## 2. ADENYLYL CYCLASE AND CAMP PHOSPHODIESTERASE

cAMP is referred to as a second messenger because changes in its concentration reflect changes in the concentration of the hormone (the first messenger). When a hormone binds and adenylyl cyclase is activated, it synthesizes cAMP from adenosine triphosphate (ATP). cAMP is hydrolyzed to AMP by cAMP phosphodiesterase, which also resides in the plasma membrane (Fig. 11.18). The concentration of cAMP and other second messengers is kept at very low levels in cells by balancing the activity of these two enzymes so that cAMP levels can change rapidly when hormone levels change. Some hormones change the concentration of cAMP by targeting the phosphodiesterase enzyme rather than adenylyl cyclase. For example, insulin lowers cAMP levels by causing phosphodiesterase activation.

cAMP exerts diverse effects in cells. It is an allosteric activator of protein kinase A (see Chapter 9, section III.B.3), which is a serine/threonine protein kinase that phosphorylates a large number of metabolic enzymes, thereby providing a rapid response to hormones such as glucagon and epinephrine. The catalytic subunits of protein kinase A also enter the nucleus and phosphorylate a gene-specific transcription factor called CREB (cyclic AMP response element-binding protein). Thus, cAMP also activates a slower response pathway, gene transcription. In other cell types, cAMP directly activates ligand-gated channels.

## 3. PHOSPHATIDYLINOSITOL SIGNALING BY HEPTAHELICAL RECEPTORS

Certain heptahelical receptors bind the q isoform of the G $\alpha$  subunit (G $\alpha_q$ ), which activates the target enzyme phospholipase C $_{\beta}$  (see Fig. 11.12). When activated, phospholipase C $_{\beta}$  hydrolyzes the membrane lipid phosphatidyl inositol bis phosphate (PI-4,5-bisP) into two second messengers, diacylglycerol (DAG) and 1,4,5-inositol trisphosphate (IP<sub>3</sub>). IP<sub>3</sub> has a binding site in the sarcoplasmic reticulum and the endoplasmic reticulum that stimulates the release of Ca<sup>2+</sup> (Fig. 11.19). Ca<sup>2+</sup> activates enzymes containing the calcium-calmodulin subunit, including a protein kinase. Diacylglycerol, which remains in the membrane, activates protein kinase C, which then propagates the response by phosphorylating target proteins.

## F. Changes in Response to Signals

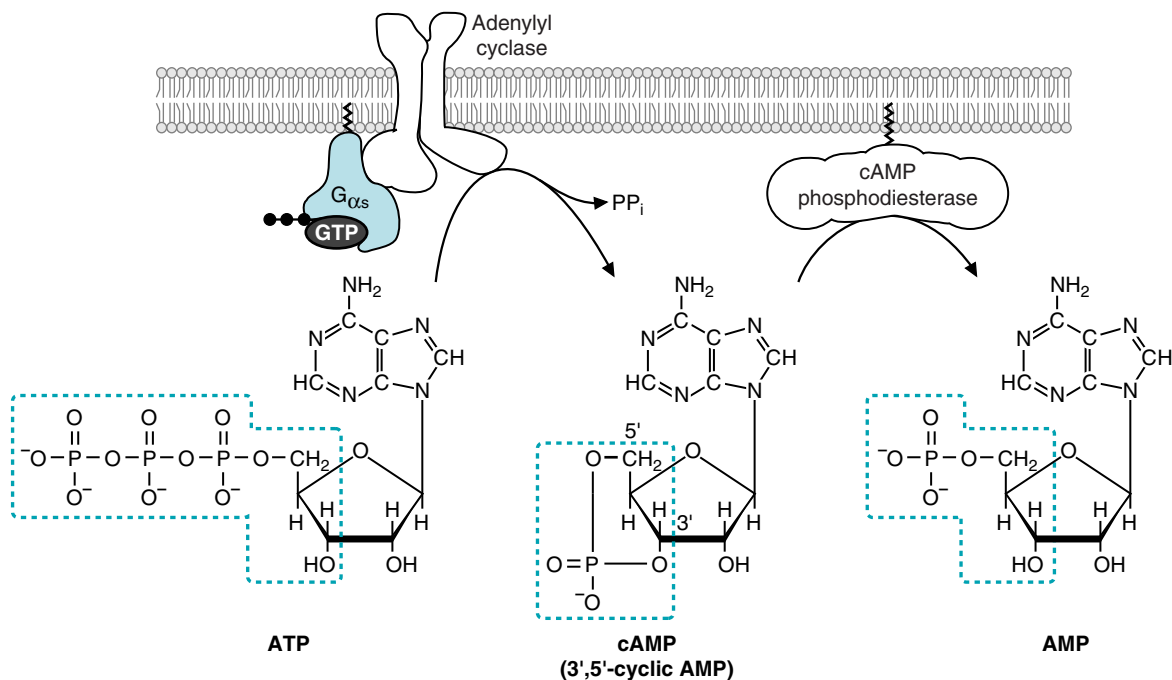
Tissues vary in their ability to respond to a message through changes in receptor activity or number. Many receptors contain intracellular phosphorylation sites that alter their ability to transmit signals. Receptor number is also varied through



**Dennis Veere** was hospitalized for dehydration resulting from cholera toxin (see Chapter 10). Cholera A toxin was absorbed into the intestinal mucosal cells, where it was processed and complexed with Arf (ADP-ribosylation factor), a small G protein normally involved in vesicular transport. Cholera A toxin is an NAD-glycohydrolase, which cleaves NAD and transfers the ADP ribose portion to other proteins. It ADP-ribosylates the G $\alpha_s$  subunit of heterotrimeric G proteins, thereby inhibiting their GTPase activity. As a consequence, they remain actively bound to adenylyl cyclase, resulting in increased production of cAMP. The CFTR channel is activated, resulting in secretion of chloride ion and Na<sup>+</sup> ion into the intestinal lumen. The ion secretion is followed by loss of water, resulting in vomiting and watery diarrhea.



Some signaling pathways cross from the MAP kinase pathway to phosphorylate CREB, and all heterotrimeric G protein pathways diverge to include a route to the MAP kinase pathway. These types of complex interconnections in signaling pathways are sometimes called hormone cross-talk.



**Fig. 11.18.** Formation and cleavage of the cyclic phosphodiester bond in cAMP. When activated by G $\alpha_s$ , adenyl cyclase converts ATP to 3',5'-cyclic AMP + PP $_i$ . cAMP phosphodiesterase hydrolyzes cAMP to AMP.



In myasthenia gravis, increased endocytosis and degradation of acetylcholine receptors lead to a signal transduction pathway that decreases synthesis of new receptors. Thus, downregulation of acetylcholine receptors is part of this disease.

down-regulation. After a hormone binds to the receptor, the hormone–receptor complex may be taken into the cell by the process of endocytosis in clathrin-coated pits (see Chapter 10, Section III.B.1.) The receptors may be degraded or recycled back to the cell surface. This internalization of receptors decreases the number available on the surface under conditions of constant high hormone levels when more of the receptors are occupied by hormones and results in decreased synthesis of new receptors. Hence, it is called down-regulation.

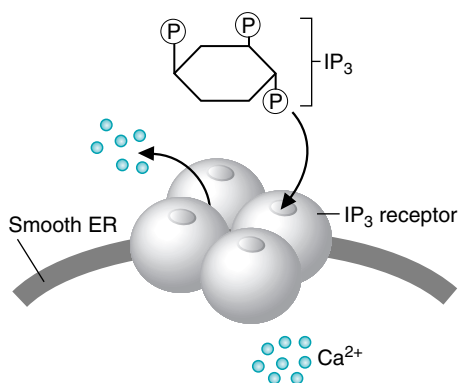
#### IV. SIGNAL TERMINATION

Some signals, such as those that modify the metabolic responses of cells or transmit neural impulses, need to turn off rapidly when the hormone is no longer being produced. Other signals, such as those that stimulate proliferation, turn off more slowly. In contrast, signals regulating differentiation may persist throughout our lifetime. Many chronic diseases are caused by failure to terminate a response at the appropriate time.

The first level of termination is the chemical messenger itself (Fig.11.20). When the stimulus is no longer applied to the secreting cell, the messenger is no longer secreted, and existing messenger is catabolized. For example, many polypeptide hormones such as insulin are taken up into the liver and degraded. Termination of the acetylcholine signal by acetylcholinesterase has already been mentioned.

Within each pathway of signal transduction, the signal may be turned off at specific steps. The receptor might be desensitized to the messenger by phosphorylation. G proteins, both monomeric and heterotrimeric, automatically terminate messages as they hydrolyze GTP. Termination also can be achieved through degradation of the second messenger (e.g., phosphodiesterase cleavage of cAMP). Each of these terminating processes is also highly regulated.

Another important pathway for reversing the message is through protein phosphatases, enzymes that reverse the action of kinases by removing phosphate groups



**Fig. 11.19.** IP $_3$  signaling calcium release from the endoplasmic reticulum.

from proteins. Specific tyrosine or serine/threonine phosphatases (enzymes that remove the phosphate group from specific proteins) exist for all of the sites phosphorylated by signal transduction kinases. Some receptors are even protein phosphatases.

### CLINICAL COMMENTS



**Mya Sthenia.** Mya Sthenia has myasthenia gravis, an autoimmune disease caused by the production of antibodies directed against the nicotinic acetylcholine receptor in skeletal muscles. The diagnosis is made by history (presence of typical muscular symptoms), physical examination (presence of inability to do specific repetitive muscular activity over time), and tests such as the inhibition of acetylcholinesterase activity. The diagnosis can be further confirmed with an electromyogram (EMG) showing a partial blockade of ion flux across muscular membranes and a diagnostic procedure involving repetitive electrical nerve stimulation.



**Ann O'Rexia.** Anorexia nervosa presents as a distorted visual self-image often associated with compulsive exercise. Although Ann has been gaining weight, she is still relatively low on stored fuels needed to sustain the metabolic requirements of exercise. Her prolonged starvation has resulted in release of the steroid hormone cortisol and the polypeptide hormone glucagon, whereas levels of the polypeptide hormone insulin have decreased. Cortisol activates transcription of genes for some of the enzymes of gluconeogenesis (the synthesis of glucose from amino acids and other precursors; see Chapter 3.) Glucagon binds to heptahelical receptors in liver and adipose tissue and, working through cAMP and protein kinase A, activates many enzymes involved in fasting fuel metabolism. Insulin, which is released when she drinks her high-energy supplement, works through a specialized tyrosine kinase receptor to promote fuel storage. Epinephrine, a catecholamine released when she exercises, promotes fuel mobilization.

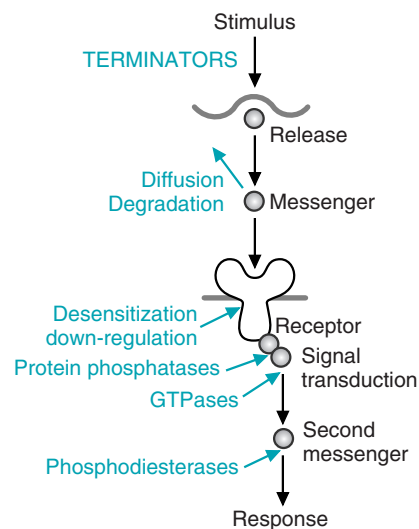


**Dennis Veere.** In the emergency room, Dennis received intravenous rehydration therapy (normal saline [0.9% NaCl]) and oral hydration therapy with a glucose-electrolyte solution to increase his glucose-dependent  $\text{Na}^+$  uptake from the intestinal lumen (see Chapter 10). Dennis quickly recovered from his bout of cholera. Cholera is self-limiting, possibly because the bacteria remain in the intestine, where they are washed out of the system by the diffuse watery diarrhea. Over the past three years, **Percy Veere** has persevered through the death of his wife and the subsequent calamities of his grandson Dennis “the Menace” Veere, including salicylate poisoning, suspected malathion poisoning, and now cholera. Mr. Veere decided to send his grandson home for the remainder of the summer.

### BIOCHEMICAL COMMENTS

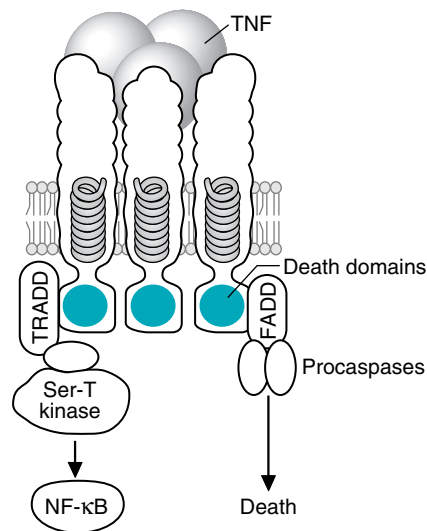


**Death domain receptors.** The cytokine TNF (tumor necrosis factor) uses a type of receptor called the death domain receptor (Fig. 11.21). These receptors function as a trimer when they bind TNF (which is also a trimer). On TNF binding, an inhibitory protein called the “silencer of death” is released from the receptor. The receptor then binds and activates several adaptor proteins. One adaptor protein, FADD (fas-associated death domain), recruits and activates the zymogen form of a proteolytic enzyme called caspase. Caspases



**Fig. 11.20.** Sites of signal termination. Processes that terminate signals are shown in blue.

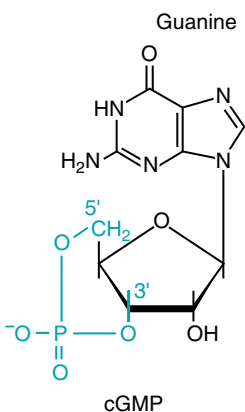




**Fig. 11.21.** Death domain receptors. The portion of the receptor shown in blue is called the death domain because it binds adaptor proteins that initiate different signaling pathways leading to cell death. The adaptor protein FADD forms a scaffold on which proteolytic procaspases cleave each other, thereby initiating a death pathway. The adaptor protein TRADD binds a protein that binds a serine-threonine kinase (Ser-T kinase) that initiates another signaling pathway leading to activation of the transcription factor NF- $\kappa$ B.



Caspases are proteolytic enzymes that have a critical role in programmed cell death (also called apoptosis) (see Chapter 16). Caspases are present as latent zymogens until their auto-proteolysis (self-cleavage) is activated by “death signals” to the receptor complex. Once activated, they work systematically to dismantle a cell through degrading a wide variety of proteins, such as DNA repair enzymes and cellular structural proteins.



**Fig. 11.22.** Structure of 3',5'-cyclic GMP (cGMP).

initiate a signal transduction pathway leading to apoptosis (programmed cell death) (see Chapter 18). Another adaptor protein, TRADD (TNF receptor-associated death domain), initiates signaling pathways that lead to activation of the gene-specific transcription factors Jun and NF- $\kappa$ B (nuclear factor- $\kappa$ B). Through these pathways, TNF mediates cell-specific responses, such as cell growth and death, the inflammatory response, and immune function.



**Guanylyl cyclase receptors.** Guanylyl cyclase receptors convert GTP to the second messenger 3',5' cyclic GMP (cGMP), which is analogous to cAMP (Fig. 11.22). Like cAMP, cGMP is degraded by a membrane-bound phosphodiesterase. Elevated cGMP activates protein kinase G, which then phosphorylates target proteins to propagate the response.

One type of guanylyl cyclase exists in the cytoplasm and is a receptor for nitric oxide (NO), a neurotransmitter/neurohormone. NO is a lipophilic gas that is able to diffuse into the cell. This receptor thus is an exception to the rule that intracellular receptors are gene transcription factors. The other type of guanylyl cyclase receptor is a membrane-spanning receptor in the plasma membrane with an external binding domain for a signal molecule (e.g., natriuretic peptide).

### Suggested References

Signaling Pathways. The May 31, 2002 issue of *Science*, Vol 296, provides a 2–3-page synopsis of many signaling pathways. The articles most relevant to the pathways discussed in this chapter are:

- Aaronson DA, Horvath CM. A road map for those who don't know JAK-STAT. *Science* 2002;296:1653–1655.
- Cantley LC. The phosphoinositide 3-kinase pathway. *Science* 2002;296:1655–1657.
- Attisano L, Wrana, JL. Signal transduction by the TGF- $\beta$  superfamily. *Science* 2002;296:1646–1647.
- Neves SR, Ram PT, Iyengar R. G protein pathways. *Science* 2002;296:1636–1639.
- Chen Q, Goeddel DV. TNF-R1 signaling: A beautiful pathway. *Science* 2002;296:1634–1635.



## REVIEW QUESTIONS—CHAPTER 11

1. Which of the following is a general characteristic of all chemical messengers?
  - (A) They are secreted by one cell, enter the blood, and act on a distant target cell.
  - (B) To achieve a coordinated response, each messenger is secreted by several types of cells.
  - (C) Each messenger binds to a specific protein receptor in a target cell.
  - (D) Chemical messengers must enter cells to transmit their message.
  - (E) Chemical messengers are metabolized to intracellular second messengers to transmit their message.
2. Which of the following is a characteristic of chemical messengers that bind to intracellular transcription factor receptors?
  - (A) They are usually cytokines or polypeptide hormones.
  - (B) They are usually small molecule neurotransmitters.
  - (C) They exert rapid actions in cells.
  - (D) They are transported through the blood bound to proteins.
  - (E) They are always present in high concentrations in the blood.

Use the following case history for questions 3 and 4. To answer this question, you do not need to know more about parathyroid hormone or pseudohypoparathyroidism than the information given.

Pseudohypoparathyroidism is a heritable disorder caused by target organ unresponsiveness to parathyroid hormone (a polypeptide hormone secreted by the parathyroid gland). One of the mutations causing this disease occurs in the gene encoding  $G_s\alpha$  in certain cells.

3. The receptor for parathyroid hormone is most likely
  - (A) an intracellular transcription factor.
  - (B) a cytoplasmic guanylyl cyclase.
  - (C) a receptor that must be endocytosed in clathrin-coated pits to transmit its signal.
  - (D) a heptahelical receptor.
  - (E) a tyrosine kinase receptor.
4. This mutation most likely
  - (A) is a gain-of-function mutation.
  - (B) decreases the GTPase activity of the  $G_{\alpha s}$  subunit.
  - (C) decreases synthesis of cAMP in response to parathyroid hormone.
  - (D) decreases generation of  $IP_3$  in response to parathyroid hormone.
  - (E) decreases synthesis of phosphatidylinositol 3,4,5-trisphosphate in response to parathyroid hormone.
5. SH2 domains on proteins are specific for which of the following sites?
  - (A) Certain sequences of amino acids containing a phosphotyrosine residue
  - (B) PI-3,4,5 trisphosphate in the membrane
  - (C) GTP-activated Ras
  - (D)  $Ca^{2+}$ -calmodulin
  - (E) Receptor domains containing phosphoserine residues



# Gene Expression and the Synthesis of Proteins

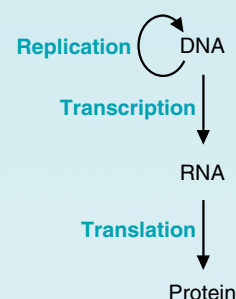
In the middle of the 20th century, DNA was identified as the genetic material, and its structure was determined. Using this knowledge, researchers then discovered the mechanisms by which genetic information is inherited and expressed. During the last quarter of the 20th century, our understanding of this critical area of science, known as molecular biology, grew at an increasingly rapid pace. We now have techniques to probe the human genome that will completely revolutionize the way medicine is practiced in the 21st century.

The genome of a cell consists of all its genetic information, encoded in DNA (deoxyribonucleic acid). In eukaryotes, DNA is located mainly in nuclei, but small amounts are also found in mitochondria. Nuclear genes are packaged in chromosomes that contain DNA and protein in tightly coiled structures (Chapter 12).

The molecular mechanism of inheritance involves a process known as replication, in which the strands of parental DNA serve as templates for the synthesis of DNA copies (Fig. 1) (Chapter 13). After DNA replication, cells divide, and these DNA copies are passed to daughter cells. Alterations in genetic material occur by recombination (the exchange of genetic material between chromosomes) and by mutation (the result of chemical changes that alter DNA). DNA repair mechanisms correct much of this damage, but, nevertheless, many gene alterations are passed to daughter cells.

The expression of genes within cells requires two processes: transcription and translation (see Fig. 1) (Chapters 14 and 15). DNA is transcribed to produce ribonucleic acid (RNA). Three major types of RNA are transcribed from DNA and subsequently participate in the process of translation (The synthesis of proteins). Messenger RNA (mRNA) carries the genetic information from the nucleus to the cytoplasm, where translation occurs on ribosomes, structures containing proteins complexed with ribosomal RNA (rRNA). Transfer RNA (tRNA) carries individual amino acids to the ribosomes, where they are joined in peptide linkage to form proteins. During translation, the sequence of nucleic acid bases in mRNA is read in sets of three (each set of three bases constitutes a codon). The sequence of codons in the mRNA dictates the sequence of amino acids in the protein. Proteins function in cell structure, signaling, and catalysis and, therefore, determine the appearance and behavior of cells and the organism as a whole. The regulation of gene expression (Chapter 16) determines which proteins are synthesized and the amount synthesized at any time, thus allowing cells to undergo development and differentiation and to respond to changing environmental conditions.

Research in molecular biology has produced a host of techniques, known collectively as recombinant DNA technology, biotechnology, or genetic engineering, that can be used for the diagnosis and treatment of disease (Chapter 17). These techniques can detect a number of genetic diseases that previously could only be diagnosed after symptoms appeared. Diagnosis of these diseases can now be made with considerable accuracy even before birth, and carriers of these diseases also can be identified.



**Fig. 1.** Replication, transcription, and translation. Replication: DNA serves as a template for producing DNA copies. Transcription: DNA serves as a template for the synthesis of RNA. Translation: RNA provides the information for the synthesis of proteins.



Many drugs used in medicine to treat bacterial infections are targeted to interfere with their ability to synthesize RNA and proteins. Thus, medical students need to know the basics of bacterial DNA replication, RNA synthesis, and protein synthesis.



Ethical dilemmas have come along with technological advances in molecular biology. Consider the case of a patient with a mild case of ornithine transcarbamoylase deficiency, a urea cycle defect that, if untreated, leads to elevated ammonia levels and nervous system dysfunction. The patient was being effectively treated by dietary restriction of protein. However, in 1999, he was treated with a common virus carrying the normal gene for ornithine transcarbamoylase. The patient developed a severe immune response to the virus and died as a result of the treatment. This case history raises the issues of appropriate patient consent, appropriate criteria to be included in this type of study, and the types of diseases for which gene therapy is appropriate. These are issues that you, the student, will be facing as you enter your practice of medicine.



Tumors may be benign or malignant. A tumor is malignant if it invades locally or if cells break away from the tumor and travel to other parts of the body, where they establish new growths (a process called metastasis), resulting in destruction of the tissues they invade. Many of the drugs used to treat malignant tumors are directed toward inhibition of DNA replication. These chemotherapeutic drugs are more toxic to cancer cells than normal cells, because the cancer cells divide more rapidly. However, such drugs also may inhibit normal rapidly dividing cells, such as the cells of the bone marrow (causing a decrease in white blood cell count) or cells in the hair follicles (resulting in hair loss during chemotherapy).

With recent developments in the field of gene therapy, diseases that for centuries have been considered hopeless are now potentially curable. Much of the therapy for these diseases is currently experimental. However, during the 21st century, physicians may be using genetic engineering techniques routinely for both the diagnosis and treatment of their patients.

Replication and cell division are highly regulated processes in the human. Cancer is a group of diseases in which a cell in the body has been transformed and begins to grow and divide out of control (Chapter 18). It results from multiple mutations or changes in DNA structure in the genes that activate cell growth, called proto-oncogenes, and those that ensure that DNA replication and repair are normal, called growth suppressor or tumor suppressor genes. Mutations that activate proto-oncogenes to oncogenes disturb the regulation of the cell cycle and the rate of cell proliferation. Mutations disrupting tumor suppressor genes lead to an increased incidence of these proto-oncogene-activating mutations. Such mutations may be inherited, causing a predisposition to a type of cancer. They also may arise from DNA replication or copying errors that remain uncorrected, from chemicals or radiation that damages DNA, from translocation of pieces of chromosomes from one chromosome to another during replication, or from incorporation of viral encoded DNA into the genome.



## 12 Structure of the Nucleic Acids

**Nucleotides in DNA and RNA.** *Nucleotides* are the monomeric units of the nucleic acids, DNA (deoxyribonucleic acid) and RNA (ribonucleic acid). Each nucleotide consists of a heterocyclic **nitrogenous base**, a **sugar**, and **phosphate**. DNA contains the purine bases **adenine** (A) and **guanine** (G) and the pyrimidine bases **cytosine** (C) and **thymine** (T). RNA contains A, G, and C, but it has **uracil** (U) instead of thymine. In DNA, the sugar is **deoxyribose**, whereas in RNA it is **ribose**.

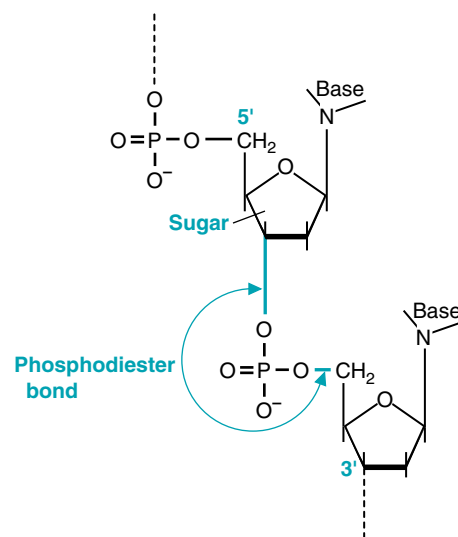
**Polynucleotides** such as DNA and RNA are linear sequences of nucleotides linked by **3'-to 5'-phosphodiester bonds** between the sugars (Fig. 12.1). The bases of the nucleotides can interact with other bases or with proteins.

**DNA Structure.** Genetic information is encoded by the sequence of different nucleotide bases in DNA. DNA is **double-stranded**; it contains **two antiparallel polynucleotide strands**. The two strands are joined by hydrogen bonding between their bases to form **base-pairs**. **Adenine** pairs with **thymine**, and **guanine** pairs with **cytosine**. The two DNA strands run in **opposite directions**. One strand runs 5' to 3', and the other strand runs 3' to 5'. The two DNA strands wind around each other, forming a **double helix**.

Transcription of a gene generates a **single-stranded RNA** that is identical in nucleotide sequence to one of the strands of the duplex DNA. The three major types of RNA are **messenger RNA** (mRNA), **ribosomal RNA** (rRNA), and **transfer RNA** (tRNA).

**RNA Structures.** mRNAs contain the nucleotide sequence that is converted into the amino acid sequence of a protein in the process of translation. Eukaryotic mRNA has a structure known as a **cap** at the 5'-end, a sequence of adenine nucleotides (a **poly(A) tail**) at the 3'-end, and a **coding region** in the center containing **codons** that dictate the sequence of amino acids in a protein or relay a signal. Each codon in the genetic code is a different sequence of three nucleotides.

rRNAs and tRNAs are part of the apparatus for protein synthesis, but do not encode proteins. rRNA has **extensive internal base pairing** and complexes with proteins to form **ribonucleoprotein particles** called **ribosomes**. The ribosomes bind mRNA and tRNAs during translation. Each tRNA binds and **activates a specific amino acid** for insertion into the polypeptide chain and therefore has a somewhat different nucleotide sequence than other tRNAs. A unique trinucleotide sequence on each tRNA called an **anticodon** binds to a complementary codon on the mRNA, thereby ensuring insertion of the correct amino acid. In spite of their differences, all tRNAs contain a number of unusual nucleotides and assume a similar **cloverleaf** structure.



**Fig. 12.1.** Structure of a polynucleotide. The 5'-carbon of the top sugar and the 3'-carbon of the bottom sugar are indicated.



## THE WAITING ROOM



**Ivy Sharer** is a 26-year-old intravenous (IV) drug abuser who admitted to sharing unsterile needles with another addict for several years. Five months before presenting to the hospital emergency department with soaking night sweats, she experienced a 3-week course of a flu-like syndrome with fever, malaise, and muscle aches. Four months ago, she noted generalized lymph node enlargement associated with chills, anorexia, and diarrhea, which led to a 22-lb weight loss. Tests were positive for human immunodeficiency virus (HIV). Because her symptoms indicated that she now had the acquired immunodeficiency syndrome (AIDS), a multidrug regimen including zidovudine (ZDV), formerly called azidothymidine (AZT), was initiated.



An adenoma is a mass of rapidly proliferating cells, called a neoplasm (neo = new; plasm = growth), that is formed from epithelial cells growing into a glandlike structure. The cells lining all the external and internal organs are epithelial cells, and most human tumors are adenocarcinomas. Adenomatous polyps are adenomas that grow into the lumen of the colon or rectum. The term *malignant* applied to a neoplasm refers to invasive unregulated growth.

**Colin Tuma** has an adenocarcinoma, which is a malignant adenoma that has started to grow through the wall of the colon into surrounding tissues. Cells from adenocarcinomas can break away and spread through the blood or lymph to other parts of the body, where they form “colony” tumors. This process is called metastasis.



**Colin Tuma** had intestinal polyps at age 45, which were removed via a colonoscope. However, he did not return for annual colonoscopic examinations as instructed. At age 56, he reappeared, complaining of tar-colored stools (melena), which are caused by intestinal bleeding. The source of the blood loss was an adenocarcinoma growing from a colonic polyp of the large intestine. At surgery, it was found that the tumor had invaded the gut wall and perforated the visceral peritoneum. Several pericolic lymph nodes contained cancer cells, and several small nodules of metastatic cancer were found in the liver. After resection of the tumor, the oncologist began treatment with 5-fluorouracil (5-FU) combined with other chemotherapeutic agents.



**Agneu (“neu”) Moania** complains to his physician of a fever and cough. His cough produces thick yellow-brown sputum. A stain of his sputum shows many Gram-positive, bullet-shaped diplococci. A sputum culture confirms that he has pneumonia, a respiratory infection caused by *Streptococcus pneumoniae*, which is sensitive to penicillin, erythromycin, tetracycline, and other antibiotics. Because of a history of penicillin allergy, he is started on oral erythromycin therapy.



DNA is a double-stranded molecule that forms base pairs (bp) between strands. The bp designation is often used to indicate the size of a DNA molecule. For example, in a stretch of DNA 200 bp long, both strands are included with 200 bases in each strand, for a total of 400 bases.

## I. DNA STRUCTURE

### A. Location of DNA

DNA and RNA serve as the genetic material for prokaryotic and eukaryotic cells, for viruses, and for plasmids, each of which stores it in a different arrangement or location. In prokaryotes, DNA is not separated from the rest of the cellular contents. In eukaryotes, however, DNA is located in the nucleus, where it is separated from the rest of the cell by the nuclear envelope (see Fig. 10.20). Eukaryotic DNA is bound to proteins, forming a complex called chromatin. During interphase (when cells are not dividing), some of the chromatin is diffuse (euchromatin) and some is dense (heterochromatin), but no distinct structures can be observed. However, before mitosis (when cells divide), the DNA is replicated, resulting in two identical chromosomes called sister chromatids. During metaphase (a period in mitosis), these condense into discrete, visible chromosomes.

Less than 0.1% of the total DNA in a cell is present in mitochondria. The genetic information in a mitochondrion is encoded in less than 20,000 base pairs of DNA; the information in a human haploid nucleus (i.e., an egg or a sperm cell) is encoded in approximately  $3 \times 10^9$  (3 billion) base pairs. The DNA and protein synthesizing systems in mitochondria more closely resemble the systems in bacteria, which do

not have membrane-enclosed organelles, than those in the eukaryotic nucleus and cytoplasm. It has been suggested that mitochondria were derived from ancient bacterial invaders of primordial eukaryotic cells.

Viruses are small infectious particles consisting of a DNA or RNA genome (but not both), proteins required for pathogenesis or replication, and a protein coat. They lack, however, complete systems for replication, transcription, and translation and, consequently, viruses must invade other cells and commandeer their DNA, RNA, and protein-synthesizing machinery to reproduce. Both eukaryotes and prokaryotes can be infected by viruses. Viruses that infect bacteria are known as bacteriophage (or more simply as phage).

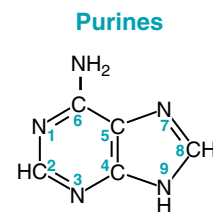
Plasmids are small, circular DNA molecules that can enter bacteria and replicate autonomously, that is, outside the host genome. In contrast to viruses, plasmids are not infectious; they do not convert their host cells into factories devoted to plasmid production. Genetic engineers use plasmids as tools for transfer of foreign genes into bacteria because segments of DNA can readily be incorporated into plasmids.

## B. Determination of the Structure of DNA

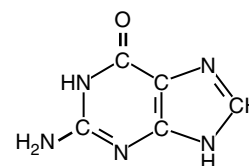
In 1865, Frederick Meischer first isolated DNA, obtaining it from pus scraped from surgical bandages. Initially, scientists speculated that DNA was a cellular storage form for inorganic phosphate, an important but unexciting function that did not spark widespread interest in determining its structure. In fact, the details of DNA structure were not fully determined until 1953, almost 90 years after it had first been isolated, but only 9 years after it had been identified as the genetic material.

Early in the 20th century, the bases of DNA were identified as the purines adenine (A) and guanine (G), and the pyrimidines cytosine (C) and thymine (T) (Fig. 12.2). The sugar was found to be deoxyribose, a derivative of ribose, lacking a hydroxyl group on carbon 2 (Fig. 12.3).

Nucleotides, composed of a base, a sugar, and phosphate, were found to be the monomeric units of the nucleic acids (Table 12.1). In nucleosides, the nitrogenous base is linked by an *N*-glycosidic bond to the anomeric carbon of the sugar, either ribose or deoxyribose. A nucleotide is a nucleoside with an inorganic phosphate attached to a 5'-hydroxyl group of the sugar in ester linkage (Fig. 12.4). The names and abbreviations of nucleotides specify the base, the sugar, and the number of phosphates attached (MP, **mon**ophosphate; DP, **dip**hosphate; TP, **triphosphate**). In deoxynucleotides, the prefix "d" precedes the abbreviation. For example, GDP is guanosine diphosphate (the base guanine attached to a ribose that has two phosphate groups) and dATP is deoxyadenosine triphosphate (the base adenine attached to a deoxyribose with three phosphate groups).

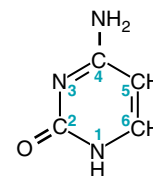


**Adenine (A)**

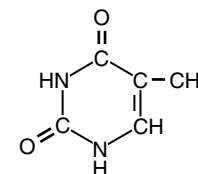


**Guanine (G)**

## Pyrimidines



**Cytosine (C)**



**Thymine (T)**

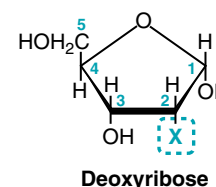
**Fig. 12.2.** Purine and pyrimidine bases in DNA.

**Table 12.1. Names of Bases and Their Corresponding Nucleosides<sup>a</sup>**

| Base             | Nucleoside           |
|------------------|----------------------|
| Adenine (A)      | Adenosine            |
| Guanine (G)      | Guanosine            |
| Cytosine (C)     | Cytidine             |
| Thymine (T)      | Thymidine            |
| Uracil (U)       | Uridine              |
| Hypoxanthine (I) | Inosine <sup>b</sup> |

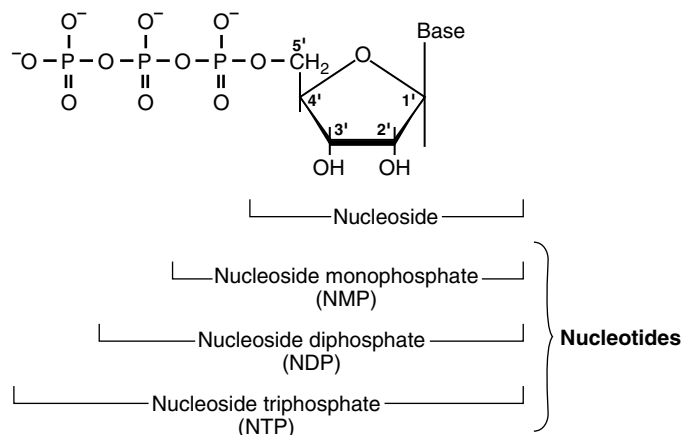
<sup>a</sup> If the sugar is deoxyribose rather than ribose, the nucleoside has "deoxy" as a prefix (e.g., deoxyadenosine). Nucleotides are given the name of the nucleoside plus mono, di, or triphosphate (e.g., adenosine triphosphate or deoxyadenosine triphosphate).

<sup>b</sup> The base hypoxanthine is not found in DNA but is produced during degradation of the purine bases. It is found in certain tRNA molecules. Its nucleoside, inosine, is produced during synthesis of the purine nucleotides (see Chapter 41).



**Deoxyribose**

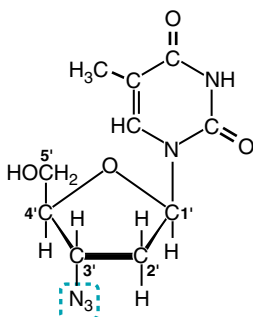
**Fig. 12.3.** Deoxyribose and ribose, the sugars of DNA and RNA. The carbon atoms are numbered from 1 to 5. When the sugar is attached to a base, the carbon atoms are numbered from 1' to 5' to distinguish it from the base. In deoxyribose the X = H; in ribose the X = OH.



**Fig. 12.4.** Nucleoside and nucleotide structures. Shown with ribose as the sugar. The corresponding deoxyribonucleotides are abbreviated dNMP, dNDP, and dNTP. N = any base (A, G, C, U, or T).



After **Ivy Sharer** was diagnosed with AIDS, she was treated with a mixture of drugs including zidovudine (ZDV), formerly called AZT. This drug is an analog of the thymine nucleotide found in DNA (the modified group is shown in the dashed box). ZDV is phosphorylated in the body by the kinases that normally phosphorylate nucleosides and nucleotides. As the viral DNA chain is being synthesized in a human cell, ZDV is then added to the growing 3'-end by viral reverse transcriptase. However, ZDV lacks a 3' OH group and, therefore, no additional nucleotides can be attached through a 5'→3' bond. Thus, chain elongation of the DNA is terminated. Reverse transcriptase has a higher affinity for ZDV than does normal human cellular DNA polymerases, enabling the drug to target viral replication more specifically than cellular replication.



In 1944, after Oswald Avery's experiments establishing DNA as the genetic material were published, interest in determining the structure of DNA intensified. Digestion with enzymes of known specificity proved that inorganic phosphate joined the nucleotide monomers, forming a phosphodiester bond between the 3'-carbon of one sugar and the 5'-carbon of the next sugar along the polynucleotide chain (Fig. 12.5). Another key to DNA structure was provided by Erwin Chargaff. He analyzed the base composition of DNA from various sources and concluded that, on a molar basis, the amount of adenine was always equal to the amount of thymine, and the amount of guanine was equal to the amount of cytosine.

During this era, James Watson and Francis Crick joined forces and, using the x-ray diffraction data of Maurice Wilkins and Rosalind Franklin, incorporated the available information into a model for DNA structure. In 1953, they published a brief paper, describing DNA as a double helix consisting of two polynucleotide strands joined by pairing between the bases (adenine with thymine and guanine with cytosine). The model of base-pairing they proposed is the basis of modern molecular biology.

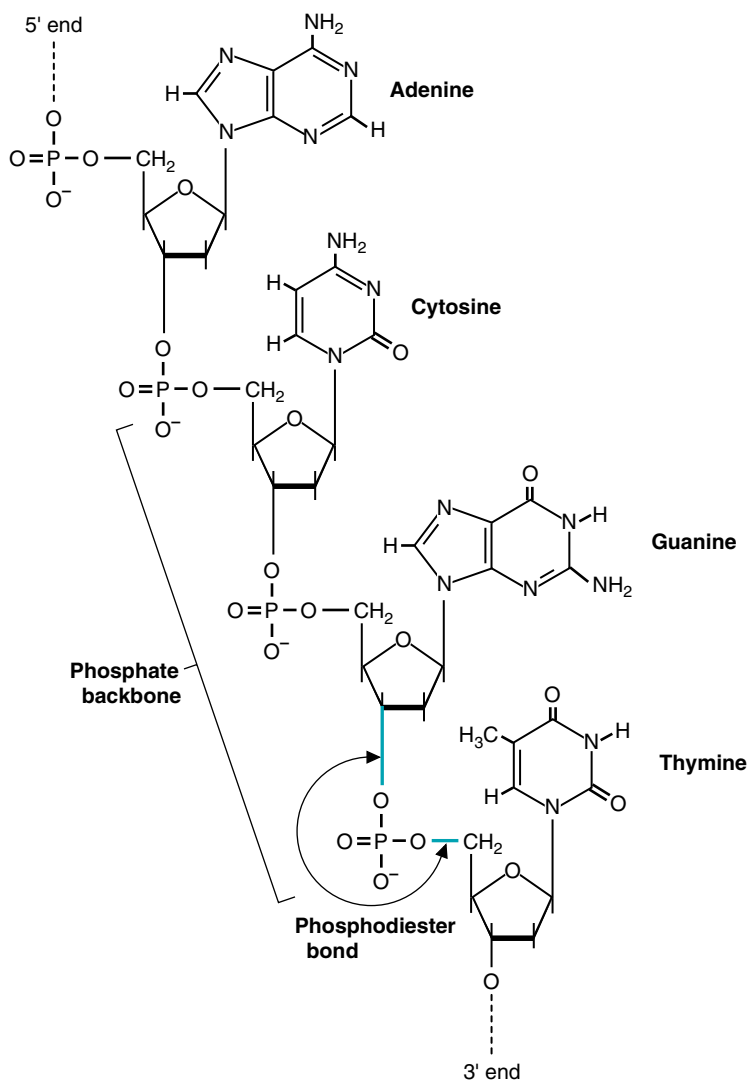
### C. Concept of Base-Pairing

As proposed by Watson and Crick, each DNA molecule consists of two polynucleotide chains joined by hydrogen bonds between the bases. In each base pair, a purine on one strand forms hydrogen bonds with a pyrimidine on the other strand. In one type of base pair, adenine on one strand pairs with thymine on the other strand (Fig. 12.6). This base pair is stabilized by two hydrogen bonds. The other base pair, formed between guanine and cytosine, is stabilized by three hydrogen bonds. As a consequence of base-pairing, the two strands of DNA are complementary, that is, adenine on one strand corresponds to thymine on the other strand, and guanine corresponds to cytosine.

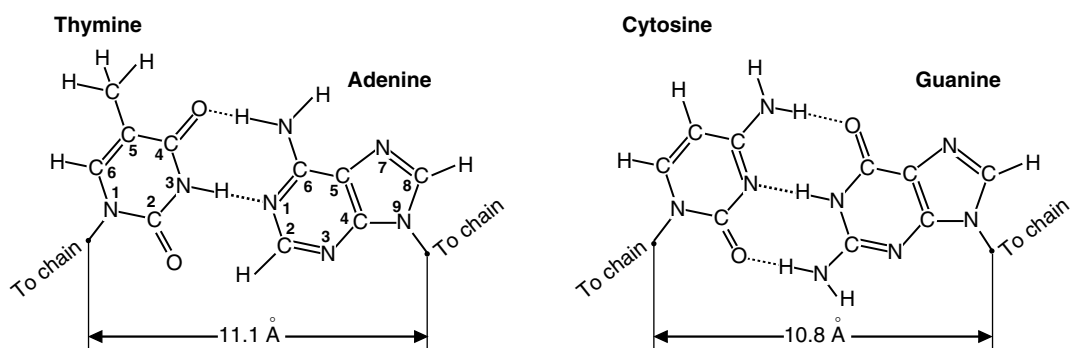
The concept of base-pairing proved to be essential for determining the mechanism of DNA replication (in which the copies of DNA are produced that are distributed to daughter cells) and the mechanisms of transcription and translation (in



Watson and Crick's one-page paper, published in 1953, contained little more than 900 words. However it triggered a major revolution in the biologic sciences and produced the conceptual foundation for the discipline of molecular biology.



**Fig. 12.5.** A segment of a polynucleotide chain of DNA. The dashes at the 5'- and 3'-ends indicate that the molecule contains more nucleotides than are shown.



**Fig. 12.6.** Base pairs of DNA. Note that the pyrimidine bases are “flipped over” from the positions in which they are usually shown (see Fig. 12.5). The bases must be in this orientation to form base pairs. The dotted lines indicate hydrogen bonds between the bases. Although the hydrogen bonds hold the bases and thus the two DNA strands together, they are weaker than covalent bonds and allow the DNA strands to separate during replication and transcription.





"It has not escaped our notice that the specific pairing we have postulated immediately suggests a possible copying mechanism for the genetic material." J.D. Watson and F.H.C. Crick, *Nature*, April 25, 1953.

which mRNA is produced from genes and used to direct the process of protein synthesis). Obviously, as Watson and Crick suggested, base-pairing allows one strand of DNA to serve as a template for the synthesis of the other strand (Fig. 12.7). Base-pairing also allows a strand of DNA to serve as a template for the synthesis of a complementary strand of RNA.

## D. DNA Strands Are Antiparallel

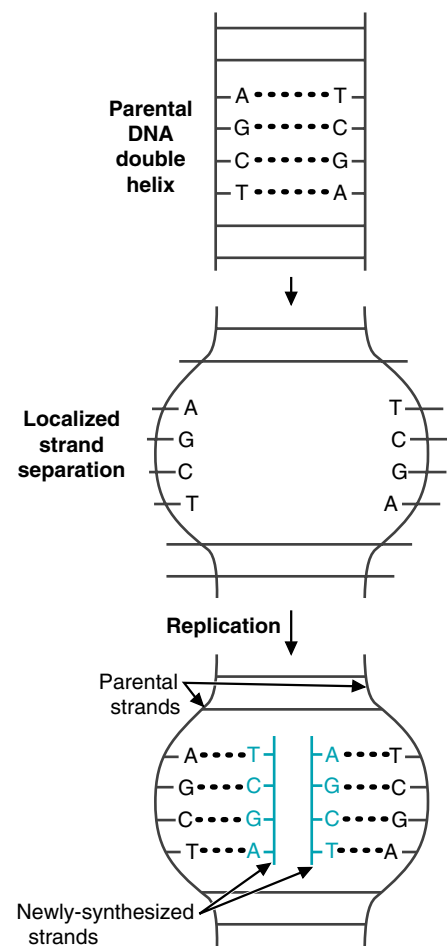
As concluded by Watson and Crick, the two complementary strands of DNA run in opposite directions. On one strand, the 5'-carbon of the sugar is above the 3'-carbon (Fig. 12.8). This strand is said to run in a 5' to 3' direction. On the other strand, the 3'-carbon is above the 5'-carbon. This strand is said to run in a 3' to 5' direction. Thus, the strands are antiparallel (that is, they run in opposite directions.) This concept of directionality of nucleic acid strands is essential for understanding the mechanisms of replication and transcription.

## E. The Double Helix

Because each base pair contains a purine bonded to a pyrimidine, the strands are equidistant from each other throughout. If two strands that are equidistant from each other are twisted at the top and the bottom, they form a double helix (Fig. 12.9). In the double helix of DNA, the base pairs that join the two strands are stacked like a spiral staircase along the central axis of the molecule. The electrons of the adjacent base pairs interact, generating stacking forces that, in addition to the hydrogen bonding of the base pairs, help to stabilize the helix.

The phosphate groups of the sugar-phosphate backbones are on the outside of the helix (see Fig. 12.9). Each phosphate has two oxygen atoms forming the phosphodiester bonds that link adjacent sugars. However, the third -OH group on the phosphate is free and dissociates a hydrogen ion at physiologic pH. Therefore, each DNA helix has negative charges coating its surface that facilitate the binding of specific proteins.

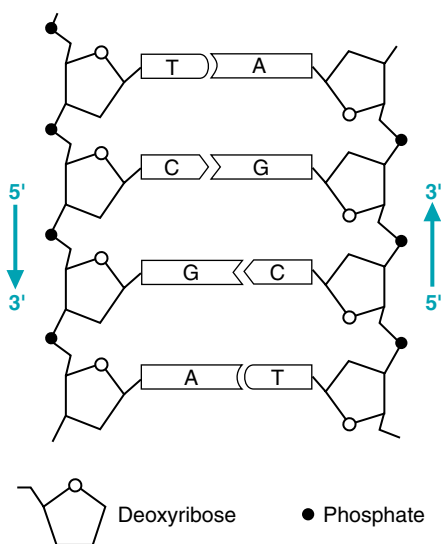
The helix contains grooves of alternating size, known as the major and minor grooves (see Fig. 12.9). The bases in these grooves are exposed and therefore can interact with proteins or other molecules.



**Fig. 12.7.** DNA strands serve as templates. During replication, the strands of the helix separate in a localized region. Each parental strand serves as a template for the synthesis of a new DNA strand.



Multi-drug regimens used to treat cancers (e.g., lymphomas) sometimes include the drug doxorubicin (adriamycin). It is a natural product with a complex multi-ring structure that intercalates or slips in between the stacked base pairs of DNA and inhibits replication and transcription.

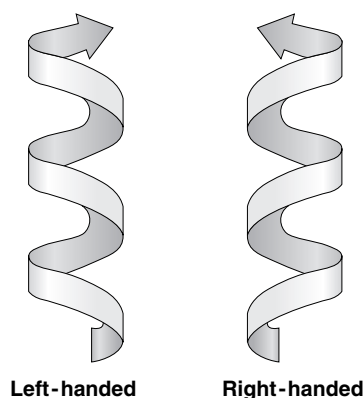


**Fig. 12.8.** Antiparallel strands of DNA. For the strand on the left, the 5'-carbon of each sugar is above the 3'-carbon, so it runs 5' to 3'. For the strand on the right, the 3'-carbon of each sugar is above the 5'-carbon, so it runs 3' to 5'.

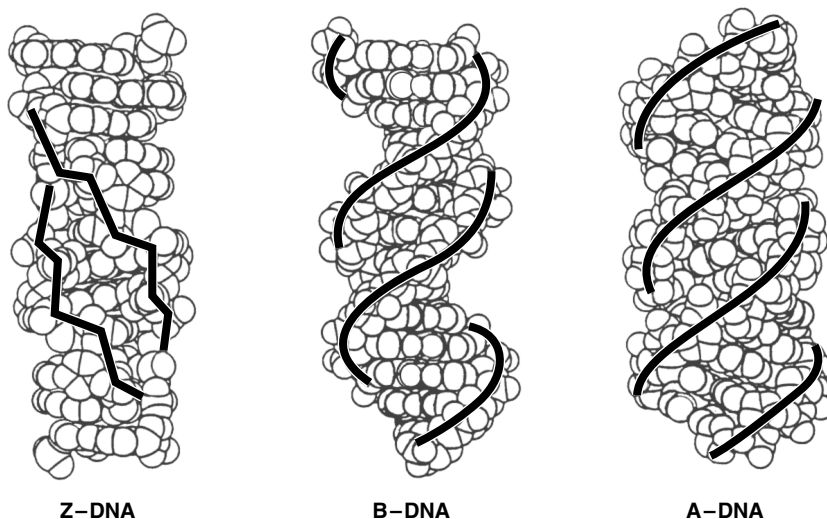
Watson and Crick described the B form of DNA, a right-handed helix, containing 3.4 Å between base pairs and 10.4 base pairs per turn. Although this form predominates *in vivo*, other forms also occur (Fig. 12.10). The A form, which predominates in DNA-RNA hybrids, is similar to the B form, but is more compact (2.3 Å between base pairs and 11 base pairs per turn). In the Z form, the bases of the two DNA strands are positioned toward the periphery of a left-handed helix. There are 3.8 Å between base pairs and 12 base pairs per turn in Z DNA. This form of the helix was designated “Z” because, in each strand, a line connecting the phosphates “zigs” and “zags.”

## F. Characteristics of DNA

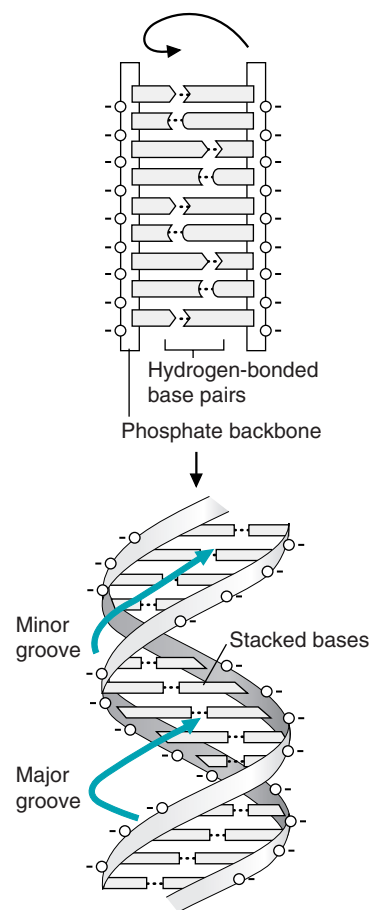
Both alkali and heat cause the two strands of the DNA helix to separate (denature). Many techniques employed to study DNA or to produce recombinant DNA molecules make use of this property. Although alkali causes the two strands of DNA to



If you look up through the bottom of a helix along the central axis and the helix spirals away from you in a clockwise direction (toward the arrowhead in the drawing), it is a right-handed helix. If it spirals away from you in a counterclockwise direction, it is a left-handed helix.



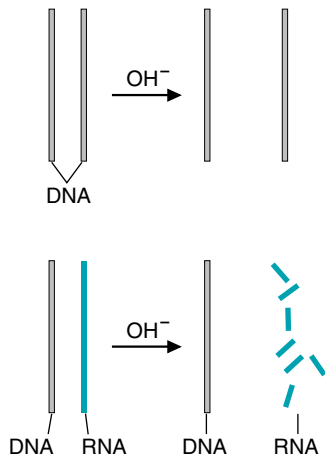
**Fig. 12.10.** Z, B, and A forms of DNA. The solid black lines connect one phosphate group to the next. Modified from Saenger W. *Principles of nucleic acid structure*. New York: Springer Verlag, 1984:257–286.



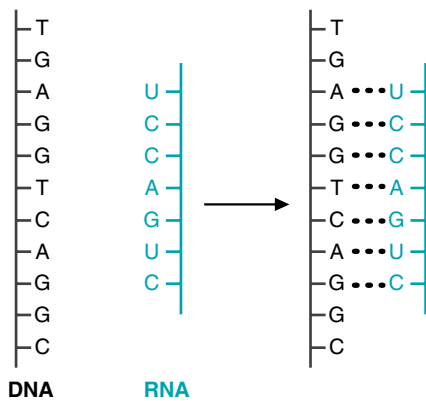
**Fig. 12.9.** Two DNA strands twist to form a double helix. The distance between the two phosphodiester backbones, shown with a ribbon, is about 11 Å. The hydrogen-bonded base pairs, shown in blue, create stacking forces with adjacent base pairs. Each phosphate group contains one negatively charged oxygen atom that provides the phosphodiester backbone with a negative charge. Because of the twisting of the helix, grooves are formed along the surface, the larger one being the major groove, and the smaller one the minor groove.



Heating and cooling cycles are used to separate and reanneal DNA strands in the polymerase chain reaction (PCR), a technique for obtaining large quantities of DNA from very small samples for research or for clinical and forensic testing (see Chapter 17).



**Fig. 12.11.** Effect of alkali on DNA and RNA. DNA strands stay intact, but they separate. RNA strands are degraded to nucleotides.



**Fig. 12.12.** Hybridization of DNA and complementary RNA.



If histones contain large amounts of arginine and lysine, will their net charge be positive or negative?

separate, it does not break the phosphodiester bonds (Fig. 12.11). In contrast, the phosphodiester bonds of RNA are cleaved by alkali. Therefore, alkali is used to remove RNA from DNA and to separate DNA strands before, or after, electrophoresis on polyacrylamide or agarose gels.

Heat alone converts double-stranded DNA to single-stranded DNA. The separation of strands is called melting, and the temperature at which 50% of the DNA is separated is called the  $T_m$ . If the temperature is slowly decreased, complementary single strands can realign and base-pair, re-forming a double helix essentially identical to the original DNA. This process is known as renaturation, reannealing, or hybridization. The process by which a single-stranded DNA anneals with complementary strands of RNA is also called hybridization (Fig. 12.12). Hybridization is used extensively in research and clinical testing.

## II. STRUCTURE OF CHROMOSOMES

### A. Size of DNA Molecules

A prokaryotic cell generally contains a single chromosome composed of double-stranded DNA that forms a circle. These circular DNA molecules are extremely large. The entire chromosome of the bacterium *Escherichia coli*, composed of a single, circular double-stranded DNA molecule, contains over  $4 \times 10^6$  base pairs. Its molecular weight is over  $2,500 \times 10^6$  g/mol (compared to the molecular weight for a glucose molecule of 180 g/mol). If this molecule were linear, its length would measure almost 2 mm.

DNA from eukaryotic cells is approximately 1,000 times larger than that from bacterial cells. In eukaryotes, each chromosome contains one continuous, linear DNA helix. The DNA of the longest human chromosome is over 7 cm in length. In fact, if the DNA from all 46 chromosomes in a diploid human cell were placed end to end, our total DNA would span a distance of about 2 m (over 6 feet). Our total DNA contains about  $6 \times 10^9$  base pairs.

### B. Packing of DNA

DNA molecules require special packaging to enable them to reside within cells because the molecules are so large. In *E. coli*, the circular DNA is supercoiled and attached to an RNA-protein core. Packaging of eukaryotic DNA is much more complex because it is larger and must be contained within the nucleus of the eukaryotic cell. Eukaryotic DNA binds to an equal weight of histones, which are small basic proteins containing large amounts of arginine and lysine. The complex of DNA and proteins is called chromatin. The organization of eukaryotic DNA into chromatin is essential for controlling transcription, as well as for packaging. When chromatin is extracted from cells, it has the appearance of beads on a string (Fig. 12.13). The beads with DNA protruding from each end are known as nucleosomes, and the beads themselves are known as nucleosome cores (Fig. 12.14). Two molecules of each of four histone classes (histones H2A, H2B, H3, and H4) form the center of the core around which approximately 140 base pairs of double-stranded DNA are wound. The DNA wrapped around the nucleosome core is continuous and joins one



DNA consists of a double helix, with the two strands of DNA wrapping around each other to form a helical structure. To compact, the DNA molecule coils about itself to form a structure called a supercoil. A telephone cord, which connects the handpiece to the phone, displays supercoiling when the coiled cord wraps about itself. When the strands of a DNA molecule separate and unwind over a small local region (which occurs during DNA replication), supercoils are introduced into the remaining portion of the molecule, thereby increasing stress on this portion. Enzymes known as topoisomerases relieve this stress so that unwinding of the DNA strands can occur.

nucleosome core to the next. The DNA joining the cores is complexed with the fifth type of histone, H1. Further compaction of chromatin occurs as the strings of nucleosomes wind into helical, tubular coils called solenoid structures.

Although complexes of DNA and histones form the nucleosomal substructures of chromatin, other types of proteins are also associated with DNA in the nucleus. These proteins were given the unimaginative name of “non-histone chromosomal proteins.” The cells of different tissues contain different amounts and types of these proteins, which include enzymes that act on DNA and factors that regulate transcription.

### C. The Human Genome

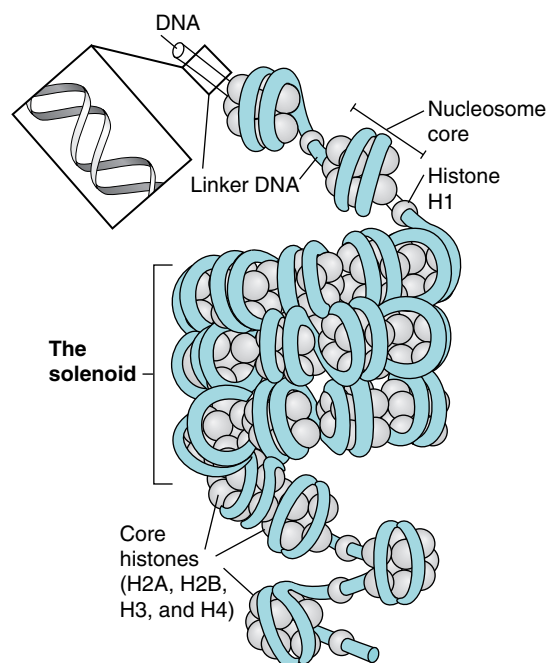
The genome, or total genetic content, of a human haploid cell (a sperm or an egg) is distributed in 23 chromosomes. Haploid cells contain one copy of each chromosome. The haploid egg and haploid sperm cells combine to form the diploid zygote, which continues to divide to form our other cells (mitosis), which are diploid. Diploid cells thus contain 22 pairs of autosomal chromosomes, with each pair composed of two homologous chromosomes containing a similar series of genes (Fig. 12.15). In addition to the autosomal chromosomes, each diploid cell has two sex chromosomes, designated X and Y. A female has two X chromosomes, and a male has one X and one Y chromosome. The total number of chromosomes per diploid cell is 46.

Genes are arranged linearly along each chromosome. A gene, in genetic terms, is the fundamental unit of heredity. In structural terms, a gene encompasses the DNA sequence encoding the structural components of the gene product (whether it be a polypeptide chain or RNA molecule) along with the DNA sequences adjacent to the 5' end of the gene which regulates its expression. A genetic locus is a specific position or location on a chromosome. Each gene on a chromosome in a diploid cell is matched by an alternate version of the gene at the same genetic locus on the homologous chromosome (Fig. 12.16). These alternate versions of a gene are called alleles. We thus have two alleles of each

**A:** At physiologic pH, arginine and lysine carry positive charges on their side chains; therefore, histones have a net positive charge. The arginine and lysine residues are clustered in regions of the histone molecules. These positively charged regions of the histones interact with the negatively charged DNA phosphate groups.

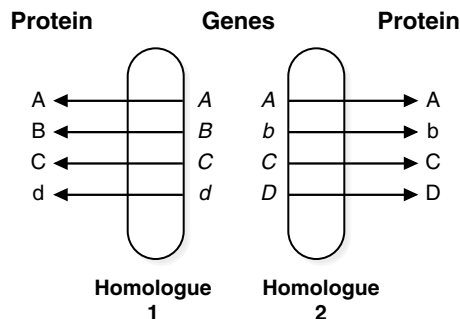


**Fig. 12.13.** Chromatin showing “beads on a string” structure.

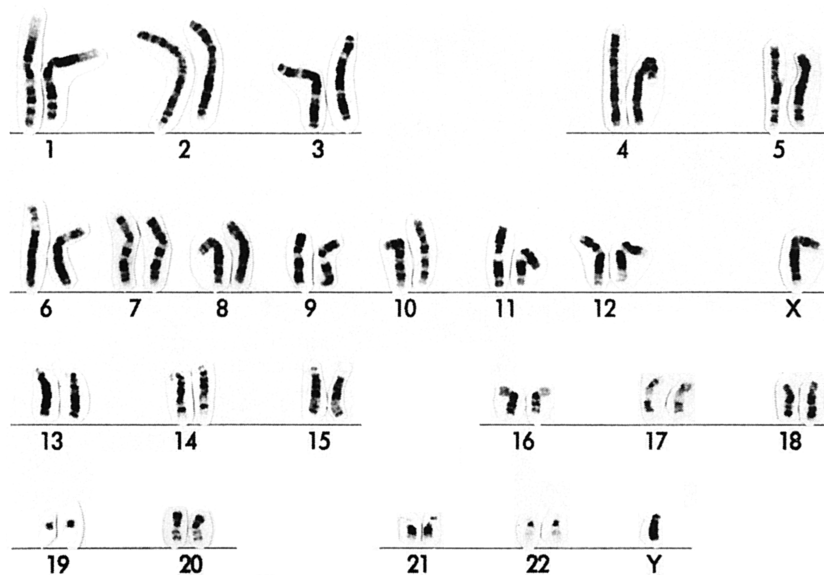


**Fig. 12.14.** A polynucleosome, indicating the histone cores and linker DNA.





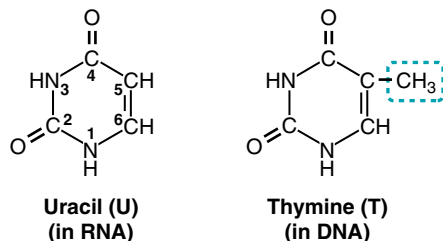
**Fig. 12.16.** Homologous chromosomes and their protein products. A set of homologous chromosomes is shown diagrammatically. (Of course, during interphase when they are producing their protein products, they cannot be visualized as discrete entities.) Four genes are shown as examples on each homologue. The genes of the homologues are alleles (e.g., *AA*, *Bb*, *CC*, *dD*). They may be identical (e.g., *AA*, *CC*), or they may differ (e.g., *Bb*, *dD*) in DNA sequence. Thus the corresponding protein products may be identical or they may differ in amino acid sequence.



**Fig. 12.15.** Human chromosomes from a male diploid cell. Each diploid cell contains 22 pairs of autosomes (the numbered chromosomes 1–22) plus one X and one Y. Each female diploid cell contains two X chromosomes. Each haploid cell contains chromosomes 1 through 22 plus either an X or a Y. From Gelehrter TD, Collins FS, Ginsburg D. *Principles of Human Genetics*, 2nd Ed. Baltimore: Williams & Wilkins, 1998.



**Will Sichel** has sickle cell anemia (see Chapters 6 and 7). He has two alleles for the  $\beta$ -globin gene that both generate the mutated form of hemoglobin, HbS. His younger sister Amanda, a carrier for sickle cell trait, has one normal allele (that produces HbA) and one that produces HbS. A carrier would theoretically be expected to produce HbA:HbS in a 50:50 ratio. However, what is generally seen in electrophoresis is 60:40 ratio of HbA:HbS. Dramatic deviations from this ratio imply the occurrence of an additional hemoglobin mutation (e.g., thalassemia).



**Fig. 12.17.** Comparison of the structures of uracil and thymine. They differ in structure only by a methyl group, outlined in blue.

gene, one from our mother and one from our father. If the alleles are identical in base sequence, we are homozygous for this gene. If the alleles differ, we are heterozygous for this gene and may produce two versions of the encoded protein that differ somewhat in primary structure.

The genomes of prokaryotic and eukaryotic cells differ in size. The genome of the bacterium *E. coli* contains approximately 3,000 genes. All of this bacterial DNA has a function; it either codes for proteins, rRNA, and tRNA, or it serves to regulate the synthesis of these gene products. In contrast, the genome of the human haploid cell contains between 30,000 and 50,000 genes, 10 to 15 times the number in *E. coli*. The function of most of this extra DNA has not been determined (an issue considered in more detail in Chapter 15).

### III. STRUCTURE OF RNA

#### A. General Features of RNA

RNA is similar to DNA. Like DNA, it is composed of nucleotides joined by 3'- to 5'-phosphodiester bonds, the purine bases adenine and guanine, and the pyrimidine base cytosine. However, its other pyrimidine base is uracil rather than thymine. Uracil and thymine are identical bases except that thymine has a methyl group at position 5 of the ring (Fig. 12.17). In RNA, the sugar is ribose, which contains a hydroxyl group on the 2'-carbon (see Fig 12.3. The prime refers to the position on the ribose ring). (The presence of this hydroxyl group allows RNA to be cleaved to its constituent nucleotides in alkaline solutions.)

RNA chains are usually single-stranded and lack the continuous helical structure of double-stranded DNA. However, RNA still has considerable secondary and tertiary structure because base pairs can form in regions where the strand loops back on itself. As in DNA, pairing between the bases is complementary and antiparallel. But in RNA, adenine pairs with uracil rather than thymine (Fig. 12.18). Base-pairing in RNA can be extensive, and the irregular looped structures generated are



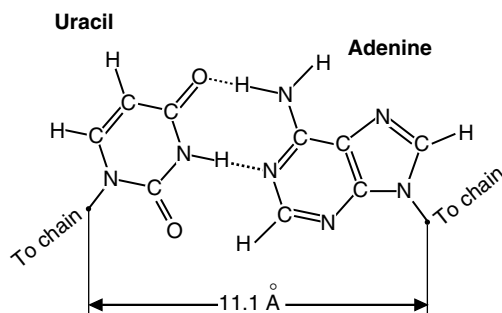


Fig. 12.18. A uracil–adenine base pair in RNA.

important for the binding of molecules, such as enzymes, that interact with specific regions of the RNA.

The three major types of RNA (mRNA, rRNA, and tRNA) participate directly in the process of protein synthesis. Other less abundant RNAs are involved in replication or in the processing of RNA, that is, in the conversion of RNA precursors to their mature forms.

Some RNA molecules are capable of catalyzing reactions. Thus, RNA, as well as protein, can have enzymatic activity. Certain rRNA precursors can remove internal segments of themselves, splicing the remaining fragments together. Because this RNA is changed by the reaction that it catalyzes, it is not truly an enzyme and therefore has been termed a “ribozyme.” Other RNAs act as true catalysts, serving as ribonucleases that cleave other RNA molecules or as a peptidyl transferase, the enzyme in protein synthesis that catalyzes the formation of peptide bonds.

## B. Structure of mRNA

Each **mRNA** molecule contains a nucleotide sequence that is converted into the amino acid sequence of a polypeptide chain in the process of translation. In eukaryotes, messenger RNA (mRNA) is transcribed from protein-coding genes as a long primary transcript that is processed in the nucleus to form mRNA. The various processing intermediates, which are mRNA precursors, are called pre-mRNA or hnRNA (**h**eterogeneous **n**uclear RNA). mRNA travels through nuclear pores to the cytoplasm, where it binds to ribosomes and tRNAs and directs the sequential insertion of the appropriate amino acids into a polypeptide chain.

Eukaryotic mRNA consists of a leader sequence at the 5′ end, a coding region, and a trailer sequence at the 3′ end (Fig 12.19). The leader sequence begins with a guanosine cap structure at its 5′ end. The coding region begins with a trinucleotide start codon that signals the beginning of translation, followed by the trinucleotide codons for amino acids, and ends at a termination signal. The trailer terminates at its 5′ end with a poly(A) tail that may be up to 200 nucleotides long. Most of the leader sequence, all of the coding region, and most of the trailer are formed by transcription of the complementary nucleotide sequence in DNA. However, the terminal guanosine in the cap structure and the poly(A) tail do not have complementary sequences; they are added posttranscriptionally.

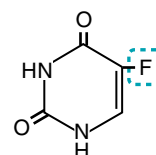
## C. Structure of rRNA

Ribosomes are subcellular ribonucleoprotein complexes on which protein synthesis occurs. Different types of ribosomes are found in prokaryotes and in the cytoplasm and mitochondria of eukaryotic cells (Fig. 12.20). Prokaryotic ribosomes contain three types of rRNA molecules with sedimentation coefficients of 16, 23, and 5S. The 30S ribosomal subunit contains the 16S rRNA complexed with proteins, and

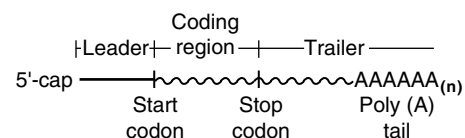
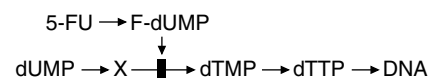


**Colin Tuma** is being treated with 5-fluorouracil (5-FU), a pyrimidine base similar to uracil and thymine.

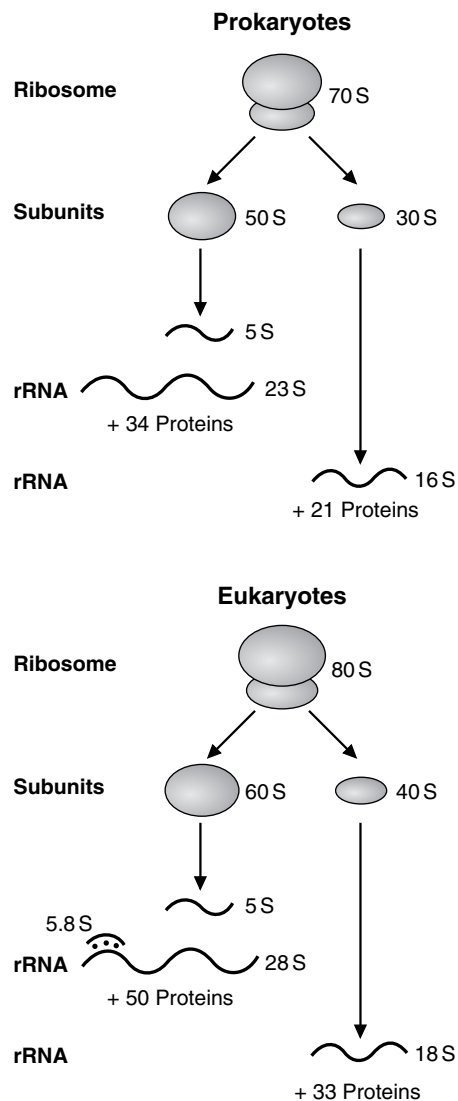
5-FU inhibits the synthesis of the thymine nucleotides required for DNA replication. Thymine is normally produced by a reaction catalyzed by thymidylate synthase, an enzyme that converts deoxyuridine monophosphate (dUMP) to deoxythymidine monophosphate (dTMP). 5-FU is converted in the body to F-dUMP, which binds tightly to thymidylate synthase in a transition state complex and inhibits the reaction (recall that thymine is 5-methyl uracil). Thus, thymine nucleotides cannot be generated for DNA synthesis, and the rate of cell proliferation decreases.



**5-Fluorouracil,**  
an analogue of uracil or thymine



**Fig. 12.19.** The regions of eukaryotic mRNA. The wavy line indicates the polynucleotide chain of the mRNA and the As constituting the poly(A) tail. The 5′-cap consists of a guanosine residue linked at its 5′ hydroxyl group to three phosphates, which are linked to the 5′-hydroxyl group of the next nucleotide in the RNA chain. The start and stop codons represent where protein synthesis is initiated and terminated from this mRNA.



**Fig. 12.20.** Comparison of prokaryotic and eukaryotic ribosomes. The cytoplasmic ribosomes of eukaryotes are shown. Mitochondrial ribosomes are similar to prokaryotic ribosomes, but they are smaller (55S rather than 70S).



Erythromycin, the antibiotic used to treat *Neu Moania*, inhibits protein synthesis on prokaryotic ribosomes, but not on eukaryotic ribosomes. It binds to the 50S ribosomal subunit, which is absent in eukaryotes. Therefore, it will selectively inhibit bacterial growth. However, because mitochondrial ribosomes are similar to those of bacteria, mitochondrial protein synthesis can also be inhibited. This fact is important in understanding some of the side effects of antibiotics that work by inhibiting bacterial protein synthesis.



A sedimentation coefficient is a measure of the rate of sedimentation of a macromolecule in a high-speed centrifuge (an ultracentrifuge). It is expressed in Svedberg units (S). Although larger macromolecules generally have higher sedimentation coefficients than do smaller macromolecules, sedimentation coefficients are not additive. Because frictional forces acting on the surface of a macromolecule slow its migration through the solvent, the rate of sedimentation depends not only on the density of the macromolecule, but also on its shape.

the 50S ribosomal subunit contains the 23S and 5S rRNAs complexed with proteins. The 30S and 50S ribosomal subunits join to form the 70S ribosome, which participates in protein synthesis.

Cytoplasmic ribosomes in eukaryotes contain four types of rRNA molecules of 18, 28, 5, and 5.8S. The 40S ribosomal subunit contains the 18S rRNA complexed with proteins, and the 60S ribosomal subunit contains the 28, 5, and 5.8S rRNAs complexed with proteins. In the cytoplasm, the 40S and 60S ribosomal subunits combine to form the 80S ribosomes that participate in protein synthesis.

Mitochondrial ribosomes, with a sedimentation coefficient of 55S, are smaller than cytoplasmic ribosomes. Their properties are similar to those of the 70S ribosomes of bacteria.

rRNAs contain many loops and exhibit extensive base-pairing in the regions between the loops (Fig. 12.21). The sequences of the rRNAs of the smaller ribosomal subunits exhibit secondary structures that are common to many different genera.

#### D. Structure of tRNA

During protein synthesis, tRNA molecules carry amino acids to ribosomes and ensure that they are incorporated into the appropriate positions in the growing polypeptide chain (Fig. 12.22). This is done through base-pairing of three bases of the tRNA (the anticodon) with the three base codons within the coding region of the mRNA. Therefore, cells contain at least 20 different tRNA molecules that differ somewhat in nucleotide sequence, one for each of the amino acids found in proteins. Many amino acids have more than one tRNA.

tRNA molecules contain not only the usual nucleotides, but also derivatives of these nucleotides that are produced by posttranscriptional modifications. In eukaryotic cells, 10 to 20% of the nucleotides of tRNA are modified. Most tRNA molecules contain ribothymidine (T), in which a methyl group is added to uridine to form ribothymidine. They also contain dihydrouridine (D), in which one of the double bonds of the base is reduced; and pseudouridine ( $\Psi$ ), in which uracil is attached to ribose by a carbon-carbon bond rather than a nitrogen-carbon bond (see Chapter 14). The base at the 5'-end of the anticodon of tRNA is frequently modified.

tRNA molecules are rather small compared with both mRNA and the large rRNA molecules. On average, tRNA molecules contain approximately 80 nucleotides and have a sedimentation coefficient of 4S. Because of their small size and high content of modified nucleotides, tRNAs were the first nucleic acids to be sequenced. Since 1965 when Robert Holley deduced the structure of the first tRNA, the nucleotide sequences of many different tRNAs have been determined. Although their primary sequences differ, all tRNA molecules can form a structure resembling a cloverleaf (discussed in more detail in Chapter 14).

#### E. Other Types of RNA

In addition to the three major types of RNA described above, other RNAs are present in cells. These RNAs include the oligonucleotides that serve as primers for DNA replication and the RNAs in the small nuclear ribonucleoproteins (snRNPs or snurps) that are involved in the splicing and modification reactions that occur during the maturation of RNA precursors (see Chapter 14).

## CLINICAL COMMENTS



**Ivy Sharer.** Ivy Sharer's clinical course was typical for the development of full-blown AIDS, in this case caused by the use of needles contaminated with HIV. The progressive immunologic deterioration that accompanies this disease ultimately results in life-threatening opportunistic infections with fungi (e.g., *Candida*, cryptococcus), other viruses (e.g., cytomegalovirus, herpes simplex), and bacteria (e.g., *Mycobacterium*, *Pneumocystis carinii*, *Salmonella*). The immunologic incompetence also frequently results in the development of certain neoplasms (e.g., Kaposi's sarcoma, non-Hodgkin's lymphoma) as well as meningitis, neuropathies, and neuropsychiatric disorders causing cognitive dysfunction. Although recent advances in drug therapy can slow the course of the disease, no cure is yet available.



**Colin Tuma.** Colin Tuma's original benign adenomatous polyp was located in the ascending colon, where 10% of large bowel cancers eventually arise. Because Mr. Tuma's father died of a cancer of the colon, his physician had warned him that his risk for developing colon cancer was three times higher than for the general population. Unfortunately, Mr. Tuma neglected to have his annual colonoscopic examinations as prescribed, and he developed an adenocarcinoma that metastasized.

The most malignant characteristic of neoplasms is their ability to metastasize, that is, form a new neoplasm at a noncontiguous site. The initial site of metastases for a tumor is usually at the first capillary bed encountered by the malignant cells once they are released. Thus, cells from tumors of the gastrointestinal tract often pass through the portal vein to the liver, which is Colin Tuma's site of metastasis. Because his adenocarcinoma has metastasized, there is little hope of eradicating it, and his therapy with 5-FU is palliative (directed toward reducing the severity of the disease and alleviation of the symptoms without actually curing the disease.)



**Agneu Moania.** Neu Moania's infection was treated with erythromycin, a macrolide antibiotic. Because this agent can inhibit mitochondrial protein synthesis in eukaryotic cells, it has the potential to alter host cell function, leading to such side effects as epigastric distress, diarrhea, and, infrequently, cholestatic jaundice.

## BIOCHEMICAL COMMENTS

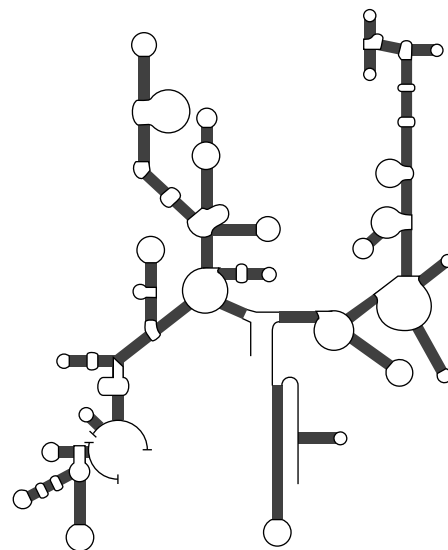


**Retroviruses.** RNA also serves as the genome for certain types of viruses, including retroviruses (e.g., the human immunodeficiency virus [HIV] that causes AIDS). Viruses must invade host cells to reproduce. They are not capable of reproducing independently. Some viruses that are pathogenic to

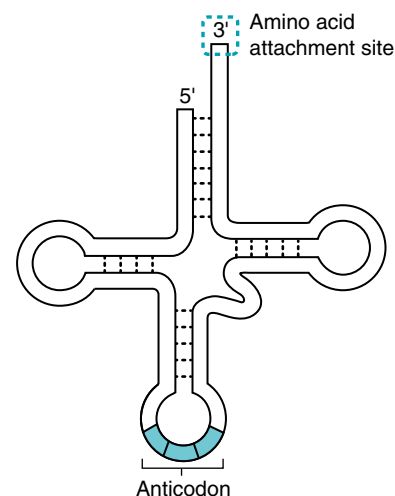


**Colin Tuma** completed his first course of intravenous 5-fluorouracil (5-FU) in the hospital. He tolerated the therapy with only mild anorexia and diarrhea and with only a mild leukopenia (a decreased white blood cell count; leuko = white). Thirty days after the completion of the initial course, these symptoms abated and he started his second course of chemotherapy with 5-FU as an out-patient.

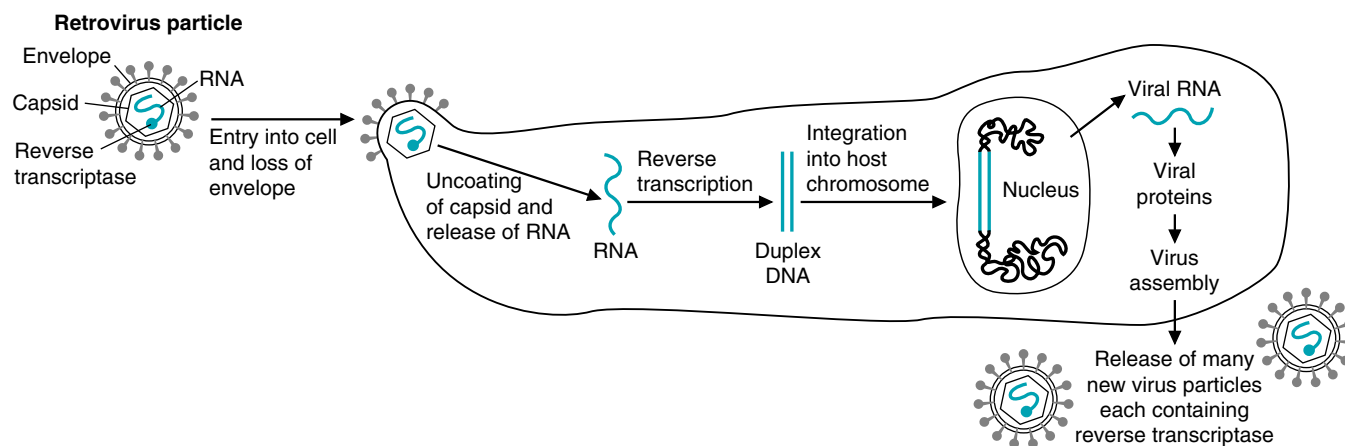
Because 5-FU inhibits synthesis of thymine, DNA synthesis is affected in all cells in the human body that are rapidly dividing, such as the cells in the bone marrow that produce leukocytes and the mucosal cells lining the intestines. Inhibition of DNA synthesis in rapidly dividing cells contributes to the side effects of 5-FU and many other chemotherapeutic drugs.



**Fig. 12.21.** Secondary structure of the portion of 16S-type ribosomal RNA that is common to many species. Darkened areas are base-paired. Circles are unpaired loops. Reproduced with permission, from *Annu Rev Biochem.* 1984;53:137. © 1984, by Annual Reviews, Inc.



**Fig. 12.22.** The typical cloverleaf structure of tRNA.



**Fig. 12.23.** The life cycle of a retrovirus. The virus contains two identical RNA strands, only one of which is shown for clarity. After penetrating the plasma membrane, the single-stranded viral RNA genome is reverse-transcribed to a double-stranded DNA form. The viral DNA migrates to the nucleus and integrates into the chromosomal DNA, where it is transcribed to form a viral RNA transcript. The viral transcript can form the viral RNA genome for progeny viruses, or can be translated to generate viral structural proteins.

humans contain DNA as their genetic material. Others contain RNA as their genetic material.

Some viruses that contain an RNA genome are known as retroviruses. HIV, the human immunodeficiency virus, is the retrovirus that causes AIDS (Fig 12.23). It invades cells of the immune system and prevents the affected individual from mounting an adequate immune response to combat infections.

According to the “central dogma” proposed by Francis Crick, information flows from DNA to RNA to proteins. For the most part, this concept holds true. However, retroviruses provide one violation of this rule. When retroviruses invade cells, their RNA genome is transcribed to produce a DNA copy. The enzyme that catalyzes this process is encoded in the viral RNA and is known as reverse transcriptase. This DNA copy integrates into the genome of the infected cell, and enzymes of the host cell are used to produce many copies of the viral RNA, as well as viral proteins, which can be packaged into new viral particles.

### Suggested Readings

Watson JD. *The Double Helix*. New York: Atheneum, 1968.

Watson JD, Crick FHC. Molecular structure of nucleic acids: a structure for deoxyribose nucleic acid. *Nature* 1953;171:737–738.

American Cancer Society Web site (<http://www.cancer.org>): Colorectal cancer. This web site provides information on the cause, treatment and prevention of a number of different cancers, and links to additional resources.

The Human Genome Project Web site ([http://www.nhgri.nih.gov/About\\_NHGRI/Der/Elsi/](http://www.nhgri.nih.gov/About_NHGRI/Der/Elsi/)): This site discusses the ethical issues involved in gene therapy protocols.

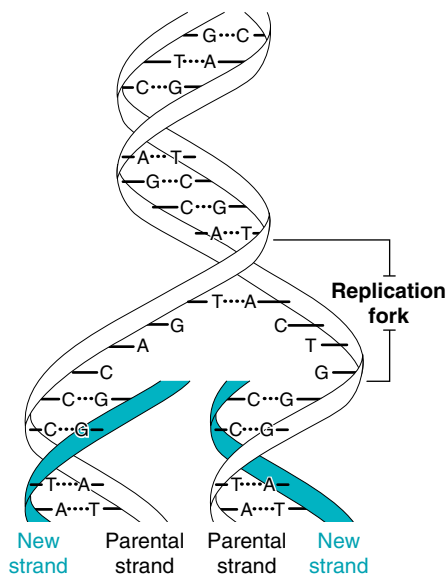


## REVIEW QUESTIONS—CHAPTER 12

- For the following DNA sequence, determine the sequence and direction of the complementary strand.  
 $5' - \text{ATCGATCGATCGATCG} - 3'$   
(A)  $5' - \text{ATCTATCGATCGATCG} - 3'$   
(B)  $3' - \text{ATCGATCGATCGATCG} - 5'$   
(C)  $5' - \text{CGAUCGAUCAUCGAU} - 3'$   
(D)  $5' - \text{CGATCGATCGATCGAT} - 3'$   
(E)  $3' - \text{CGATCGATCGATCGAT} - 5'$
- If the DNA strand shown below serves as a template for the synthesis of RNA, which of the following choices gives the sequence and direction of the RNA?  
 $5' - \text{GCTATGCATCGTGATCGAATTGCGT} - 3'$   
(A)  $5' - \text{ACGCAATTCGATCACGATGCATAGC} - 3'$   
(B)  $5' - \text{UGCGUUAAGCUAGUGCUACGUAUCG} - 3'$   
(C)  $5' - \text{ACGCAAUUCGAUCACGAUGCAUAGC} - 3'$   
(D)  $5' - \text{CGAUACGUAGCACUAGCUUAACGCA} - 3'$   
(E)  $5' - \text{GCTATGCATCGTGATCGAATTGCGT} - 3'$
- In DNA, the bond between the deoxyribose sugar and the phosphate is which of the following?  
(A) A polar bond  
(B) An ionic bond  
(C) A hydrogen bond  
(D) A covalent bond  
(E) A van der Waals bond
- How many double-stranded DNA molecules 8 base pairs long are theoretically possible?  
(A) 12  
(B) 32  
(C) 64  
(D) 256  
(E) 65,536
- The backbone of a DNA strand is composed of which of the following?  
(A) Sugars and bases  
(B) Phosphates and sugars  
(C) Bases and phosphates  
(D) Nucleotides and sugars  
(E) Phosphates and nucleosides



# 13 Synthesis of DNA



**Fig. 13.1.** A replicating DNA helix. The parental strands separate at the replication fork. Each parental strand serves as a template for the synthesis of a new strand.

DNA synthesis occurs by the process of **replication**. During replication, each of the two parental strands of DNA serves as a **template** for the synthesis of a complementary strand. Thus, each DNA molecule generated by the replication process contains one intact parental strand and one newly synthesized strand (Fig. 13.1). In eukaryotes, **DNA replication** occurs during the **S phase** of the **cell cycle**, which is followed by the  $G_2$  phase. The cell **divides** during the next phase (**M**), and each daughter cell receives an exact copy of the DNA of the parent cell.

**The Replication Fork.** In both prokaryotes and eukaryotes, the site at which replication is occurring at any given moment is called the **replication fork**. As replication proceeds, the two parental strands separate in front of the fork. Behind the fork, each newly synthesized strand of DNA base-pairs with its complementary parental template strand. A complex of proteins is involved in replication.

**Helicases** and **topoisomerases** unwind the parental strands, and **single-strand binding proteins** prevent them from reannealing.

The major enzyme involved in replication is a **DNA polymerase** that copies each parental template strand in the 3' to 5' direction, producing new strands in a 5' to 3' direction. **Deoxyribonucleoside triphosphates** serve as the precursors. One strand of newly synthesized DNA grows continuously, whereas the other strand is synthesized discontinuously in short segments known as **Okazaki fragments**. These fragments are subsequently joined by **DNA ligase**.

**Initiation.** **DNA polymerase cannot initiate** the synthesis of new strands. Therefore, a short **primer** is produced, which contains ribonucleotides (RNA). DNA polymerase can add deoxyribonucleotides to the 3' end of this primer. This RNA primer is subsequently removed and replaced by deoxyribonucleotides.

**Telomeres.** The ends of linear chromosomes are called telomeres. The enzyme **telomerase**, an RNA-dependant DNA polymerase that carries its own RNA template, is required for their replication.

**Errors and Repair.** **Errors** occurring **during replication** could lead to deleterious **mutations**. However, many errors are corrected by enzyme activities associated with the complex at the replication fork. The **error rate** is thus kept at a very **low** level.

**Damage** to DNA molecules also causes mutations. **Repair mechanisms** correct DNA damage, usually by removing and replacing the damaged region. The intact, undamaged strand serves as a template for the DNA polymerase involved in the repair process.

**Recombination.** Although cells have mechanisms to correct replication errors and to repair DNA damage, some genetic change is desirable. It produces new proteins or variations of proteins that may increase the survival rate of the species. Genetic change is produced by unrepaired mutations and by a mechanism known as **recombination** in which portions of chromosomes are exchanged.



## THE WAITING ROOM



**Ivy Sharer** is having difficulty complying with her multidrug regimen. She often forgets to take her pills. When she returns for a checkup, she asks whether such a large number of pills are really necessary for treatment of acquired immunodeficiency syndrome (AIDS).



**Di Abietes** responded to treatment for her diabetes mellitus but subsequently developed a low-grade fever, an increase in urinary urgency and frequency, and burning at the urethral opening with urination (dysuria). A urinalysis showed a large number of white blood cells and many Gram-negative bacilli. A urine culture indicated many colonies of *Escherichia coli*, which is sensitive to several antibiotics, including the quinolone norfloxacin.



**Melvin (Mel) Anoma** is a 46-year-old man who noted a superficial, brownish-black, 5-mm nodule with irregular borders in the skin on his chest. He was scheduled for outpatient surgery, at which time a wide excision biopsy was performed. (In an excision biopsy, the complete mole is removed and biopsy performed). Examination of the nodule indicated histologic changes characteristic of a malignant melanoma reaching a thickness of only 0.7 mm from the skin surface (Stage I).



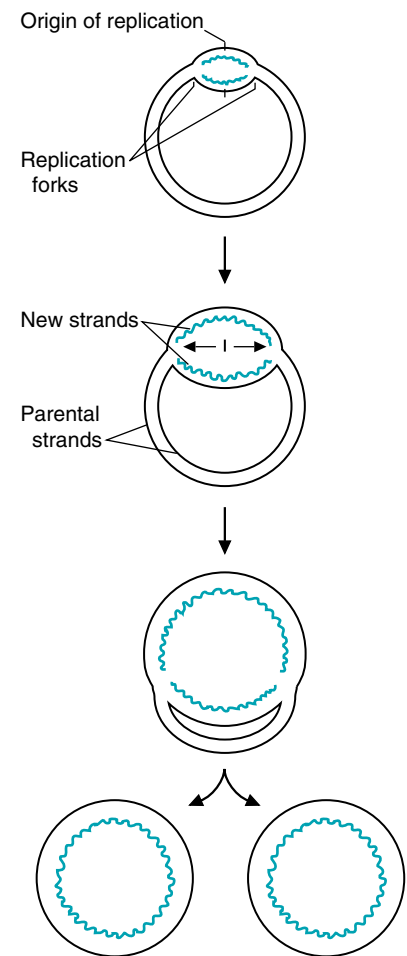
**Nick O'Tyne** is a 62-year-old electrician who has smoked two packs of cigarettes a day for 40 years. He recently noted that his chronic cough had gotten worse. His physician ordered a chest radiograph, which showed a 2-cm nodule in the upper lobe of the right lung. Cytologic study of the sputum by Papanicolaou technique showed cells consistent with the presence of a well-differentiated adenocarcinoma of the lung.

## I. DNA SYNTHESIS IN PROKARYOTES

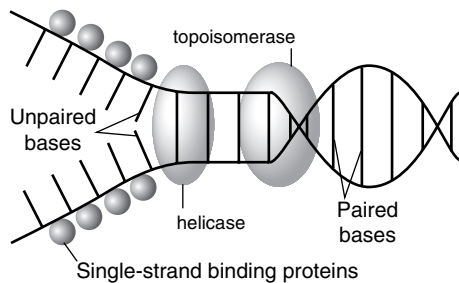
The basic features of the mechanism of DNA replication are illustrated by the processes occurring in the bacterium *E. coli*. This bacillus grows symbiotically in the human colon. It has been extensively studied and serves as a model for the more complex and, consequently, less well-understood processes that occur in eukaryotic cells.

### A. Replication Is Bidirectional

Replication of the circular, double-stranded DNA of the chromosome of *E. coli* begins with the binding of approximately 30 molecules of the protein DnaA at a single point of origin, designated *oriC*, where the DNA coils around the DnaA core (Fig. 13.2). With the assistance of other proteins (e.g., a helicase, gyrase, and single-stranded binding protein), the two parental strands separate within this region, and both strands are copied simultaneously. Synthesis begins at the origin and occurs at two replication forks that move away from the origin bidirectionally (in both directions at the same time). Replication ends on the other side of the chromosome at a termination point. One round of synthesis, involving the incorporation of over 4 million nucleotides in each new strand of DNA, is completed in approximately 40 minutes. However, a second round of synthesis can begin at the origin before the first round is finished. These multiple initiations of replication allow bacterial multiplication to occur much more quickly than the time it takes to complete a single round of replication.



**Fig. 13.2.** Bidirectional replication of a circular chromosome. Replication begins at the point of origin (*oriC*) and proceeds in both directions at the same time.



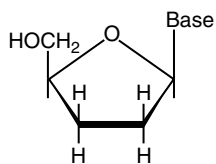
**Fig. 13.3.** Proteins involved in separating and unwinding parental DNA strands at the replication fork in prokaryotes.



**Di Abietes'** urinary tract infection was treated with norfloxacin, a fluorinated member of the quinolone family. This group of drugs inhibits bacterial DNA gyrase, a topoisomerase that unwinds the closed circular bacterial DNA helix ahead of the replication fork, and thus inhibits bacterial DNA synthesis. Because eukaryotic cells have linear DNA and do not contain DNA gyrase, they are not affected by quinolones.



One of the drugs used to treat **Ivy Sharer** was didanosine (ddI). It is a dideoxynucleoside, an example of which is shown below.



**A dideoxynucleoside**

The dideoxynucleosides do not have a hydroxyl group on either the 2'- or 3'-carbon. They can be converted to dideoxynucleoside triphosphates in cells and, like ZDV, terminate chain growth when incorporated into DNA. In the case of the dideoxynucleosides, chain termination results from the absence of a hydroxyl group on the 3' carbon. The HIV virus mutates very rapidly (mostly because reverse transcriptase lacks 3' to 5' exonuclease activity, the proofreading activity) and frequently develops resistance to one or more of these drugs. Therefore, it is recommended that AIDS patients take a number of drugs, including more than one reverse transcriptase inhibitor.

## B. Replication Is Semiconservative

Each daughter chromosome contains one of the parental DNA strands and one newly synthesized, complementary strand. Therefore, replication is said to be semi-conservative; i.e., the parental strands are conserved but are no longer together. Each one is paired with a newly synthesized strand (see Figs. 13.1 and 13.2).

## C. Unwinding of Parental Strands

Replication requires separation of the parental DNA strands and unwinding of the helix ahead of the replication fork. Helicases (DnaB) separate the DNA strands and unwind the parental duplex. Single-strand binding proteins prevent the strands from reassociating and protect them from enzymes that cleave single-stranded DNA (Fig. 13.3). Topoisomerases, enzymes that can break phosphodiester bonds and rejoin them, relieve the supercoiling of the parental duplex caused by unwinding. DNA gyrase is a major topoisomerase in bacterial cells.

## D. Action of DNA Polymerase

Enzymes that catalyze the synthesis of DNA are known as DNA polymerases. *E. coli* has three DNA polymerases, Pol I, Pol II, and Pol III. Pol III is the major replicative enzyme (Table 13.1). All DNA polymerases that have been studied copy a DNA template strand in its 3' to 5' direction, producing a new strand in the 5' to 3' direction (Fig. 13.4). Deoxyribonucleoside triphosphates (dATP, dGTP, dCTP, and dTTP) serve as substrates for the addition of nucleotides to the growing chain.

The incoming nucleotide forms a base pair with its complementary nucleotide on the template strand. Then an ester bond is formed between the first 5'-phosphate of the incoming nucleotide and the free 3'-hydroxyl group at the end of the growing chain. Pyrophosphate is released. The release of pyrophosphate and its subsequent cleavage by a pyrophosphatase provide the energy that drives the polymerization process.

DNA polymerases that catalyze the synthesis of new strands during replication exhibit a feature called processivity. They remain bound to the parental template strand while continuing to "process" down the chain, rather than dissociating and reassociating as each nucleotide is added. Consequently, synthesis is much more rapid than it would be with an enzyme that was not processive.

## E. Elimination of Base-Pairing Errors

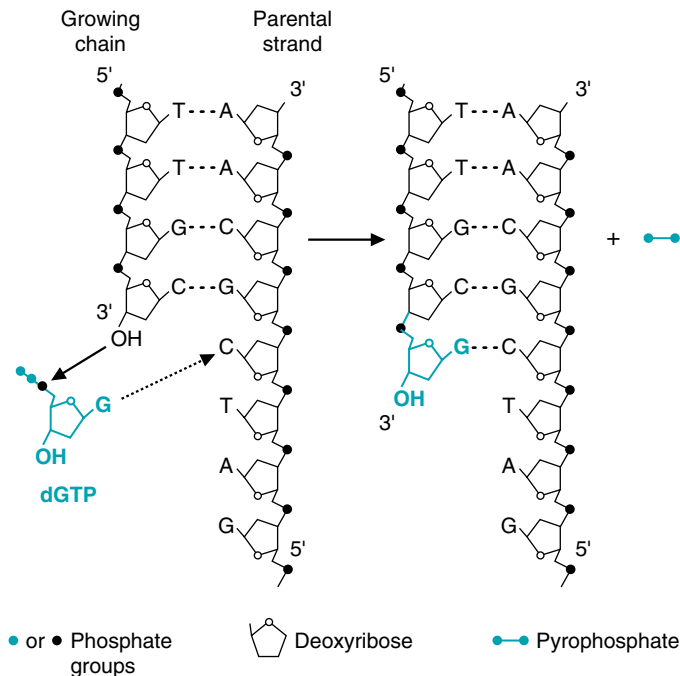
In *E. coli*, the replicative enzyme Pol III also performs a proofreading or editing function. This enzyme has 3'-5'-exonuclease activity in addition to its polymerase activity (see Table 13.1). If the nucleotide at the end of the growing chain is incorrectly base-paired with the template strand, Pol III removes this nucleotide before continuing to lengthen the growing chain. This proofreading activity eliminates

**Table 13.1. Functions of Bacterial DNA Polymerases**

| Polymerases | Functions <sup>a</sup>                                                                                        | Exonuclease Activity <sup>b</sup> |
|-------------|---------------------------------------------------------------------------------------------------------------|-----------------------------------|
| Pol I       | Filling of gap after removal of RNA primer<br>DNA repair<br>Removal of RNA primer in conjunction with RNase H | 5' to 3' and 3' to 5'             |
| Pol II      | DNA repair                                                                                                    | 3' to 5'                          |
| Pol III     | Replication – synthesis of DNA                                                                                | 3' to 5'                          |

<sup>a</sup> Synthesis of new DNA strands always occurs 5' to 3'

<sup>b</sup> Exonucleases remove nucleotides from DNA strands and act at the 5' end (cleaving 5' to 3') or at the 3' end (cleaving 3' to 5').



**Fig. 13.4.** Action of DNA polymerase. Deoxyribonucleoside triphosphates serve as precursors (substrates) used by DNA polymerase to lengthen the DNA chain. DNA polymerase copies the DNA template strand in the 3' to 5' direction. The new strand grows 5' to 3'.

most base-pairing errors as they occur. Only about one base pair in a million is mismatched in the final DNA product; the error rate is about  $10^{-6}$ . If this proofreading activity is experimentally removed from the enzyme, the error rate increases to about  $10^{-3}$ .

After replication, other mechanisms replace mismatched bases that escaped proofreading so that the fidelity of DNA replication is very high. The two processes of proofreading and postreplication mismatch repair result in an overall error rate of about  $10^{-10}$ , that is, less than one mismatched base pair in 10 billion.

## F. Function of RNA Primers

DNA polymerase cannot initiate the synthesis of new strands; it requires the presence of a free 3'-OH group to function. Therefore, a primer is required to supply the free 3'-OH group. This primer is an RNA oligonucleotide. It is synthesized in a 5' to 3' direction by an RNA polymerase (primase) that copies the DNA template strand. DNA polymerase initially adds a deoxyribonucleotide to the 3'-hydroxyl group of the primer and then continues adding deoxyribonucleotides to the 3'-end of the growing strand (Fig. 13.5).

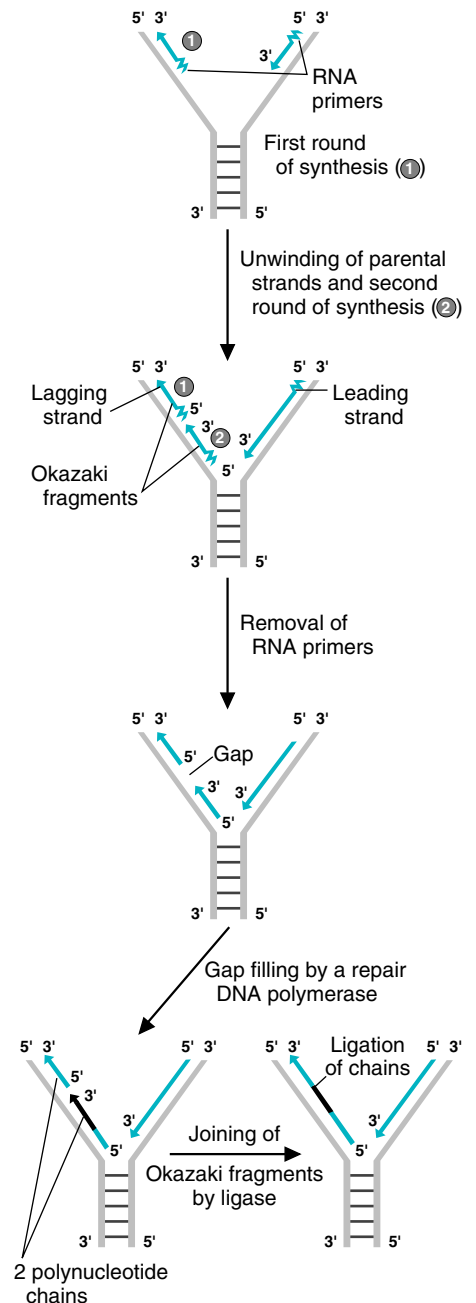
## G. DNA Synthesis at the Replication Fork

Both parental strands are copied at the same time in the direction of the replication fork, an observation difficult to reconcile with the known activity of DNA polymerase, which can produce chains only in a 5' to 3' direction. Because the parental strands run in opposite directions relative to each other, synthesis should occur in a 5' to 3' direction **toward** the fork on one template strand and in a 5' to 3' direction **away** from the fork on the other template strand.

Okazaki resolved this dilemma by showing that synthesis on one strand, called the leading strand, is continuous in the 5' to 3' direction toward the fork. The other



Nucleases cleave phosphodiester bonds. Endonucleases cleave bonds within polynucleotide chains. Exonucleases cleave bonds at the ends of polynucleotide chains, removing one nucleotide at a time. 5'-Exonucleases start at the 5'-end of the chain and cleave 5' to 3'. 3'-Exonucleases start at the 3'-end and cleave 3' to 5'.



**Fig. 13.5.** Synthesis of DNA at the replication fork. (See Fig. 13.6 for the ligation reaction.)

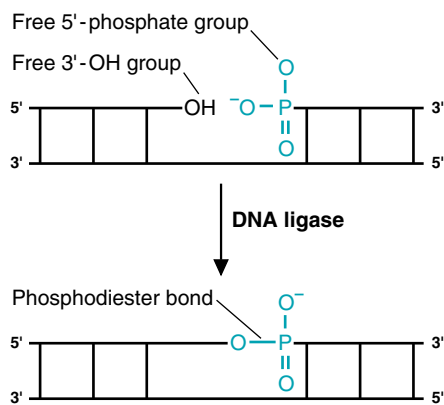


RNase H is a ribonuclease that specifically degrades RNA from an RNA-DNA hybrid. The HIV virus (see Chapter 12) converts an RNA genome to a double-stranded DNA copy using the enzyme reverse transcriptase. An intermediate in the conversion of the single-stranded RNA genome to double-stranded DNA is an RNA-DNA hybrid. To remove the RNA so a double-stranded DNA molecule can be made, reverse transcriptase contains RNase H activity. Scientists are currently attempting to target the RNase H activity of reverse transcriptase with inhibitors as a means of controlling spread of the virus.

strand, called the lagging strand, is synthesized discontinuously in short fragments (see Fig. 13.5). These fragments, named for Okazaki, are produced in a 5' to 3' direction (away from the fork), but then joined together so that, overall, synthesis proceeds toward the replication fork.

## H. Function of DNA Ligase

As replication progresses, the RNA primers are removed from Okazaki fragments, probably by the combined action of DNA polymerase I (Pol I, using its 5' → 3' exonuclease activity) and RNase H. Pol I fills in the gaps produced by removal of the primers. Because DNA polymerases cannot join two polynucleotide chains together, an additional enzyme, DNA ligase, is required to perform this function. The 3'-hydroxyl group at the end of one fragment is ligated to the phosphate group at the 5'-end of the next fragment (Fig. 13.6).



**Fig. 13.6.** Action of DNA ligase. Two polynucleotide chains, one with a free 3'-OH group and one with a free 5'-phosphate group, are joined by DNA ligase, which forms a phosphodiester bond.

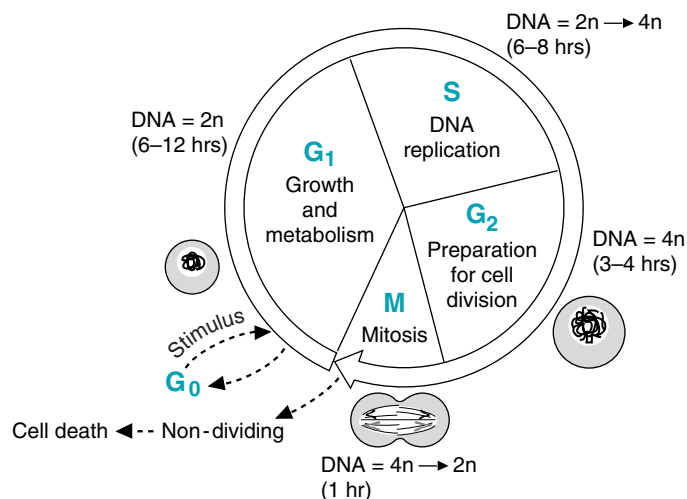
## II. DNA SYNTHESIS IN EUKARYOTES

The process of replication in eukaryotes is similar to that in prokaryotes. Differences in the processes are related mainly to the vastly larger amount of DNA in eukaryotic cells (over 1,000 times the amount in *E. coli*) and the association of eukaryotic DNA with histones in nucleosomes. Enzymes with DNA polymerase, primase, ligase, helicase, and topoisomerase activity are all present in eukaryotes, although these enzymes differ in some respects from those of prokaryotes.

### A. Eukaryotic Cell Cycle

The cell cycle of eukaryotes consists of four phases (Fig. 13.7). The first three phases ( $G_1$ , S, and  $G_2$ ) constitute interphase. Cells spend most of their time in these three phases, carrying out their normal metabolic activities. The fourth phase is mitosis, the process of cell division. This phase is very brief.

The first phase of the cell cycle,  $G_1$  (the first “gap” phase), is the most variable in length. Late in  $G_1$ , the cells prepare to duplicate their chromosomes (e.g., by producing nucleotide precursors). In the second or S phase, DNA replicates. Nucleosomes disassemble as the replication forks advance. Throughout S phase, the synthesis of histones and other proteins associated with DNA is markedly increased.



**Fig. 13.7.** Eukaryotic cell cycle. The times given for the length of each phase are for cells growing in culture.



The amount of DNA and histones both double, and chromosomes are duplicated. Histones complex with DNA, and nucleosomes are formed very rapidly behind the advancing replication forks.

During the third phase of the cell cycle,  $G_2$  (the second “gap” phase), the cells prepare to divide and synthesize tubulin for construction of the microtubules of the spindle apparatus. Finally, division occurs in the brief mitotic or M phase.

After mitosis, some cells reenter  $G_1$ , repeatedly going through the phases of the cell cycle and dividing. Other cells leave the cycle after mitosis, never to divide again, or they enter an extended  $G_1$  phase (sometimes called  $G_0$ ), in which they remain for long periods. On the appropriate signal, cells in  $G_0$  are stimulated to reenter the cycle and divide.

## B. Points of Origin for Replication

In contrast to bacterial chromosomes (see section I.A. of this chapter), eukaryotic chromosomes have multiple points of origin at which replication begins. “Bubbles” appear at these points on the chromosomes. At each end of a bubble, a replication fork forms; thus, each bubble has two forks. DNA synthesis occurs at each of these forks, as illustrated in Figure 13.8. As the bubbles enlarge, they eventually merge, and replication is completed. Because eukaryotic chromosomes contain multiple points of origin of replication (and, thus, multiple replicons—units of replication), duplication of such large chromosomes can occur within a few hours.

## C. Eukaryotic DNA polymerases

At least nine DNA polymerases exist in eukaryotic cells ( $\alpha$ ,  $\beta$ ,  $\gamma$ ,  $\delta$ ,  $\epsilon$ ,  $\zeta$ ,  $\kappa$ ,  $\eta$ , and  $\iota$ ) (Table 13.2). Polymerase  $\delta$  (pol  $\delta$ ) is the major replicative enzyme. Pol  $\alpha$  and pol  $\epsilon$  are also involved in replication. Polymerases  $\beta$  and  $\epsilon$ , as well as pol  $\alpha$ , appear to be involved in DNA repair. Polymerase  $\gamma$  is located in mitochondria and replicates the DNA of this organelle. Polymerases  $\zeta$ ,  $\kappa$ ,  $\eta$  and  $\iota$ , which lack  $3' \rightarrow 5'$  exonuclease activity, are used when DNA is damaged.

## D. The Eukaryotic Replication Complex

Many proteins bind at or near the replication fork and participate in the process of duplicating DNA (Fig. 13.9)(Table 13.3). Polymerase  $\delta$  (pol  $\delta$ ) is the major replica-



HeLa cells, derived from a human cervical carcinoma, are rapidly dividing cells that can be grown in culture flasks. Their cell cycle is approximately 20 hours. Only 1 hour of this time is spent in mitosis.

Cells within the body that divide less frequently, such as liver cells, spend days, weeks, or months in interphase before going through a brief mitotic phase.



Although Prometheus was chained to a rock as punishment for his theft of fire from the gods, and a vulture pecked at his liver each day, he survived. Can you guess why?



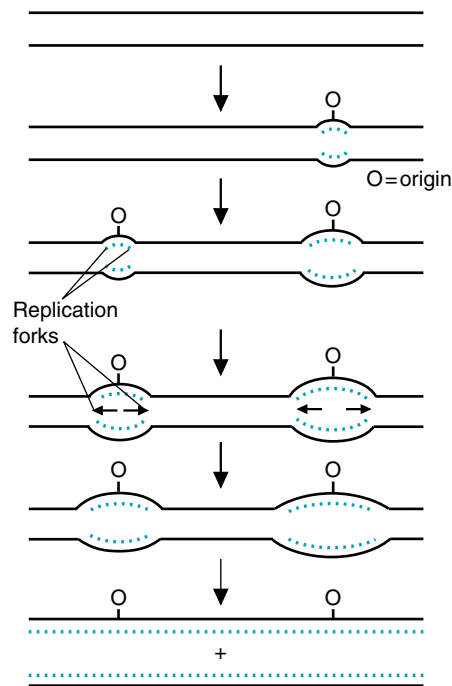
In the human body, many cells cycle frequently, e.g., hair follicles, skin cells, and cells of the duodenal crypts. Other cells, such as the precursors of red blood cells, divide a number of times, then lose their nuclei and leave the cell cycle to form mature red blood cells. These cells transport oxygen and carbon dioxide between the lungs and other tissues for about 120 days, then die. Other cells are normally quiescent (in  $G_0$ ). However, they can be stimulated to divide. In many instances, the stimuli are growth factors or hormones (e.g., mammary alveolar cells and uterine cells). In the case of liver cells, the stimulus is produced by death of some of the cells.

**Table 13.2. Functions of Eukaryotic DNA Polymerases**

| Polymerase     | Functions <sup>a</sup>                                                                | Exonuclease Activity |
|----------------|---------------------------------------------------------------------------------------|----------------------|
| Pol $\alpha$   | Replication (in a complex with primase and aids in starting the primer)<br>DNA repair | None                 |
| Pol $\beta$    | DNA repair exclusively                                                                | None                 |
| Pol $\gamma$   | DNA replication in mitochondria                                                       | $3' \rightarrow 5'$  |
| Pol $\delta$   | Replication (processive DNA synthesis on leading and lagging strands)<br>DNA repair   | $3' \rightarrow 5'$  |
| Pol $\epsilon$ | Replication (in some tissues takes the place of Pol $\delta$ )<br>DNA repair          | $3' \rightarrow 5'$  |
| Pol $\kappa$   | DNA repair (bypass polymerase) <sup>b</sup>                                           | None                 |
| Pol $\eta$     | DNA repair (bypass polymerase)                                                        | None                 |
| Pol $\zeta$    | DNA repair (bypass polymerase)                                                        | None                 |
| Pol $\iota$    | DNA repair (bypass polymerase)                                                        | None                 |

<sup>a</sup> Synthesis of new DNA strands always occurs  $5' \rightarrow 3'$ .

<sup>b</sup> Bypass polymerase are able to “bypass” areas of DNA damage and continue DNA replication. Some enzymes are error-free and insert the correct bases; other enzymes are error prone and insert random bases.



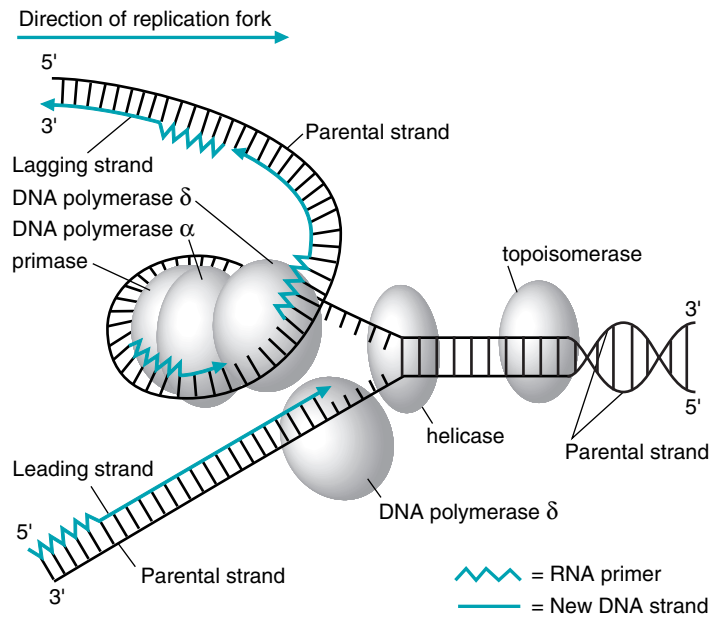
**Fig. 13.8.** Replication of a eukaryotic chromosome. Synthesis is bidirectional from each point of origin (O) and semiconservative, each daughter DNA helix contains one intact parental strand (solid line) and one newly synthesized strand (dashed line).



Okazaki fragments are much smaller in eukaryotes than in prokaryotes (about 200 nucleotides vs. 1,000 to 2,000). Because the size of eukaryotic Okazaki fragments are equivalent to the size of the DNA found in nucleosomes, it seems likely that one nucleosome at a time may release its DNA for replication.



Liver cells are in  $G_0$ . Up to 90% of the human liver can be removed. The remaining liver cells are stimulated to re-enter the cell cycle and divide, regenerating a mass equivalent to the original mass of the liver within a few weeks. The myth of Prometheus indicates that the capacity of the liver to regenerate was recognized even in ancient times.



**Fig. 13.9.** Replication complex in eukaryotes. The lagging strand is shown looped around the replication complex. Single-strand binding proteins (not shown) are bound to the unpaired, single-stranded DNA. Other proteins also participate in this complex (see text).

tive enzyme. However, before it acts, a primase associated with polymerase  $\alpha$  (pol  $\delta$ ) produces an RNA primer (approximately 10 nucleotides in length). Then pol  $\alpha$  adds about 20 deoxyribonucleotides to this RNA and dissociates from the template, because of the low processivity of pol  $\alpha$ . On the leading strand, pol  $\delta$  adds deoxyribonucleotides to this RNA-DNA primer, continuously producing this strand. Pol  $\delta$  is a highly processive enzyme.

The lagging strand is produced from a series of Okazaki fragments (see Fig. 13.5). Synthesis of each Okazaki fragment is initiated by pol  $\alpha$  and its associated primase, as described above. After pol  $\alpha$  dissociates, pol  $\delta$  adds deoxyribonucleotides to the primer, producing an Okazaki fragment. Pol  $\delta$  stops synthesizing one fragment when it reaches the start of the previously synthesized Okazaki fragment (see Fig. 13.5). The primer of the previously synthesized Okazaki fragment is

**Table 13.3 Major proteins involved in replication**

|                                |                                                                                                                                                    |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| DNA polymerases                | - add nucleotides to a strand growing 5' → 3', copying a DNA template 3' → 5'                                                                      |
| Primase                        | - synthesizes RNA primers                                                                                                                          |
| Helicases                      | - Separate parental DNA strands, i.e., unwind the double helix                                                                                     |
| Single-strand binding proteins | - Prevent single strands of DNA from reassociating                                                                                                 |
| Topoisomerases                 | - Relieve torsional strain on parental duplex caused by unwinding                                                                                  |
| Enzymes that remove primers    | - RNase H – hydrolyzes RNA of DNA-RNA hybrids                                                                                                      |
|                                | - Flap endonucleases 1 (FEN1) – recognizes “flap” (Unannealed portion of RNA) near 5'-end of primer and cleaves downstream in DNA region of primer |
| DNA ligase                     | - Joins, by forming a phosphodiester bond, two adjacent DNA strands that are bound to the same template                                            |

removed by flap endonuclease 1 (FEN1) and Rnase H. The gap left by the primer is filled by a DNA polymerase that uses the parental DNA strand as its template and the newly synthesized Okazaki fragment as its primer. DNA ligase subsequently joins the Okazaki fragments together (see Fig. 13.6).

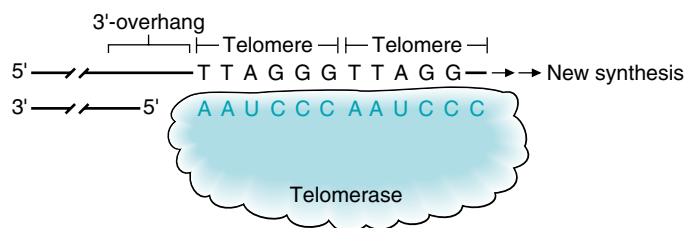
Obviously, eukaryotic replication requires many proteins. The complexity of the fork and the fact that it is not completely understood limits the detail shown in Figure 13.9. One protein not shown is proliferating cell nuclear antigen (PCNA), which is involved in organizing and orchestrating the replication process.

Additional activities that occur during replication include proofreading and DNA repair. Pol  $\delta$ , which is part of the replication complex, has the  $3' \rightarrow 5'$ -exonuclease activity required for proofreading. Enzymes that catalyze repair of mismatched bases are also present (see section III.B.3 of this chapter). Consequently, eukaryotic replication occurs with high fidelity; approximately one mispairing occurs for every  $10^9$  to  $10^{12}$  nucleotides incorporated into growing DNA chains.

### E. Replication of the Ends of Chromosomes

Eukaryotic chromosomes are linear, and the ends of the chromosomes are called telomeres. As DNA replication approaches the end of the chromosome, a problem develops in the lagging strand (Fig. 13.10). Either primase cannot lay down a primer at the very end of the chromosome, or, after DNA replication is complete, the RNA at the end of the chromosome is degraded. Consequently, the newly synthesized strand is shorter at the 5' end, and there is a 3'-overhang in the DNA strand being replicated. If the chromosome became shorter with each successive replication, genes would be lost. How is this problem solved?

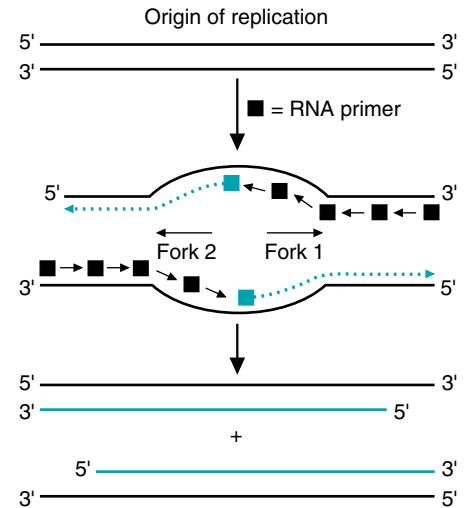
The 3' overhang is lengthened by the addition of telomeres so that primase can bind and synthesize the complementary strand. Telomeres consist of a repeating sequence of bases (TTAGGG for humans), which may be repeated thousands of times. The enzyme telomerase contains both proteins and RNA and acts as an RNA-dependent DNA polymerase (just like reverse transcriptase). The RNA within telomerase contains the complementary copy of the repeating sequence in the telomeres and can base pair with the existing 3'-overhang (Fig. 13.11). The polymerase activity of telomerase then uses the existing 3'-hydroxyl group of the overhang as a primer, and its own RNA as a template, and synthesizes new DNA that lengthens the 3'-end of the DNA strand. The telomerase moves down the DNA toward the new 3'-end and repeats the process a number of times. When the 3'-overhang is sufficiently long, primase binds, and synthesis of the complementary strand is completed. The 3' overhang can also form a complicated structure with telomere binding proteins to protect the ends of the chromosomes from damage and nuclease attack once they have been lengthened.



**Fig. 13.11.** Telomerase action. The RNA present in telomerase base-pairs with the overhanging 3'-end of telomeres and extends it by acting both as a template and a reverse transcriptase. After copying a small number of repeats, the complex moves down to the 3'-end of the overhang and repeats the process.



Proliferating cell nuclear antigen (PCNA) is used clinically as a diagnostic marker for proliferating cells.



**Fig. 13.10.** The “end-replication” problem in linear chromosomes. After replication, the telomeres have 3'-overhangs. When these molecules are replicated, chromosome shortening will result. The figure depicts a linear chromosome with one origin of replication. At the origin, two replication forks are generated, each moving in the opposite direction, labeled as Fork I and Fork II. As Fork I moves to the right, the bottom strand is read in the  $3'$  to  $5'$  direction, which means it is the template for the leading strand. The newly synthesized DNA complementary to the upper strand at Fork 1 will be the lagging strand. Now consider Fork 2. As this replication fork moves to the left, the upper strand is read in the  $3'$  to  $5'$  direction, so the newly synthesized DNA complementary to this strand will be the leading strand. For this fork, the newly synthesized DNA complementary to the bottom strand will be the lagging strand.



An inability to replicate telomeres has been linked to cell aging and death. Many somatic cells do not express telomerase; when placed in culture they survive a fixed number of population doublings, enter senescence, and then die. Analysis has shown significant telomere shortening in those cells. In contrast, stem cells do express telomerase and appear to have an infinite lifetime in culture. Research is underway to understand the role of telomeres in cell aging, growth, and cancer.



**Nick O'Tyne** has been smoking for 40 years in spite of the warnings on cigarette packs that this habit can be dangerous and even deadly. The burning of tobacco, and, for that matter, the burning of any organic material, produces many different carcinogens, such as benzo[a]pyrene. These carcinogens coat the airways and lungs. They can cross cell membranes and interact with DNA, causing damage to bases that interferes with normal base pairing. If these DNA lesions cannot be repaired or if they are not repaired rapidly enough, a permanent mutation can be produced when the cells replicate. Some mutations are silent, whereas other mutations can lead to abnormal cell growth, and cancer results.



Melanomas develop from exposure of the skin to the ultraviolet rays of the sun. The ultraviolet radiation causes pyrimidine dimers to form in DNA. Mutations may result that produce melanomas, appearing as dark brown growths on the skin.

Fortunately, **Mel Anoma's** malignant skin lesion was discovered at an early stage. Because there was no evidence of cancer in the margins of the resected mass, full recovery was expected. However, lifelong surveillance for return of the melanoma was planned.



Pyrimidine dimers, most commonly thymine dimers, can be repaired by photoreactivating enzymes that cleave the bonds between the bases by using energy from visible light. In this process, nucleotides are not removed from the damaged DNA. This repair process is used by bacteria and might serve as a very minor repair mechanism in human cells.

### III. DNA REPAIR

#### A. Actions of Mutagens

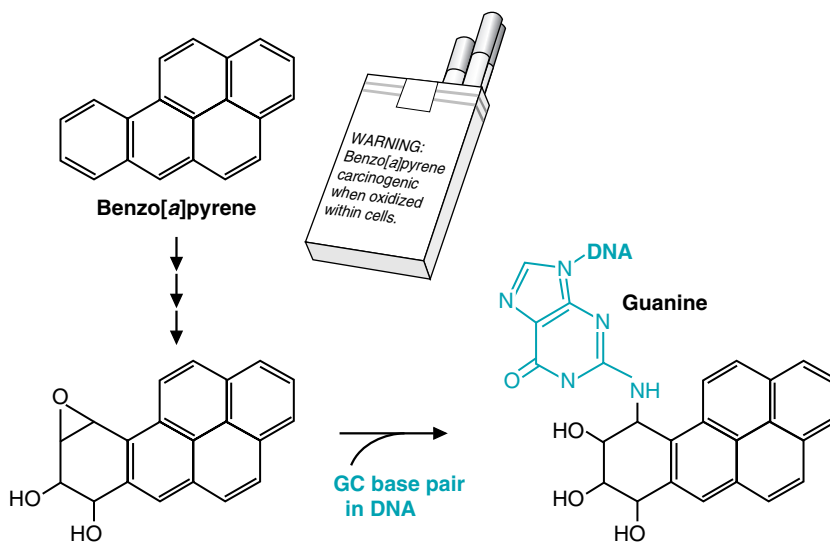
Despite proofreading and mismatch repair during replication, some mismatched bases do persist. Additional problems may arise from DNA damaged by mutagens, chemicals produced in cells, inhaled, or absorbed from the environment that cause mutations. Mutagens that cause normal cells to become cancer cells are known as carcinogens. Unfortunately, mismatching of bases and DNA damage produce thousands of potentially mutagenic lesions in each cell every day. Without repair mechanisms, we could not survive these assaults on our genes.

DNA damage can be caused by radiation and by chemicals (Fig. 13.12). These agents can directly affect the DNA or they can act indirectly. For example, x-rays, a type of ionizing radiation, act indirectly to damage DNA by exciting water in the cell and generating the hydroxyl radical, which reacts with DNA, thereby altering the structure of the bases or cleaving the DNA strands.

While exposure to x-rays is infrequent, it is more difficult to avoid exposure to cigarette smoke and virtually impossible to avoid exposure to sunlight. Cigarette smoke contains carcinogens such as the aromatic polycyclic hydrocarbon benzo[a]pyrene (see Fig. 13.12). When this compound is oxidized by cellular enzymes, which normally act to make foreign compounds more water soluble and easy to excrete, it becomes capable of forming bulky adducts with guanine residues in DNA. Ultraviolet rays from the sun, which also produce distortions in the DNA helix, excite adjacent pyrimidine bases on DNA strands, causing them to form covalent dimers (Fig. 13.13).

#### B. Repair Mechanisms

The mechanisms used for the repair of DNA have many similarities (Fig. 13.14). First, a distortion in the DNA helix is recognized, and the region containing the distortion is removed. The gap in the damaged strand is replaced by the action of a DNA polymerase that uses the intact, undamaged strand as a template. Finally, a ligase seals the nick in the strand that has undergone repair.



**Fig. 13.12.** Oxidation of benzo[a]pyrene and covalent binding to DNA. Benzo[a]pyrene is not carcinogenic until it is oxidized within cells. Then it can covalently bind to guanine residues in DNA, interrupting hydrogen bonding in G-C base pairs and producing distortions of the helix.

### 1. NUCLEOTIDE EXCISION REPAIR

Nucleotide excision repair involves local distortions of the DNA helix, such as mismatched bases or bulky adducts (e.g., oxidized benzo[*a*]pyrene) (Fig. 13.15, see also Fig. 13.14). Endonucleases cleave the abnormal chain and remove the distorted region. The gap is then filled by a DNA polymerase that adds deoxyribonucleotides, one at a time, to the 3'-end of the cleaved DNA, using the intact, complementary DNA strand as a template. The newly synthesized segment is joined to the 5'-end of the remainder of the original DNA strand by a DNA ligase.

### 2. BASE EXCISION REPAIR

DNA glycosylases recognize small distortions in DNA involving lesions caused by damage to a single base (e.g., the conversion of cytosine to uracil). A glycosylase cleaves the *N*-glycosidic bond that joins the damaged base to deoxyribose (see Fig. 13.15). The sugar-phosphate backbone of the DNA now lacks a base at this site (known as an apurinic or apyrimidinic site, or an AP site). Then an AP endonuclease cleaves the sugar-phosphate strand at this site. Subsequently, the same types of enzymes involved in other types of repair mechanisms restore this region to normal.

### 3. MISMATCH REPAIR

Mismatched bases (bases that do not form normal Watson-Crick base pairs) are recognized by enzymes of the mismatch repair system. Because neither of the bases in a mismatch is damaged, these repair enzymes must be able to determine which base of the mispair to correct.

The mismatch repair enzyme complex acts during replication when an incorrect, but normal base (i.e., A, G, C, or T) is incorporated into the growing chain (Fig. 13.16). In bacteria, parental DNA strands contain methyl groups on adenine bases in specific sequences. During replication, the newly synthesized strands are not immediately methylated. Before methylation occurs, the proteins involved in mismatch repair can distinguish parental from newly synthesized strands. A region of the new, unmethylated strand, containing the mismatched base, is removed and replaced.

Human enzymes also can distinguish parental from newly synthesized strands and repair mismatches. However, the mechanisms have not yet been as clearly defined as those in bacteria.



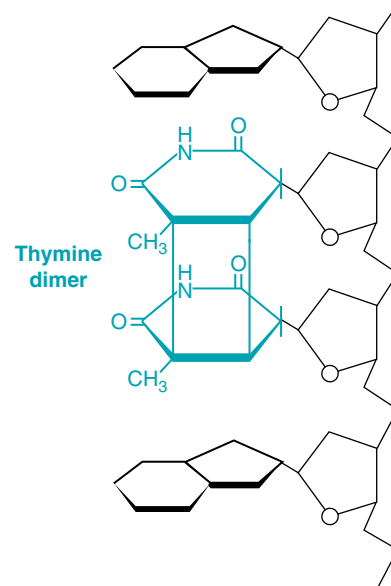
Pyrimidine dimers occur frequently in the skin. Usually repair mechanisms correct this damage, and cancer rarely occurs. However, in individuals with xeroderma pigmentosum, cancers are extremely common. These individuals have defects in their DNA repair systems. The first defect to be identified was a deficiency of the endonuclease involved in removal of pyrimidine dimers from DNA. Because of the inability to repair DNA, the frequency of mutation increases. A cancer develops once proto-oncogenes or tumor suppressor genes mutate. By scrupulously avoiding light, these individuals can reduce the number of skin cancers that develop.



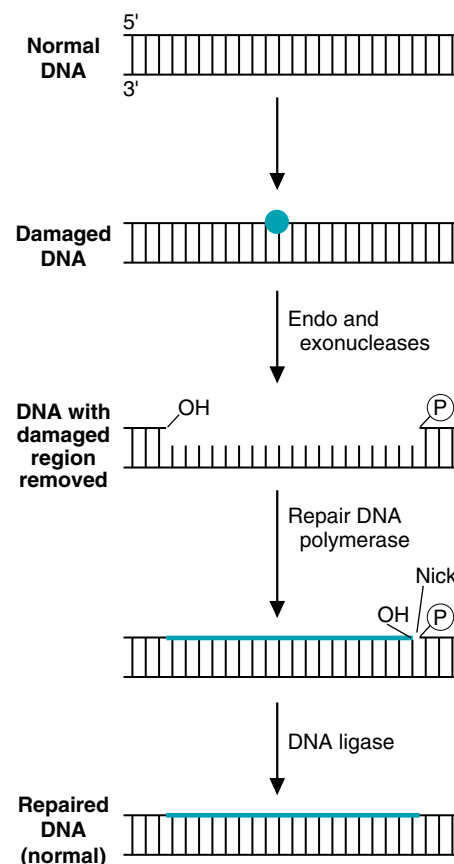
Spontaneous deamination occurs frequently in human DNA and converts cytosine bases to uracil. This base is not normally found in DNA and is potentially harmful because U pairs with A, forming U-A base pairs instead of the normal C-G pairs. To prevent this change from occurring, a uracil *N*-glycosylase removes uracil, and it is replaced by a cytosine via base excision repair.



Hereditary nonpolyposis colorectal cancer (a human cancer that does not arise from intestinal polyps) is caused by mutations in genes for proteins involved in mismatch repair (hMSH1, hMSH2, hPMS1, or hPMS2). The inability to repair mismatches increases the mutation frequency, resulting in cancers from mutations in growth regulatory genes.

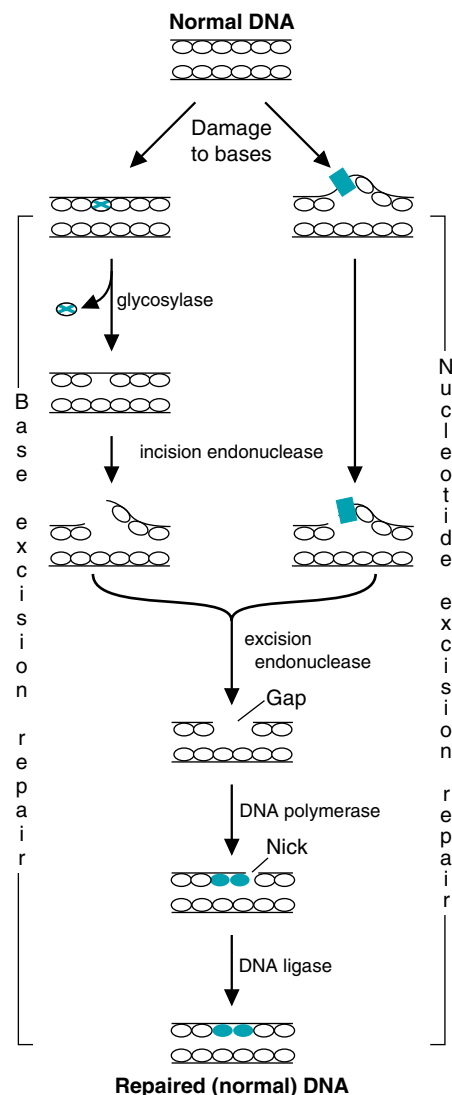


**Fig. 13.13.** A thymine dimer in a DNA strand. Ultraviolet light can cause two adjacent pyrimidines to form a covalent dimer.

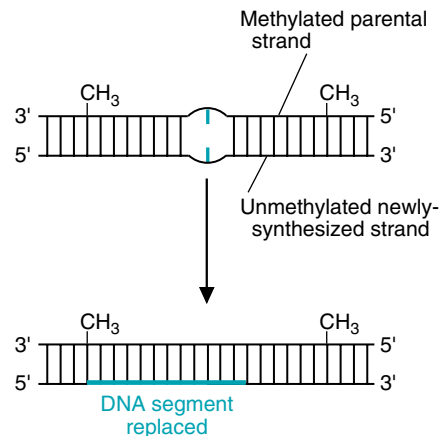


**Fig. 13.14.** Common steps in DNA repair mechanisms.





**Fig. 13.15.** Types of damage and various repair mechanisms. In base excision repair, the glycosylase cleaves the glycosidic bond between the altered base (shown with an X) and ribose. In nucleotide excision repair, the entire nucleotide is removed at once. The gap formed by the incision (cut) and excision (removal) endonucleases is usually several nucleotides wider than the one shown.



**Fig. 13.16.** Mismatch repair. Normal, undamaged but mismatched bases bind proteins of the mismatch repair system. In bacteria, these proteins recognize the older, parental strand because it is methylated and replace a segment of newly synthesized (and unmethylated) DNA containing the mismatched base. The mechanism for distinguishing between parental and newly synthesized strands in humans is not as well understood.

#### 4. TRANSCRIPTION-COUPLED REPAIR

Genes that are actively transcribed to produce mRNA are preferentially repaired. The RNA polymerase that is transcribing a gene (see Chapter 14 for a description of the process) stalls when it encounters a damaged region of the DNA template. Excision repair proteins are attracted to this site and repair the damaged region. Subsequently, RNA polymerase can resume transcription.

### IV. GENETIC REARRANGEMENTS

The exchange of segments between DNA molecules occurs quite frequently and is responsible for genetic alterations that can have beneficial or devastating consequences for the affected individuals and, in some instances, for their offspring. The DNA segments that are exchanged may be homologous (that is, of very similar sequence) or they may be totally unrelated. The size of these segments can range from a few nucleotides to tens of thousands and can include many different genes or portions of genes. Many of the enzymes involved in these exchanges are the same as or similar to those used for replication and repair and include endonucleases, exonucleases, unwinding enzymes, topoisomerases, DNA polymerases, and ligases.

One type of genetic rearrangement that has been observed for many years is “crossing-over” between homologous chromosomes during meiosis. Another type occurs in stem cells as they differentiate into lymphocytes. Segments of the genes of stem cells are rearranged so that the mature cell is capable of producing only a single type of antibody (see Chapter 16). Other types of genetic exchanges involve transposable elements (transposons) that can move from one site in the genome to another or produce copies that can be inserted into new sites. Translocations occur when chromosomes break and portions randomly become joined to other chromosomes, producing gross changes that can be observed under the light microscope. Genetic exchanges can even occur between species, for example, when foreign DNA is inserted into the human genome as a result of viral infection.

#### A. General or Homologous Recombination

Various models, supported by experimental evidence, have been proposed for the mechanism of recombination between homologous DNA sequences. Although

these mechanisms are complex, a simplified scheme for one type of recombination is presented in Figure 13.17.

Initially, two homologous chromosomes or segments of double-helical (duplex) DNA that have very similar, but not necessarily identical, sequences become aligned (see Fig. 13.17). One strand of one duplex is nicked by an enzyme and invades the other DNA duplex, base-pairing with a region of complementary sequence. The match between the sequences does not have to be perfect, but a significant number of bases must pair so that the strand displaced from its partner can form a displacement (D) loop. This D loop is nicked, and the displaced strand now base-pairs with the former partner of the invading strand. Ligation occurs, and a Holliday structure is generated (see Fig. 13.17). The branch point of the Holliday structure can migrate and may move many thousands of nucleotides from its original position. The Holliday structure, named for the scientist who discovered it, is finally cleaved and then religated, forming two chromosomes that have exchanged segments. In addition to enzymes similar to those used in DNA replication, enzymes for strand invasion, branch migration, and cleavage of the Holliday structure are required.

## B. Translocations

Breaks in chromosomes, caused by agents such as x-rays or chemical carcinogens, can result in gross chromosomal rearrangements (Fig. 13.18). If the free ends of the DNA at the break point reseat with the free ends of a different broken chromosome, a translocation is produced. These exchanges of large portions of chromosomes can have deleterious effects and are frequently observed in cancer cells.

## C. Transposable Elements

Movable (or transposable) genetic elements, “jumping genes,” were first observed by Barbara McClintock in the 1940s. Her work, initially greeted with skepticism, was ultimately accepted, and she was awarded the Nobel Prize in 1983.

Transposons are segments of DNA that can move from their original position in the genome to a new location (Fig. 13.19). They are found in all organisms. Transposons contain the gene for an enzyme called a transposase, which is involved in cleaving the transposon from the genome and moving it from one location to another.

Retroposons are similar to transposons except that they involve an RNA intermediate. Reverse transcriptase (see below) makes a single-stranded DNA copy of the RNA. A double-stranded DNA is then produced that is inserted into the genome at a new location.

## V. REVERSE TRANSCRIPTASE

Reverse transcriptase is an enzyme that uses a single-stranded RNA template and makes a DNA copy (Fig. 13.20). The RNA template can be transcribed from DNA by RNA polymerase or obtained from another source, such as an RNA virus. The DNA copy of the RNA produced by reverse transcriptase is known as complementary DNA (because it is complementary to the RNA template), or cDNA. Retroviruses (RNA viruses) contain a reverse transcriptase, which copies the viral RNA genome. A double-stranded cDNA is produced, which can become integrated into the human genome (see Fig. 12.23). After integration, the viral genes may be inactive, or they may be transcribed, sometimes causing diseases such as AIDS or cancer (see Chapter 18).

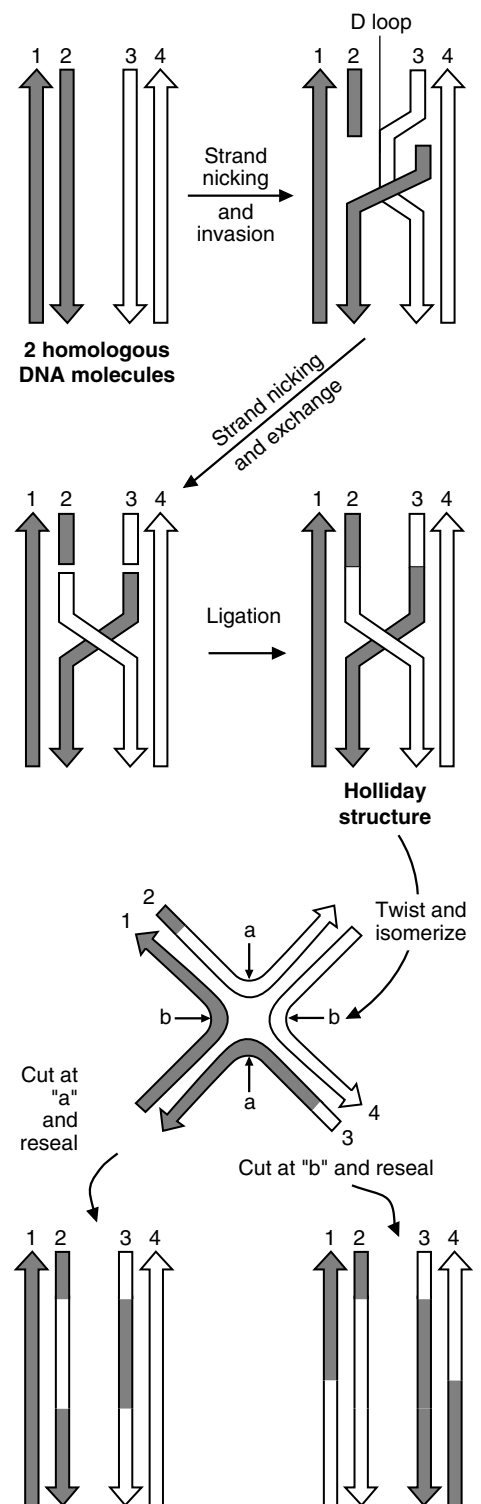
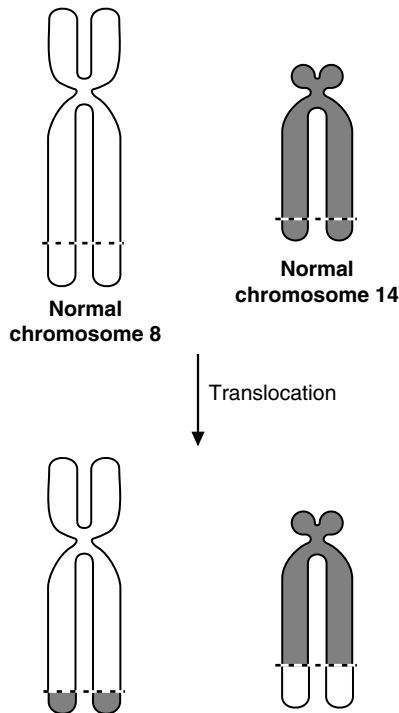
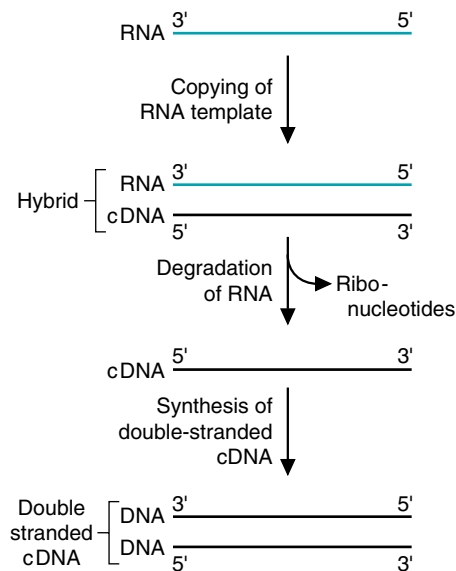


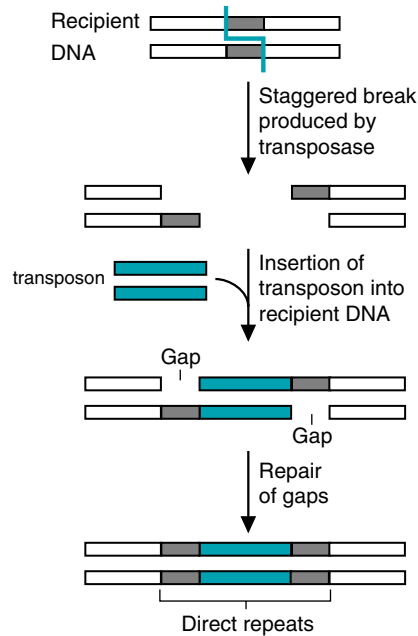
Fig. 13.17. Key steps in recombination.



**Fig. 13.18.** A chromosomal translocation. A portion of the long arm of chromosome 8 is exchanged for a portion of the long arm of chromosome 14. This chromosomal translocation occurs in Burkitt's lymphoma.



**Fig. 13.20.** Action of reverse transcriptase. This enzyme catalyzes the production of a DNA copy from an RNA template. The RNA of a DNA-RNA hybrid is degraded, and the single DNA strand is used as a template to make double-stranded DNA. This figure is a simplified version of a more complex process.



**Fig. 13.19.** Transposons. The steps involved in transposition are shown. Direct repeats are regions of DNA that have the same base sequence in the 5' → 3' direction.

## CLINICAL COMMENTS



**Ivy Sharer.** Ivy Sharer contracted AIDS when she used needles contaminated with HIV to inject drugs intravenously. Intravenous drug abusers account for 15 to 20% of new AIDS cases in the United States. HIV mutates rapidly, and, therefore, current treatment involves a combination of drugs that affect different aspects of its life cycle. This multidrug therapy lowers the viral titer (the number of viral particles found in a given volume of blood), sometimes to undetectable levels. However, if treatment is not followed carefully (i.e., if the patient is not “compliant”), the titer increases rapidly. Therefore, Ivy’s physician emphasized that she must carefully follow her drug regimen.



**Di Abietes.** Di Abietes’ poorly controlled diabetes mellitus predisposed her to a urinary tract infection because glucose in the urine serves as a “culture medium” for bacterial growth. The kidney glomerulotubular unit reabsorbs filtered glucose so that normally, the urine is glucose-free. However, when serum blood glucose levels exceed 175 to 185 mg/dL (the tubular threshold for glucose), the capacity for reabsorption is exceeded. In Ms. Abietes’ case, blood glucose levels frequently exceed this threshold.



**Mel Anoma.** The average person has 20 moles on the body surface; however, only seven people of every 100,000 develop a malignant melanoma. The incidence of malignant melanoma, however, is rising rapidly. Because 35 to 40% of patients with malignant melanoma die as a result of this cancer, the physician’s decision to perform a biopsy on a pigmented mole with an irregular border and variation of color probably saved **Mel Anoma’s** life.



**Nick O'Tyne.** Lung cancer currently accounts for one fifth of all cancers in men and one tenth in women. The overall 5-year survival rate is still less than 15%. For those who smoke two or more packs of cigarettes daily, as does

**Nick O'Tyne**, the death rate is 265 per 100,000 population. Thankfully, cigarette smoking has declined in the United States. Whereas 50% of men and 32% of women smoked in 1965, these figures have currently fallen to 26% and 24%, respectively.

## BIOCHEMICAL COMMENTS



**Chemical carcinogens and tumor promoters.** Once it was realized that a number of chemicals react with DNA, leading to mutations, and that mutations may lead to tumor formation, toxicologists searched for chemicals with the ability to cause skin tumors in rats. To test a chemical, it was applied to an area of shaved skin on the back of a rat to see whether a tumor would develop. The chemicals fell into two groups: Group I, the initiators, caused mutations in DNA and group II, the promoters, greatly enhanced the probability that cells that had been previously initiated would develop a tumor. Compounds falling into group II activate a protein kinase (protein kinase C), which is usually only transiently activated when normal cells are stimulated to grow. Tumor promoters, however, are long-lived, and lead to an enhanced activation of protein kinase C.

Is there an easy way to determine whether a compound is an initiator? Bruce Ames developed a rapid and simple test to determine whether chemicals are mutagens. The basic test uses bacteria that have a mutation in a gene necessary for histidine biosynthesis and require histidine for growth. The bacteria are treated with the test chemicals and the number that can grow in the absence of histidine measured. Bacteria that no longer require histidine for growth must have acquired a second, chemically induced mutation that opposed the inactivating, original mutation. In this fashion, the ability of chemicals to alter DNA can be determined. Because many chemicals (such as benz[*a*]pyrene) do not become carcinogenic in humans until they are metabolized by the liver, the Ames test also involves incubating the chemical with a liver extract, and then testing the metabolized extract for DNA-modifying activity.

## Suggested References

- Hanawalt P. Transcription-coupled repair and human disease. *Science* 1994;266:1957–1958.  
Kunkel T. DNA replication fidelity. *J Biol Chem* 1992;267:18251–18254.  
Modrich P. Mismatch repair, genetic stability, and cancer. *Science* 1994;266:1959–1960.  
Sancar A. Mechanisms of DNA excision repair. *Science* 1994;266:1954–1956.  
Waga S, Stillman B. The DNA replication fork in eukaryotic cells. *Annu Rev Biochem* 1998;67:721–751.



## REVIEW QUESTIONS—CHAPTER 13

1. Reverse transcriptase, an RNA-dependant DNA polymerase, differs from DNA polymerase  $\delta$  by which of the following?
  - (A) Synthesizes DNA in the 5' to 3' direction
  - (B) Contains 3' to 5' exonuclease activity
  - (C) Follows Watson-Crick base pair rules
  - (D) Synthesizes DNA in the 3' to 5' direction
  - (E) Can insert inosine into a growing DNA chain

2. If a 1,000-kilobase fragment of DNA has 10 evenly spaced and symmetric replication origins and DNA polymerase moves at 1 kilobase per second, how many seconds will it take to produce two daughter molecules (ignore potential problems at the ends of this linear piece of DNA)? Assume that the 10 origins are evenly spaced from each other, but not from the ends of the chromosome.
- (A) 20
  - (B) 30
  - (C) 40
  - (D) 50
  - (E) 100
3. Primase is not required during DNA repair processes because of which of the following?
- (A) All of the primase is associated with replication origins.
  - (B) RNA would be highly mutagenic at a repair site.
  - (C) DNA polymerase I does not require a primer.
  - (D) DNA polymerase III does not require a primer.
  - (E) DNA polymerase I or III can use any 3' -OH for elongation.
4. Which of the following enzymes is required to actively enhance the separation of DNA strands during replication?
- (A) Helicase
  - (B) 3' to 5' exonuclease
  - (C) DNA ligase
  - (D) Primase
  - (E) AP endonuclease
5. The key mechanistic failure in patients with xeroderma pigmentosum involves which of the following?
- (A) Mutation in the primase gene
  - (B) Inability to excise a section of the UV-damaged DNA
  - (C) Mutation of one of the mismatch repair components
  - (D) Inability to synthesize DNA across the damaged region
  - (E) Loss of proofreading capacity



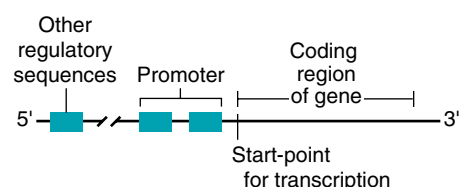
# 14 Transcription: Synthesis of RNA

Synthesis of RNA from a **DNA template** is called **transcription**. Genes are transcribed by enzymes called **RNA polymerases** that generate a **single-stranded RNA** identical in sequence (with the exception of U in place of T) to one of the strands of the double-stranded DNA. The DNA strand that directs the sequence of nucleotides in the RNA by **complementary base-pairing** is the template strand. The RNA strand that is initially generated is **the primary transcript**. The DNA **template is copied** in the **3' to 5'** direction, and the **RNA transcript is synthesized** in the **5' to 3'** direction. RNA polymerases differ from DNA polymerases in that they can **initiate** the **synthesis** of new strands in the absence of a primer.

In addition to catalyzing the polymerization of **ribonucleotides**, RNA polymerases must be able to recognize the appropriate gene to transcribe, the appropriate strand of the double-stranded DNA to copy, and the **startpoint** of transcription (Fig. 14.1). Specific sequences on DNA, called **promoters**, determine where the RNA polymerase binds and how frequently it initiates transcription. Other regulatory sequences, such as **promoter-proximal elements** and **enhancers**, also affect the frequency of transcription.

In **bacteria**, a **single RNA polymerase** produces the primary transcript precursors for all three major classes of RNA: messenger RNA (mRNA), ribosomal RNA (rRNA), and transfer RNA (tRNA). Because bacteria do not contain nuclei, ribosomes bind to mRNA as it is being transcribed, and protein synthesis occurs simultaneously with transcription.

**Eukaryotic genes** are transcribed in the nucleus by **three different RNA polymerases**, each principally responsible for one of the major classes of RNA. The primary transcripts are **modified** and **trimmed** to produce the mature RNAs. The precursors of **mRNA** (called **pre-mRNA**) have a **guanosine “cap”** added at the 5'-end and a **poly(A) “tail”** at the 3'-end. **Exons**, which contain the coding sequences for the proteins, are separated in **pre-mRNA** by **introns**, regions that have no coding function. During **splicing reactions**, introns are removed and the exons connected to form the mature mRNA. In eukaryotes, tRNA and rRNA precursors are also modified and trimmed, although not as extensively as pre-mRNA.



**Fig. 14.1.** Regions of a gene. A gene is a segment of DNA that functions as a unit to generate an RNA product or, through the processes of transcription and translation, a polypeptide chain. The transcribed region of a gene contains the template for synthesis of an RNA, which begins at the startpoint. A gene also includes regions of DNA that regulate production of the encoded product, such as a promoter region. In a structural gene, the transcribed region contains the coding sequences that dictate the amino acid sequence of a polypeptide chain.



## THE WAITING ROOM



**Anne Niemick** is a 4-year-old girl of Mediterranean ancestry whose height and body weight are below the 20th percentile for girls of her age. She is listless, tires easily, and complains of loss of appetite and shortness of breath on exertion. A dull pain has been present in her right upper quadrant for



The thalassemias are a heterogeneous group of hereditary anemias that constitute the most common gene disorder in the world, with a carrier rate of almost 7%. The disease was first discovered in countries around the Mediterranean Sea and was named for the Greek word “thalassa” meaning “sea”. However, it is also present in areas extending into India and China that are near the equator.

The thalassemia syndromes are caused by mutations that decrease or abolish the synthesis of the  $\alpha$  or  $\beta$  chains in the adult hemoglobin A tetramer. Individual syndromes are named according to the chain whose synthesis is affected and the severity of the deficiency. Thus, in  $\beta^0$  thalassemia, the superscript 0 denotes none of the  $\beta$  chain is present; in  $\beta^+$  thalassemia, the + denotes a partial reduction in the synthesis of the  $\beta$  chain. More than 170 different mutations have been identified that cause  $\beta$  thalassemia; most of these interfere with the transcription of  $\beta$ -globin mRNA or its processing or translation.

the last 3 months. Her complexion is slate-gray and she appears pale. Initial laboratory studies indicate a severe anemia (decreased red blood cell count) with a hemoglobin of 6.2 g/dL (reference range, 12–16). A battery of additional hematologic tests shows that Anne has  $\beta^+$ -thalassemia, intermediate type.



**Ivy Sharer**, a patient with AIDS (see Chapters 12 and 13), has developed a cough that produces a gray, slightly blood-tinged sputum. A chest radiograph indicates infiltrates in the cavities of both upper lung fields (cavitary infiltrates). A stain of sputum shows the presence of acid-fast bacilli, suggesting a diagnosis of pulmonary tuberculosis caused by *Mycobacterium tuberculosis*.



**Amanda Tin** picked mushrooms in a wooded area near her home. A few hours after eating one small mushroom, she experienced mild nausea and diarrhea. She brought a mushroom with her to the hospital emergency room. A poison expert identified it as *Amanita phalloides* (the “death cap”). These mushrooms contain the toxin  $\alpha$ -amanitin.

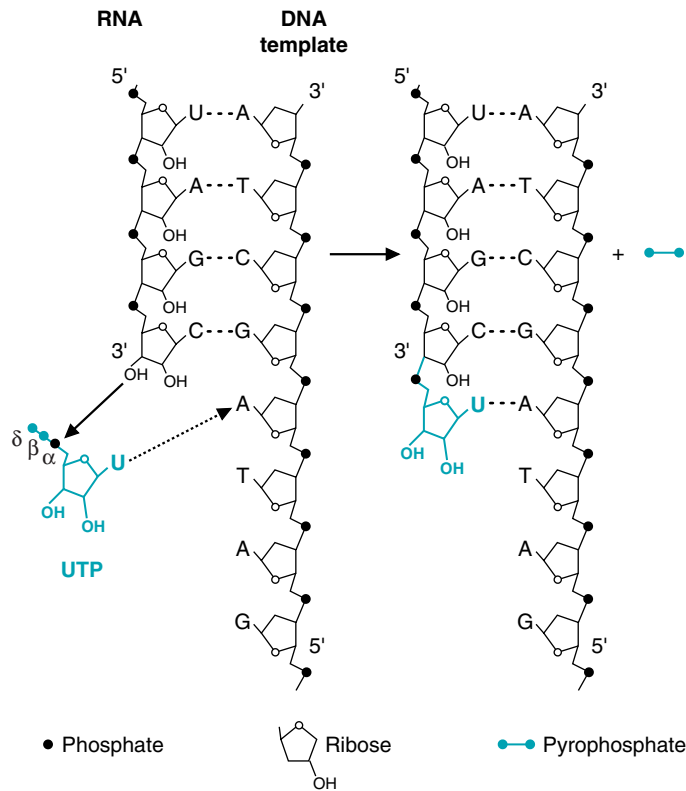


**Sis Lupus**, a 28-year-old computer programmer, notes increasing fatigue, pleuritic chest pain, and a nonproductive cough. In addition, she complains of joint pains, especially in her hands. A rash on both cheeks and the bridge of her nose (“butterfly rash”) has been present for the last 6 months. Initial laboratory studies indicate a subnormal white blood cell count and a mild reduction in hemoglobin. Tests result in a diagnosis of systemic lupus erythematosus (SLE) (frequently called lupus).

## I. ACTION OF RNA POLYMERASE

Transcription, the synthesis of RNA from a DNA template, is carried out by RNA polymerases (Fig. 14.2). Like DNA polymerases, RNA polymerases catalyze the formation of ester bonds between nucleotides that base-pair with the complementary nucleotides on the DNA template. Unlike DNA polymerases, RNA polymerases can initiate the synthesis of new chains in the absence of primers. They also lack the 3' to 5' exonuclease activity found in DNA polymerases. A strand of DNA serves as the template for RNA synthesis and is copied in the 3' to 5' direction. Synthesis of the new RNA molecule occurs in the 5' to 3' direction. The ribonucleoside triphosphates ATP, GTP, CTP, and UTP serve as the precursors. Each nucleotide base sequentially pairs with the complementary deoxyribonucleotide base on the DNA template (A, G, C, and U pair with T, C, G and A, respectively). The polymerase forms an ester bond between the  $\alpha$ -phosphate on the ribose 5'-hydroxyl of the nucleotide precursor and the ribose 3'-hydroxyl at the end of the growing RNA chain. The cleavage of a high-energy phosphate bond in the nucleotide triphosphate and release of pyrophosphate (from the  $\beta$  and  $\gamma$  phosphates) provides the energy for this polymerization reaction. Subsequent cleavage of the pyrophosphate by a pyrophosphatase also helps to drive the polymerization reaction forward by removing a product.

RNA polymerases must be able to recognize the startpoint for transcription of each gene and the appropriate strand of DNA to use as a template. They also must be sensitive to signals that reflect the need for the gene product and control the frequency of transcription. A region of regulatory sequences called the promoter, usually contiguous with the transcribed region, controls the binding of RNA polymerase to DNA and identifies the startpoint (see Fig. 14.1). The frequency of transcription is controlled by regulatory sequences within the promoter, nearby the promoter (promoter-proximal elements), and by other regulatory sequences, such as enhancers, that may be located at considerable distances, sometimes thousands of nucleotides, from the startpoint. Both the promoter-proximal elements, and the enhancers interact with proteins that stabilize RNA polymerase binding to the promoter.



**Fig. 14.2.** RNA synthesis. The  $\alpha$ -phosphate from the added nucleotide (shown in black) connects the ribosyl groups.

## II. TYPES OF RNA POLYMERASES

Bacterial cells have a single RNA polymerase that transcribes DNA to generate all of the different types of RNA (mRNA, rRNA, and tRNA). The RNA polymerase of *Escherichia coli* contains four subunits ( $\alpha_2\beta\beta'$ ), which form the core enzyme. Another protein called a  $\sigma$  (sigma) factor binds the core enzyme and directs binding of RNA polymerase to specific promoter regions of the DNA template. The  $\sigma$  factor dissociates shortly after transcription begins. *E. coli* has a number of different  $\sigma$  factors that recognize the promoter regions of different groups of genes. The major  $\sigma$  factor is  $\sigma^{70}$ , a designation related to its molecular weight of 70,000 Daltons.

In contrast to prokaryotes, eukaryotic cells have three RNA polymerases (Table 14.1). Polymerase I produces most of the rRNAs, polymerase II produces mRNA, and polymerase III produces small RNAs, such as tRNA and 5S rRNA. All of these RNA polymerases have the same mechanism of action. However, they recognize different types of promoters.

### A. Sequences of Genes

Double-stranded DNA consists of a coding strand and a template strand (Fig. 14.3). The DNA template strand is the strand that is actually used by RNA polymerase during the process of transcription. It is complementary and antiparallel both to the coding (nontemplate) strand of the DNA and to the RNA transcript produced from the template. Thus, the coding strand of the DNA is identical in base sequence and direction to the RNA transcript, except, of course, that wherever this DNA strand contains a T, the RNA transcript contains a U. By convention, the



Patients with acquired immune deficiency syndrome (AIDS) frequently develop tuberculosis. After **Ivy Sharer's** sputum stain suggested that she had tuberculosis, a multidrug antituberculous regimen, which includes an antibiotic of the rifamycin family (rifampin), was begun. A culture of her sputum was taken to confirm the diagnosis.

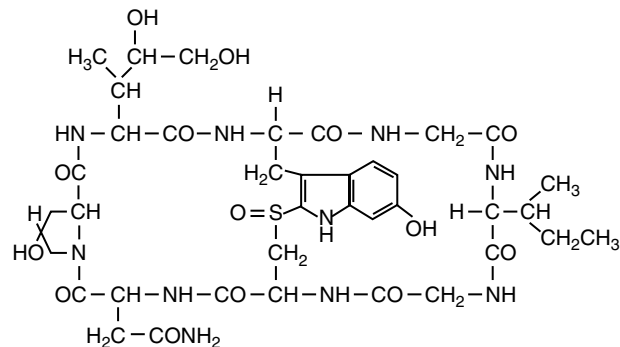
Rifampin inhibits bacterial RNA polymerase, selectively killing the bacteria that cause the infection. The nuclear RNA polymerase from eukaryotic cells is not affected. Although rifampin can inhibit the synthesis of mitochondrial RNA, the concentration required is considerably higher than that used for treatment of tuberculosis.

**Table 14.1** Products of Eukaryotic RNA Polymerases

|                     |                            |
|---------------------|----------------------------|
| RNA polymerase I:   | RNA                        |
| RNA polymerase II:  | mRNA                       |
| RNA polymerase III: | tRNA +<br>other small RNAs |



The mushrooms picked by **Amanda Tin** contained  $\alpha$ -amanitin, an inhibitor of eukaryotic RNA polymerases. It is particularly effective at blocking the action of RNA polymerase II. This toxin initially causes gastrointestinal disturbances, then electrolyte imbalance and fever, followed by liver and kidney dysfunction. Between 40 and 90% of the individuals who ingest  $\alpha$ -amanitin die within a few days.



$\alpha$ -Amanitin

nucleotide sequence of a gene is represented by the letters of the nitrogenous bases of the coding strand of the DNA duplex. It is written from left to right in the 5' to 3' direction.

During translation, mRNA is read 5' to 3' in sets of three bases, called codons, that determine the amino acid sequence of the protein (see Fig. 14.3). Thus, the base sequence of the coding strand of the DNA can be used to determine the amino acid sequence of the protein. For this reason, when gene sequences are given, they refer to the coding strand.

A gene consists of the transcribed region and the regions that regulate transcription of the gene (e.g., promoter and enhancer regions) (Fig. 14.4). The base in the coding strand of the gene serving as the startpoint for transcription is numbered +1. This nucleotide corresponds to the first nucleotide incorporated into the RNA at the 5'-end of the transcript. Subsequent nucleotides within the transcribed region of the gene are numbered +2, +3, etc., toward the 3'-end of the gene. Untranscribed sequences to the left of the startpoint, known as the 5'-flanking region of the gene, are numbered -1, -2, -3, etc., starting with the nucleotide (-1) immediately to the left of the startpoint (+1) and moving from right to left. By analogy to a river, the sequences to the left of the startpoint are said to be upstream from the startpoint and those to the right are said to be downstream.

## B. Recognition of Genes by RNA Polymerase

For genes to be expressed, RNA polymerase must recognize the appropriate point to start transcription and the strand of the DNA to transcribe (the template strand).

**DNA coding strand** 5'-ATGCCAGTAGGCCACTTGTCA-3'

**DNA template strand** 3'-TACGGTCATCCGGTGAACAGT-5'

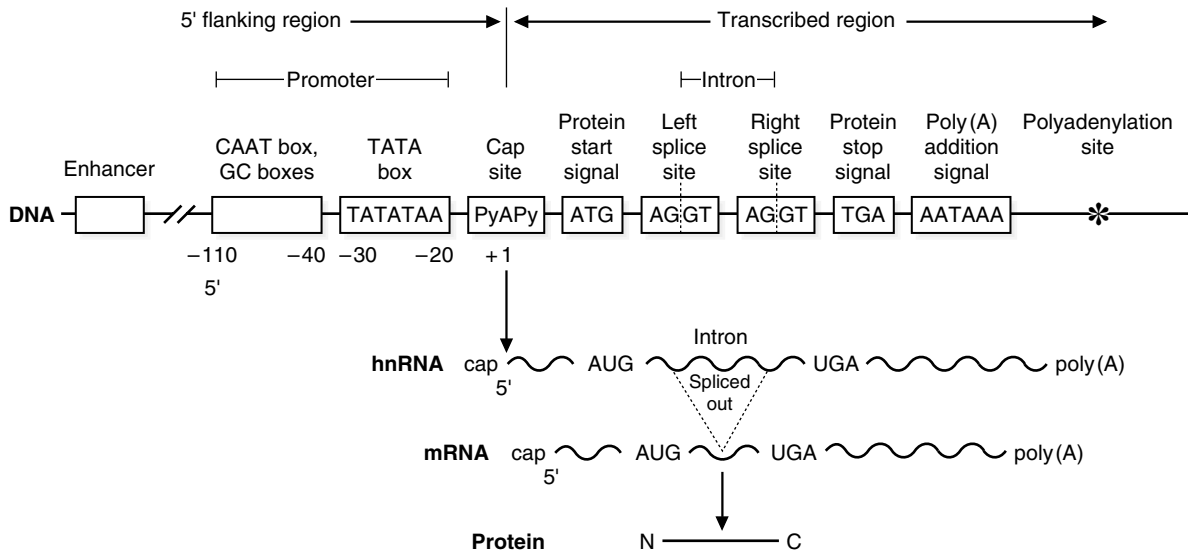
**mRNA** 5'-AUG CCA GUA GGC CAC UUG UCA-3'

**Protein** N-Met-Pro-Val-Gly-His-Leu-Ser-C



The two strands of DNA are antiparallel, with complementary nucleotides at each position. Thus, each strand would produce a different mRNA, resulting in different codons for amino acids and a different protein product. Therefore, it is critical that RNA polymerase transcribe the correct strand.

**Fig. 14.3.** Relationship between the coding strand of DNA, the DNA template strand, the mRNA transcript, and the protein produced from the gene. The bases in mRNA are used in sets of three (called codons) to specify the order of the amino acids inserted into the growing polypeptide chain during the process of translation (see Chapter 15).



**Fig. 14.4.** A schematic view of a eukaryotic gene, and steps required to produce a protein product. The gene consists of promoter and transcribed regions. The transcribed region contains introns, which do not contain coding sequence for proteins, and exons, which do carry the coding sequences for proteins. The first RNA form produced is heterogenous nuclear RNA (hn RNA), which contains both intronic and exonic sequences. The hnRNA is modified such that a cap is added at the 5' end (cap site), and a poly-A tail added to the 3' end. The introns are removed (a process called splicing) to produce the mature mRNA, which leaves the nucleus to direct protein synthesis in the cytoplasm. Py is pyrimidine (C or T).

RNA polymerase also must recognize which genes to transcribe because transcribed genes are only a small fraction of the total DNA. The genes that are transcribed differ from one type of cell to another and change with changes in physiologic conditions. The signals in DNA that RNA polymerase recognizes are called promoters. Promoters are sequences in DNA (often composed of smaller sequences called boxes or elements) that determine the startpoint and the frequency of transcription. Because they are located on the same molecule of DNA and near the gene they regulate, they are said to be *cis* acting (i.e., “*cis*” refers to acting on the same side). Proteins that bind to these DNA sequences and facilitate or prevent the binding of RNA polymerase are said to be *trans* acting.

### C. Promoter Regions of Genes for mRNA

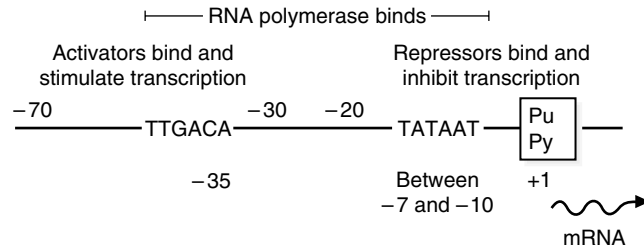
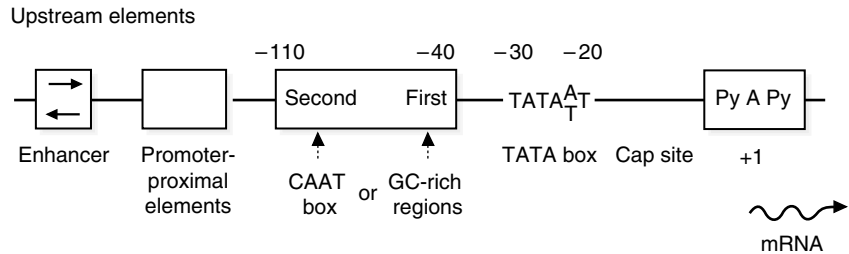
The binding of RNA polymerase and the subsequent initiation of gene transcription involves a number of consensus sequences in the promoter regions of the gene (Fig. 14.5). A consensus sequence is the sequence most commonly found in a given region when many genes are examined. In both prokaryotes and eukaryotes, an adenine- and thymine-rich consensus sequence in the promoter determines the startpoint of transcription by binding proteins that facilitate the binding of RNA polymerase. In the prokaryote *E. coli*, this consensus sequence is TATAAT, which is known as the TATA or Pribnow box. It is centered about  $-10$  and is recognized by the sigma factor  $\sigma^{70}$ . A similar sequence in the  $-25$  region of eukaryotic genes has a consensus sequence of TATA(A/T)A. (The (A/T) in the fifth position indicates that either A or T occurs with equal frequency.) This eukaryotic sequence is also known as a TATA box, but is sometimes named the Hogness or Hogness-Goldberg box after its discoverers. Other consensus sequences involved in binding of RNA polymerase are found further upstream in the promoter region (see Fig. 14.5). Bacterial promoters contain a sequence TTGACA in the  $-35$  region. Eukaryotes frequently have CAAT boxes and GC-rich sequences in the region between  $-40$  and  $-110$ . Eukaryotic genes also contain promoter-proximal elements (in the region of  $-100$



**Anne Niemick** has a  $\beta^+$  thalassemia classified clinically as  $\beta$ -thalassemia intermedia. She produces an intermediate amount of functional  $\beta$  globin chains (her hemoglobin is 6.2 g/dL; normal is 12–16).  $\beta$ -thalassemia intermedia is usually the result of two different mutations (one that mildly affects the rate of synthesis of  $\beta$ -globin and one severely affecting its rate of synthesis), or, less frequently, homozygosity for a mild mutation in the rate of synthesis, or a complex combination of mutations. For example, mutations within the promoter region of the  $\beta$ -globin gene could result in a significantly decreased rate of  $\beta$ -globin synthesis in an individual who is homozygous for the allele, without completely abolishing synthesis of the protein.

Two of the point mutations that result in a  $\beta^+$  phenotype are within the TATA box (A  $\rightarrow$  G or A  $\rightarrow$  C in the  $-28$  to  $-31$  region). These mutations reduce the accuracy of the startpoint of transcription so that only 20 to 25% of the normal amount of  $\beta$ -globin is synthesized. Other mutations that also reduce the frequency of  $\beta$ -globin transcription have been observed further upstream in the promoter region ( $-87$  C  $\rightarrow$  G and  $-88$  C  $\rightarrow$  T).

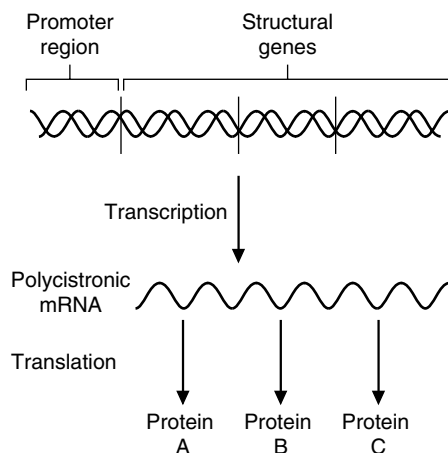


**Prokaryotic promoters****Eukaryotic promoters**

**Fig. 14.5.** Prokaryotic and eukaryotic promoters. The promoter-proximal region contains binding sites for transcription factors which that can accelerate the rate at which RNA polymerase binds to the promoter. Pu = purine; Py = pyrimidine.



What property of an AT-rich region of a DNA double helix makes it suitable to serve as a recognition site for the startpoint of transcription?



**Fig. 14.6.** Bacterial operon. A cistron encodes a single polypeptide chain. In bacteria, a single promoter may control transcription of an operon containing many cistrons. A single polycistronic mRNA is transcribed. Its translation produces a number of polypeptide chains.

to  $-200$ ), which are sites that bind other gene regulatory proteins. Genes vary in the number of such sequences present.

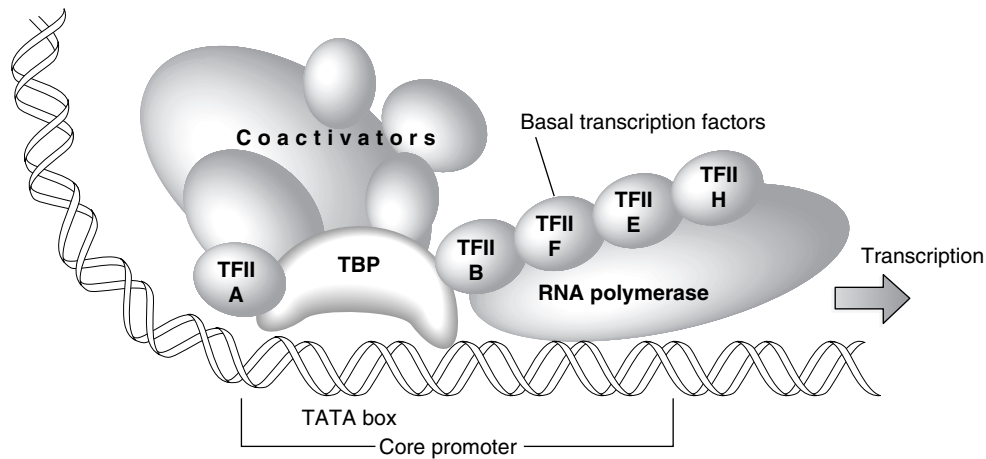
In bacteria, a number of protein-producing genes may be linked together and controlled by a single promoter. This genetic unit is called an operon (Fig. 14.6). Proteins bind to the promoter and either inhibit or facilitate transcription of the operon. Repressors are proteins that bind to a region in the promoter known as the operator and inhibit transcription by preventing the binding of RNA polymerase to DNA. Activators are proteins that stimulate transcription by binding within the  $-35$  region or upstream from it, facilitating the binding of RNA polymerase. (Operons are described in more detail in Chapter 16.)

In eukaryotes, proteins known as general transcription factors (or basal factors) bind to the TATA box and facilitate the binding of RNA polymerase II, the polymerase that transcribes mRNA (Fig. 14.7). This binding process involves at least six basal transcription factors (labeled as TFIIs, transcription factors for RNA polymerase II). The TATA-binding protein (TBP), which is a component of TFIID, initially binds to the TATA box. TFIID consists of both the TBP and a number of transcriptional coactivators. TFIIA and TFIIB interact with TBP. RNA polymerase II binds to the complex of transcription factors and to DNA, and is aligned at the startpoint for transcription. TFIIE, TFIIIF, and TFIIH subsequently bind, cleaving adenosine triphosphate (ATP), and transcription of the gene is initiated.

With only these transcription (or basal) factors and RNA polymerase II attached (the basal transcription complex), the gene is transcribed at a low or basal rate.



TFIIH plays a number of roles in both transcription and DNA repair. In both processes, it acts as an ATP-dependent DNA helicase, unwinding DNA for either transcription or repair to occur. Two of the forms of xeroderma pigmentosum (XPB and XPD; see Chapter 13) arise from mutations within two different helicase subunits of TFIIH. TFIIH also contains a kinase activity, and RNA polymerase II is phosphorylated by this factor during certain phases of transcription.



**Fig. 14.7.** Transcription apparatus. The TATA-binding protein (TBP), a component of TFIID, binds to the TATA box. Transcription factors TFII A and B bind to TBP. RNA polymerase binds, then TFII E, F, and H bind. This complex can transcribe at a basal level. Some coactivator proteins are present as a component of TFIID, and these can bind to other regulatory DNA binding proteins (called specific transcription factors or transcriptional activators).

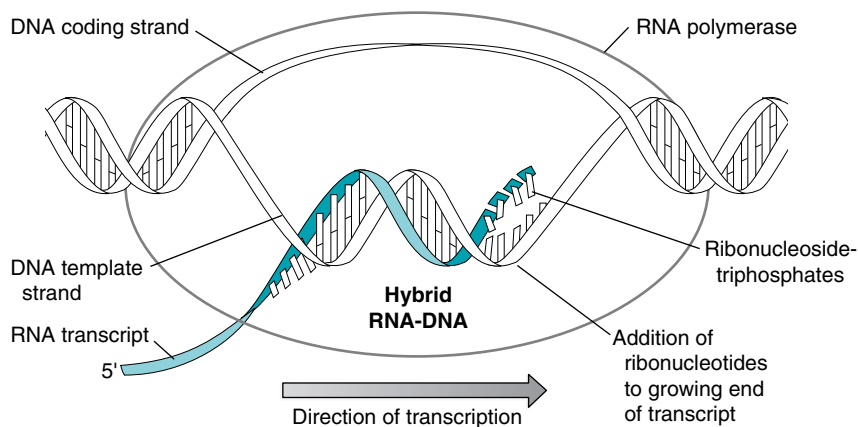
The rate of transcription can be further increased by binding of other regulatory DNA binding proteins to additional gene regulatory sequences (such as the promoter proximal or enhancer regions). These regulatory DNA binding proteins are called gene-specific transcription factors (or transactivators) because they are specific to the gene involved (see Chapter 16). They interact with coactivators in the basal transcription complex.



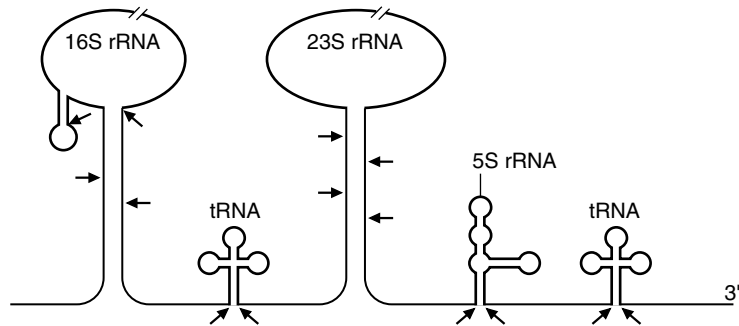
In regions in which DNA is being transcribed, the two strands of the DNA must be separated. AT base pairs in DNA are joined by only two hydrogen bonds, whereas GC pairs have three hydrogen bonds. Therefore, in AT-rich regions of DNA, the two strands can be separated more readily than in regions that contain GC base pairs.

### III. TRANSCRIPTION OF BACTERIAL GENES

In bacteria, binding of RNA polymerase with a  $\sigma$  factor to the promoter region of DNA causes the two DNA strands to unwind and separate within a region approximately 10 to 20 nucleotides in length. As the polymerase transcribes the DNA, the untranscribed region of the helix continues to separate, whereas the transcribed region of the DNA template rejoins its DNA partner (Fig. 14.8). The sigma factor is released when the growing RNA chain is approximately 10 nucleotides long. The elongation reactions continue until the RNA polymerase encounters a transcription termination signal. One type of termination signal involves the formation of a hairpin loop in the transcript, preceding a number of U residues. The second type of



**Fig. 14.8.** Mechanism of transcription.



**Fig. 14.9.** Bacterial rRNA and tRNA transcripts. One large precursor is cleaved (at arrows) to produce 16S, 23S, and 5S rRNA and some tRNAs.

mechanism for termination involves the binding of a protein, the rho factor, which causes release of the RNA transcript from the template. The signal for both termination processes is the sequence of bases in the newly synthesized RNA.

A cistron is a region of DNA that encodes a single polypeptide chain. In bacteria, mRNA is usually generated from an operon as a polycistronic transcript (one that contains the information to produce a number of different proteins). The polycistronic transcript is translated as it is being transcribed. This transcript is not modified and trimmed, and it does not contain introns (regions within the coding sequence of a transcript that are removed before translation occurs). Several different proteins are produced during translation of the polycistronic transcript, one from each cistron (see Fig. 14.6).

In prokaryotes, rRNA is produced as a single, long transcript that is cleaved to produce the 16S, 23S, and 5S ribosomal RNAs. tRNA is also cleaved from larger transcripts (Fig. 14.9). One of the cleavage enzymes, RNase P, is a protein containing an RNA molecule. This RNA actually catalyzes the cleavage reaction.

#### IV. TRANSCRIPTION OF EUKARYOTIC GENES

The process of transcription in eukaryotes is similar to that in prokaryotes. RNA polymerase binds to the transcription factor complex in the promoter region and to the DNA, the helix unwinds within a region near the startpoint of transcription, DNA strand separation occurs, synthesis of the RNA transcript is initiated, and the RNA transcript is elongated, copying the DNA template. The DNA strands separate as the polymerase approaches and rejoin as the polymerase passes.

One of the major differences between eukaryotes and prokaryotes is that eukaryotes have more elaborate mechanisms for processing the transcripts, particularly the precursors of mRNA (pre-mRNA). Eukaryotes also have three polymerases, rather than just the one present in prokaryotes. Other differences include the facts that eukaryotic mRNA usually contains the coding information for only one polypeptide chain and that eukaryotic RNA is transcribed in the nucleus and migrates to the cytoplasm where translation occurs.

##### A. Synthesis of Eukaryotic mRNA

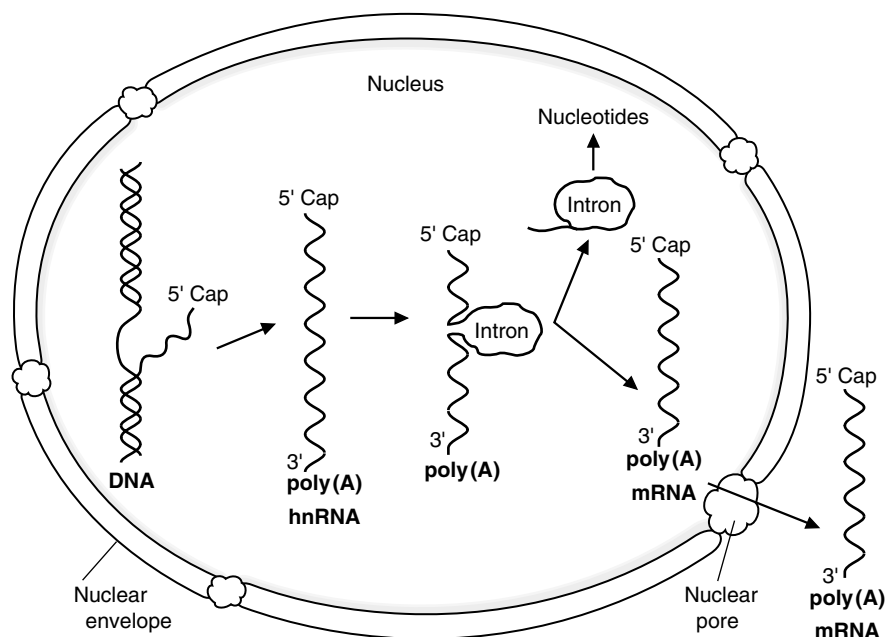
In eukaryotes, extensive processing of the primary transcript occurs before the mature mRNA is formed and can migrate to the cytosol, where it is translated into a protein product. RNA polymerase II synthesizes a large primary transcript from the template strand that is capped at the 5' end as it is transcribed (Fig. 14.10). The transcript also rapidly acquires a poly(A) tail at the 3' end. Pre-mRNAs thus contain untranslated regions at both the 5' and 3' ends (the leader and trailing sequences, respectively). These untranslated regions are retained in the mature



Although each eukaryotic mRNA only codes for one polypeptide chain, a number of proteins can contain different polypeptide chains or have multiple active sites, allowing the protein to catalyze more than one reaction.



The terms hnRNA (heterogeneous nuclear RNA) and pre-mRNA are both used to denote mRNA precursors. The term hnRNA was originally applied to a pool of RNA molecules in the nucleus that were rapidly synthesized and varied greatly in size. These RNA molecules are now known to be the mRNA precursors that vary greatly in size because they contain exons that encode different sizes of polypeptide chains and introns that vary in amount and size.



**Fig. 14.10.** Overview of mRNA synthesis. Transcription produces hnRNA from the DNA template. hnRNA processing involves addition of a 5'-cap and a poly(A) tail and splicing to join exons and remove introns. The product, mRNA, migrates to the cytoplasm, where it will direct protein synthesis.

mRNA. The coding region of the pre-mRNA, which begins with the start codon for protein synthesis and ends with the stop codon, contains both exons and introns. Exons consist of the nucleotide codons that dictate the amino acid sequence of the eventual protein product. Between the exons, interspersing regions called introns contain nucleotide sequences that are removed by splicing reactions to form the mature RNA. The mature RNA thus contains a leader sequence (that includes the cap), a coding region comprising exons, and a tailing sequence that includes the poly(A) tail.

This mature mRNA complexes with the poly(A) binding protein and other proteins. It travels through pores in the nuclear envelope into the cytoplasm. There it combines with ribosomes and directs the incorporation of amino acids into proteins.

### 1. TRANSCRIPTION AND CAPPING OF mRNA

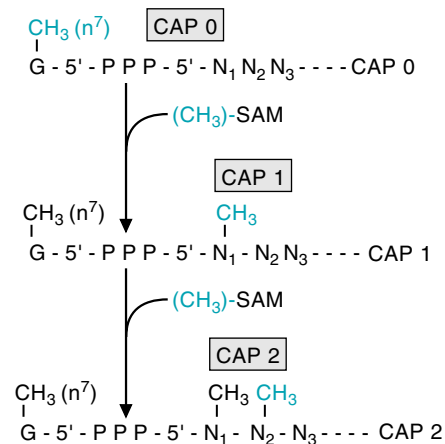
“Capping” of the primary transcript synthesized by RNA polymerase II occurs at its 5'-end as it is being transcribed (Fig.14.11). The 5'-terminal, the initial nucleotide of the transcript, is a pyrimidine with three phosphate groups attached to the 5'-hydroxyl of the ribose. To form the cap, the terminal triphosphate loses one phosphate, forming a 5'-diphosphate. The  $\beta$ -phosphate of the diphosphate then attacks the  $\alpha$ -phosphate of GTP, liberating pyrophosphate, and forming an unusual 5' to 5' triphosphate linkage. A methyl group is transferred from S-adenosylmethionine (SAM), a universal methyl donor, to position 7 of the added guanine ring. Methylation also occurs on the ribose 2'-hydroxyl group in the terminal nucleotide to which the cap is attached, and sometimes the 2'-hydroxyl group of the adjacent nucleotide ribose. This cap “seals” the 5' end of the primary transcript and decreases the rate of degradation. It also serves as a recognition site for the binding of the mature mRNA to a ribosome at the initiation of protein synthesis.

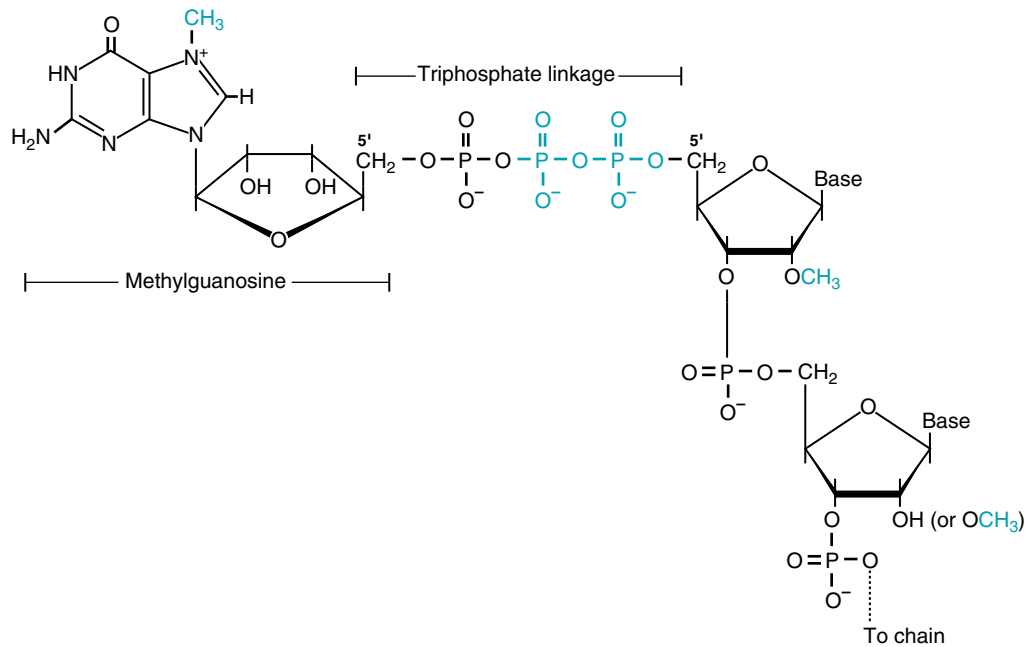


Once SAM (S-adenosylmethionine) donates its methyl group, it must be regenerated by reactions that require the vitamins folate and B<sub>12</sub>. Thus formation of mRNA is also one of the processes affected by a deficiency of these vitamins.



There are three different types of methyl caps, shown in blue: CAP0 refers to the methylated guanosine (on the nitrogen at the seven position, N<sup>7</sup>) added in the 5' to 5' linkage to the mRNA; CAP1 refers to CAP0 with the addition of a methyl to the 2' carbon of ribose on the nucleotide (N<sub>1</sub>) at the 5' end of the chain; and CAP2 refers to CAP1 with the addition of another 2' methyl group to the next nucleotide (N<sub>2</sub>). The methyl groups are donated by S-adenosylmethionine (SAM).

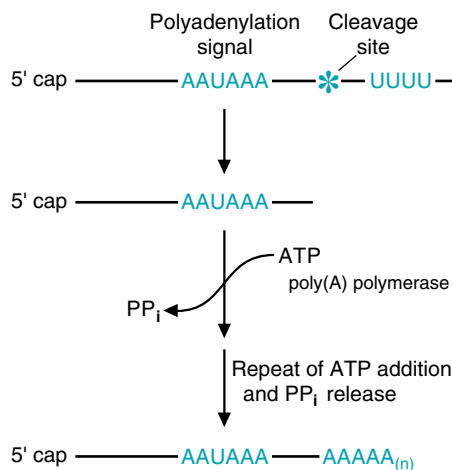




**Fig. 14.11.** The cap structure in eukaryotic mRNA. The phosphates in blue originated from the original RNA transcript; the phosphate in black comes from GTP.

## 2. ADDITION OF A POLY(A) TAIL

After the RNA polymerase transcribes the stop codon for protein translation, it passes a sequence called the polyadenylation signal (AAUAAA) (Fig. 14.12). It continues past the polyadenylation signal until it reaches an unknown, and possibly unspecific, termination signal many nucleotides later. However, as the primary transcript is released from the RNA polymerase elongation complex, an enzyme complex binds to the polyadenylation signal and cleaves the primary transcript approximately 10 to 20 nucleotides downstream, thereby forming the 3' end. After this cleavage, a poly(A) tail that can be over 200 nucleotides in length is added to the 3'-end. Thus, there is no poly(dT) sequence in the DNA template that corresponds to this tail; it is added posttranscriptionally. ATP serves as the precursor for the sequential addition of the adenine nucleotides. They are added one at a time, with poly(A) polymerase catalyzing each addition. The poly(A) tail is a protein binding site that protects the mRNA from degradation.



**Fig. 14.12.** Synthesis of the poly(A) tail. As RNA polymerase continues to transcribe the DNA, enzymes cleave the transcript (hnRNA) at a point 10–20 nucleotides beyond an AAUAAA sequence, just before a run of Us (or Gs). Approximately 250 adenine nucleotides are then added to the 3'-end of the hnRNA, one at a time, by poly(A) polymerase.



Within a few days of initiation of treatment for tuberculosis, laboratory staining results of **Ivy Sharer's** sputum confirmed the diagnosis of pulmonary tuberculosis caused by *M. tuberculosis*. Therefore, the multidrug therapy, which included the antibiotic rifampin, was continued. Rifampin binds to the RNA polymerases of several bacteria. *M. tuberculosis* rapidly develops resistance to rifampin through mutations that result in an RNA polymerase that cannot bind the complex structure. Simultaneous treatment with a drug that works through a different mechanism decreases the selective advantage of the mutation and the rate at which resistance develops.



The presence of a poly (A) tail on eukaryotic mRNA allows this form of RNA to be easily separated from the more abundant rRNA. After extracting all of the RNA from a cell, the total RNA is applied to a column of beads to which oligo-dT has been covalently attached. As the mRNA flows through the column, its poly (A) tail will base pair with the oligo-dT, and the mRNA will become bound to the column. All other types of RNA will flow through the column and not bind to the beads. The bound mRNA can then be eluted from the column by changing the ionic strength of the buffer.

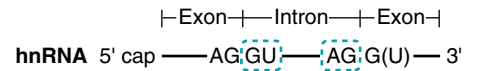


### 3. REMOVAL OF INTRONS

Eukaryotic pre-mRNA transcripts contain regions known as exons and introns. Exons appear in the mature mRNA; introns are removed from the transcript and are not found in the mature mRNA (see Fig. 14.10). Introns, therefore, do not contribute to the amino acid sequence of the protein. Some genes contain 50 or more introns. These introns are carefully removed from the pre-mRNA transcript and the exons spliced together, so that the appropriate protein is produced from the gene.

The consensus sequences at the intron/exon boundaries of the pre-mRNA are AGGU (AGGT in the DNA). The sequences vary to some extent on the exon side of the boundaries, but almost all introns begin with a 5' GU and end with a 3' AG (Fig. 14.13). These intron sequences at the left splice site and the right splice site are, therefore, invariant. Because every 5' GU and 3' AG combination does not result in a functional splice site, clearly other features within the exon or intron help to define the appropriate splice sites. These features, which are currently being identified, involve the presence of positive- and negative-acting *cis*-regulatory sequences within the intron.

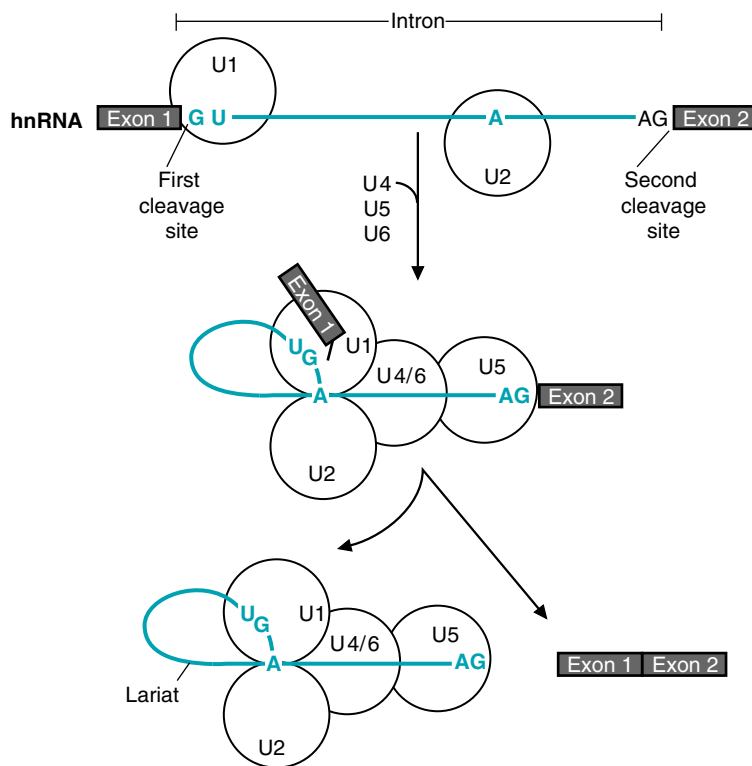
A complex structure known as a spliceosome ensures that exons are spliced together with great accuracy (Fig. 14.14). Small nuclear ribonucleoproteins



**Fig. 14.13.** Splice junctions in hnRNA. The intron sequences shown in blue dashed boxes are invariant. They always appear at this position in introns. The sequences on the exon side of the splice sites are more variable.



**Anne Niemick** has  $\beta^+$ -thalassemia (enough of the  $\beta$ -chain is produced to maintain blood hemoglobin levels above 6.0 g/dL). One mutation resulting in  $\beta^+$ -thalassemia is a point mutation (AATAAA  $\rightarrow$  AACAAA) that changes the sequence in hnRNA at the polyadenylation signal site from AAUAAA to AACAAA. Homozygous individuals with this mutation produce only one-tenth the amount of normal  $\beta$ -globin mRNA.



**Fig. 14.14.** Splicing process. Nuclear ribonucleoproteins (snurps  $U_1$  to  $U_6$ ) bind to the intron, causing it to form a loop. The complex is called a spliceosome. The  $U_1$  snurp binds near the first exon/intron junction, and  $U_2$  binds within the intron in a region containing an adenine nucleotide residue. Another group of snurps,  $U_4$ ,  $U_5$ , and  $U_6$ , binds to the complex, and the loop is formed. The phosphate attached to the G residue at the 5'-end of the intron forms a 2'-5' linkage with the 2'-hydroxyl group of the adenine nucleotide residue. Cleavage occurs at the end of the first exon, between the AG residues at the 3' end of the exon and the GU residues at the 5' end of the intron. The complex continues to be held in place by the spliceosome. A second cleavage occurs at the 3'-end of the intron after the AG sequence. The exons are joined together. The intron, shaped like a lariat, is released and degraded to nucleotides.



Some types of  $\beta^0$ -thalassemia (little or none of the hemoglobin  $\beta$  chain produced) are caused by homozygous mutations in the splice junction sequences at intron/exon boundaries. In some individuals, an AT replaces a GT in the gene at the 5' end of the first or second intron. Mutations also occur within the splice junction sequences at the 3'-end of introns (GT at the donor site 5'-end and AG at the acceptor site 3'-end). Mutations at either site totally abolish normal splicing and result in  $\beta^0$  thalassemia.

(snRNPs), called “snurps,” are involved in formation of the spliceosome. Because snurps are rich in uracil, they are identified by numbers preceded by a U.

Exons frequently code for separate functional or structural domains of proteins. Proteins with similar functional regions (e.g., ATP or NAD binding regions) frequently have similar domains, although their overall structure and amino acid sequence is quite different. A process known as exon shuffling has probably occurred throughout evolution, allowing new proteins to develop with functions similar to those of other proteins.

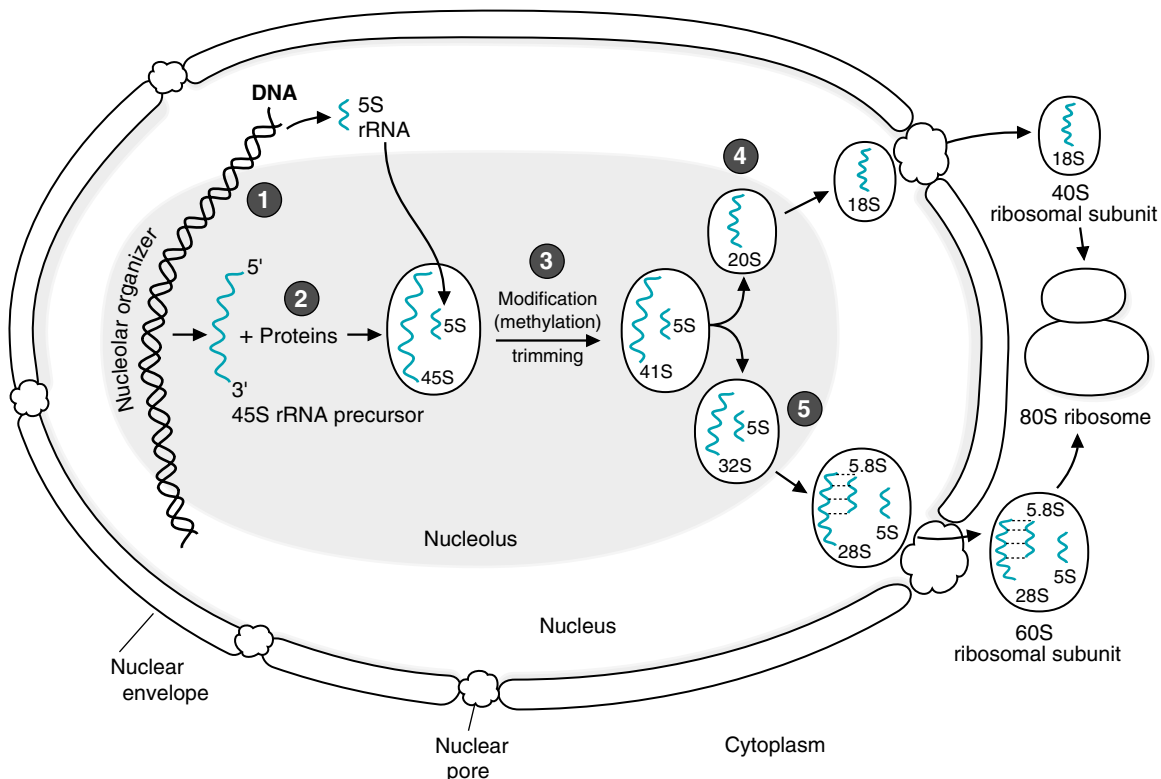
## B. Synthesis of Eukaryotic rRNA

Ribosomal RNAs (rRNAs) form the ribonucleoprotein complexes on which protein synthesis occurs. In eukaryotes, the rRNA gene exists as many copies in the nucleolar organizer region of the nucleus (Fig. 14.15, circle 1). Each gene produces a large, 45S transcript that is cleaved to produce the 18S, 28S, and 5.8S rRNAs. Approximately 1,000 copies of this gene are present in the human genome. The genes are linked in tan-



Systemic lupus erythematosus is an autoimmune disease characterized by a particular spectrum of autoantibodies against many cellular components, including chromatin, ribonucleoprotein, and cell membrane phospholipids. In this disorder, the body makes these antibodies against its own components. snRNPs are one of the targets of these antibodies. In fact, snRNPs were discovered as a result of studies using antibodies obtained from patients with SLE.

Tests were performed on **Sis Lupus's** blood to detect elevated levels of antinuclear antibodies (including antibodies to DNA, antibodies to histone, antibodies to ribonucleoproteins, and antibodies to nuclear antigens). The tests were strongly positive and, in conjunction with her symptoms, led to a diagnosis of systemic lupus erythematosus (SLE).



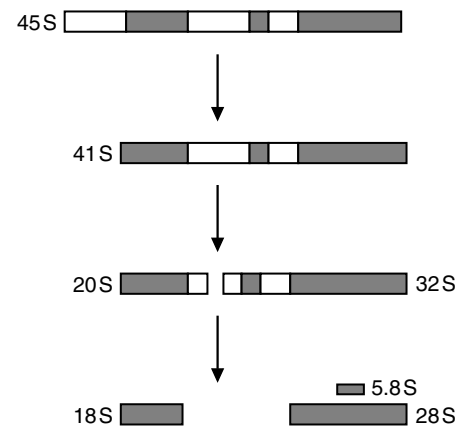
**Fig. 14.15.** rRNA and ribosome synthesis. The 5S rRNA is transcribed in the nucleoplasm and moves into the nucleolus. The other rRNAs are transcribed from DNA and mature in the nucleolus, forming the 40S and 60S ribosomal subunits, which migrate to the cytoplasm.

dem, separated by spacer regions that contain the termination signal for one gene and the promoter for the next. Promoters for rRNA genes are located in the 5'-flanking region of the genes and extend into the region surrounding the startpoint. rRNA genes caught in the act of transcription by electron micrographs show that many RNA polymerase I molecules can be attached to a gene at any given time, all moving toward the 3' end as the 45S rRNA precursors are synthesized.

As the 45S rRNA precursors are released from the DNA, they complex with proteins, forming ribonucleoprotein particles that generate the granular regions of the nucleolus (see Fig. 14.15, circle 2). Processing of the transcript occurs in the granular regions. 5S rRNA, produced by RNA polymerase III from genes located outside the nucleolus in the nucleoplasm, migrates into the nucleolus and joins the ribonucleoprotein particles.

One to two percent of the nucleotides of the 45S precursor become methylated, primarily on the 2'-hydroxyl groups of ribose moieties (see Fig. 14.15, circle 3). These methyl groups may serve as markers for cleavage of the 45S precursors and are conserved in the mature rRNA. A series of cleavages in the 45S transcripts occur to produce the mature rRNAs (Fig. 14.16).

In the production of cytoplasmic ribosomes in human cells, one portion of the 45S rRNA precursor becomes the 18S rRNA that, complexed with proteins, forms the small 40S ribosomal subunit (Fig. 14.15, circle 4). Another segment of the precursor folds back on itself and is cleaved, forming 28S rRNA, hydrogen-bonded to the 5.8S rRNA. The 5S rRNA, transcribed from nonnucleolar genes, and a number of proteins complex with the 28S and 5.8S rRNAs to form the 60S ribosomal subunit (Fig. 14.15, circle 5). The ribosomal subunits migrate through the nuclear pores. In the cytoplasm, the 40S and 60S ribosomal subunits interact with mRNA, forming the 80S ribosomes on which protein synthesis occurs.



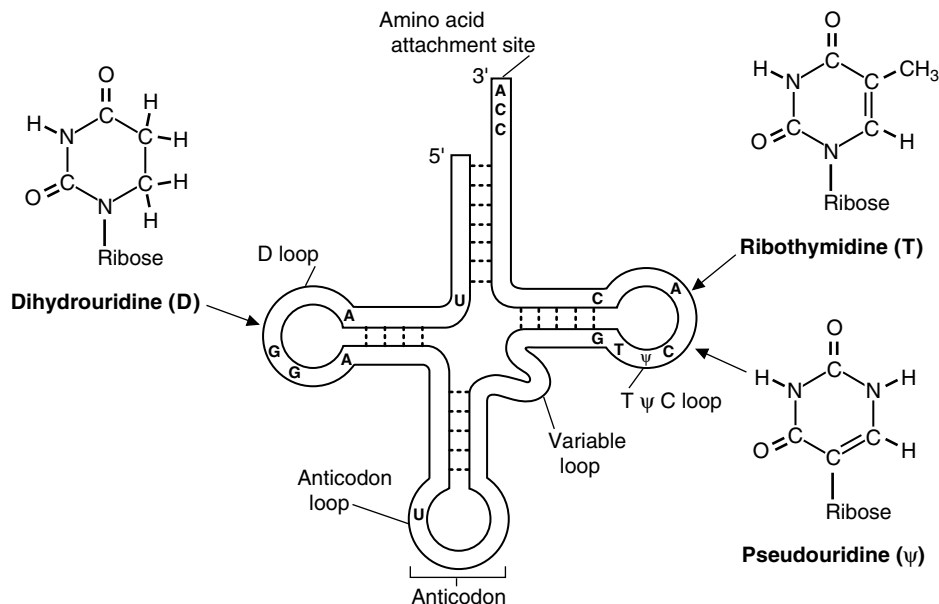
**Fig. 14.16** Maturation of the 45S rRNA precursor. The clear regions are removed, and the shaded regions become the mature rRNAs. (The 5S rRNA is not produced from this precursor.)



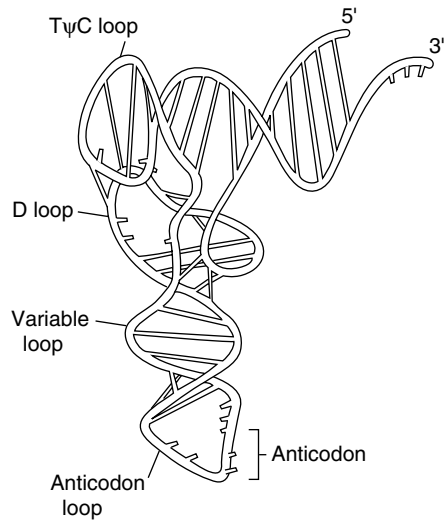
Some rRNA precursors contain introns within the regions that become the mature rRNAs. These introns are removed by the rRNA precursors themselves, which undergo self-splicing reactions. The rRNA precursors that catalyze their own splicing are known as ribozymes (see Chapter 12, III.A).

### C. Synthesis of Eukaryotic tRNA

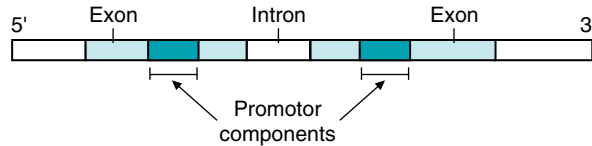
A transfer RNA has one binding site for a specific sequence of three nucleotides in mRNA (the anticodon site) and another binding site for the encoded amino acid. tRNAs thus ensure that the genetic code is translated into the correct sequence of



**Fig. 14.17** The tRNA cloverleaf. Bases that commonly occur in a particular position are indicated by letters. Base-pairing in stem regions is indicated by lines between the strands. The locations of the modified bases dihydrouridine (D), ribothymidine (T), and pseudouridine (Ψ) are indicated.



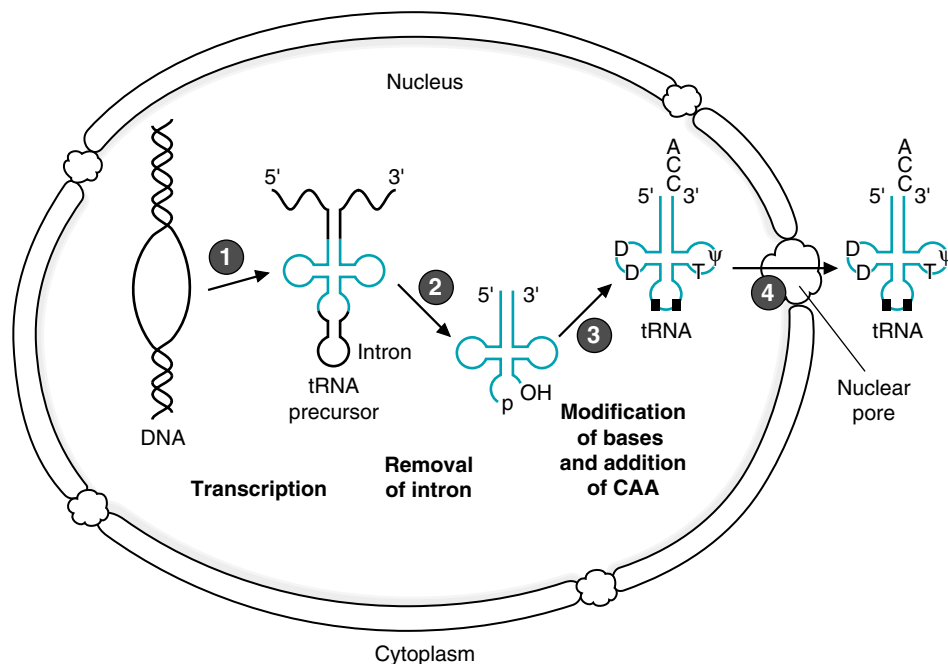
**Fig. 14.18.** The three-dimensional folding of tRNA. (Reprinted with permission from Kim SH, et al. *Science* 1974;185:436. Copyright 1974 American Association for the Advancement of Science.)



**Fig. 14.19.** Promoter for tRNA transcription. The segments of the genes from which the mature tRNA is produced are indicated by the light blue color. The two regions of the promoter lie within these segments, and are indicated by the dark blue color.

amino acids. At least 20 types of tRNAs occur in cells, one for every amino acid that is incorporated into growing polypeptide chains during the synthesis of proteins. tRNAs have a cloverleaf structure that folds into a three-dimensional L shape, and contains a number of bases that are modified posttranscriptionally (Fig. 14.17). The loop closest to the 5'-end is known as the D-loop because it contains dihydrouridine (D). The second, or anticodon, loop contains the trinucleotide anticodon that base-pairs with the codon on mRNA. The third loop (the TΨC loop) contains both ribothymidine (T) and pseudouridine (Ψ). A fourth loop, known as the variable loop because it varies in size, is frequently found between the anticodon and TΨC loops. Base-pairing occurs in the stem regions of tRNA, and a three-nucleotide sequence (e.g., CCA) at the 3'-end is the attachment site for the specific amino acid carried by each tRNA. Different tRNAs bind different amino acids. The three-dimensional structure of tRNA has been determined and is shown in Figure 14.18. tRNA is produced by RNA polymerase III, which recognizes a split promoter within the transcribed region of the gene (Fig. 14.19). One segment of the promoter is located between +8 and +19. A second segment is 30 to 60 base pairs downstream from the first.

tRNA precursors of approximately 100 nucleotides in length are generated. (Fig. 14.20, circle 1). The pre-tRNA assumes a cloverleaf shape and is subsequently cleaved at the 5'- and 3'-ends (see Fig. 14.20, circle 2). The enzyme that acts at the



**Fig. 14.20.** Overview of tRNA synthesis. D, T, Ψ, and ■ indicate modified bases. D = dihydrouracil, T = ribothymine, Ψ = pseudouridine, ■ = other modified bases

5'-end is RNase P, similar to the RNase P of bacteria. Both enzymes contain a small RNA (M1) that has catalytic activity and serves as an endonuclease. Some tRNA precursors contain introns that are removed by endonucleases. To close the opening, a 2'- or 3'-phosphate group from one end is ligated to a 5'-hydroxyl on the other end by an RNA ligase.

The bases are modified at the same time the endonucleolytic cleavage reactions are occurring (see Fig. 14.20, circle 3). Three modifications occur in most tRNAs: (1) Uracil is methylated by *S*-adenosylmethionine (SAM) to form thymine; (2) one of the double bonds of uracil is reduced to form dihydrouracil; and (3) a uracil residue (attached to ribose by an *N*-glycosidic bond) is rotated to form pseudouridine, which contains uracil linked to ribose by a carbon-carbon bond. (see Fig. 14.17). Other, less common but more complex, modifications also occur and involve bases other than uracil. Of particular note is the deamination of adenosine to form the base inosine.

The final step in forming the mature tRNA is the addition of a CCA sequence at its 3'-end (see Fig. 14.20, circle 4). These nucleotides are added one at a time by nucleotidyltransferase. The tRNA then migrates to the cytoplasm. The terminal adenosine at the 3'-end is the site at which the specific amino acid for each tRNA is bound and activated for incorporation into a protein.

## V. DIFFERENCES IN SIZE BETWEEN EUKARYOTIC AND PROKARYOTIC DNA

### A. Human Cells Are Diploid

Except for the germ cells, most normal human cells are diploid. Therefore, they contain two copies of each chromosome, and each chromosome contains genes that are alleles of the genes on the homologous chromosome. Because one chromosome in each set of homologous chromosomes is obtained from each parent, the alleles can be identical, containing the same DNA sequence, or they can differ. A diploid human cell contains 2,000 times more DNA than the genome of the bacterium in the haploid *E. coli* cell (approximately  $4 \times 10^6$  base pairs).

### B. Human Genes Contain Introns

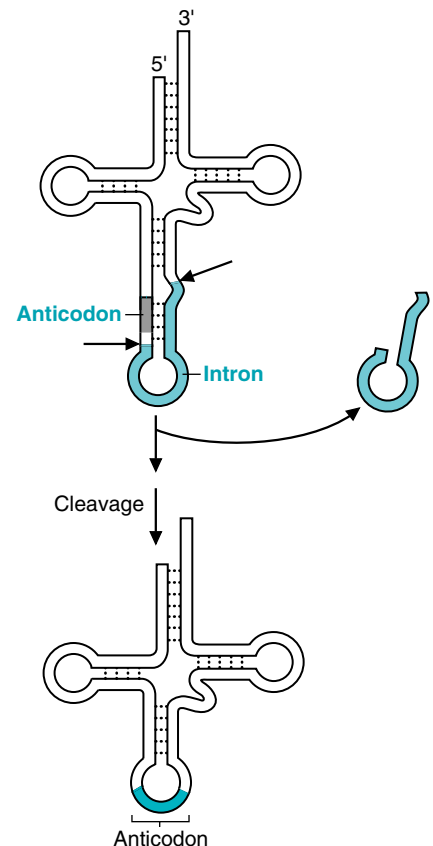
Eukaryotic introns contribute to the DNA size difference between bacteria and human cells. In eukaryotic genes, introns (noncoding regions) occur within sequences that code for proteins. Consequently, the primary transcript (heterogeneous nuclear RNA or hnRNA) averages roughly 10 times longer than the mature mRNA produced by removal of the introns. In contrast, bacterial genes do not contain introns.

### C. Repetitive Sequences in Eukaryotic DNA

Although diploidy and introns account for some of the difference between the DNA content of humans and bacteria, a large difference remains that is related to the greater complexity of the human organism. Bacterial cells have a single copy of each gene, called unique DNA, and they contain very little DNA that does not produce functional products. Eukaryotic cells contain substantial amounts of DNA that does not code for functional products (i.e., proteins or rRNA and tRNA). In addition, some genes that encode functional products are present in multiple copies,



The removal of introns in pre-tRNA has been most extensively studied in yeast. The intron, less than 20 nucleotides long, is located within the anticodon loop at the 3'-end. Removal of the intron by endonucleases leaves the two halves of the tRNA, which remain held together by hydrogen bonds between base pairs in the stem regions. The opening is closed by an RNA ligase.



**Q:** Calculate the number of different proteins, 300 amino acids in length, which could be produced from the *E. coli* genome ( $4 \times 10^6$  base pairs of DNA).



A number of differences between eukaryotes and prokaryotes affect the processes of replication, transcription, and translation, in addition to the content of their DNA. Eukaryotic DNA is complexed with histones, and prokaryotic DNA is not. In eukaryotic cells, the process of transcription, which occurs in the nucleus, is separated by the nuclear envelope from the process of translation (protein synthesis from the mRNA template), which occurs in the cytoplasm. Because prokaryotes lack nuclei, the processes of transcription and translation occur simultaneously. Transcription of bacterial DNA requires only one promoter per operon. In contrast, human DNA requires one promoter for each gene.





Complexity may explain some of the differences between the DNA content of bacteria and humans. But an extension of this line of reasoning would lead to the conclusion that frogs are more complex than humans, because frogs have 8 feet of DNA per diploid nucleus compared to the 6 feet in a human cell. Logic, or perhaps vanity, suggests that the amount of DNA per cell does not necessarily reflect the complexity of the organism. One of the features of frog DNA that may explain its length is that frogs have more repetitive DNA than humans. More than 75% of the frog genome is in the moderately and highly repetitive category, whereas only about 35% of the human genome is repetitive.



Four million base pairs contain  $4 \times 10^6/3$  or 1.33 million codons. If each protein contained approximately 300 amino acids, *E. coli* could produce about 4,000 different proteins ( $1.33 \times 10^6/300$ ).

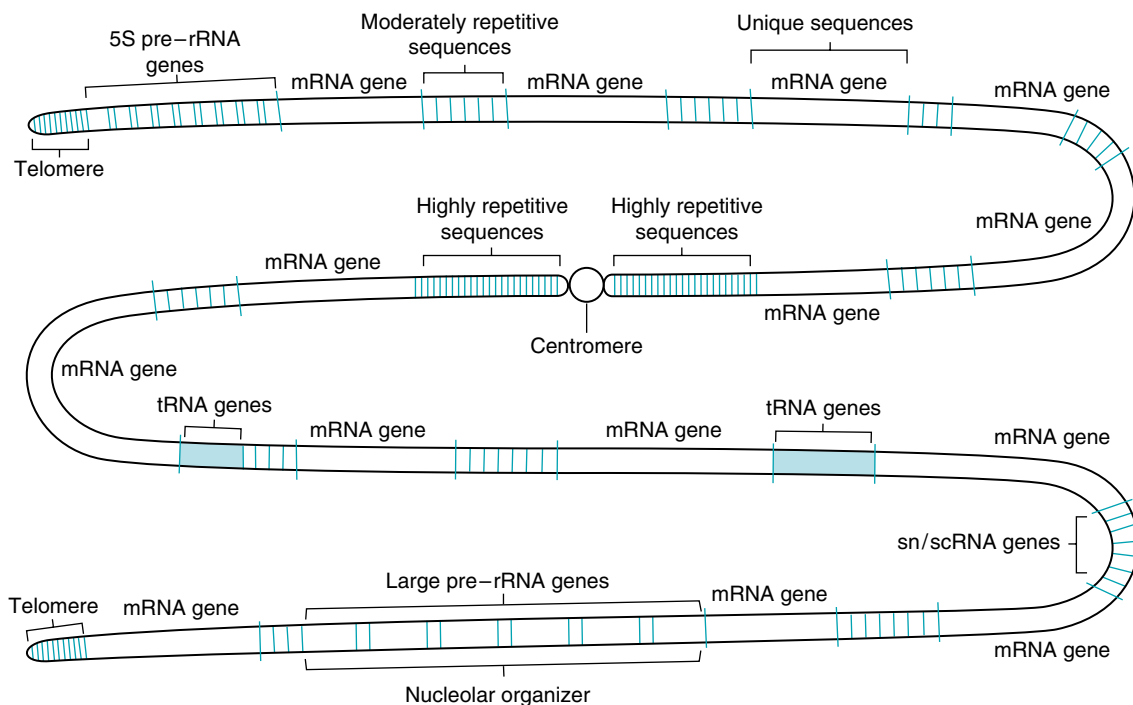
called highly repetitive or moderately repetitive DNA. Approximately 64% of the DNA in the human genome is unique, consisting of DNA sequences present in one or a very few copies in the genome (Fig. 14.21). Unique DNA sequences are transcribed to generate mRNA, which is translated to produce proteins.

Highly repetitive DNA consists of sequences approximately 6 to 100 base pairs in length that are present in hundreds of thousands to millions of copies, clustered within a few locations in the genome (see Fig. 14.21). It occurs in centromeres (which join sister chromatids during mitosis) and in telomeres (the ends of chromosomes). This DNA represents approximately 10% of the human genome. It is not transcribed.

Moderately repetitive DNA is present in a few to tens of thousands of copies in the genome (see Fig. 14.21). This fraction constitutes approximately 25% of the human genome. It contains DNA that is functional and transcribed to produce rRNA, tRNA, and also some mRNA. The histone genes, present in a few hundred copies in the genome, belong to this class. Moderately repetitive DNA also includes some gene sequences that are functional but not transcribed. Promoters and enhancers (which are involved in regulating gene expression) are examples of gene sequences in this category. Other groups of moderately repetitive gene sequences that have been found in the human are called the Alu sequences (approximately 300 base pairs in length). Alu



Alu sequences in DNA were named for the enzyme Alu (obtained from *Arthrobacter luteus*), that which is able to cleave them. Alu sequences make up 6 to 8% of the human genome. In some cases of familial hypercholesterolemia, homologous recombination is believed to have occurred between two Alu repeats, resulting in a large deletion in the low-density lipoprotein (LDL) receptor gene. The LDL receptor mediates uptake of the cholesterol-containing LDL particle into many cell types and, in the absence of functional LDL receptors, blood cholesterol levels are elevated. Patients who are homozygous for this mutation may die from cardiac disease as early as in their second or third decade of life.



**Fig. 14.21.** Distribution of unique, moderately repetitive, and highly repetitive sequences in a hypothetical human chromosome. Unique genes encode mRNA. These genes occur in single copies. The genes for the large rRNA and the tRNA precursors occur in multiple copies that are clustered in the genome. The large rRNA genes form the nucleolar organizer. Moderately repetitive sequences are dispersed throughout the genome, and highly repetitive sequences are clustered around the centromere and at the ends of the chromosome (the telomeres). From Wolfe SL. Mol Cell Biol 1993:761.

**Table 14.2. Differences between Eukaryotes and Prokaryotes**

|                 | Eukaryotes (human)                          | Prokaryotes (E. coli)      |
|-----------------|---------------------------------------------|----------------------------|
| Nucleus         | Yes                                         | No                         |
| Chromosomes     |                                             |                            |
| Number          | 23 per haploid cell                         | 1 per haploid cell         |
| DNA             | Linear                                      | Circular                   |
| Histones        | Yes                                         | No                         |
| Genome          |                                             |                            |
| Diploid         | Somatic cells                               | No                         |
| Haploid         | Germ cells                                  | All cells                  |
| Size            | $3 \times 10^9$ base pairs per haploid cell | $4 \times 10^6$ base pairs |
| Genes           |                                             |                            |
| Unique          | 64%                                         | 100%                       |
| Repetitive      |                                             |                            |
| Moderately      | 25%                                         | None                       |
| Highly          | 10%                                         | None                       |
| Operons         | No                                          | Yes                        |
| mRNA            |                                             |                            |
| Polycistronic   | No                                          | Yes                        |
| Introns (hnRNA) | Yes                                         | No                         |
| Translation     | Separate from transcription                 | Coupled with transcription |

sequences are also examples of SINEs (Short INterspersed Elements). The LINE sequences (Long INterspersed Elements) are 6,000 to 7,000 base pairs in length. The function of the Alu and LINE sequences has not been determined.

Major differences between prokaryotic and eukaryotic DNA and RNA are summarized in Table 14.2.



LINE (Long INterspersed Elements) make up about 5% of the human genome. In some patients with hemophilia (a disease in which blood does not clot normally), a LINE sequence has been inserted into exon 14 of the gene for Factor VIII, a protein of the blood-clotting system. The insertion of the LINE sequence leads to the production of a nonfunctional protein.

## CLINICAL COMMENTS



**Anne Niemick.** Patients with  $\beta^+$ -thalassemia who maintain their hemoglobin levels above 6.0 to 7.0 g/dL are usually classified as having thalassemia intermedia. In the  $\beta$ -thalassemias, the  $\alpha$  chains of adult hemoglobin A ( $\alpha_2\beta_2$ ) continue to be synthesized at a normal rate. These chains accumulate in the bone marrow, where the red blood cells are synthesized in erythropoiesis (generation of red blood cells). The accumulation of  $\alpha$  chains diminishes erythropoiesis, resulting in an anemia. Individuals who are homozygous for a severe mutation require constant transfusions.

Individuals with thalassemia intermedia, such as **Anne Niemick**, could have inherited two different defective alleles, one from each parent. One parent may be a “silent” carrier, with one normal allele and one mildly affected allele. This parent produces enough functional  $\beta$ -globin, so no clinical symptoms of thalassemia appear. (However, they generally have a somewhat decreased amount of hemoglobin, resulting in microcytic hypochromic red blood cells.) When this parent contributes the mildly defective allele and the other heterozygous parent contributes a more severely defective allele,



The mutations that cause the thalassemias affect the synthesis of either the  $\alpha$ - or the  $\beta$ -chains of adult hemoglobin, causing an anemia. They are classified by the chain affected ( $\alpha$ - or  $\beta$ -) and by the amount of chain synthesized (0 for no synthesis and + for synthesis of some functional chains). They are also classified as major, intermediate, or minor, according to the severity of the clinical disorder.  $\beta$ -thalassemia major (also called homozygous  $\beta$ -thalassemia) is a clinically severe disorder requiring frequent blood transfusions. It is caused by the inheritance of two alleles for a severe mutation. In  $\beta$ -thalassemia intermedia, the patient exhibits a less severe clinical phenotype and is able to maintain hemoglobin levels above 6 g/dL. It is usually the result of two different mild mutations or homozygosity for a mild mutation.  $\beta$ -thalassemia minor (also known as  $\beta$ -thalassemia trait) is a heterozygous disorder involving a single mutation that is often clinically asymptomatic.

During embryonic and fetal life, the  $\beta$ -chain is replaced by the  $\epsilon$  and  $\gamma$  chains. As a result, patients with severe mutations in the  $\alpha$ -chain tend to die in utero, whereas those with mutations in the  $\beta$  chains exhibit symptoms postnatally, as hemoglobin F is normally replaced with adult hemoglobin A.

thalassemia intermedia occurs in the child. The child is thus heterozygous for two different defective alleles.



**Ivy Sharer.** Ivy Sharer was treated with a multidrug regimen for tuberculosis because the microbes that cause the disease frequently become resistant to the individual drugs. Rifampin in combination with the drug isoniazid (which affects metabolism of vitamin B<sub>6</sub> in the pathogenic bacteria) is usually effective, but months of treatment are required.

Just as bacteria can become resistant to drugs, so can HIV. A great concern to physicians treating patients with AIDS is the appearance of resistant strains of HIV-1 in patients taking a single drug, such as ZDV (zidovudine), for 6 months or more. Therefore, multidrug regimens are used that include other nucleoside reverse transcriptase inhibitors, such as dideoxyinosine (didanosine, formerly called ddI) and dideoxycytidine (zalcitabine, formerly called ddC). The multidrug therapy often includes nonnucleoside inhibitors (e.g., efavirenz), which acts allosterically on reverse transcriptase, and protease inhibitors (e.g., indinavir), which prevent the HIV polypeptide from being cleaved into its mature products (see Biochemical Comments).



Recent studies have indicated that a failure to properly dispose of cellular debris, a normal byproduct of cell death, may lead to the induction of autoantibodies directed against chromatin in patients with SLE. Normal cells have a finite lifetime, and are programmed to die (apoptosis) through a distinct biochemical mechanism. One of the steps in this mechanism is the stepwise degradation of cellular DNA (and other cellular components). If the normal intracellular components are exposed to the immune system, auto-antibodies against them may be generated. The enzyme in cells that degrades DNA is deoxyribonuclease I (DNase I), and individuals with SLE have reduced serum activity levels of DNase I compared with individuals who do not have the disease. Through an understanding of the molecular mechanism whereby autoantibodies are generated, it may be possible to develop therapies to combat this disorder.



**Amanda Tin.** The toxin  $\alpha$ -amanitin is capable of causing irreversible hepatocellular and renal dysfunction through inhibition of mammalian RNA polymerases. Fortunately, **Amanda Tin's** toxicity proved mild. She developed only gastrointestinal symptoms and slight changes in her hepatic and renal function, which returned to normal within a few weeks. Treatment was primarily supportive, with fluid and electrolyte replacement for that lost through the gastrointestinal tract. No effective antidote is available for the *A. phalloides* toxin.



**Sis Lupus.** SLE is a multisystem disease of unknown origin characterized by inflammation related to the presence of autoantibodies in the blood. These autoantibodies react with antigens normally found in the nucleus, cytoplasm, and plasma membrane of the cell. Such “self” antigen–antibody (autoimmune) interactions initiate an inflammatory cascade that produces the broad symptom profile of multiorgan dysfunction found in **Sis Lupus**.

Pharmacological therapy for SLE, directed at immunosuppression, includes high doses of corticosteroids, which suppress the immune response. In refractory cases that do not respond to corticosteroids, cytotoxic drugs inhibiting the synthesis of antibody-producing cells by the bone marrow are also used to cause immunosuppression.

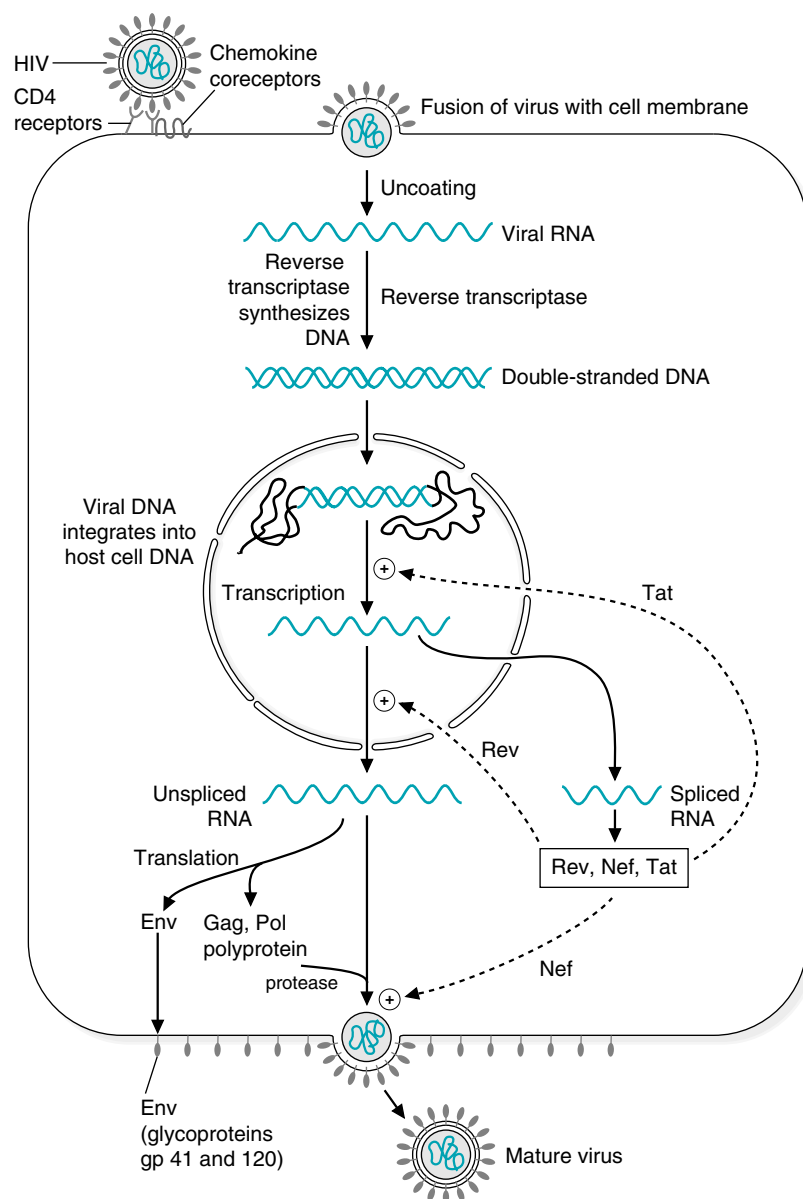


The reverse transcriptase that copies the viral genome has a high error rate because it does not have proofreading capability. Because incorporation of mismatched bases leads to evolution of the virus, new populations develop rapidly in response to changes in the environment. Mutations in the viral reverse transcriptase gene cause the enzyme to become resistant to drugs such as ZDV. Current vaccines developed experimentally against the gp120 surface protein may not be effective because this protein also mutates. As a consequence of the rapid mutation rate of HIV, current treatments for AIDS have limited effectiveness, and a cure has proved elusive.

## BIOCHEMICAL COMMENTS

**Production of the Virus That Causes AIDS.** AIDS is caused by the human immuno-deficiency virus (HIV). Two forms of the virus have been discovered, HIV-1, which is prevalent in industrialized countries, and HIV-2, which is prevalent in certain regions of Africa. Eight to ten years or more can elapse between the initial infection and development of the full-blown syndrome.

Proteins in the viral coat bind to membrane receptors (named CD4) of helper T lymphocytes, a class of cells involved in the immune response. Subsequently, conformational changes occur that allow the viral coat proteins to bind to a chemokine coreceptor in the cell membrane. The lipid in the viral coat then fuses with the cell membrane, and the viral core enters the cell, releasing its RNA and enzymes (including the reverse transcriptase) by a process called “uncoating.” Reverse transcriptase uses the viral RNA as a template to produce a single-stranded DNA copy, which then serves as a template for synthesis of a double-stranded DNA. An integrase enzyme, also carried by the virus, enables this DNA to integrate into the host cell genome as a provirus (Fig. 14.22).



**Fig. 14.22.** Infection of a host cell by HIV. The HIV virus particle binds to the CD4 receptor and a chemokine coreceptor in the host cell membrane. The virus enters the cell and uncoats, releasing its RNA and proteins. The viral enzyme reverse transcriptase produces a double-stranded DNA copy that is integrated into the host cell genome. HIV is now a provirus. Transcripts of the viral DNA are spliced and translated to produce the proteins Tat, Rev, and Nef. Tat stimulates transcription of the viral DNA, and Rev causes the viral RNA transcripts to leave the nucleus unspliced. The unspliced RNA serves as the viral genome and also codes for the proteins of the viral core and envelope. The envelope proteins (gp41 and gp120, which are derived from the env protein) enter the cell membrane. The viral core proteins are synthesized as a polyprotein, which is cleaved by a protease as the viral particles form and bud from the cell membrane. The particles carry membrane lipid as a coat that contains gp41 and gp120. Nef indirectly aids in the assembly of viral particles. Pol is the reverse transcriptase produced from the viral RNA. ⊕ = stimulates.

In the initial stage of transcription of the provirus, the transcript is spliced, and three proteins, Nef, Tat, and Rev, are produced. Tat stimulates transcription of the viral genes. As Rev accumulates, it allows unspliced viral RNA to leave the nucleus and to produce proteins of the viral envelope and viral core, including reverse transcriptase. Two of the envelope glycoproteins (gp41 and gp120, which are derived from the env gene product) form a complex that embeds in the cell membrane. The other proteins, which are translated as a polypeptide and cleaved by the viral protease, combine with the full-length viral RNA to form core viral particles, which bud from the cell membrane. Thus, the virus obtains its lipid coat from the host cell membrane, and the coat contains the viral proteins gp41 and gp120. These surface proteins of the virus bind to CD4 receptors on other human helper T lymphocytes, and the infection spreads.

In an uninfected person, helper T lymphocytes usually number approximately 1,000/mL. Infection with HIV causes the number of these cells to decrease, which results in a deficiency of the immune system. When the number of T lymphocytes drops below 200/mL, the disease is in an advanced stage, and opportunistic infections, such as tuberculosis, occur. Although macrophages and dendritic cells lack CD4 receptors, they can also become infected with HIV and can carry the virus to the central nervous system.

The most effective means of combating HIV infection involves the use of drugs that inhibit the viral reverse transcriptase or the viral protease. However, these drugs only hold the infection at bay; they do not effect a cure.



Drugs currently used to treat AIDS act on the viral reverse transcriptase or the protease (see Fig. 14.22). The nonnucleoside drugs (e.g., efavirenz) bind to reverse transcriptase and inhibit its action. The nucleoside analogs (e.g., ZDV) add to the 3' end of the growing DNA transcript produced by reverse transcriptase and prevent further elongation. The protease inhibitors (e.g., indinavir) bind to the protease and prevent it from cleaving the polypeptide.

### Suggested References

A more complete coverage of transcription can be found in:

Lewin B. Chapter 5: mRNA. In: *Genes VII*. Oxford: Oxford University Press, 2000:119–138.

Lewin B. Chapter 9: mRNA and Transcription. In: *Genes VII*. Oxford: Oxford University Press, 2000:232–272.

Lewin, B. Chapter 20: Initiation of transcription. In: *Genes VII*. Oxford: Oxford University Press, 2000:649–684.

Tjian R. Molecular machines that control genes. *Sci Am* 1995;272:54–61.

References covering diseases discussed in this chapter:

Clumeck N. Choosing the best initial therapy for HIV-1 infection. *N Engl J Med* 1999;341:1925–1926. [See also the accompanying articles.]

Frankel AD, Young JAT. HIV-1: Fifteen proteins and an RNA. *Annu Rev Biochem* 1998;67:1–25.

Weatherall DJ, Clegg JB, Higgs DR, Wood WG. The hemoglobinopathies. Vol. III. In: Scriver CR, Beaudet AL, Sly WS, Valle B., eds. *The Metabolic and Molecular Bases of Inherited Disease*. 8th Ed. New York: McGraw Hill, 2001: 4571–4636.

Walport MJ. Lupus, DNase and defective disposal of cellular debris. *Nature Genetics* 2000;25:135–136.



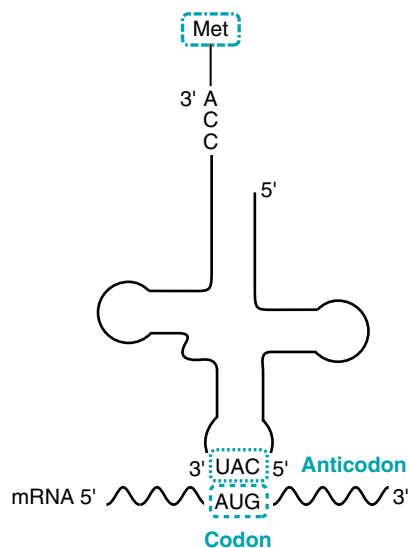
### REVIEW QUESTIONS—CHAPTER 14

- The short transcript AUCCGUACG would be derived from which of the following template DNA sequences? (Note all sequences are written from 5' to 3')
  - ATCCGTACG
  - CGTACGGAT
  - AUCCGUACG
  - TAGGCATGC
  - GCATGCCTA



2. Given that the  $LD_{50}$  (the dose at which 50% of the recipients die) of amanitin is 0.1 mg per kg body weight, and that the average mushroom contains 7 mg amanitin, how many mushrooms must be consumed by Amanda Tin (50 kg body weight) to be above the  $LD_{50}$ ?
- (A) 1
  - (B) 2
  - (C) 3
  - (D) 4
  - (E) 5
3. Which of the following eukaryotic DNA control sequences does not need to be in a fixed location, and is most responsible for high rates of transcription of particular genes?
- (A) Promoter
  - (B) Promoter-proximal element
  - (C) Enhancer
  - (D) Operator
  - (E) Splice donor site
4. Which of the following is true of both eukaryotic and prokaryotic gene expression?
- (A) After transcription, a 3' poly A tail and a 5' cap are added to mRNA.
  - (B) Translation of mRNA can begin before transcription is complete.
  - (C) mRNA is synthesized in the 3' to 5' direction.
  - (D) RNA polymerase binds at a promoter region upstream of the gene.
  - (E) Mature mRNA is always precisely co-linear to the gene from which it was transcribed.
5. In a segment of a transcribed gene, the nontemplate strand of DNA has the following sequence:  
5'...AGCTCACTG...3'
- What will be the corresponding sequence in the RNA produced from this segment of the gene?
- (A) CAGUGAGCU
  - (B) AGCUCACUG
  - (C) CAGTGAGCT
  - (D) UCGAGUGAC
  - (E) GTCACCTCGA

# 15 Translation: Synthesis of Proteins



**Fig. 15.1.** Binding of tRNA to a codon on mRNA. The tRNA contains an amino acid at its 3'-end that corresponds to the codon on mRNA with which the anticodon of the tRNA can base-pair. Note that the codon-anticodon pairing is complementary and antiparallel.

Proteins are produced by the process of **translation**, which occurs on **ribosomes** and is directed by **mRNA**. The genetic message encoded in DNA is first transcribed into mRNA, and the **nucleotide sequence** in the coding region of the mRNA is then translated into the **amino acid sequence** of the protein.

**Translation of the code.** The portion of mRNA that specifies the amino acid sequence of the protein is read in **codons**, which are **sets of three nucleotides** that specify individual amino acids (Fig 15.1). The codons on mRNA are read sequentially in the 5' to 3' direction, starting with the 5'-AUG (or “**start**” codon) that specifies **methionine** and sets the **reading frame** and ending with a 3'-**termination** (or “**stop**”) codon (UAG, UGA, or UAA). The protein is synthesized from its **N-terminus** to its **C-terminus**.

Each amino acid is carried to the ribosome by an **aminoacyl-tRNA** (i.e., a tRNA with an amino acid covalently attached). **Base-pairing** between the **anti-codon** of the tRNA and the **codon** on the mRNA ensures that each amino acid is inserted into the growing polypeptide at the appropriate position.

**Synthesis of the protein. Initiation** involves formation of a complex containing the initial **methionyl-tRNA** bound to the AUG “start” codon of the mRNA and to the “P” site of the **ribosome**. It requires **GTP** and proteins known as **eukaryotic initiation factors** (eIFs).

**Elongation** of the polypeptide involves **three steps**: (a) **binding** of an **aminoacyl-tRNA** to the “A” site on the ribosome where it base-pairs with the second codon on the mRNA; (b) **formation** of a **peptide bond** between the first and second amino acids; and (c) **translocation**, movement of the mRNA relative to the ribosome, so that the third mRNA codon moves into the “A” site. These three **elongation steps** are **repeated** until a **termination** codon aligns with the site on the ribosome where the next aminoacyl-tRNA would normally bind. **Release factors** bind instead, causing the completed protein to be released from the ribosome.

After one ribosome binds and moves along the mRNA, translating the polypeptide, another ribosome can bind and begin translation. The complex of a single mRNA with multiple ribosomes is known as a **polysome**.

**Folding and modification and targeting of the protein.** **Folding** of the polypeptide into its three-dimensional configuration occurs as the polypeptide is being translated. This process involves proteins called **chaperones**. **Modification** of amino acid residues in a protein occurs during or after translation. Proteins synthesized on **cytosolic ribosomes** are released into the cytosol or transported into mitochondria, peroxisomes, and nucleus. Proteins synthesized on ribosomes attached to the **rough endoplasmic reticulum** (RER) are destined for lysosomes, cell membranes, or secretion from the cell. These proteins are transferred to the **Golgi complex**, where they are modified and **targeted** to their ultimate locations.



## THE WAITING ROOM



**Anne Niemick**, a 4-year old patient with  $\beta^+$ -thalassemia intermedia (see Chapter 14), showed no improvement in her symptoms at her second visit. Her hemoglobin level was 7.0 g/dL (reference range for females = 12–16 g/dL).



**Jay Sakz** is a 9-month-old male infant of Ashkenazi Jewish parentage. His growth and development were normal until age 5 months, when he began to exhibit mild, generalized muscle weakness. By 7 months, he had poor head control, slowed development of motor skills, and was increasingly inattentive to his surroundings. His parents also noted unusual eye movements and staring episodes. On careful examination of his retinae, his pediatrician observed a “cherry-red” spot within a pale macula. The physician suspected Tay-Sachs disease and sent whole blood samples to the molecular biology-genetics laboratory.



**Neu Moania** returned to his physician’s office after 1 week of erythromycin therapy (see Chapter 12). The sputum sample from his previous visit had been cultured. The results confirmed that his respiratory infection was caused by *Streptococcus pneumoniae* and that the organism was sensitive to penicillin, macrolides (e.g., erythromycin, clarithromycin), tetracycline, and other antibiotics.



**Erna Nemdy**, a 25-year-old junior medical student ( “earn an M.D.”), brings her healthy 4-month-old daughter, **Beverly**, to the pediatrician for her second diphtheria, pertussis, tetanus (DPT-2) immunization. Erna tells the doctor that her great, great aunt had died of diphtheria during an epidemic many years ago.



The results of tests performed in the molecular biology laboratory show that **Jay Sakz** has an insertion in exon 11 of the  $\beta$ -chain of the hexosaminidase A gene, the most common mutation found in patients of Ashkenazi Jewish background who have Tay-Sachs disease.

## I. THE GENETIC CODE

Transcription, the transfer of the genetic message from DNA to RNA, and translation, the transfer of the genetic message from the nucleotide language of nucleic acids to the amino acid language of proteins, both depend on base-pairing. In the late 1950s and early 1960s, molecular biologists attempting to decipher the process of translation recognized two problems. The first involved decoding the relationship between the “language” of the nucleic acids and the “language” of the proteins, and the second involved determining the molecular mechanism by which translation between these two languages occurs.

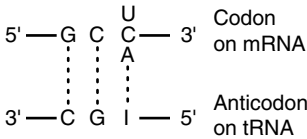
Twenty different amino acids are commonly incorporated into proteins, and, therefore, the protein “alphabet” has 20 characters. The nucleic acid alphabet, however, has only four characters, corresponding to the four nucleotides of mRNA (A, G, C, and U). If two nucleotides constituted the code for an amino acid, then only  $4^2$  or 16 amino acids could be specified. Therefore, the number of nucleotides that code for an amino acid has to be three, providing  $4^3$  or 64 possible combinations or “codons,” more than required, but not overly excessive.

Scientists set out to determine the specific codons for each amino acid. In 1961, Marshall Nirenberg produced the first crack in the genetic code (the collection of codons that specify all the amino acids found in proteins). He showed that poly(U), a polynucleotide in which all the bases are uracil, produced polyphenylalanine in a cell-free protein-synthesizing system. Thus, UUU must be

A. Codons for alanine



B. Base pairing of alanine codons with anticodon



**Fig. 15.2.** Base-pairing of codons for alanine with 5'-IGC-3'. A. Note that the variation is in the third base. B. The first three of these codons can pair with a tRNA that contains the anticodon 5'-IGC-3'. Hypoxanthine (I) is an unusual base found in tRNA that can form base pairs with U, C, or A. It is formed by the deamination of adenine.



Hypoxanthine is the base attached to ribose in the nucleoside inosine. The single-letter abbreviation for hypoxanthine is “I,” in reference to the nucleoside inosine. (In other cases, the first letter of the base is also the first letter of the nucleoside and the single letter abbreviation. For example, A is the base adenine and the nucleoside adenosine.)

the codon for phenylalanine. As a result of experiments using synthetic polynucleotides in place of mRNA, other codons were identified.

The pioneering molecular biologists recognized that, because amino acids cannot bind directly to the sets of three nucleotides that form their codons, “adapters” are required. The “adapters” were found to be tRNA molecules. Each tRNA molecule contains an anticodon and covalently binds a specific amino acid at its 3'-end (see Chapters 12 and 14). The anticodon of a tRNA molecule is a set of three nucleotides that can interact with a codon on mRNA (Fig. 15.2). To interact, the codon and anticodon must be complementary (i.e., they must be able to form base pairs in an antiparallel orientation). Thus, the anticodon of a tRNA serves as the link between an mRNA codon and the amino acid that the codon specifies.

Obviously, each codon present within mRNA must correspond to a specific amino acid. Nirenberg found that trinucleotides of known base sequence could bind to ribosomes and induce the binding of specific aminoacyl-tRNAs (i.e., tRNAs with amino acids covalently attached). As a result of these and the earlier experiments, the relationship between all 64 codons and the amino acids they specify (the entire genetic code) was determined by the mid-1960s (Table 15.1).

Three of the 64 possible codons (UGA, UAG, and UAA) terminate protein synthesis and are known as “stop” or nonsense codons. The remaining 61 codons specify amino acids. Two amino acids each have only one codon (AUG = methionine; UGG = tryptophan). The remaining amino acids have multiple codons.

A. The Code Is Degenerate, But Unambiguous

Because many amino acids are specified by more than one codon, the genetic code is described as “degenerate,” which means that an amino acid may have more than one codon. However, each codon specifies only one amino acid, and the genetic code is, thus, unambiguous.

Inspection of a codon table shows that in most instances of multiple codons for a single amino acid, the variation occurs in the third base of the codon (see Table 15.1). Crick noted that the pairing between the 3'-base of the codon and the 5'-base of the anticodon does not always follow the strict base-pairing rules that he and Watson had previously discovered (i.e., A pairs with U, and G with C). This observation resulted in the “wobble” hypothesis.

At the third base of the codon (the 3'-position of the codon and the 5'-position of the anticodon), the base pairs can “wobble”, e.g., G can pair with U; and A, G, or U can pair with the unusual base hypoxanthine (I) found in tRNA. Thus, three of

**Table 15.1. The Genetic Code**

| First Base | Second Base |     |      | Third Base |      |
|------------|-------------|-----|------|------------|------|
| (5')       | U           | C   | A    | G          | (3') |
| U          | Phe         | Ser | Tyr  | Cys        | U    |
|            | Phe         | Ser | Tyr  | Cys        | C    |
|            | Leu         | Ser | Stop | Stop       | A    |
|            | Leu         | Ser | Stop | Trp        | G    |
| C          | Leu         | Pro | His  | Arg        | U    |
|            | Leu         | Pro | His  | Arg        | C    |
|            | Leu         | Pro | Gln  | Arg        | A    |
|            | Leu         | Pro | Gln  | Arg        | G    |
| A          | Ile         | Thr | Asn  | Ser        | U    |
|            | Ile         | Thr | Asn  | Ser        | C    |
|            | Ile         | Thr | Lys  | Arg        | A    |
|            | Met         | Thr | Lys  | Arg        | G    |
| G          | Val         | Ala | Asp  | Gly        | U    |
|            | Val         | Ala | Asp  | Gly        | C    |
|            | Val         | Ala | Glu  | Gly        | A    |
|            | Val         | Ala | Glu  | Gly        | G    |

**Fig. 15.3.** Reading frame of mRNA. A. For any given mRNA sequence, there are three possible reading frames (1, 2, and 3). B. An AUG near the 5'-end of the mRNA (the start codon) sets the reading frame for translation of a protein from the mRNA. The codons are read in linear order, starting with this AUG. (The other potential reading frames are not used. They would give proteins with different amino acid sequences.)



**Table 15.2. Types of Mutations**

| Type      | Description                                    | Example                 |
|-----------|------------------------------------------------|-------------------------|
| Point     | A single base change                           |                         |
| Silent    | A change that specifies the same amino acid    | CGA → CGG<br>Arg → Arg  |
| Missense  | A change that specifies a different amino acid | CGA → CCA<br>Arg → Pro  |
| Nonsense  | A change that produces a stop codon            | CGA → UGA<br>Arg → Stop |
| Insertion | An addition of one or more bases               |                         |
| Deletion  | A loss of one or more bases                    |                         |



Sickle cell anemia is caused by a missense mutation. In each of the alleles for  $\beta$ -globin, **Will Sichel's** DNA has a single base change (see Chapter 6). In the sickle cell gene, GTG replaces the normal GAG. Thus, in the mRNA, the codon GUG replaces GAG and a valine residue replaces a glutamate residue in the protein.



One type of thalassemia is caused by a nonsense mutation. Codon 17 of the  $\beta$ -globin chain is changed from UGG to UGA. This change results in the conversion of a codon for a tryptophan residue to a stop codon.

Is it likely that **Anne Niemick** has this mutation in codon 17?



Some types of thalassemia are caused by deletions in the globin genes. Patients have been studied who have large deletions in either the 5' or the 3' coding region of the  $\beta$ -globin gene, removing almost one third of the DNA sequence.

Is it likely that **Anne Niemick** has a large deletion in the  $\beta$ -globin gene?

## A. Point Mutations

Point mutations occur when only one base in DNA is altered, producing a change in a single base of an mRNA codon. There are three basic types of point mutations: silent mutations, missense mutations, and nonsense mutations. Point mutations are said to be “silent” when they do not affect the amino acid sequence of the protein. For example, a codon change from CGA to CGG does not affect the protein because both of these codons specify arginine (see Table 15.1). In missense mutations, one amino acid in the protein is replaced by a different amino acid. For example, a change from CGA to CCA causes arginine to be replaced by proline. A “nonsense” mutation causes the premature termination of a polypeptide chain. For example, a codon change from CGA to UGA causes a codon for arginine to be replaced by a stop codon and synthesis of the mutant protein terminates at this point.

## B. Insertions, Deletions, and Frameshift Mutations

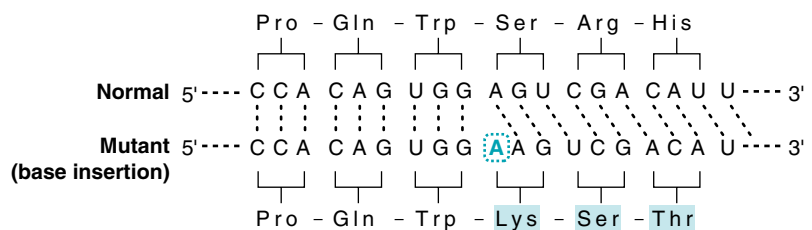
An insertion occurs when one or more nucleotides are added to DNA. If the insertion does not generate a stop codon, a protein with more amino acids than normal could be produced.

When one or more nucleotides are removed from DNA, the mutation is known as a deletion. If the deletion does not affect the normal start and stop codons, a protein with fewer than the normal number of amino acids could be produced.

A frameshift mutation occurs when the number of inserted or deleted nucleotides is not a multiple of three (Fig. 15.4). The reading frame shifts at the point where the insertion or deletion begins. Beyond that point, the amino acid sequence of the protein translated from the mRNA differs from the normal protein.

## IV. FORMATION OF AMINOACYL-tRNA

A tRNA that contains an amino acid covalently attached to its 3'-end is called an aminoacyl-tRNA and is said to be “charged.” Aminoacyl-tRNAs are named both for the amino acid and the tRNA that carries the amino acid. For example, the tRNA



**Fig. 15.4.** A frameshift mutation. The insertion of a single nucleotide (the A in the dotted box) causes the reading frame to shift, so that the amino acid sequence of the protein translated from the mRNA is different after the point of insertion. A similar effect can result from the insertion or deletion of nucleotides if the number inserted or deleted is not a multiple of 3.

for alanine ( $\text{tRNA}^{\text{Ala}}$ ) acquires alanine to become alanyl- $\text{tRNA}^{\text{Ala}}$ . A particular tRNA recognizes only the AUG start codon that initiates protein synthesis and not other AUG codons that specify insertion of methionine within the polypeptide chain. This initiator methionyl- $\text{tRNA}^{\text{Met}}$  is denoted by the subscript “i” in methionyl- $\text{tRNA}_i^{\text{Met}}$ .

Amino acids are attached to their tRNAs by highly specific enzymes known as aminoacyl-tRNA synthetases. Twenty different synthetases exist, one for each amino acid. Each synthetase recognizes a particular amino acid and all of the tRNAs that carry that amino acid.

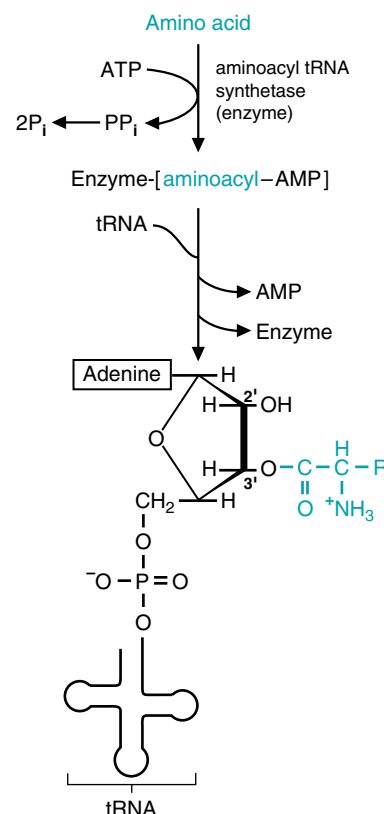
The formation of the ester bond that links the amino acid to the tRNA by an aminoacyl-tRNA synthetase is an energy-requiring process that occurs in two steps. The amino acid is activated in the first step when its carboxyl group reacts with adenosine triphosphate (ATP) to form an enzyme/aminoacyl-AMP complex and pyrophosphate (Fig. 15.5). The cleavage of a high-energy bond of ATP in this reaction provides energy, and the subsequent cleavage of pyrophosphate by a pyrophosphatase helps to drive the reaction by removing one of the products. In the second step, the activated amino acid is transferred to the 2'- or 3'-hydroxyl group (depending on the type of aminoacyl-tRNA synthetase catalyzing the reaction) of the 3' terminal A residue of the tRNA, and AMP is released (recall that all tRNAs have a CCA added to their 3' end posttranscriptionally). The energy in the aminoacyl-tRNA ester bond is subsequently used in the formation of a peptide bond during the process of protein synthesis.

Some aminoacyl-tRNA synthetases use the anticodon of the tRNA as a recognition site as they attach the amino acid to the hydroxyl group at the 3'-end of the tRNA (Fig. 15.6). However, other synthetases do not use the anticodon but recognize only bases located at other positions in the tRNA. Nevertheless, insertion of the amino acid into a growing polypeptide chain depends solely on the bases of the anticodon, through complementary base-pairing with the mRNA codon.

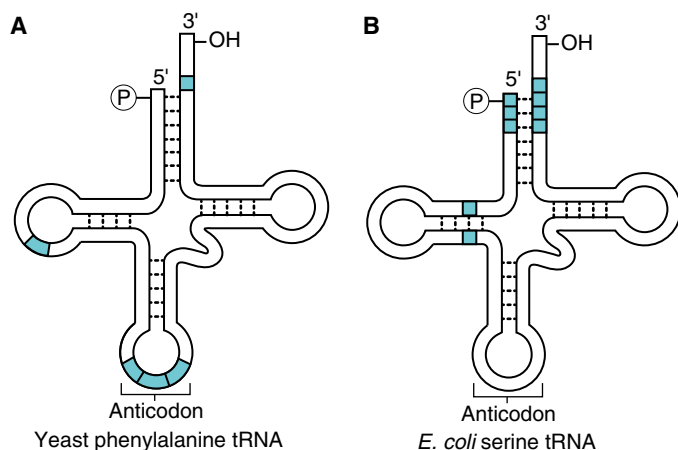
## V. PROCESS OF TRANSLATION

Translation of a protein involves three steps: initiation, elongation, and termination. It begins with the formation of the initiation complex. Subsequently, synthesis of the polypeptide occurs by a series of elongation steps that are repeated as each

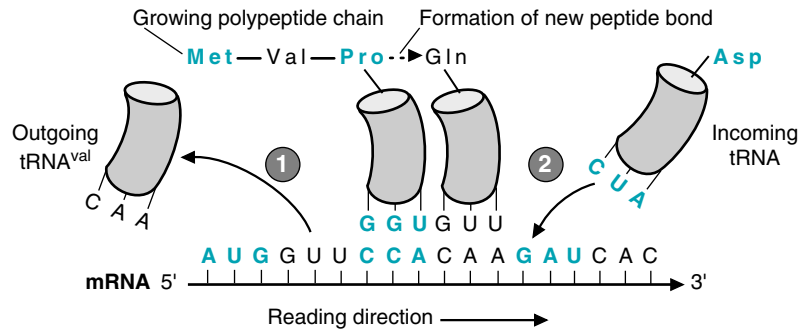
**A:** A nonsense mutation at codon 17 would cause premature termination of translation. A nonfunctional peptide containing only 16 amino acids would result, producing a  $\beta^0$ -thalassemia if the mutation occurred in both alleles. A large deletion in the coding region of the gene could also produce a truncated protein. If **Anne Niemick** has a nonsense mutation or a large deletion, it could only be in one allele. The mutation in the other allele must be milder, since she produces some normal  $\beta$ -globin. Her hemoglobin is 7 g/dL, typical of thalassemia intermedia (a  $\beta^+$ -thalassemia).



**Fig. 15.5.** Formation of aminoacyl-tRNA. The amino acid is first activated by reacting with ATP. The amino acid is then transferred from the aminoacyl-AMP to tRNA.



**Fig. 15.6.** Some aminoacyl-tRNA synthetase recognition sites on tRNA. Each aminoacyl-tRNA synthetase is specific for one tRNA, which it “recognizes” by binding the sequences of nucleotides called the recognition sites, shown in blue. In some cases, the anticodon is a recognition site; in others, it is not. This is true for human tRNAs as well as those shown here.



**Fig. 15.7.** Overview of the process of translation. 1. Once a tRNA has donated its amino acid to the growing polypeptide chain, it is released from the mRNA. 2. A new aminoacyl-tRNA binds to the correct codon in the mRNA to donate its amino acid to the growing polypeptide chain.

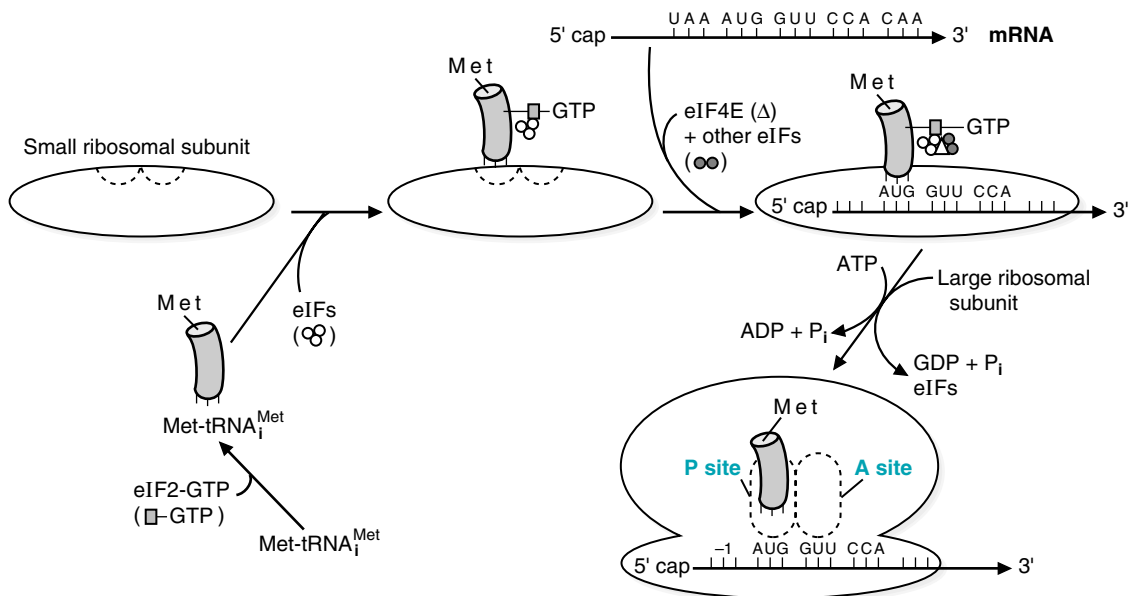
amino acid is added to the growing chain (Fig. 15.7). Termination occurs where the mRNA contains an in-frame stop codon, and the completed polypeptide chain is released.

### A. Initiation of Translation



Eukaryotic initiation factor 2 (eIF2) and also elongation factor 1 (EF1) are types of heterotrimeric G proteins (see Chapter 11). They dramatically change their conformation and actively form complexes when they bind GTP but become inactive and dissociate when they hydrolyze this GTP to GDP. GTP can then displace the bound GDP to reactivate the initiation factor eIF2 or the elongation factor EF1.

In eukaryotes, initiation of translation involves formation of a complex composed of methionyl-tRNA<sup>Met</sup>, mRNA, and a ribosome (Fig. 15.8). Methionyl-tRNA<sup>Met</sup> (also known as Met-tRNA<sub>i</sub><sup>Met</sup>) initially forms a complex with the protein eukaryotic initiation factor 2 (eIF2), which binds GTP. This complex then binds to the small (40S) ribosomal subunit. The cap at the 5'-end of the mRNA binds an initiation factor known as the cap binding protein (CBP). CBP contains a number of subunits, including eIF4E. Several other eIFs join, and the mRNA then binds to the eIFs-Met-tRNA<sub>i</sub><sup>Met</sup> – 40S ribosome complex. In a reaction requiring hydrolysis of ATP (due to the helicase activity of an eIF subunit), this complex unwinds a hairpin loop in the mRNA and scans the mRNA until it locates the AUG start codon (usually the



**Fig. 15.8.** Initiation of protein synthesis. P site = peptidyl site on the ribosome; A site = aminoacyl site on the ribosome (The A and P sites or portions of them are indicated by dashed lines); eIF = eukaryotic initiation factor.

**Table 15.3. Differences between Eukaryotes and Prokaryotes in the Initiation of Protein Synthesis**

|                                            | Eukaryotes                                                                                                                               | Prokaryotes                                                                                    |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Binding of mRNA to small ribosomal subunit | Cap at 5' end of mRNA binds eIFs and 40S ribosomal subunit containing $\text{tRNA}_i^{\text{Met}}$ . mRNA is scanned for AUG start codon | Shine-Dalgarno sequence upstream of initiating AUG binds to complementary sequence in 16S rRNA |
| First amino acid                           | Methionine                                                                                                                               | Formyl-methionine                                                                              |
| Initiation factors                         | eIFs (12 or more)                                                                                                                        | IFs (3)                                                                                        |
| Ribosomes                                  | 80S<br>(40S and 60S subunits)                                                                                                            | 70S<br>(30S and 50S subunits)                                                                  |

first AUG). GTP is hydrolyzed, the initiation factors are released, and the large ribosomal (60S) subunit binds. The ribosome is now complete. It contains one small and one large subunit, and has two binding sites for tRNA, known as the P (peptidyl) and A (aminoacyl) sites. During initiation,  $\text{Met-tRNA}_i^{\text{Met}}$  binds to the ribosome at the P site.

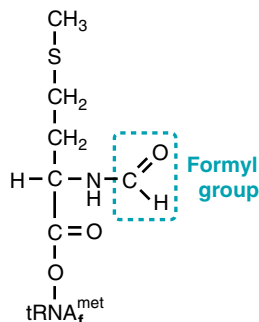
The initiation process differs for prokaryotes and eukaryotes (Table 15.3). In bacteria, the initiating methionyl-tRNA is formylated, producing a formyl-methionyl- $\text{tRNA}_i^{\text{Met}}$  that participates in formation of the initiation complex (Fig. 15.9). Only three initiation factors (IFs) are required to generate this complex in prokaryotes, compared with the dozen or more required by eukaryotes. The ribosomes also differ in size. Prokaryotes have 70S ribosomes, composed of 30S and 50S subunits, and eukaryotes have 80S ribosomes, composed of 40S and 60S subunits. Unlike eukaryotic mRNA, bacterial mRNA is not capped. Identification of the initiating AUG triplet in prokaryotes occurs when a sequence in the mRNA (known as the Shine–Dalgarno sequence) binds to a complementary sequence near the 3'-end of the 16S rRNA of the small ribosomal subunit.

## B. Elongation of Polypeptide Chains

After the initiation complex is formed, addition of each amino acid to the growing polypeptide chain involves binding of an aminoacyl-tRNA to the A site on the ribosome, formation of a peptide bond, and translocation of the peptidyl-tRNA to the P site (Fig. 15.10). The peptidyl-tRNA contains the growing polypeptide chain.

### 1. BINDING OF AMINOACYL-tRNA TO THE A SITE

When  $\text{Met-tRNA}_i$  (or a peptidyl-tRNA) is bound to the P site, the mRNA codon in the A site determines which aminoacyl-tRNA will bind to that site. An aminoacyl-tRNA binds when its anticodon is antiparallel and complementary to the mRNA codon. In eukaryotes, the incoming aminoacyl-tRNA first combines with elongation



**Fig. 15.9.** Bacterial tRNA containing formyl-methionine. The initial methionine is not formylated in eukaryotic protein synthesis.



Insulin, an anabolic hormone, stimulates general protein synthesis by activating the initiation factor eIF4E. Normally, eIF4E is bound to an inhibitor protein, 4E binding protein (4E-BP). When insulin binds to its cell surface receptor, it initiates an intracellular sequence of events resulting in phosphorylation of 4E-BP. Phosphorylated 4E-BP no longer binds to eIF4E, and eIF4E is now free to participate in the initiation of protein synthesis.



eIF2 is a regulator of the initiation step in protein synthesis. When it is phosphorylated, it is inactive, and protein synthesis cannot begin. Conditions such as starvation, heat shock, and viral infection result in phosphorylation of eIF2 by a specific kinase.

The regulation of globin synthesis by heme in reticulocytes illustrates the role of eIF2 in regulation of translation. Reticulocytes, which are the precursors of red blood cells, synthesize the oxygen-carrying hemoglobin molecules from the globin polypeptide chains and the Fe-binding pigment, heme. In the absence of heme, the rate of initiation of globin synthesis decreases. Heme acts by inhibiting the phosphorylation of the initiation factor eIF2. Thus, eIF2 is active in the presence of heme and globin synthesis is initiated.

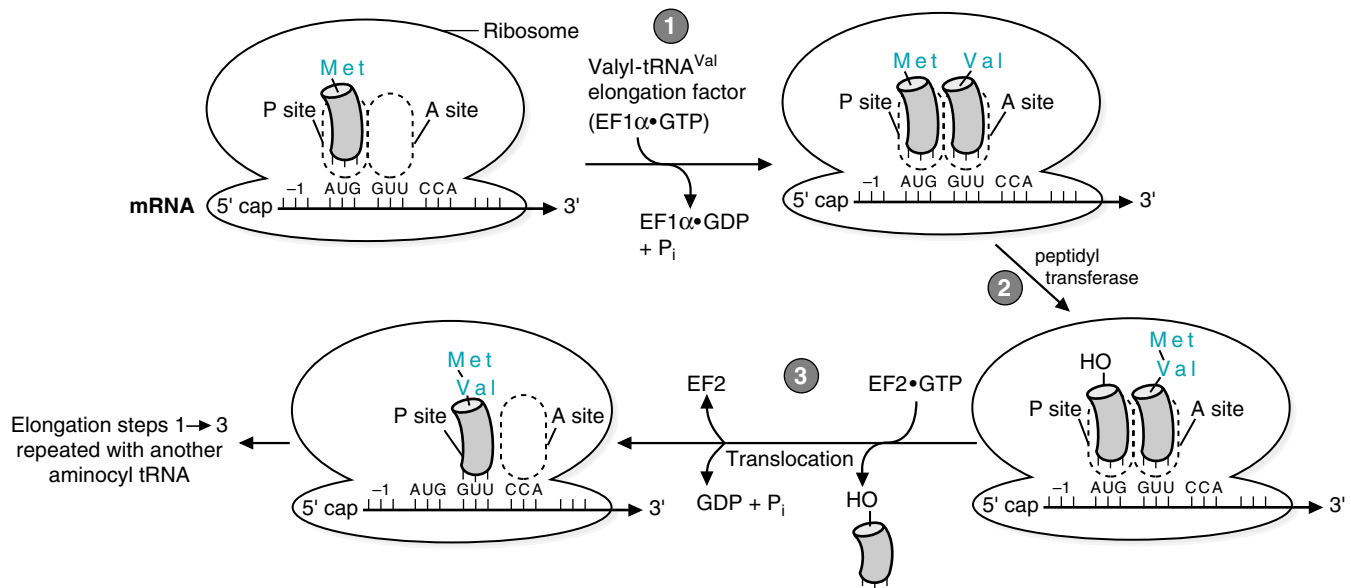


Many antibiotics that are used to combat bacterial infections in humans take advantage of the differences between the mechanisms for protein synthesis in prokaryotes and eukaryotes. For example, streptomycin binds to the 30S ribosomal subunit of prokaryotes. It interferes with initiation of protein synthesis and causes misreading of mRNA.

Although the bacterium causing his infection was sensitive to streptomycin, this drug was not used to treat **Neu Moania** because it can cause permanent hearing loss. Its use is, therefore, confined mainly to the treatment of tuberculosis or other infections that do not respond adequately to other antibiotics.

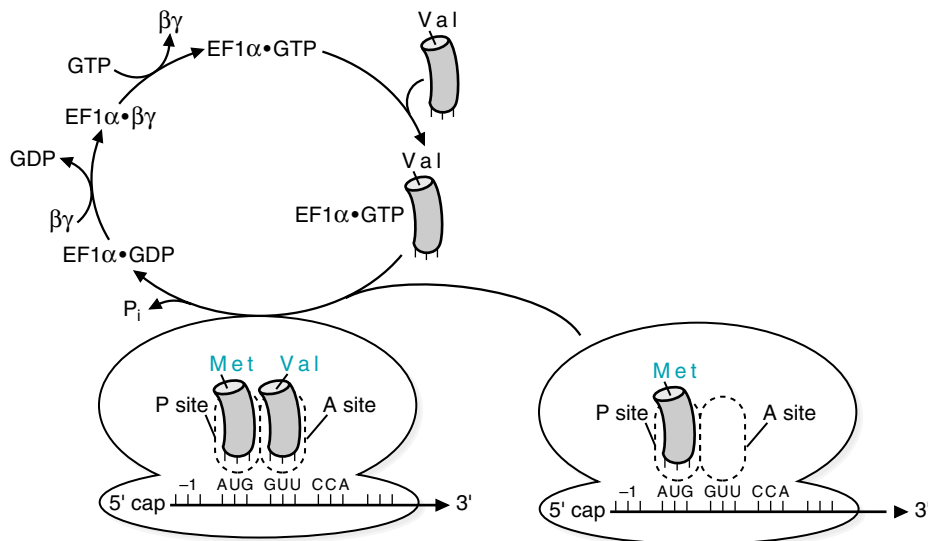


The antibiotic tetracycline binds to the 30S ribosomal subunit of prokaryotes and inhibits binding of aminoacyl-tRNA to the A site of the ribosome. The bacterium causing **Neu Moania's** infection was found to be sensitive to tetracycline.



**Fig. 15.10.** Elongation of a polypeptide chain. 1. Binding of valyl-tRNA<sup>Val</sup> to the A site. 2. Formation of a peptide bond. 3. Translocation. After step 3, step 1 is repeated using the aminoacyl-tRNA for the new codon in the A site. Steps 2 and 3 follow. These three steps keep repeating until termination occurs. (In prokaryotes, an exit site called the E site binds the t-RNA after it is displaced from the P site). EF = elongation factor.

factor EF1α containing bound GTP before binding to the mRNA–ribosome complex (EF1α is the GTP-binding α subunit of a heterotrimeric G protein, which is activated for association with other proteins when it contains GTP; see Chapter 11). When the aminoacyl-tRNA-EF1α-GTP complex binds to the A site, GTP is hydrolyzed to GDP. This prompts dissociation of EF1α-GDP from the aminoacyl-tRNA ribosomal complex, thereby allowing protein synthesis to continue (Fig. 15.11).



**Fig. 15.11.** Recycling of EF1 in eukaryotes. Note that EF1 is a heterotrimeric G protein. Its α-subunit binds GTP and activates the process whereby an aminoacyl-tRNA binds to the A site of the ribosome. GTP is hydrolyzed, and EF1α binds to the EF1βγ subunits, releasing GDP. GTP binds to the α subunit, the βγ subunits are released, and EF1α GTP is ready for another round. In prokaryotes, EF1α is EF-Tu and the protein corresponding to βγ is EF-Ts.



The free EF1 $\alpha$ -GDP reassociates with the EF1 $\beta\gamma$  subunits, and GDP is released. Subsequently, GTP binds and the  $\beta\gamma$  subunits dissociate. Thus, EF1 $\alpha$ -GTP is ready to bind another aminoacyl-tRNA molecule.

The process of elongation is very similar in prokaryotes, except that the corresponding factor for EF1 $\alpha$  is named EF-Tu and the associating elongation factors are called EF-Ts instead of EF1 $\beta\gamma$ .

## 2. FORMATION OF A PEPTIDE BOND

In the first round of elongation, the amino acid on the tRNA in the A site forms a peptide bond with the methionine on the tRNA in the P site. In subsequent rounds of elongation, the amino acid on the tRNA in the A site forms a peptide bond with the peptide on the tRNA in the P site (see Fig. 15.10). Peptidyltransferase, which is not a protein but the rRNA of the large ribosomal subunit, catalyzes the formation of the peptide bond. The tRNA in the A site now contains the growing polypeptide chain, and the tRNA in the P site is uncharged (i.e., it no longer contains an amino acid or peptide).

## 3. TRANSLOCATION

Translocation in eukaryotes involves another G protein, elongation factor EF2 (EF-G in prokaryotes) that complexes with GTP and binds to the ribosome, causing a conformational change that moves the mRNA and its base-paired tRNAs with respect to the ribosome. The uncharged tRNA moves from the P site and is released from the ribosome. The peptidyl-tRNA moves into the P site, and the next codon of the mRNA occupies the A site. During translocation, GTP is hydrolyzed to GDP, which is released from the ribosome along with the elongation factor (see Fig. 15.10).

## C. Termination of Translation

The three elongation steps are repeated until a termination (stop) codon moves into the A site on the ribosome. Because no tRNAs with anticodons that can pair with stop codons normally exist in cells, release factors bind to the ribosome instead, causing peptidyltransferase to hydrolyze the bond between the peptide chain and tRNA. The newly synthesized polypeptide is released from the ribosome, which dissociates into its individual subunits, releasing the mRNA.



Currently, some parents are not having their children immunized. The decrease in the incidence of infectious disease in the United States has led to complacency. Other parents who lack health insurance cannot afford to have their children vaccinated. Anyone who remembers the summertime fear of poliomyelitis in the 1940s and 1950s realizes that immunizations are a blessing that should be available to all children.



Protein synthesis requires a considerable amount of energy. Formation of each aminoacyl-tRNA requires the equivalent of two high-energy phosphate bonds because ATP is converted to AMP and pyrophosphate, which is cleaved to form two inorganic phosphates. As each amino acid is added to the growing peptide chain, two GTPs are hydrolyzed, one at the step involving EF1 and the second at the translocation step. Thus, four high-energy bonds are cleaved for each amino acid of the polypeptide. In addition, energy is required for initiation of synthesis of a polypeptide chain and for synthesis from nucleotide triphosphate precursors of the mRNA, tRNA, and rRNA involved in translation.



The macrolide antibiotics (e.g., erythromycin, clarithromycin) bind to the 50S ribosomal subunit of bacteria and inhibit translocation. Clarithromycin was used to treat **Neu Moania** because he had taken it previously without difficulty. It has less serious side effects than many other antibiotics and is used as an alternative drug in patients, such as **Mr. Moania**, who are allergic to penicillin. After 1 week of therapy, **Mr. Moania** recovered from his infection.

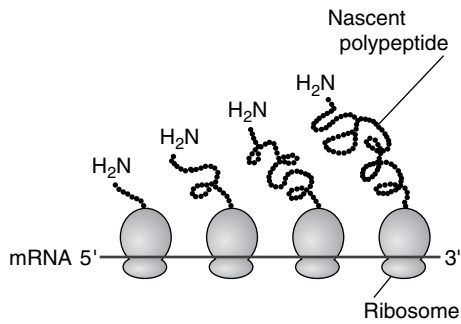


Chloramphenicol is an antibiotic that interferes with the peptidyltransferase activity of the 50S ribosomal subunit of bacteria. It was not used to treat **Neu Moania** because it is very toxic to humans, partly because of its effect on mitochondrial protein synthesis.



Diphtheria is a highly contagious disease caused by a toxin secreted by the bacterium *Corynebacterium diphtheriae*. Although the toxin is a protein, it is not produced by a bacterial gene, but by a gene brought into the bacterial cell by an infecting bacteriophage.

Diphtheria toxin is composed of two protein subunits. The B subunit binds to a cell surface receptor, facilitating the entry of the A subunit into the cell. In the cell, the A subunit catalyzes a reaction in which the ADP-ribose (ADPR) portion of NAD is transferred to EF2 (ADP-ribosylation). In this reaction, the ADPR is covalently attached to a post-translationally modified histidine residue, known as diphthamide. ADP-ribosylation of EF2 inhibits protein synthesis, leading to cell death. Children, including **Erna Nemdy's** daughter, are usually immunized against this often fatal disease at an early age.



**Fig. 15.12.** A polysome. The complex of mRNA and multiple ribosomes, each of which is producing a polypeptide chain, is called a polysome.

## VI. POLYSOMES

As one ribosome moves along the mRNA, producing a polypeptide chain, a second ribosome can bind to the vacant 5'-end of the mRNA. Many ribosomes can simultaneously translate a single mRNA, forming a complex known as a polysome (Fig. 15.12). A single ribosome covers approximately 80 nucleotides of mRNA. Therefore, ribosomes are positioned on mRNA at intervals of approximately 100 nucleotides. The growing polypeptide chains attached to the ribosomes become longer as each ribosome moves from the 5'-end toward the 3'-end of the mRNA.

## VII. PROCESSING OF PROTEINS

Nascent polypeptide chains (i.e., polypeptides that are in the process of being synthesized) are processed. As they are being produced, they travel through a tunnel in the ribosome, which can hold roughly 30 amino acid residues. As polymerization of the chain progresses, the amino acid residues at the N-terminal end begin to emerge from this protected region within the ribosome and to fold and refold into the three-dimensional conformation of the polypeptide. Proteins bind to the nascent polypeptide and mediate the folding process. These mediators are called chaperones (see Chapter 7) because they prevent improper interactions from occurring. Disulfide bond formation between cysteine residues is catalyzed by disulfide isomerases and may also be involved in producing the three-dimensional structure of the polypeptide.

## VIII. POSTTRANSLATIONAL MODIFICATIONS

After proteins emerge from the ribosome, they may undergo posttranslational modifications. The initial methionine is removed by specific proteases; methionine is not the N-terminal amino acid of all proteins. Subsequently, other specific cleavages also may occur that convert proteins to more active forms (e.g., the conversion of proinsulin to insulin). In addition, amino acid residues within the peptide chain can be enzymatically modified to alter the activity or stability of the proteins, direct it to a subcellular compartment, or prepare it for secretion from the cell.

Amino acid residues are enzymatically modified by the addition of various types of functional groups. (Box 15.1) For example, the N-terminal amino acid is sometimes acetylated, and methyl groups can be added to lysine residues. These changes alter the charge on the protein. Proline and lysine residues can be modified by hydroxylation. In collagen, hydroxylations lead to stabilization of the protein. Carboxylations are important, especially for the function of proteins involved in blood coagulation. Formation of  $\gamma$ -carboxylglutamate allows these proteins to chelate  $\text{Ca}^{2+}$ , a step in clot formation. Fatty acids or other hydrophobic groups (e.g., prenyl groups) anchor the protein in membranes. An ADP-ribose group can be transferred from  $\text{NAD}^+$  to certain proteins. The addition and removal of phosphate groups (which bind covalently to serine, threonine, or tyrosine residues) serve to regulate the activity of many proteins (e.g., the enzymes of glycogen degradation and regulators of gene transcription.) Glycosylation, the addition of carbohydrate groups, is a common modification that occurs mainly on proteins that are destined to be secreted or incorporated into lysosomes or cellular membranes.

### Box 15.1 Posttranslational Modifications of Proteins

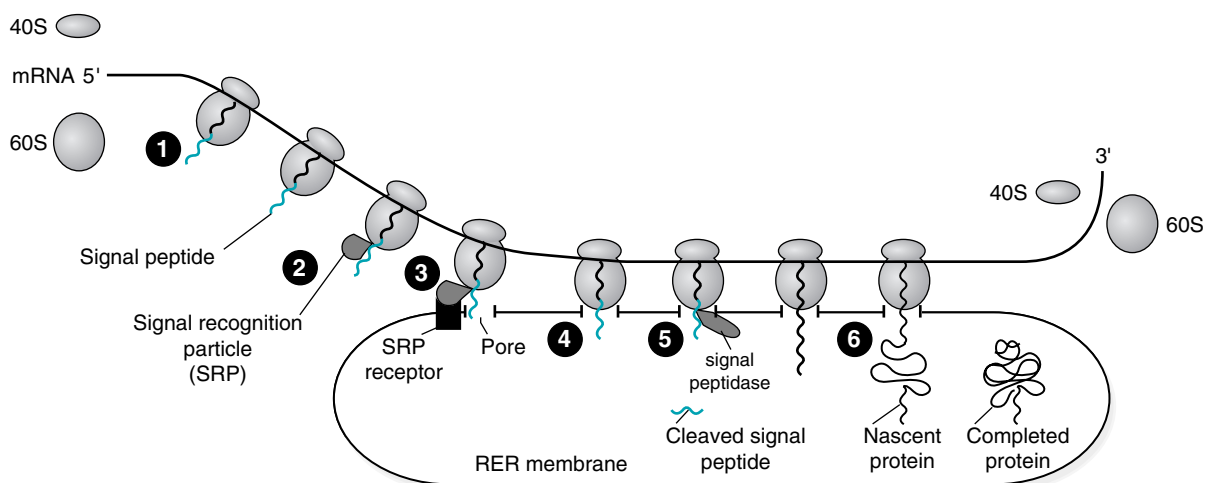
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Acetylation  
ADP-ribosylation  
Carboxylation  
Fatty acylation  
Glycosylation  
Hydroxylation  
Methylation  
Phosphorylation  
Prenylation

---

## IX. TARGETING OF PROTEINS TO SUBCELLULAR AND EXTRACELLULAR LOCATIONS

Many proteins are synthesized on polysomes in the cytosol. After they are released from ribosomes, they remain in the cytosol, where they carry out their functions. Other proteins synthesized on cytosolic ribosomes enter organelles,



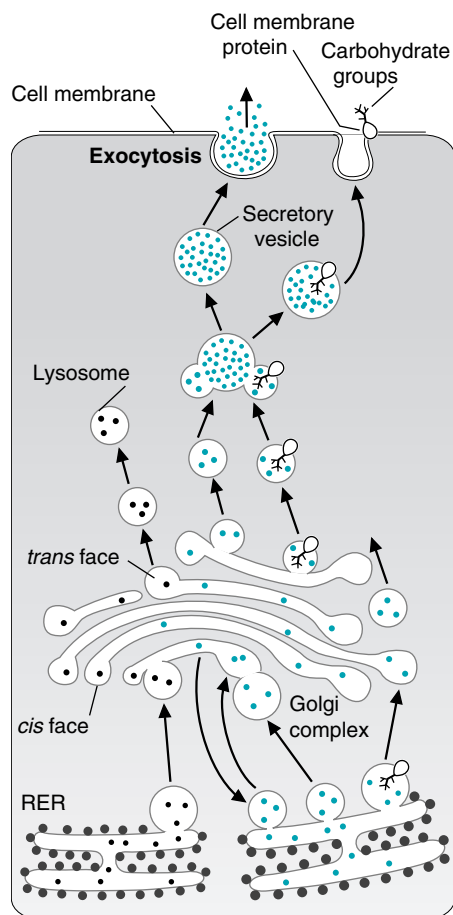
**Fig. 15.13.** Synthesis of proteins on the RER. **1.** Translation of the protein begins in the cytosol. **2.** As the signal peptide emerges from the ribosome, a signal recognition particle (SRP) binds to it and to the ribosome and inhibits further synthesis of the protein. **3.** The SRP binds to the SRP receptor in the RER membrane, docking the ribosome on the RER. **4.** The SRP is released and protein synthesis resumes. **5.** As the signal peptide moves through a pore into the RER, a signal peptidase removes the signal peptide. **6.** Synthesis of the nascent protein continues, and the completed protein is released into the lumen of the RER.

such as mitochondria or nuclei. These proteins contain amino acid sequences called targeting sequences or signal sequences that facilitate their transport into a certain organelle. Another group of proteins are synthesized on ribosomes bound to the RER. These proteins are destined for secretion or for incorporation into various subcellular organelles (e.g., lysosomes, endoplasmic reticulum [ER], Golgi complex) or cellular membranes, including the plasma membrane. Proteins that enter the RER as they are being synthesized have signal peptides near their N-termini that do not have a common amino acid sequence. However, they do contain a number of hydrophobic residues and are 15 to 30 amino acids in length (Fig. 15.13). A signal recognition particle (SRP) binds to the ribosome and to the signal peptide as the nascent polypeptide emerges from the tunnel in the ribosome, and translation ceases. When the SRP subsequently binds to an SRP receptor (docking protein) on the RER, translation resumes, and the polypeptide begins to enter the lumen of the RER. The signal peptide is removed by the signal peptidase, and the remainder of the newly synthesized protein enters the lumen of the RER. These proteins are transferred in small vesicles to the Golgi complex.

The Golgi complex serves to process the proteins it receives from the RER and to sort them so that they are delivered to their appropriate destinations (Fig. 15.14). Processing, which can be initiated in the endoplasmic reticulum, involves glycosylation, the addition of carbohydrate groups, and modification of existing carbohydrate chains. Sorting signals permit delivery of proteins to their target locations. For example, glycosylation of enzymes destined to become lysosomal enzymes results in the presence of a mannose 6-phosphate residue on an oligosaccharide attached to the enzyme. This residue is recognized by the mannose 6-phosphate receptor protein, which incorporates the enzyme into a clathrin-coated vesicle. The vesicle travels to endosomes, and is eventually incorporated into lysosomes. Other proteins containing a KDEL (lys-asg-glu-leu) sequence at their carboxyl terminal are returned to the ER from the Golgi. Proteins with hydrophobic regions can embed in various membranes. Some proteins, whose sorting signals have not yet been determined, enter secretory vesicles and travel to the cell membrane, where they are secreted by the process of exocytosis.



I-cell disease (Mucopolidosis II) is a disorder of protein targeting. Lysosomal proteins are not sorted properly from the Golgi to the lysosomes, and lysosomal enzymes end up secreted from the cell. This is because of a mutation in the enzyme N-acetylglucosamine phosphotransferase, which is a required first step for attaching the lysosomal targeting signal, mannose-6-phosphate, to lysosomal proteins. Thus, lysosomal proteins cannot be targeted to the lysosomes, and these organelles become clogged with materials that cannot be digested, destroying overall lysosomal function. This leads to a lysosomal storage disease of severe consequence, with death before the age of 8.



**Fig. 15.14.** Fate of proteins synthesized on the RER. Proteins synthesized on ribosomes attached to the ER travel in vesicles to the *cis* face of the Golgi complex. After the membranes fuse, the proteins enter the Golgi complex. Structural features of the proteins determine their fate. Some remain in the Golgi complex, and some return to the RER. Others bud from the *trans* face of the Golgi complex in vesicles. These vesicles can become lysosomes or secretory vesicles, depending on their contents. Secretory proteins are released from the cell when secretory vesicles fuse with the cell membrane (exocytosis). Proteins with hydrophobic regions embedded in the membrane of secretory vesicles become cell membrane proteins. See Chapter 10 for descriptions of the endoplasmic reticulum, Golgi complex, lysosomes, and the cell membrane, and also for an explanation of the process of exocytosis.

## CLINICAL COMMENTS



**Ann Niemick** has a  $\beta^+$  thalassemia classified clinically as  $\beta$ -thalassemia intermedia. She produces an intermediate amount of functional  $\beta$  globin chains (her hemoglobin is 7 g/dL; normal is 12–16). In  $\beta^0$ -thalassemia, little or none of the hemoglobin  $\beta$  chain is produced.  $\beta$ -thalassemia intermedia is usually the result of two different mutations (one that mildly affects the rate of synthesis of  $\beta$ -globin and one severely affecting its rate of synthesis), or, less frequently, homozygosity for a mild mutation in the rate of synthesis, or a complex combination of mutations. The mutations that cause the thalassemias have been studied extensively and are summarized in Table 15.4. For each of these mutations, you should now be able to explain whether it is most likely to result in a  $\beta^+$  or  $\beta^0$  thalassemia.



**Jay Sakz.** The molecular biology genetics laboratory's report on Jay Sakz's white blood cells indicated that he had a deficiency of hexosaminidase A caused by a defect in the gene encoding the  $\alpha$  subunit of this enzyme (variant B, Tay-Sachs disease). Hexosaminidases are lysosomal enzymes necessary for the normal degradation of glycosphingolipids, such as the gangliosides. Gangliosides are found in high concentrations in neural ganglia, although they are produced in many areas of the nervous system. When the activity of these degradative enzymes is absent or subnormal, partially degraded gangliosides accumulate in lysosomes in various cells of the central nervous system, causing a wide array of neurologic disorders known collectively as gangliosidoses.

**Table 15.4** Some Examples of Mutations in  $\beta$ -Thalassemia

| Type of Mutation                           | Phenotype     | Origin                |
|--------------------------------------------|---------------|-----------------------|
| Nonsense                                   |               |                       |
| Codon 17 (A $\rightarrow$ T)               | $\beta^0$     | Chinese               |
| Codon 39 (C $\rightarrow$ T)               | $\beta^0$     | Mediterranean         |
| Codon 121 (A $\rightarrow$ T)              | $\beta^0$     | Polish                |
| Frameshift                                 |               |                       |
| Codon 6 (-1 bp)                            | $\beta^0$     | Mediterranean         |
| Codon 16 (-1 bp)                           | $\beta^0$     | Asian Indian          |
| Codon 41/42 (-4 bp)                        | $\beta^0$     | Asian Indian, Chinese |
| Codon 71/72 (+1 bp)                        | $\beta^0$     | Chinese               |
| Promoter                                   |               |                       |
| Position -88 (C $\rightarrow$ T)           | $\beta^+$     | African American      |
| Position -31 (A $\rightarrow$ G)           | $\beta^+$     | Japanese              |
| Position -28 (A $\rightarrow$ C)           | $\beta^+$     | Kurdish               |
| Cap Site                                   |               |                       |
| Position +1 (A $\rightarrow$ C)            | $\beta^+$     | Asian Indian          |
| Splice Junction                            |               |                       |
| Intron 1, position 1 (G $\rightarrow$ A)   | $\beta^0$     | Mediterranean         |
| Intron 1, 3'-end (-25 bp)                  | $\beta^0$     | Asian Indian          |
| Intron 2, position 1 (G $\rightarrow$ A)   | $\beta^0$     | Mediterranean         |
| Intron 2, 3'-end (A $\rightarrow$ G)       | $\beta^0$     | African American      |
| Intron, internal                           |               |                       |
| Intron 1, position 5 (G $\rightarrow$ T)   | $\beta^+$     | Mediterranean         |
| Intron 1, position 6 (T $\rightarrow$ C)   | $\beta^+$     | Mediterranean         |
| Intron 2, position 110 (G $\rightarrow$ A) | $\beta^+$     | Mediterranean         |
| Intron 2, position 654 (C $\rightarrow$ T) | $\beta^0$     | Chinese               |
| Intron 2, position 745 (C $\rightarrow$ G) | $\beta^+$     | Mediterranean         |
| Exon, internal                             |               |                       |
| Codon 24 (T $\rightarrow$ A)               | $\beta^+$     | African American      |
| Codon 26 (G $\rightarrow$ A)               | $\beta^E$     | Southeast Asian       |
| Codon 27 (G $\rightarrow$ T)               | $\beta^{Kno}$ | Mediterranean         |
| RNA cleavage/polyadenylation               |               |                       |
| AATAAA $\pm$ AACAAA                        | $\beta^+$     | African American      |

Data from Scriver CR, et al. The Metabolic and Molecular Basis of Inherited Diseases, Vol III. New York: McGraw-Hill, 1995:3456–3457.

When the enzyme deficiency is severe, symptoms appear within the first 3–5 months of life. Eventually symptoms include upper and lower motor neuron deficits, visual difficulties that can progress to blindness, seizures, and increasing cognitive dysfunction. By the second year of life, the patient may regress into a completely vegetative state, often succumbing to bronchopneumonia caused by aspiration or an inability to cough.



**Erna Nemdy.** With the availability of diphtheria toxoid as part of the almost universal DPT immunization practices in the United States, fatalities due to infection by the Gram-positive bacillus *C. diphtheriae* are rare. Most children, as is the case with Erna Nemdy's daughter Beverly, are immunized. In unimmunized individuals, however, symptoms are caused by a bacterial exotoxin encoded by a phage that infects the bacterial cells. The toxin enters human cells, inhibiting protein synthesis and, ultimately, causing cell death. Complications related to cardiac and nervous system involvement are the major cause of morbidity and mortality. Patients for whom a definitive diagnosis of diphtheria is established are treated with equine diphtheria antitoxin.

## BIOCHEMICAL COMMENTS



**Antibiotics That Inhibit Protein Synthesis.** The processes of translation on bacterial ribosomes and on the cytoplasmic ribosomes of eukaryotic cells have many similarities, but there are a number of subtle differences. Antibiotics act at steps at which these differences occur, and different antibiotics target each of the major steps of protein synthesis (Table 15.5). Therefore, these compounds can be used selectively to prevent bacterial protein synthesis and inhibit bacterial proliferation, while having little or no effect on human cells. Caution must be exercised in their use, however, because some of the antibiotics affect human mitochondria, which have a protein-synthesizing system similar to that of bacteria. Another problem with these drugs is that bacteria can become resistant to their action. Mutations in genes encoding the proteins or RNA of bacterial ribosomes can cause resistance. Resistance also results when bacteria take up plasmids carrying genes for inactivation of the antibiotic. Because of the widespread and often indiscriminate use of antibiotics, strains of bacteria are rapidly developing that are resistant to all known antibiotics.



**Streptomycin.** Streptomycin inhibits initiation by binding to three proteins and probably the 16S rRNA of the 30S ribosomal subunit of bacteria. Abnormal initiation complexes, known as streptomycin monosomes, accumulate. Streptomycin can also cause misreading of mRNA, resulting in premature termination of translation or in the incorporation of incorrect amino acids into polypeptide chains that already have been initiated. The use of this antibiotic is limited because it causes ototoxicity that can result in loss of hearing.

**Table 15.5 Inhibitors of Protein Synthesis in Prokaryotes**

| Antibiotic      | Mode of Action                                                                                                                               |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Streptomycin    | Binds to the 30S ribosomal subunit of prokaryotes, thereby preventing formation of the initiation complex. It also causes misreading of mRNA |
| Tetracycline    | Binds to the 30S ribosomal subunit and inhibits binding of aminoacyl-tRNA to the A site                                                      |
| Chloramphenicol | Binds to the 50S ribosomal subunit and inhibits peptidyltransferase                                                                          |
| Erythromycin    | Binds to the 50S ribosomal subunit and prevents translocation                                                                                |





**Tetracycline.** Tetracycline binds to the 30S ribosomal subunit of bacteria and prevents an aminoacyl-tRNA from binding to the A site on the ribosome. This effect of the drug is reversible; thus, when the drug is removed, bacteria resume protein synthesis and growth, resulting in a rekindling of the infection. Furthermore, tetracycline is not absorbed well from the intestine, and its concentration can become elevated in the contents of the gut, leading to changes in the intestinal flora. Because it has been used to treat human infections and added to animal feed to prevent animal infections, humans have had extensive exposure to tetracycline. As a result, resistant strains of bacteria have developed.



**Chloramphenicol.** Chloramphenicol binds to the 50S ribosomal subunit of bacteria and prevents binding of the amino acid portion of the aminoacyl-tRNA, effectively inhibiting peptidyltransferase action. This antibiotic is used only for certain extremely serious infections, such as meningitis and typhoid fever. Chloramphenicol readily enters human mitochondria, where it inhibits protein synthesis. Cells of the bone marrow often fail to develop in patients treated with chloramphenicol, and use of this antibiotic has been linked to fatal blood dyscrasias, including an aplastic anemia.



**Erythromycin.** Erythromycin and the other macrolide antibiotics bind to the 50S ribosomal subunit of bacteria near the binding site for chloramphenicol. They prevent the translocation step, the movement of the peptidyl-tRNA from the “A” to the “P” site on the ribosome. Because the side effects are less severe and more readily reversible than those of many other antibiotics, the macrolides are often used to treat infections in persons who are allergic to penicillin, an antibiotic that inhibits bacterial cell wall synthesis. However, bacterial resistance to erythromycin is increasing. Therefore, its close relative, clarithromycin, is often used.

### Suggested References

#### Translation:

Pestova TV, Hellen CUT. Ribosome recruitment and scanning. *TIBS* 1999;24:85–87.

Lewin B. *Genes VII. Protein synthesis. Translation: expressing genes as proteins.* Oxford: Oxford University Press, 2000:139–166.

#### Tay-Sachs:

Gravel R, Clarke J, Kaback M, et al. The Gm2 gangliosidosis. In: Scriver C, Beaudet A, Sly W, Valle D (eds). *The Metabolic and Molecular Bases of Inherited Disease*. 8th Ed. Vol III. New York: McGraw-Hill, 2001:3827–3876.

#### Thalassemia:

Weatherall D, Clegg J, Higgs D, Wood W. The hemoglobinopathies. In: Scriver C, Beaudet A, Sly W, Valle D, eds. *The Metabolic and Molecular Bases of Inherited Disease*. 8th Ed. Vol III. New York: McGraw-Hill, 2001:4571–4636.

#### Antibiotics that inhibit protein synthesis:

Hardman JG, Limbird LE, et al. *Goodman and Gilman's The Pharmacological Basis of Therapeutics*. New York: McGraw-Hill, 1996:1103–1153.



## REVIEW QUESTIONS—CHAPTER 15

- In the read-out of the genetic code in prokaryotes, which one of the following processes acts before any of the others?
  - tRNA<sub>i</sub> alignment with mRNA
  - Termination of transcription
  - Movement of the ribosome from one codon to the next
  - Recruitment of termination factors to the A site
  - Export of mRNA from the nucleus

2. tRNA charged with cysteine can be chemically treated so that the amino acid changes its identity to alanine. If some of this charged tRNA is added to a protein-synthesizing extract that contains ALL the normal components required for translation, which of the following statements represents THE MOST LIKELY OUTCOME after adding an mRNA that has both cys and ala codons in the normal reading frame?
- (A) Cysteine would be added each time the alanine codon was translated.
  - (B) Alanine would be added each time the cysteine codon was translated.
  - (C) The protein would have a deficiency of cysteine residues.
  - (D) The protein would have a deficiency of alanine residues.
  - (E) The protein would be entirely normal.
3. The genetic code is said to be degenerate because of which of the following?
- (A) Many codons have pairs of identical bases next to each other.
  - (B) Some triplets are made up of repeating purines or pyrimidines.
  - (C) Many of the amino acids have more than one triplet code.
  - (D) There is wobble in the bond between the first base of the anticodon and the third base of the codon.
  - (E) All triplets seem to have at least one uracil.
4. Which of the following is responsible for retaining the enzyme protein disulfide isomerase in the ER?
- (A) Glycosylation
  - (B) The presence of a KDEL sequence at the C-terminal
  - (C) Retention of the signal peptide
  - (D) Phosphorylation of Ser 22
  - (E) Attachment of a fatty acid
5. The reason there are 64 possible codons is which of the following?
- (A) There are 64 aminoacyl tRNA synthetases.
  - (B) Each base is able to participate in wobbling.
  - (C) All possible reading frames can be used this way.
  - (D) There are four possible bases at each of three codon positions.
  - (E) The more codons, the faster protein synthesis can be accomplished.

# 16 Regulation of Gene Expression



The prokaryotic bacterium *Escherichia coli* can use a wide range of different nutrients to sustain growth. It accomplishes this partially by turning on and off expression of the genes required for synthesis of enzymes in different metabolic pathways. For example, enzymes encoded by the *lac* operon allow *E. coli* to use lactose for energy and the presence of lactose activates transcription of these genes. Thus, the nutrient itself is involved in regulating gene expression.

In contrast, the cells of the adult human are exposed to a nearly constant and safe environment composed of blood and interstitial fluid. Hormones maintain this nearly constant environment, in spite of changes in nutrient demand and availability, partially by regulating gene transcription. The immune system protects cells against foreign organisms, partially by controlling the expression of genes required for the immune response. However, some nutrient regulation of gene expression also occurs. For example, iron controls expression of the genes for its storage and transport proteins at the level of their mRNAs.

**Gene expression**, the generation of a protein or RNA product from a particular gene, is controlled by complex mechanisms. Normally, only a fraction of the genes in a cell are expressed at any time. Gene expression is regulated differently in prokaryotes and eukaryotes.

**Regulation of gene expression in prokaryotes.** In prokaryotes, gene expression is regulated mainly by controlling the **initiation of gene transcription**. Sets of genes encoding proteins with related functions are organized into **operons**, and each operon is under the control of a **single promoter** (or regulatory region). Regulatory proteins called **repressors** bind to the promoter and inhibit the binding of RNA polymerase (**negative control**), whereas **activator proteins** facilitate RNA polymerase binding (**positive control**). Repressors are controlled by nutrients or their metabolites, classified as **inducers** or **corepressors**. Regulation also may occur through **attenuation of transcription**.

**Eukaryotes: Regulation of gene expression at the level of DNA.** In eukaryotes, activation of a gene requires changes in the state of chromatin (**chromatin remodeling**) that are facilitated by **acetylation of histones and methylation of bases**. These changes in DNA determine which genes are available for transcription.

**Regulation of eukaryotic gene transcription.** Transcription of specific genes is regulated by proteins (called **specific transcription factors or transactivators**) that bind to **gene regulatory sequences** (called **promoter-proximal elements, response elements, or enhancers**) that activate or inhibit assembly of the basal transcription complex and RNA polymerase at the TATA box. These specific transcription factors, which may bind to DNA sequences some distance from the promoter, interact with coactivators or corepressors that bind to components of the basal transcription complex. These protein factors are said to work in “**trans**”; the DNA sequences to which they bind are said to work in “**cis**.”

**Other sites for regulation of eukaryotic gene expression.** Regulation also occurs during the **processing** of RNA, during **RNA transport** from the nucleus to the cytoplasm, and at the level of **translation** in the cytoplasm. Regulation can occur simultaneously at multiple levels for a specific gene, and many factors act in concert to stimulate or inhibit expression of a gene.



## THE WAITING ROOM



**Arlyn Foma**, a 68-year old man, complained of fatigue, loss of appetite, and a low-grade fever. An open biopsy of a lymph node indicated the presence of non-Hodgkin's lymphoma, follicular type. Computed tomography and other noninvasive procedures showed a diffuse process with bone marrow

involvement. He is receiving multidrug chemotherapy with AV/CM (doxorubicin [adriamycin], vincristine, cyclophosphamide, and methotrexate). His disease is not responding well to this regimen, and the follicular lymphoma appears to be evolving into a more aggressive process. Because recombinant interferon  $\alpha$ -2b has been reported to have synergistic or additive effects with these agents, it is added to the protocol. Although resistance to methotrexate is considered, the drug is continued as part of the combined therapeutic approach.



**Mannie Weitzels** is a 56-year-old male who complains of headaches, weight loss related to a declining appetite for food, and a decreasing tolerance for exercise. He notes discomfort and fullness in the left upper quadrant of his abdomen. On physical examination, he is noted to be pale and to have ecchymoses (bruises) on his arms and legs. His spleen is markedly enlarged.

Initial laboratory studies show a hemoglobin of 10.4 g/dL (normal = 13.5–17.5 g/dL) and a leukocyte (white blood cell) count of 86,000 cells/mm<sup>3</sup> (normal = 4,500–11,000 cells/mm<sup>3</sup>). Most of the leukocytes are granulocytes (white blood cells arising from the myeloid lineage), some of which have an “immature” appearance. The percentage of lymphocytes in the peripheral blood is decreased. A bone marrow aspiration and biopsy show the presence of an abnormal chromosome (the Philadelphia chromosome) in dividing marrow cells.



**Ann O'Rexia**, who has anorexia nervosa, has continued on an almost meat-free diet (see Chapters 1, 3, 9, and 11). She now appears emaciated and pale. Her hemoglobin is 9.7 g/dL (normal = 12–16 g/dL), her hematocrit (volume of packed red cells) is 31% (reference range for women = 36–46%), and her mean corpuscular hemoglobin (the average amount of hemoglobin per red cell) is 21 pg/cell (reference range = 26–34 pg/cell). These values indicate an anemia that is microcytic (small red cells) and hypochromic (light in color, indicating a reduced amount of hemoglobin per red cell). Her serum ferritin (the cellular storage form of iron) was also subnormal. Her plasma level of transferrin (the iron transport protein in plasma) was greater than normal, but its percent saturation with iron was below normal. This laboratory profile is consistent with changes that occur in an iron deficiency state.

## I. GENE EXPRESSION IS REGULATED FOR ADAPTATION AND DIFFERENTIATION

Although most cells of an organism contain identical sets of genes, at any given time only a small number of the total genes in each cell are expressed (that is, generate a protein or RNA product). The remaining genes are inactive. Organisms gain a number of advantages by regulating the activity of their genes. For example, both prokaryotic and eukaryotic cells adapt to changes in their environment by turning the expression of genes on and off. Because the processes of RNA transcription and protein synthesis consume a considerable amount of energy, cells conserve fuel by making proteins only when they are needed.

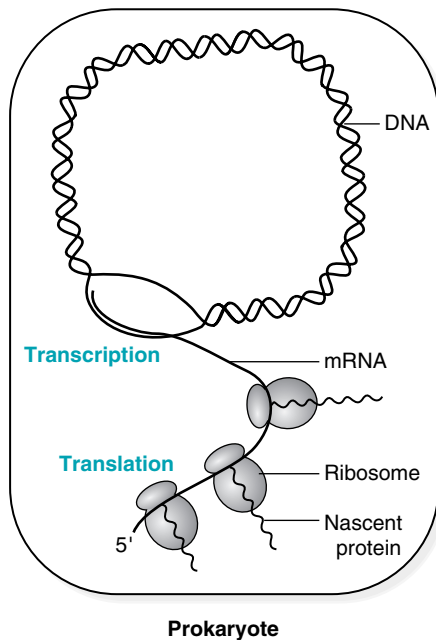
In addition to regulating gene expression to adapt to environmental changes, eukaryotic organisms alter expression of their genes during development. As a fertilized egg becomes a multicellular organism, different kinds of proteins are synthesized in varying quantities. In the human, as the child progresses through adolescence and then into adulthood, physical and physiologic changes result from variations in gene expression and, therefore, of protein synthesis. Even after an organism has reached the adult stage, regulation of gene expression enables certain cells to undergo differentiation to assume new functions.



Each of the drugs used by **Arlyn Foma** inhibits the proliferation of cancer cells in a different way. Doxorubicin (adriamycin) is a large nonpolar molecule synthesized by fungi that intercalates between DNA bases, inhibiting replication and transcription and forming DNA with single- and double-stranded breaks. Vincristine binds to tubulin and inhibits formation of the mitotic spindle, thereby preventing cell division. Cyclophosphamide is an alkylating agent that damages DNA by covalently attaching alkyl groups to DNA bases. Methotrexate is an analogue of the vitamin folate. It inhibits folate-requiring enzymes in the pathways for synthesis of thymine and purines, thereby depriving cells of precursors for DNA synthesis.



*E. coli* is a facultative anaerobe, which means that it can grow in the presence or absence of oxygen. The switch to oxygen-requiring pathways for fuel metabolism is under control of *arc*, the aerobic respiration control gene. When *arc* is activated, transcription is increased by 1,000-fold or more for enzymes in the pathways that ultimately transfer electrons to oxygen (e.g., proteins of the respiratory chain, the TCA cycle, and fatty acid oxidation). In the absence of oxygen (i.e., anaerobic, without air), these proteins are not synthesized, an energy-saving feature useful for bacteria growing in the largely anaerobic colon. Most human cells, in contrast, express constant (constitutive) levels of respiratory enzymes and die without oxygen.



**Fig. 16.1.** *E. coli* cell. In prokaryotes, DNA is not separated from the rest of the cellular contents by a nuclear envelope; therefore, simultaneous transcription and translation occur in bacteria. Once a small piece of mRNA is synthesized, ribosomes bind to the mRNA, and translation begins.

## II. REGULATION OF GENE EXPRESSION IN PROKARYOTES

Prokaryotes are single-celled organisms and, therefore, require less complex regulatory mechanisms than the multicellular eukaryotes (Fig. 16.1). The most extensively studied prokaryote is the bacterium *Escherichia coli*, an organism that thrives in the human colon, usually enjoying a symbiotic relationship with its host. Based on the size of its genome ( $4 \times 10^6$  base pairs), *E. coli* should be capable of making several thousand proteins. However, under normal growth conditions, they synthesize only about 600 to 800 different proteins. Obviously, many genes are inactive, and only those genes are expressed that generate the proteins required for growth in that particular environment.

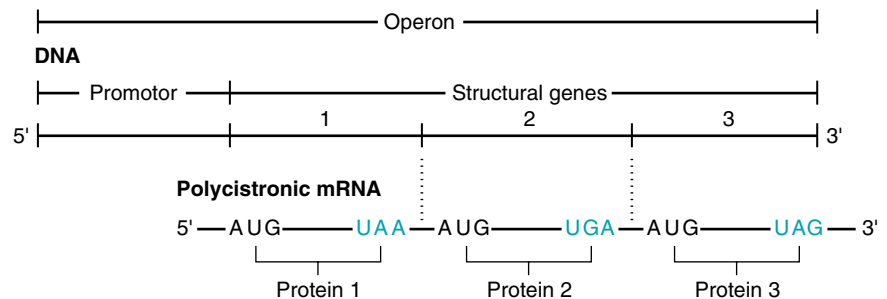
All *E. coli* cells of the same strain are morphologically similar and contain an identical circular chromosome (see Fig. 16.1). As in other prokaryotes, DNA is not complexed with histones, no nuclear envelope separates the genes from the contents of the cytoplasm, and gene transcripts do not contain introns. In fact, as mRNA is being synthesized, ribosomes bind and begin to produce proteins, so that transcription and translation occur simultaneously. The mRNAs in *E. coli* have very short half-lives and are degraded within a few minutes so that an mRNA must be constantly generated from transcription to maintain synthesis of its proteins. Thus, regulation of transcription, principally at the level of initiation, is sufficient to regulate the level of proteins within the cell.

### A. Operons

The genes encoding proteins are called structural genes. In the bacterial genome, the structural genes for proteins involved in performing a related function (such as the enzymes of a biosynthetic pathway) are often grouped sequentially into units called operons (Fig. 16.2). The genes in an operon are coordinately expressed; that is, they are either all “turned on” or all “turned off.” When an operon is expressed, all of its genes are transcribed. A single polycistronic mRNA is produced that codes for all the proteins of the operon. This polycistronic mRNA contains multiple sets of start and stop codons that allow a number of different proteins to be produced from this single transcript at the translational level. Transcription of the genes in an operon is regulated by its promoter, which is located in the operon at the 5′-end, upstream from the structural genes.

### B. Regulation of RNA Polymerase Binding by Repressors

In bacteria, the principle means of regulating gene transcription is through repressors, which are regulatory proteins that prevent the binding of RNA polymerase to the promoter and, thus, act on initiation of transcription (Fig. 16.3). In general,



**Fig. 16.2.** An operon. The structural genes of an operon are transcribed as one long polycistronic mRNA. During translation, different start (AUG) and stop (shown in blue) codons lead to a number of distinct proteins being produced from this single mRNA.



regulatory mechanisms such as repressors that work through inhibition of gene transcription are referred to as negative control, and mechanisms that work through stimulation of gene transcription are called positive control.

The repressor is encoded by a regulatory gene (see Fig. 16.3). Although this gene is considered part of the operon, it is not always located near the remainder of the operon. Its product, the repressor protein, diffuses to the promoter and binds to a region of the operon called the operator. The operator is located within the promoter or near its 3'-end, just upstream from the transcription startpoint. When a repressor is bound to the operator, the operon is not transcribed because the repressor blocks the binding of RNA polymerase to the promoter. Two regulatory mechanisms work through controlling repressors: induction (an inducer inactivates the repressor), and repression (a co-repressor is required to activate the repressor).

## 1. INDUCERS

Induction involves a small molecule, known as an inducer, which stimulates expression of the operon by binding to the repressor and changing its conformation so that it can no longer bind to the operator (Fig. 16.4). The inducer is either a nutrient or a metabolite of the nutrient. In the presence of the inducer, RNA polymerase can therefore bind to the promoter and transcribe the operon. The key to this mechanism is that *in the absence of the inducer, the repressor is active, transcription is repressed, and the genes of the operon are not expressed*.

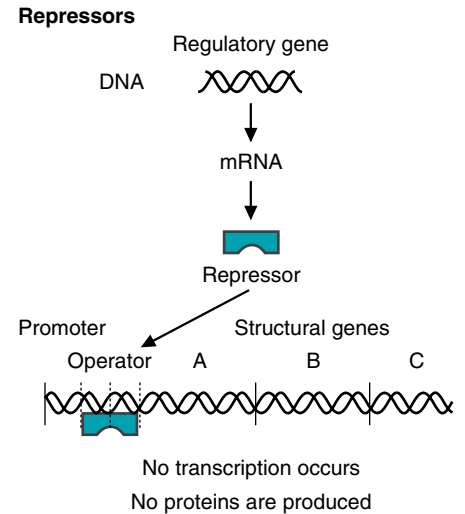
Consider, for example, induction of the *lac* operon of *E. coli* by lactose (Fig. 16.5). The enzymes for metabolizing glucose by glycolysis are produced constitutively; that is, they are constantly being made. If the milk sugar lactose is available, the cells adapt and begin to produce the three additional enzymes required for lactose metabolism, which are encoded by the *lac* operon. A metabolite of lactose (allolactose) serves as an inducer, binding to the repressor and inactivating it. Because the inactive repressor no longer binds to the operator, RNA polymerase can bind to the promoter and transcribe the structural genes of the *lac* operon, producing a polycistronic mRNA that encodes for the three additional proteins. However, the presence of glucose can prevent activation of the *lac* operon (see “Stimulation of RNA polymerase binding,” below).

## 2. COREPRESSORS

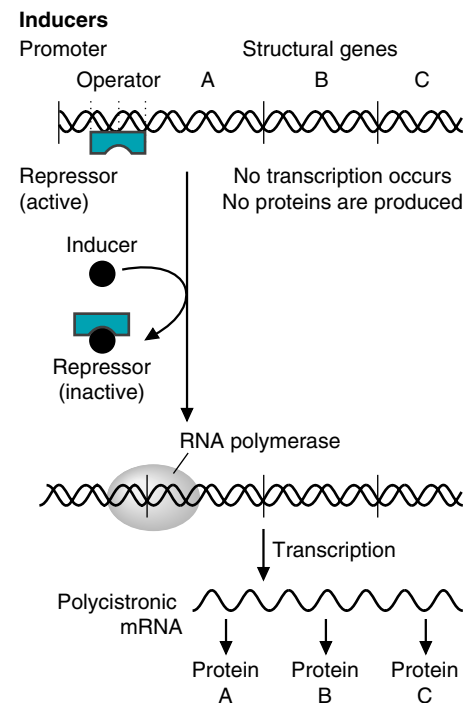
In a regulatory model called repression, the repressor is inactive until a small molecule called a corepressor (a nutrient or its metabolite) binds to the repressor, activating it (Fig. 16.6). The repressor–corepressor complex then binds to the operator, preventing binding of RNA polymerase and gene transcription. Consider, for example, the *trp* operon, which encodes the five enzymes required for the synthesis of the amino acid tryptophan. When tryptophan is available, *E. coli* cells save energy by no longer making these enzymes. Tryptophan is a corepressor that binds to the inactive repressor, causing it to change conformation and bind to the operator, thereby inhibiting transcription of the operon. *Thus, in the repression model, the repressor is inactive without a corepressor; in the induction model, the repressor is active unless an inducer is present.*



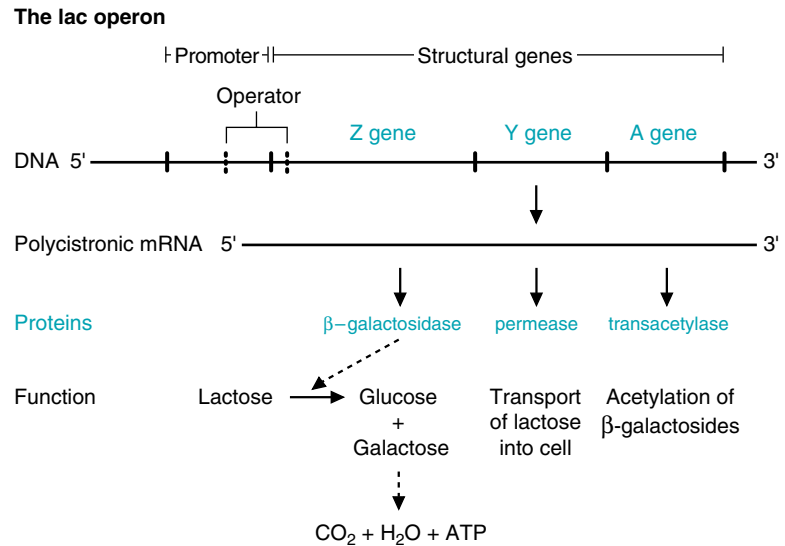
If one of the *lac* operon enzymes induced by lactose is lactose permease (which increases lactose entry into the cell), how does lactose initially get into the cell to induce these enzymes? A small amount of the permease exists even in the absence of lactose, and a few molecules of lactose enter the cell and are metabolized to allolactose, which begins the process of inducing the operon. As the amount of the permease increases, more lactose can be transported into the cell.



**Fig. 16.3.** Regulation of operons by repressors. When the repressor protein is bound to the operator, RNA polymerase cannot bind, and transcription therefore does not occur.



**Fig. 16.4.** An inducible operon. In the absence of an inducer, the repressor binds to the operator, preventing the binding of RNA polymerase. When the inducer is present, the inducer binds to the repressor, inactivating it. The inactive repressor no longer binds to the operator. Therefore, RNA polymerase can bind to the promoter region and transcribe the structural genes.

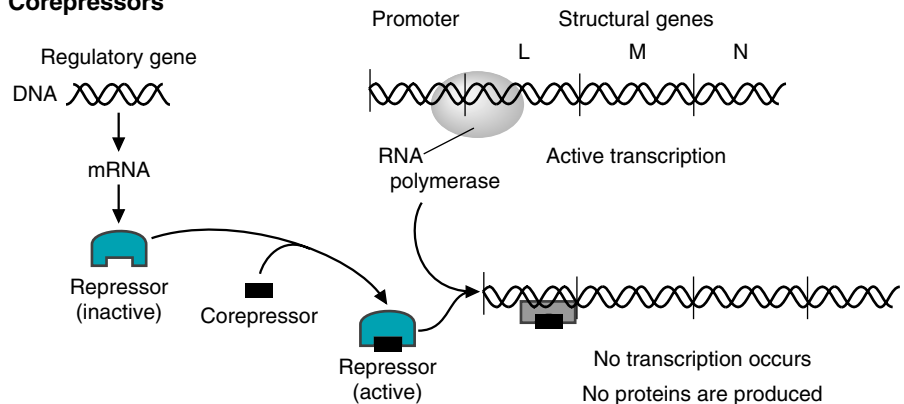


**Fig. 16.5.** The protein products of the *lac* operon. Lactose is a disaccharide that is hydrolyzed to glucose and galactose by β-galactosidase (the Z gene). Both glucose and galactose can be oxidized by the cell for energy. The permease (Y gene) enables the cell to take up lactose more readily. The A gene produces a transacetylase that acetylates β-galactosides. The function of this acetylation is not clear. The promoter binds RNA polymerase and the operator binds a repressor protein. Lactose is converted to allolactose, an inducer that binds the repressor protein and prevents it from binding to the operator. Transcription of the *lac* operon also requires activator proteins that are inactive when glucose levels are high.

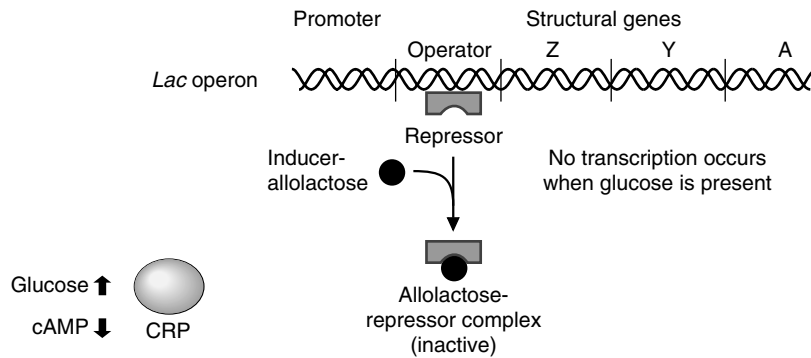
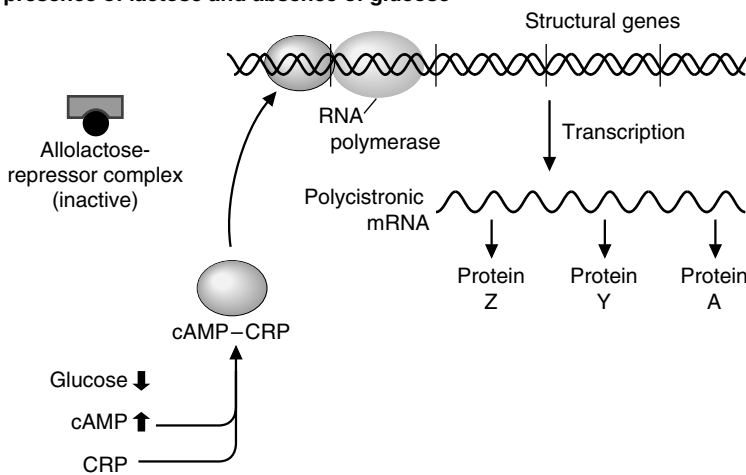
### C. Stimulation of RNA Polymerase Binding

In addition to regulating transcription by means of repressors that inhibit RNA polymerase binding to promoters (negative control), bacteria regulate transcription by means of activating proteins that bind to the promoter and stimulate the binding of RNA polymerase (positive control). Transcription of the *lac* operon, for example, can be induced by allolactose only if glucose is absent. The presence or absence of glucose is communicated to the promoter by a regulatory protein named the cyclic adenosine monophosphate (cAMP) receptor protein (CRP)

#### Corepressors



**Fig. 16.6.** A repressible operon. The repressor is inactive until a small molecule, the corepressor, binds to it. The repressor–corepressor complex binds to the operator and prevents transcription.

**A. In the presence of lactose and glucose****B. In the presence of lactose and absence of glucose**

**Fig. 16.7.** Catabolite repression of stimulatory proteins. The *lac* operon is used as an example. **A.** The inducer allolactose (a metabolite of lactose) inactivates the repressor. However, because of the absence of the required coactivator, cAMP-CRP, no transcription occurs unless glucose is absent. **B.** In the absence of glucose, cAMP levels rise. cAMP forms a complex with the cAMP receptor protein (CRP). The binding of the cAMP-CRP complex to a regulatory region of the operon permits the binding of RNA polymerase to the promoter. Now the operon is transcribed, and the proteins are produced.

(Fig. 16.7). This regulatory protein is also called a catabolite activator protein (CAP). A decrease in glucose levels increases levels of the intracellular second messenger cAMP by a mechanism that is not well understood. cAMP binds to CRP, and the cAMP-CRP complex binds to a regulatory region of the operon, stimulating binding of RNA polymerase to the promoter and transcription. When glucose is present, cAMP levels decrease, CRP assumes an inactive conformation that does not bind to the operon, and transcription is inhibited. Thus, the enzymes encoded by the *lac* operon are not produced if cells have an adequate supply of glucose, even if lactose is present at very high levels.

### D. Regulation of RNA Polymerase Binding by Sigma Factors

*E. coli* has only one RNA polymerase. Sigma factors bind to this RNA polymerase, stimulating its binding to certain sets of promoters, thus simultaneously activating transcription of several operons. The standard sigma factor in *E. coli* is

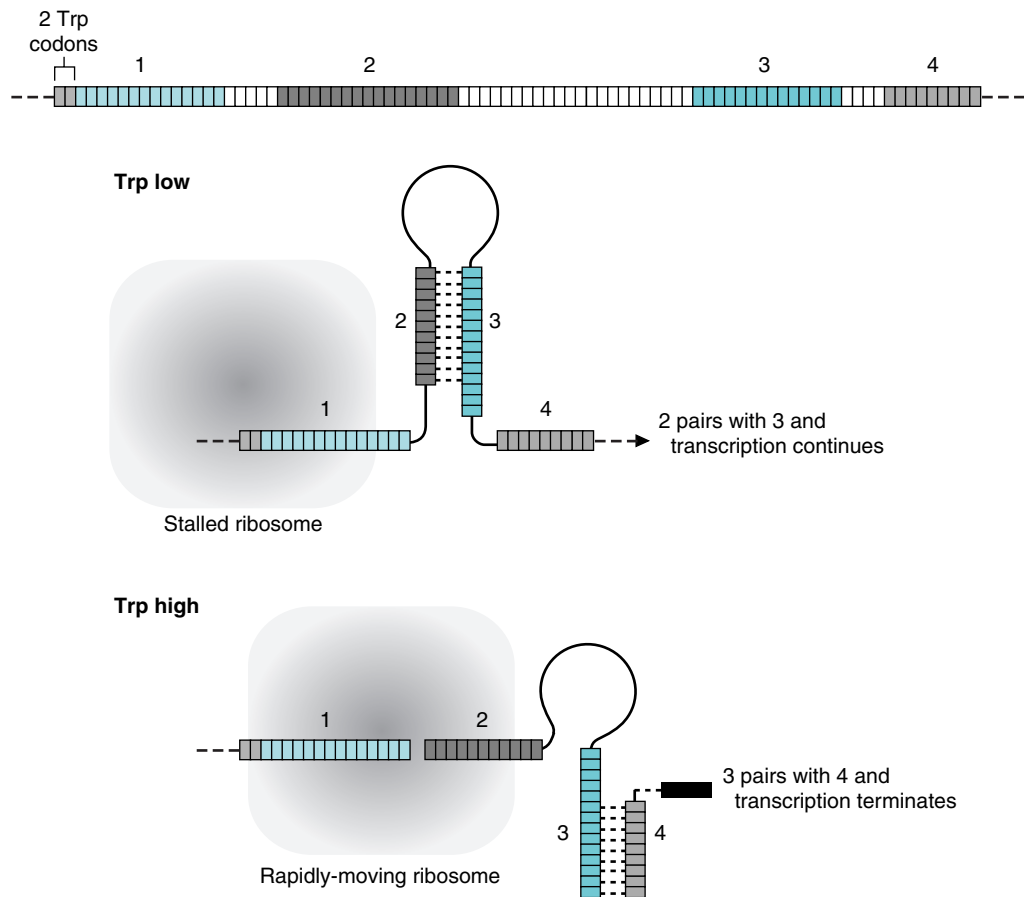


Nutrient regulation of gene expression may occur through the nutrient itself, or through a metabolite of the nutrient synthesized inside the cell. Sometimes both the nutrient and its metabolite are grouped into the term “catabolite,” as in “catabolite repression.”

$\sigma^{70}$ , a protein with a molecular weight of 70,000 daltons (see Chapter 14). Other sigma factors also exist. For example,  $\sigma^{32}$  helps RNA polymerase recognize promoters for the different operons that encode the “heat shock” proteins. Thus, increased transcription of the genes for heat shock proteins, which prevent protein denaturation at high temperatures, occurs in response to elevated temperatures.

### E. Attenuation of Transcription

Some operons are regulated by a process that interrupts (attenuates) transcription after it has been initiated (Fig. 16.8). For example, high levels of tryptophan attenuate transcription of the *E. coli trp* operon, as well as repress its transcription. As mRNA is being transcribed from the *trp* operon, ribosomes bind and rapidly begin to translate the transcript. Near the 5′-end of the transcript, there are a number of codons for tryptophan. Initially, high levels of tryptophan in the cell result in high levels of  $\text{trp-tRNA}^{\text{trp}}$  and rapid translation of the transcript. However, rapid translation generates a hairpin loop in the mRNA that serves as a termination signal for RNA polymerase, and transcription terminates. Conversely, when tryptophan levels are low, levels of  $\text{trp-tRNA}^{\text{trp}}$  are low, and ribosomes stall at codons for tryptophan. A different hairpin loop forms in the mRNA that does not terminate transcription, and the complete mRNA is transcribed.



**Fig. 16.8.** Attenuation of the *trp* operon. Sequences 2, 3, and 4 in the mRNA transcript can form base pairs (2 with 3 or 3 with 4) that generate hairpin loops. When tryptophan levels are low, the ribosome stalls at the adjacent *trp* codons in sequence 1, the 2–3 loop forms, and transcription continues. When tryptophan levels are high, translation is rapid and the ribosome blocks formation of the 2–3 loop. Under these conditions, the 3–4 loop forms and terminates transcription.

The tryptophan, histidine, leucine, phenylalanine, and threonine operons are regulated by attenuation. Repressors and activators also act on the promoters of some of these operons, allowing the levels of these amino acids to be very carefully and rapidly regulated.

### III. REGULATION OF PROTEIN SYNTHESIS IN EUKARYOTES

Multicellular eukaryotes are much more complex than single-celled prokaryotes. As the human embryo develops into a multicellular organism, different sets of genes are turned on, and different groups of proteins are produced, resulting in differentiation into morphologically distinct cell types able to perform different functions. Even beyond the reproductive age, certain cells within the organism continue to differentiate, such as those that produce antibodies in response to an infection, renew the population of red blood cells, and replace digestive cells that have been sloughed into the intestinal lumen. All of these physiologic changes are dictated by complex alterations in gene expression.

#### A. Regulation of Eukaryotic Gene Expression at Multiple Levels

Differences between eukaryotic and prokaryotic cells result in different mechanisms for regulating gene expression. DNA in eukaryotes is organized into the nucleosomes of chromatin, and genes must be in an active structure to be expressed in a cell. Furthermore, operons are not present in eukaryotes, and the genes encoding proteins that function together are usually located on different chromosomes. Thus, each gene needs its own promoter. In addition, the processes of transcription and translation are separated in eukaryotes by intracellular compartmentation (nucleus and cytosol, or endoplasmic reticulum [ER]) and by time (eukaryotic hnRNA must be processed and translocated out of the nucleus before it is translated). Thus, regulation of eukaryotic gene expression occurs at multiple levels:

- DNA and the chromosome, including chromosome remodeling and gene rearrangement
- Transcription, primarily through transcription factors affecting binding of RNA polymerase
- Processing of transcripts
- Initiation of translation and stability of mRNA

Once a gene is activated through chromatin remodeling, the major mechanism of regulating expression affects initiation of transcription at the promoter.

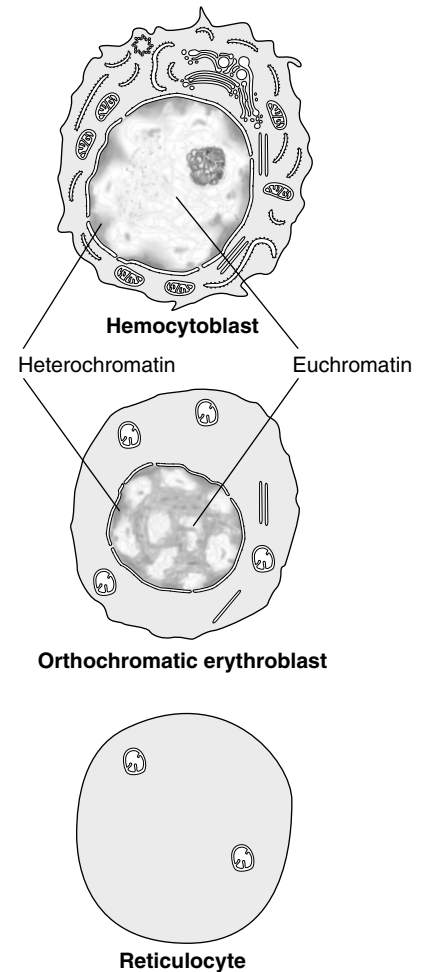
#### B. Regulation of Availability of Genes for Transcription

Once a haploid sperm and egg combine to form a diploid cell, the number of genes in human cells remains approximately the same. As cells differentiate, different genes are available for transcription. A typical nucleus contains chromatin that is condensed (heterochromatin) and chromatin that is diffuse (euchromatin) (Fig. 16.9) (see Chapter 12). The genes in heterochromatin are inactive, whereas those in euchromatin produce mRNA. Long-term changes in the activity of genes occur during development as chromatin goes from a diffuse to a condensed state or vice versa.

The cellular genome is packaged together with histones into nucleosomes, and initiation of transcription is prevented if the promoter region is part of a nucleosome. Thus, activation of a gene for transcription requires changes in the state of the chromatin, called chromatin remodeling. The availability of genes for transcription

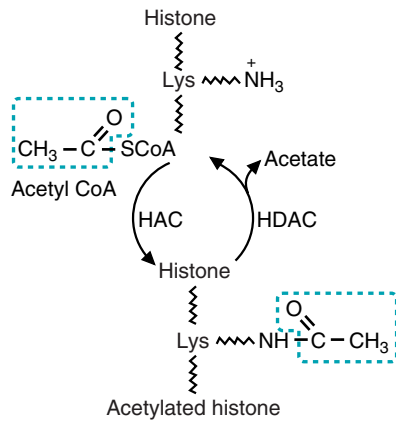


The globin chains of hemoglobin provide an example of functionally related proteins that are on different chromosomes. The gene for the  $\alpha$ -globin chain is on chromosome 16, whereas the gene for the  $\beta$ -globin chain is on chromosome 11. As a consequence of this spatial separation, each gene must have its own promoter. This situation is different from that of bacteria, in which genes encoding proteins that function together are often sequentially arranged in operons controlled by a single promoter.



**Fig. 16.9.** Inactivation of genes during development of red blood cells. Diffuse chromatin (euchromatin) is active in RNA synthesis. Condensed chromatin (heterochromatin) is inactive. As red blood cell precursors mature, their chromatin becomes more condensed. Eventually, the nucleus is extruded.





**Fig. 16.10.** Histone acetylation. Abbreviations: HAC, histone acetylase; HDAC, histone deacetylase.

also can be affected in certain cells, or under certain circumstances, by gene rearrangements, amplification, or deletion. For example, during lymphocyte maturation, genes are rearranged to produce a variety of different antibodies.

## 1. CHROMATIN REMODELING

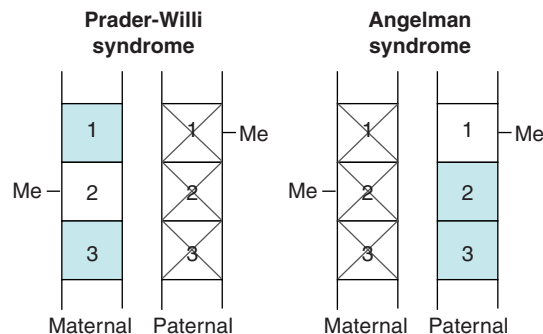
The remodeling of chromatin generally refers to displacement of the nucleosome from specific DNA sequences so that transcription of the genes in that sequence can be initiated. This occurs through two different mechanisms. The first mechanism is by an adenosine triphosphate (ATP)-driven chromatin remodeling complex, which uses energy from ATP hydrolysis to unwind certain sections of DNA from the nucleosome core. The second mechanism is by covalent modification of the histone tails through acetylation (Fig. 16.10). Histone acetyltransferases (HAT) transfer an acetyl group from acetyl CoA to lysine residues in the tail (the amino terminal ends of histones H2A, H2B, H3, and H4). This reaction removes a positive charge from the  $\epsilon$ -amino group of the lysine, thereby reducing the electrostatic interactions between the histones and the negatively charged DNA, making it easier for DNA to unwind from the histones. The acetyl groups can be removed by histone deacetylases (HDAC). Each histone has a number of lysine residues that may be acetylated and, through a complex mixing of acetylated and nonacetylated sites, different segments of DNA can be freed from the nucleosome. A number of transcription factors and co-activators also contain histone acetylase activity, which facilitates the binding of these factors to the DNA and simultaneous activation of the gene and initiation of its transcription.

## 2. METHYLATION OF DNA

Cytosine residues in DNA can be methylated to produce 5-methylcytosine. The methylated cytosines are located in GC-rich sequences (called GC-islands), which are often near or in the promoter region of a gene. In certain instances, genes that are methylated are less readily transcribed than those that are not methylated. For example, globin genes are more extensively methylated in nonerythroid cells (cells which are not a part of the erythroid, or red blood cell, lineage) than in the cells in which these genes are expressed (such as the erythroblast and reticulocyte). Methylation is a mechanism for regulating gene expression during differentiation, particularly in fetal development.



Methylation has been implicated in genomic imprinting, a process occurring during the formation of the eggs or sperm that blocks the expression of the gene in the fertilized egg. Males methylate a different set of genes than females. This sex-dependent differential methylation has been most extensively studied in two human disorders, Prader-Willi syndrome and Angelman syndrome. Both syndromes, which have very different symptoms, result from deletions of the same region of chromosome 15 (a microdeletion of less than 5 megabases in size). If the deletion is inherited from the father, Prader-Willi syndrome is seen in the child; if the deletion is inherited from the mother, Angelman's syndrome is observed. A disease occurs when a gene that is in the deleted region of one chromosome is methylated on the other chromosome. The mother methylates different genes than the father, so different genes are expressed depending on which parent transmitted the intact chromosome. For example, if genes 1, 2, and 3 are deleted in the paternal chromosome in the Prader-Willi syndrome, and gene 2 is methylated in the maternal chromosome, only genes 1 and 3 will be expressed. If in the Angelman syndrome, genes 1, 2 and 3 are deleted on the maternal chromosome and gene 1 is methylated on the paternal chromosome, only genes 2 and 3 would be expressed.



### 3. GENE REARRANGEMENT

Segments of DNA can move from one location to another in the genome, associating with each other in various ways so that different proteins are produced (Fig. 16.11). The most thoroughly studied example of gene rearrangement occurs in cells that produce antibodies. Antibodies contain two light chains and two heavy chains, each of which contains both a variable and a constant region (see Chapter 7, section V.B, Fig. 7.19). Cells called B cells make antibodies. In the precursors of B cells, hundreds of  $V_H$  sequences, approximately 20  $D_H$  sequences, and approximately 6  $J_H$  sequences are located in clusters within a long region of the chromosome. During the production of the immature B cells, a series of recombinational events occur that join one  $V_H$ , one  $D_H$ , and one  $J_H$  sequence into a single exon. This now encodes the variable region of the heavy chain of the antibody. Given the large number of immature B cells that are produced, virtually every recombinational possibility occurs, such that all VDJ combinations are represented within this cell population. Later in development, during differentiation of mature B cells, recombinational events join a VDJ sequence to one of the nine heavy chain elements. When the immune system encounters an antigen, the one immature B cell that can bind to that antigen (because of its unique manner in forming the VDJ exon) is stimulated to proliferate (clonal expansion) and to produce antibodies against the antigen.

### 4. GENE AMPLIFICATION

Gene amplification is not the usual physiologic means of regulating gene expression in normal cells, but it does occur in response to certain stimuli if the cell can obtain a growth advantage by producing large amounts of a protein. In gene amplification, certain regions of a chromosome undergo repeated cycles of DNA replication. The newly synthesized DNA is excised and forms small, unstable chromosomes called “double minutes.” The double minutes integrate into other chromosomes throughout the genome, thereby amplifying the gene in the process. Normally, gene amplification occurs through errors during DNA replication and cell division and, if the environmental conditions are correct, cells containing amplified genes may have a growth advantage over those without the amplification.



In fragile X syndrome, a GCC triplet is amplified on the 5'-side of a gene (FMR-1) associated with the disease. This gene is located on the X chromosome. The disease is named for the finding that in the absence of folic acid (which impairs nucleotide production and hence, the replication of DNA) the X chromosome develops single and double-stranded breaks in its DNA. These were termed fragile sites. It was subsequently determined that the FMR-1 gene was located in one of these fragile sites. A normal person has about 30 copies of the GCC triplet, but in affected individuals, thousands of copies can be present. This syndrome, which is a common form of inherited mental retardation, affects about 1 in 1,250 males and 1 in 2,000 females.

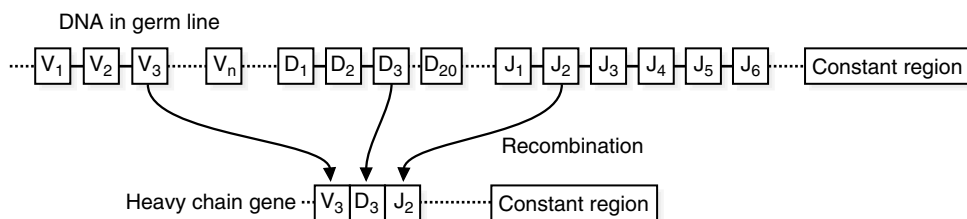


Although rearrangements of short DNA sequences are difficult to detect, microscopists have observed major rearrangements for many years. Such major rearrangements, known as translocations, can be observed in metaphase chromosomes under the microscope.

**Mannie Weitzels** has such a translocation, known as the Philadelphia chromosome because it was first observed in that city. The Philadelphia chromosome is produced by a balanced exchange between chromosomes 9 and 22.



**Arlyn Foma** has been treated with a combination of drugs that includes methotrexate, a drug that inhibits cell proliferation by inhibiting dihydrofolate reductase. Dihydrofolate reductase reduces dihydrofolate to tetrahydrofolate, a cofactor required for synthesis of thymine and purine nucleotides. Because Arlyn Foma has not been responding well, the possibility that he has become resistant to methotrexate was considered. Sometimes, rapidly dividing cancer cells treated with methotrexate amplify the gene for dihydrofolate reductase, producing hundreds of copies in the genome. These cells generate large amounts of dihydrofolate reductase, and normal doses of methotrexate are no longer adequate. Gene amplification is one of the mechanisms by which patients become resistant to a drug.



**Fig. 16.11.** Rearrangement of DNA. The heavy chain gene from which lymphocytes produce immunoglobulins is generated by combining specific segments from among a large number of potential sequences in the DNA of precursor cells. The variable and constant regions of immunoglobulins (antibodies) are described in Chapter 7.

## 5. GENE DELETIONS

With a few exceptions, the deletion of genetic material is likewise not a normal means of controlling transcription, although such deletions do result in disease. Gene deletions can occur through errors in DNA replication and cell division and are usually only noticed if a disease results. For example, various types of cancers result from the loss of a good copy of a tumor suppressor gene, leaving the cell with a mutated copy of the gene (see Chapter 18).

## C. Regulation at the Level of Transcription

The transcription of active genes is regulated by controlling assembly of the basal transcription complex containing RNA polymerase and its binding to the TATA box of the promoter (see Chapter 14). The basal transcription complex contains the TATA binding protein (TBP, a component of TFIID) and other proteins called general (basal) transcription factors (such as TFIIA, etc.) that form a complex with RNA polymerase II. Additional transcription factors that are ubiquitous to all promoters bind upstream at various sites in the promoter region. They increase the frequency of transcription and are required for a promoter to function at an adequate level. Genes that are regulated solely by these consensus elements in the promoter region are said to be constitutively expressed.

The control region of a gene also contains DNA regulatory sequences that are specific for that gene and may increase its transcription 1,000-fold or more (Fig. 16.12). Gene-specific transcription factors (also called transactivators or activators) bind to these regulatory sequences and interact with a mediator protein, such as a coactivator. By forming a loop in the DNA, coactivators interact with the basal transcription complex and can activate its assembly at the initiation site on the promoter. These DNA regulatory sequences might be some distance from the promoter and may be either upstream or downstream of the initiation site.



The terminology used to describe components of gene-specific regulation varies somewhat, depending on the system. For example, in the original terminology, DNA regulatory sequences called enhancers bound transactivators, which bound coactivators. Similarly, silencers bound corepressors. Hormones bound to hormone receptors, which bound to hormone response elements in DNA. Although these terms are still used, they are often replaced with more general terms such as “DNA regulatory sequences” and “specific transcription factors,” in recognition of the fact that many transcription factors activate one gene while inhibiting another or that a specific transcription factor may be changed from a repressor to an activator by phosphorylation.

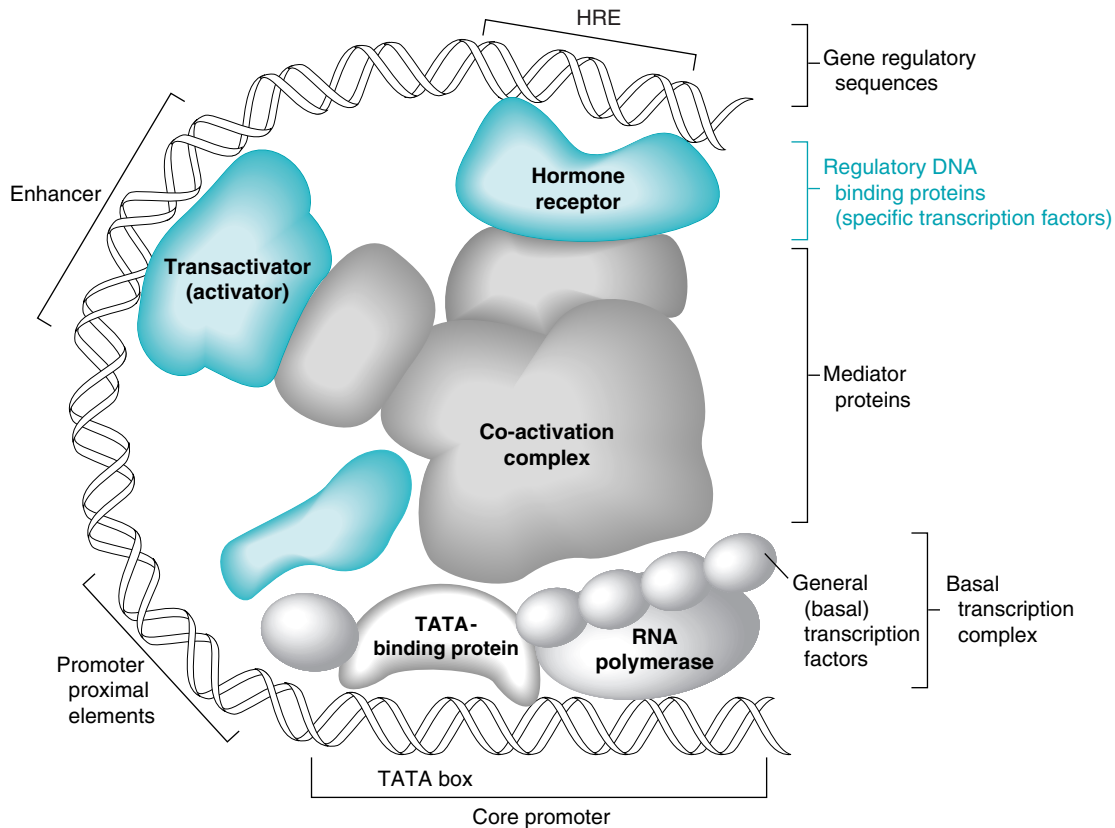


Histone acetylase activity has been associated with a number of transcription factors and co-activators. The proteins ACTR (activator of the thyroid and retinoic acid receptor) and SRC-1 (steroid receptor co-activator) are involved in activation of transcription by several ligand-bound nuclear receptors, and both contain histone acetylase activity. TAF250, a component of TFIID, also contains histone acetylase activity, as does the co-activator p300/CBP (CREB binding protein), which interacts with the transcription factor CREB.

### 1. GENE-SPECIFIC REGULATORY PROTEINS

The regulatory proteins that bind directly to DNA sequences are most often called transcription factors or gene-specific transcription factors (if it is necessary to distinguish them from the general transcription factors of the basal transcription complex). They also can be called activators (or transactivators), inducers, repressors, or nuclear receptors. In addition to their DNA-binding domain, these proteins usually have a domain that binds to mediator proteins (coactivators, corepressors, or TATA binding protein associated factors—TAFs). Coactivators, corepressors, and other mediator proteins do not bind directly to DNA but generally bind to components of the basal transcription complex and mediate its assembly at the promoter. They can be specific for a given gene transcription factor or general and bind many different gene-specific transcription factors. Certain co-activators have histone acetylase activity, and certain corepressors have histone deacetylase activity. When the appropriate interactions between the transactivators, coactivators, and the basal transcription complex occur, the rate of transcription of the gene is increased (induction).

Some regulatory DNA binding proteins inhibit (repress) transcription and may be called repressors. Repression can occur in a number of ways. A repressor bound to its specific DNA sequence may inhibit binding of an activator to its regulatory sequence. Alternately, the repressor may bind a corepressor that inhibits binding of a coactivator to the basal transcription complex. The repressor may directly bind a component of the basal transcription complex. Some steroid hormone receptors that are transcription factors bind either coactivators or corepressors, depending on whether the receptor contains bound hormone. Furthermore, a particular transcription factor may induce transcription when bound to the regulatory sequence of one gene and may repress transcription when bound to the regulatory sequence of another gene.



**Fig. 16.12.** The gene regulatory control region consists of the promoter region and additional gene regulatory sequences, including enhancers and hormone response elements (shown in blue). Gene regulatory proteins that bind directly to DNA (regulatory DNA binding proteins) are usually called specific transcription factors or transactivators; they may be either activators or repressors of the transcription of specific genes. The specific transcription factors bind mediator proteins (co-activators or corepressors) that interact with the general transcription factors of the basal transcription complex. The basal transcription complex contains RNA polymerase and associated general transcription factors (TFII factors) and binds to the TATA box of the promoter, initiating gene transcription.

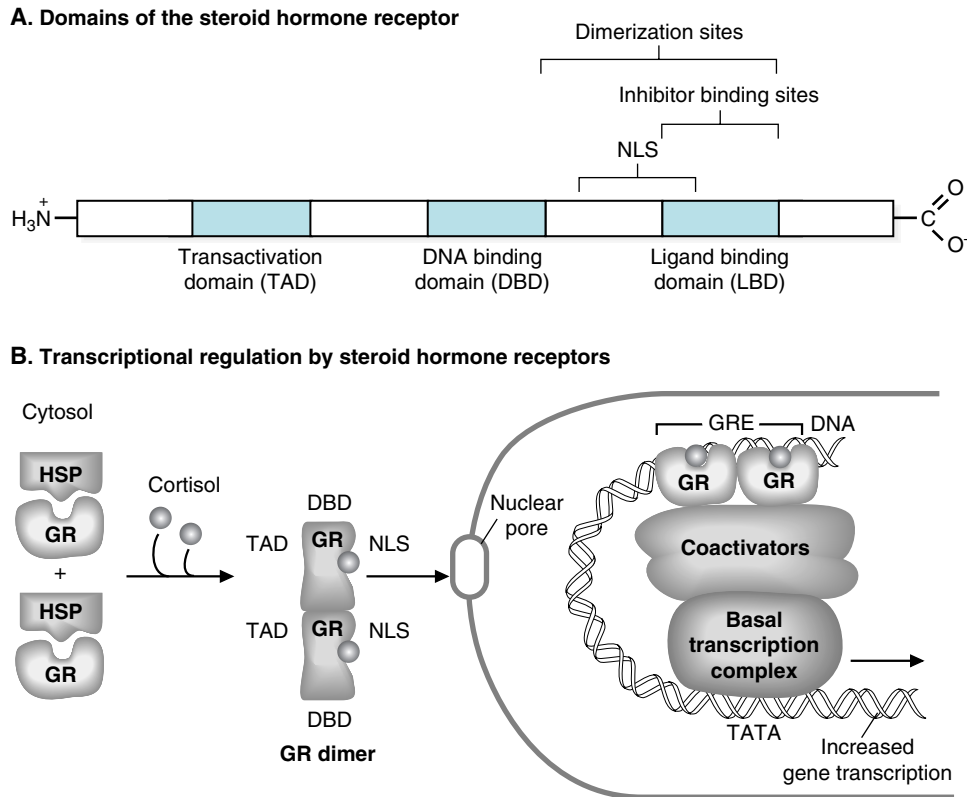
## 2. TRANSCRIPTION FACTORS THAT ARE STEROID HORMONE/THYROID HORMONE RECEPTORS

In the human, steroid hormones and other lipophilic hormones activate or inhibit transcription of specific genes through binding to nuclear receptors that are gene-specific transcription factors (Fig. 16.13A). The nuclear receptors bind to DNA regulatory sequences called hormone response elements and induce or repress transcription of target genes. The receptors contain a hormone (ligand) binding domain, a DNA binding domain, and a dimerization domain that permits two receptor molecules to bind to each other, forming characteristic homodimers or heterodimers. A transactivation domain binds the coactivator proteins that interact with the basal transcription complex. The receptors also contain a nuclear localization signal domain that directs them to the nucleus at various times after they are synthesized.

Various members of the steroid hormone/thyroid hormone receptor family work in different ways. The glucocorticoid receptor, which binds the steroid hormone cortisol, resides principally in the cytosol bound to heat shock proteins. As cortisol binds, the receptor dissociates from the heat shock proteins, exposing the nuclear localization signal (see Fig. 16.13B). The receptors form homodimers that are translocated to the nucleus, where they bind to the hormone response elements (glucocorticoid response elements–GRE) in the DNA control region of certain genes. The transactivation domains of the receptor dimers bind mediator proteins, thereby activating transcription of specific genes and inhibiting transcription of others.



In a condition known as testicular feminization, patients produce androgens (the male sex steroids), but target cells fail to respond to these steroid hormones because they lack the appropriate intracellular transcription factor receptors. Therefore, the transcription of the genes responsible for masculinization is not activated. A patient with this condition has an XY (male) karyotype (set of chromosomes) but looks like a female. External male genitalia do not develop, but testes are present, usually in the inguinal region.



**Fig. 16.13.** Steroid hormone receptors. A. Domains of the steroid hormone receptor. The transactivation domain (TAD) binds coactivators; DNA-binding domain (DBD) binds to hormone response element in DNA; ligand-binding domain (LBD) binds hormone; NLS is the nuclear localization signal; the dimerization sites are the portions of the protein involved in forming a dimer. The inhibitor binding site binds heat shock proteins and masks the nuclear localization signal. B. Transcriptional regulation by steroid hormone receptors. Additional abbreviations: HSP, heat shock proteins; GRE, glucocorticoid response element; GR, glucocorticoid receptor.

Other members of the steroid hormone/thyroid hormone family of receptors are also gene-specific transactivation factors but generally form heterodimers that constitutively bind to a DNA regulatory sequence in the absence of their hormone ligand and repress gene transcription (Fig. 16.14). For example, the thyroid hormone receptor forms a heterodimer with the retinoid X receptor (RXR) that binds to thyroid hormone response elements and to corepressors (including one with deacetylase activity), thereby inhibiting expression of certain genes. When thyroid hormone binds, the receptor dimer changes conformation, and the transactivation domain binds coactivators, thereby initiating transcription of the genes.

The RXR receptor, which binds the retinoid 9-*cis* retinoic acid, can form heterodimers with at least eight other nuclear receptors. Each heterodimer has a different DNA binding specificity. This allows the RXR to participate in the regulation of a wide variety of genes, and to regulate gene expression differently, depending on the availability of other active receptors.

### 3. STRUCTURE OF DNA BINDING PROTEINS

Several unique structural motifs have been characterized for specific transcription factors. Each of these proteins has a distinct recognition site (DNA binding domain) that binds to the bases of a specific sequence of nucleotides in DNA. Four of the best-characterized structural motifs are zinc fingers, b-zip proteins (including leucine zippers), helix-turn-helix, and helix-loop-helix.





To regulate gene transcription, two estrogen receptors combine to form a dimer that binds to a palindrome in the promoter region of certain genes (see Figs. 16.12 and 16.13). A palindrome is a sequence of bases that is identical on the antiparallel strand when read in the opposite direction. For example, the sequence ATCGCGAT base-pairs to form the sequence TAGCGCTA, which when read in the opposite direction is ATCGCGAG. Each estrogen receptor is approximately 73 amino acids long and contains two zinc fingers. Each zinc is chelated to two cysteines in an  $\alpha$ -helix and two cysteines in a  $\beta$ -sheet region. The position of the nucleotide recognition sequence in an  $\alpha$ -helix keeps the sequence in a relatively rigid conformation as it fits into the major groove of DNA. The zinc finger that lies closest to the carboxyl terminal is involved in dimerization with the second estrogen receptor, thus inverting the nucleotide recognition sequence to match the other half of the palindrome. The dimer-palindrome requirement enormously enhances the specificity of binding, and, consequently, only certain genes are affected.

Zinc finger motifs (commonly found in the DNA binding domain of steroid hormone receptors) contain a bound zinc chelated at four positions with either histidine or cysteine in a sequence of approximately 20 amino acids (Fig. 16.15). The result is a relatively small, tight, autonomously-folded domain. The zinc is required to maintain the tertiary structure of this domain. Eukaryotic transcription factors generally have two to six zinc finger motifs that function independently. At least one of the zinc fingers forms an  $\alpha$ -helix containing a nucleotide recognition signal, a sequence of amino acids that specifically fits into the major groove of DNA (Fig. 16.16A).

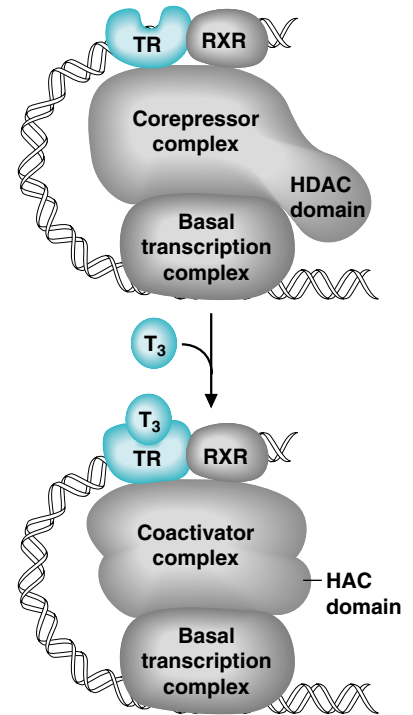
Leucine zippers also function as dimers to regulate gene transcription (see Fig. 16.16B). The leucine zipper motif is an  $\alpha$ -helix of 30 to 40 amino acid residues that contains a leucine every seven amino acids, positioned so that they align on the same side of the helix. Two helices dimerize so that the leucines of one helix align with the



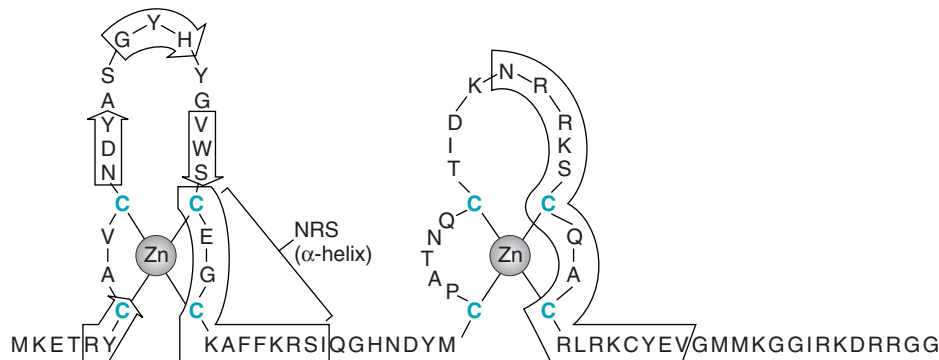
A wide variety of transcription factors contain the zinc finger motif, including the steroid hormone receptors, such as the estrogen and the glucocorticoid receptor. Other transcription factors that contain zinc finger motifs include Sp1 and polymerase III transcription factor TFIIIA (part of the basal transcription complex), which has nine zinc finger motifs.



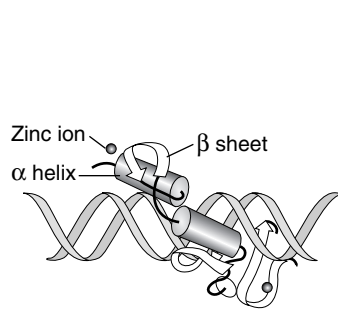
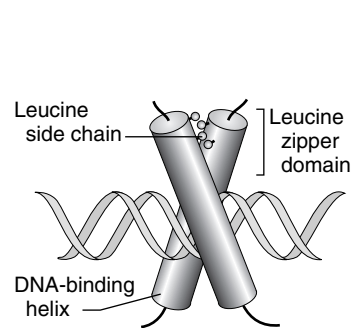
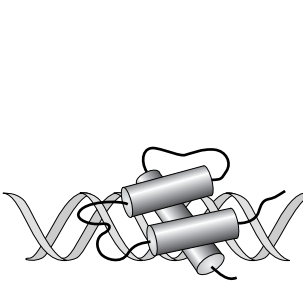
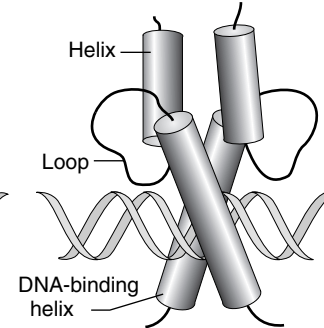
Leucine zipper transcription factors function as homodimers or heterodimers. For example, AP1 is a heterodimer whose subunits are encoded by the genes *fos* and *jun*.



**Fig. 16.14.** Activity of the thyroid hormone receptor-retinoid receptor dimer (TR-RXR) in the presence and absence of thyroid hormone ( $T_3$ ). Abbrev: HAC, histone acetylase; HDAC, histone deacetylase.



**Fig. 16.15.** Zinc fingers of the estrogen receptor. In each of the two zinc fingers, one zinc ion is coordinated with four cysteine residues, shown in blue. The region labeled  $\alpha$ -helix with NRS forms an  $\alpha$ -helix that contains a nucleotide recognition signal (NRS). This signal consists of a sequence of amino acid residues that bind to a specific base sequence in the major groove of DNA. Regions enclosed in boxed arrows participate in rigid helices.

**A. Zinc fingers****B. Leucine zipper****C. Helix-turn-helix****D. Helix-loop-helix**

**Fig. 16.16.** Interaction of DNA binding proteins with DNA. (A) Zinc finger motifs consist of an  $\alpha$ -helix and a  $\beta$  sheet in which 4 cysteine and/or histidine residues coordinately bind a zinc ion. The nucleotide recognition signal (contained within the  $\alpha$ -helix) of at least one zinc finger binds to a specific sequence of bases in the major groove of DNA. (B) Leucine zipper motifs form from two distinct polypeptide chains. Each polypeptide contains a helical region in which leucine residues are exposed on one side. These leucines form hydrophobic interactions with each other, causing dimerization. The remaining helices interact with DNA. (C) Helix-turn-helix motifs contain three (or sometimes four) helical regions, one of which binds to DNA, whereas the others lie on top and stabilize the interaction. (D) Helix-loop-helix motifs contain helical regions that bind to DNA like leucine zippers. However, their dimerization domains consist of two helices, each of which is connected to its DNA binding helix by a loop.



The helix-turn-helix motif is found in homeodomain proteins (proteins that play critical roles in the regulation of gene expression during development).



Many of the transcription factors containing the helix-loop-helix motif are involved in cellular differentiation (such as myogenin in skeletal muscle, neurogenin in neurogenesis and SCL/tal-1 in hematopoiesis, blood cell development).

other helix through hydrophobic interactions to form a coiled coil. The portions of the dimer adjacent to the zipper “grip” the DNA through basic amino acid residues (arginine and lysine) that bind to the negatively charged phosphate groups. This DNA binding portion of the molecule also contains a nucleotide recognition signal.

In the helix-turn-helix motif, one helix fits into the major groove of DNA, making most of the DNA binding contacts (see Fig. 16.16C). It is joined to a segment containing two additional helices that lie across the DNA binding helix at right angles. Thus, a very stable structure is obtained without dimerization.

Helix-loop-helix transcription factors are a fourth structural type of DNA binding protein (see Fig. 16.16D). They also function as dimers that fit around and grip DNA in a manner geometrically similar to leucine zipper proteins. The dimerization region consists of a portion of the DNA-gripping helix and a loop to another helix. Like leucine zippers, helix-loop-helix factors can function as either hetero or homodimers. These factors also contain regions of basic amino acids near the amino terminus and are also called basic helix-loop-helix (bHLH) proteins.

#### 4. REGULATION OF TRANSCRIPTION FACTORS

The activity of gene-specific transcription factors is regulated in a number of different ways. Because transcription factors need to interact with a variety of coactivators to stimulate transcription, the availability of coactivators or other mediator proteins is critical for transcription factor function. If a cell upregulates or downregulates its synthesis of coactivators, the rate of transcription can also be increased or decreased. Transcription factor activity can be modulated by changes in the amount of transcription factor synthesized (see section 5, “Multiple regulators of promoters”), by binding a stimulatory or inhibitory ligand (such as steroid hormone binding to the steroid hormone receptors), and by stimulation of nuclear entry (illustrated by the glucocorticoid receptor). The ability of a transcription factor to influence the transcription of a gene is also augmented or antagonized by the presence of other transcription factors. For example, the thyroid hormone receptor is critically dependent on the concentration of the retinoid receptor to provide a dimer partner. Another example is provided by the phosphoenol pyruvate (PEP) carboxykinase gene, which is induced or repressed by a variety of hormone-activated transcription factors (see section 5). Frequently, transcription factor activity is regulated through phosphorylation.

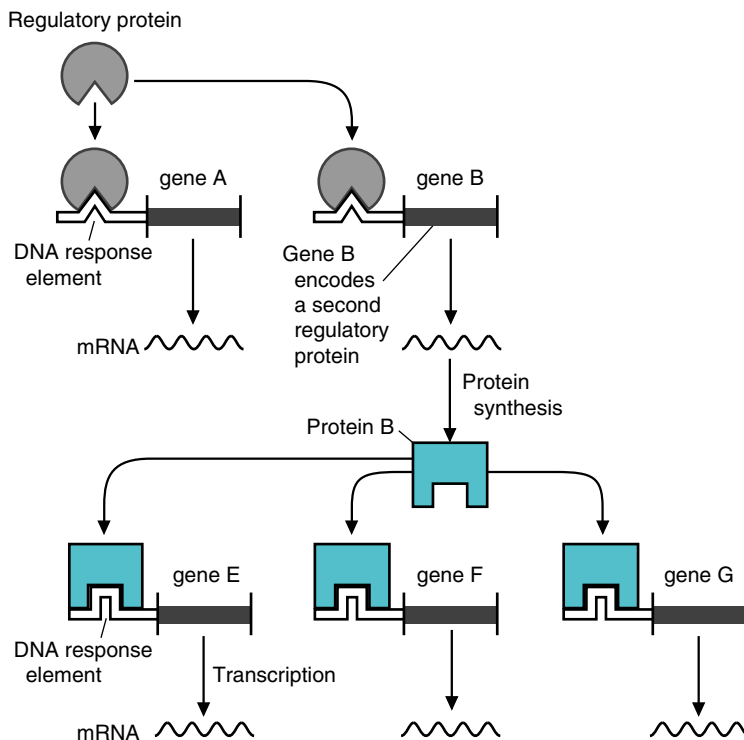
Growth factors, cytokines, polypeptide hormones, and a number of other signal molecules regulate gene transcription through phosphorylation of specific transcription factors by receptor kinases.

For example, STAT proteins are transcription factors phosphorylated by JAK-STAT receptors, and SMAD proteins are transcription factors phosphorylated by serine-threonine kinase receptors such as the transforming growth factor- $\beta$  (TGF- $\beta$ ) receptor (see Chapter 11).

Nonreceptor kinases, such as protein kinase A, also regulate transcription factors through phosphorylation. Many hormones generate the second messenger **cAMP**, which activates protein kinase A. Activated protein kinase A enters the nucleus and phosphorylates the transcription factor CREB (**cAMP response element binding protein**). CREB is constitutively bound to the DNA response element CRE (cAMP response element) and is activated by phosphorylation. Other hormone signaling pathways, such as the MAP kinase pathway, also phosphorylate CREB (as well as many other transcription factors).

## 5. MULTIPLE REGULATORS OF PROMOTERS

The same transcription factor inducer can activate transcription of many different genes if the genes each contain a common response element. Furthermore, a single inducer can activate sets of genes in an orderly, programmed manner (Fig. 16.17). The inducer initially activates one set of genes. One of the protein products of this set of genes can then act as a specific transcription factor for another set of genes. If this process is repeated, the net result is that one inducer can set off a series of events that result in the activation of many different sets of genes.



**Fig. 16.17.** Activation of sets of genes by a single inducer. Each gene in a set has a common DNA regulatory element, so one regulatory protein can activate all the genes in the set. In the example shown, the first regulatory protein stimulates the transcription of genes A and B, which have a common DNA regulatory sequence in their control regions. The protein product of gene B is itself a transcriptional activator, which in turn stimulates the transcription of genes E, F, and G, which likewise contain common response elements.



Interferons, cytokines produced by cells that have been infected with a virus, bind to the JAK-STAT family of cell surface receptors. When an interferon binds, JAK (a receptor-associated tyrosine kinase) phosphorylates STAT transcription factors bound to the receptors (see Chap. 11). The phosphorylated STAT proteins are released, dimerize, enter the nucleus, and bind to specific gene regulatory sequences. Different combinations of phosphorylated STAT proteins bind to different sequences and activate transcription of a different set of genes. One of the genes activated by interferon produces the oligonucleotide 2'-5'-oligo(A), which is an activator of a ribonuclease. This RNase degrades mRNA, thus inhibiting synthesis of the viral proteins required for its replication.

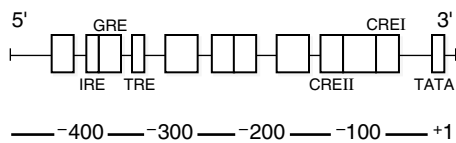
In addition to antiviral effects, interferons have antitumor effects. The mechanisms of the antitumor effects are not well understood, but are probably likewise related to stimulation of specific gene expression by STAT proteins. Interferon- $\alpha$ , produced by recombinant DNA technology, has been used to treat patients such as **Arlyn Foma** who have certain types of nodular lymphomas and patients, such as **Mannie Weitzels**, who have chronic myelogenous leukemia.



An example of a transcriptional cascade of gene activation is observed during adipocyte (fat cell) differentiation. Fibroblast-like cells can be induced to form adipocytes by the addition of dexamethasone (a steroid hormone) and insulin to the cells. These factors induce the transient expression of two similar transcription factors named C/EPB $\beta$  and C/EPB $\gamma$ . The names stand for CCAAT enhancer binding protein, and  $\beta$  and  $\gamma$  are two forms of these factors which recognize CCAAT sequences in DNA. The C/EPB transcription factors then induce the synthesis of yet another transcription factor, named the peroxisome proliferator-activated receptor  $\gamma$  (PPAR $\gamma$ ), which forms heterodimers with RXR to regulate the expression of two more transcription factors, C/EPB  $\alpha$  and STAT5. The combination of PPAR $\gamma$ , STAT5, and C/EPB $\alpha$  then leads to the expression of adipocyte-specific genes.



The enzyme PEP carboxykinase (PEPCK) is required for the liver to produce glucose from amino acids and lactate. **Ann O'Rexia**, who has an eating disorder, needs to maintain a certain blood glucose level to keep her brain functioning normally. When her blood glucose levels drop, cortisol (a glucocorticoid) and glucagon (a polypeptide hormone) are released. In the liver, glucagon increases intracellular cAMP levels, resulting in activation of protein kinase A and subsequent phosphorylation of CREB. Phosphorylated CREB binds to its response element in DNA, as does the cortisol receptor. Both transcription factors enhance transcription of the PEPCK gene (see Figure 16.17). Insulin, which is released when blood glucose levels rise after a meal, can inhibit expression of this gene.



**Fig. 16.18.** A simplified view of the regulatory region of the PEPCK gene. Boxes represent various response elements in the 5'-flanking region of the gene. Not all elements are labeled. Regulatory proteins bind to these DNA elements and stimulate or inhibit the transcription of the gene. This gene encodes the enzyme phosphoenolpyruvate carboxykinase (PEPCK), which catalyzes a reaction of gluconeogenesis (the pathway for production of glucose) in the liver. Synthesis of the enzyme is stimulated by glucagon (by a cAMP-mediated process), by glucocorticoids, and by thyroid hormone. Synthesis of PEPCK is inhibited by insulin. CRE = cAMP response element; TRE = thyroid hormone response element; GRE = glucocorticoid response element; IRE = insulin response element.

An individual gene contains many different response elements and enhancers, and genes that encode different protein products contain different combinations of response elements and enhancers. Thus, each gene does not have a single, unique protein that regulates its transcription. Rather, as different proteins are stimulated to bind to their specific response elements and enhancers in a given gene, they act cooperatively to regulate expression of that gene (Fig. 16.18). Overall, a relatively small number of response elements and enhancers and a relatively small number of regulatory proteins generate a wide variety of responses from different genes.

## D. Posttranscriptional Processing of RNA

After the gene is transcribed (i.e., posttranscription), regulation can occur during processing of the RNA transcript (hnRNA) into the mature mRNA. The use of alternative splice sites or sites for addition of the poly(A) tail (polyadenylation sites) can result in the production of different mRNAs from a single hnRNA and, consequently, in the production of different proteins from a single gene.

### 1. ALTERNATIVE SPLICING AND POLYADENYLATION SITES

Processing of the primary transcript involves the addition of a cap to the 5'-end, removal of introns, and polyadenylation (the addition of a poly(A) tail to the 3'-end) to produce the mature mRNA (see Chapter 14). In certain instances, the use of alternative splicing and polyadenylation sites causes different proteins to be produced from the same gene (Fig. 16.19). For example, genes that code for antibodies are regulated by alterations in the splicing and polyadenylation sites, in addition to undergoing gene rearrangement (Fig. 16.20). At an early stage of maturation, pre-B lymphocytes produce IgM antibodies that are bound to the cell membrane. Later, a shorter protein (IgD) is produced that no longer binds to the cell membrane, but rather is secreted from the cell.

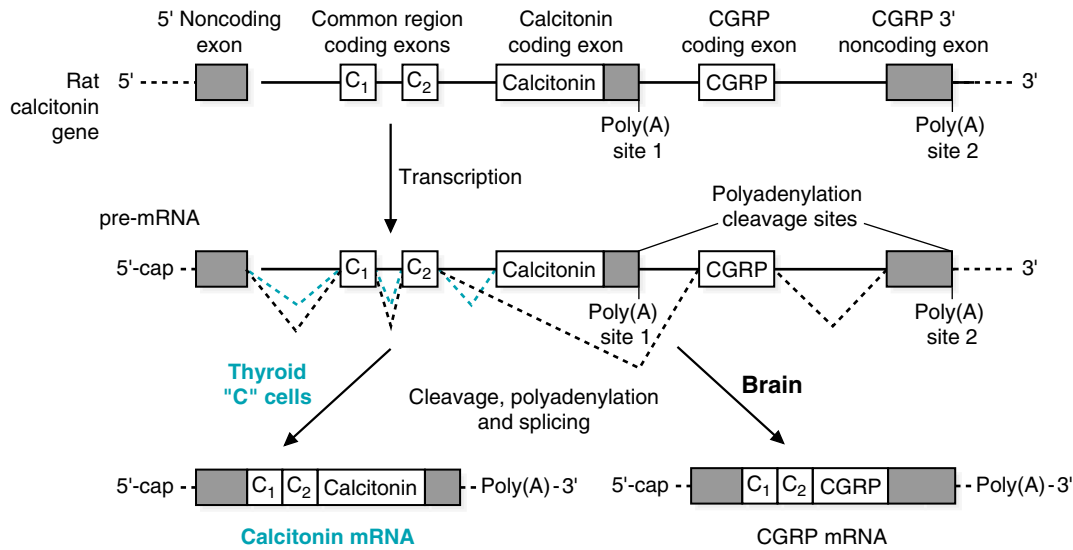
### 2. RNA EDITING

In some instances, RNA is “edited” after transcription. Although the sequence of the gene and the primary transcript (hnRNA) are the same, bases are altered or nucleotides are added or deleted after the transcript is synthesized so that the mature mRNA differs in different tissues (Fig. 16.21).

## E. Regulation at the Level of Translation and the Stability of mRNA

### 1. INITIATION OF TRANSLATION

In eukaryotes, regulation of gene transcription at the level of translation usually involves the initiation of protein synthesis by eIFs (eukaryotic initiation factors), which are regulated through mechanisms involving phosphorylation (see Chapter 15, section V.B.). For example, heme regulates translation of globin mRNA in reticulocytes by controlling the phosphorylation of eIF2 (Fig. 16.22). In reticulocytes (red blood cell precursors), globin is produced when heme levels in the cell are high but not when they are low. Because reticulocytes lack nuclei, globin synthesis must be regulated at the level of translation rather than transcription. Heme acts by preventing phosphorylation of eIF2 by a specific kinase (heme kinase) that is inactive when heme is bound. Thus, when heme levels are high, eIF2 is not phosphorylated and is active, resulting in globin synthesis. Similarly, in other cells, conditions such as starvation, heat shock, or viral infections



**Fig. 16.19.** Alternative splicing of the calcitonin gene produces an mRNA for calcitonin in thyroid cells and an mRNA for CGRP in neurons. In thyroid cells, the pre-mRNA from the calcitonin gene is processed to form an mRNA that codes for calcitonin. Cleavage occurs at poly(A) site 1, and splicing occurs along the blue dashed lines. In the brain, the pre-mRNA of this gene undergoes alternative splicing and polyadenylation to produce calcitonin gene-related protein (CGRP). Cleavage occurs at poly(A) site 2, and splicing occurs along the black dashed lines. CGRP is involved in the sensation of taste.

may result in activation of a specific kinase that phosphorylates eIF2 to an inactive form. Another example is provided by insulin, which stimulates general protein synthesis by activating the phosphorylation of an inhibitor of eIF4E, called 4E-BP. When 4E-BP is phosphorylated, it dissociates, leaving eIF4E in the active form.

A different mechanism for regulation of translation is illustrated by iron regulation of ferritin synthesis (Fig. 16.23). Ferritin, the protein involved in the storage of iron within cells, is synthesized when iron levels increase. The mRNA for ferritin has an iron response element (IRE), consisting of a hairpin loop near its 5'-end, which can bind a regulatory protein called the iron response element binding protein (IRE-BP). When IRE-BP does not contain bound iron, it binds to the IRE and prevents initiation of translation. When iron levels increase and IRE-BP binds iron, it changes to a conformation that can no longer bind to the IRE on the ferritin mRNA. Therefore, the mRNA is translated and ferritin is produced.

## F. Transport and stability of mRNA

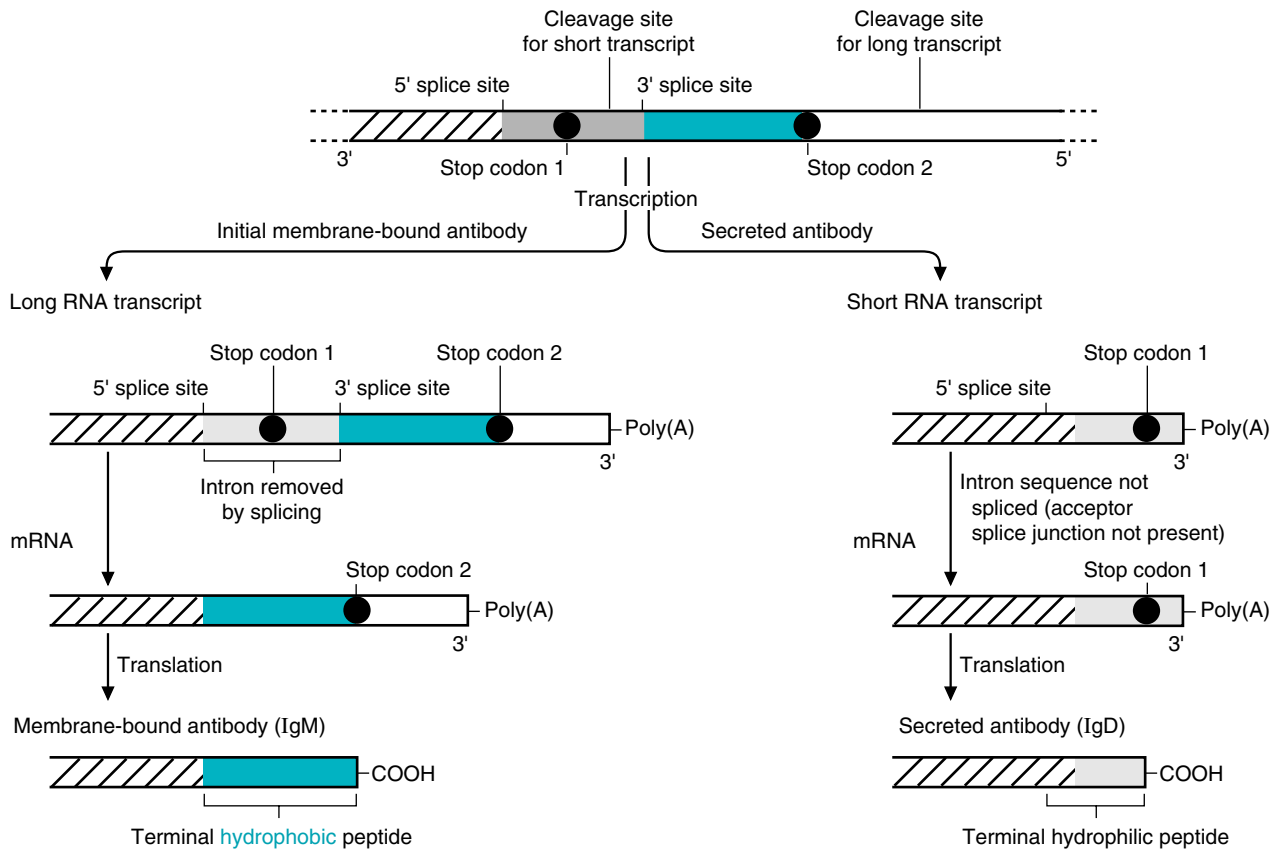
Stability of an mRNA also plays a role in regulating gene expression, because mRNAs with long half-lives can generate a greater amount of protein than can those with shorter half-lives. The mRNA of eukaryotes is relatively stable (with half-lives measured in hours to days), although it can be degraded by nucleases in the nucleus or cytoplasm before it is translated. To prevent degradation during transport from the nucleus to the cytoplasm, mRNA is bound to proteins that help to prevent its degradation. Sequences at the 3'-end of the mRNA appear to be involved in determining its half-life and binding proteins that prevent degradation. One of these is the poly(A) tail, which protects the mRNA from attack by nucleases. As mRNA ages, its poly(A) tail becomes shorter.

An example of the role of mRNA degradation in control of translation is provided by the transferrin receptor mRNA (Fig. 16.24). The transferrin receptor is a

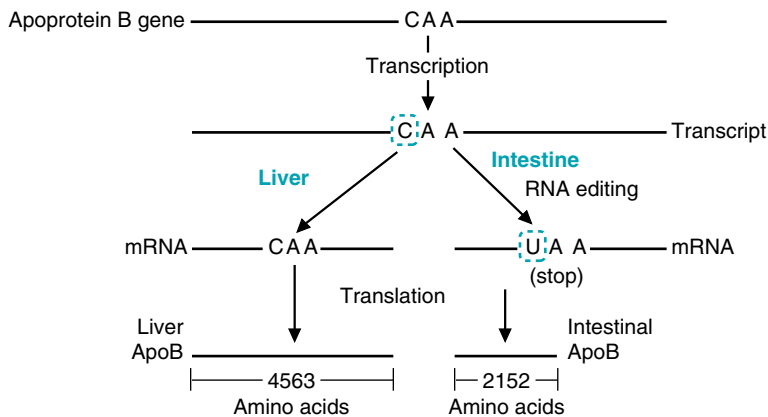


In addition to stimulating degradation of mRNA, interferon also causes eIF2 to become phosphorylated and inactive. This is a second mechanism by which interferon prevents synthesis of viral proteins.

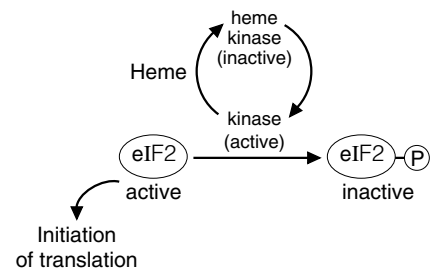




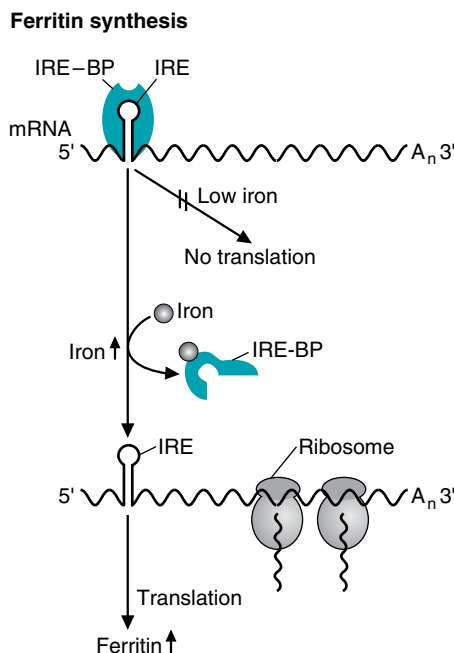
**Fig. 16.20.** Production of a membrane-bound antibody (IgM) and a smaller secreted antibody (IgD) from the same gene. Initially, the lymphocytes produce a long transcript that is cleaved and polyadenylated after the second stop codon. The intron that contains the first stop codon is removed by splicing between the 5'- and 3'-splice sites. Therefore, translation ends at the second stop codon, and the protein contains a hydrophobic exon at its C-terminal end that becomes embedded in the cell membrane. After antigen stimulation, the cells produce a shorter transcript by using a different cleavage and polyadenylation site. This transcript lacks the 3'-splice site for the intron, so the intron is not removed. In this case, translation ends at the first stop codon. The IgD antibody does not contain the hydrophobic region at its C-terminus, so it is secreted from the cell.



**Fig. 16.21.** RNA editing. In liver, the apoprotein B (ApoB) gene produces a protein that contains 4,563 amino acids. In intestinal cells, the same gene produces a protein that contains only 2,152 amino acids. Conversion of a C to a U (through deamination) in the RNA transcript generates a stop codon in the intestinal mRNA. Thus, the protein produced in the intestine (B-48) is only 48% of the length of the protein produced in the liver (B-100).

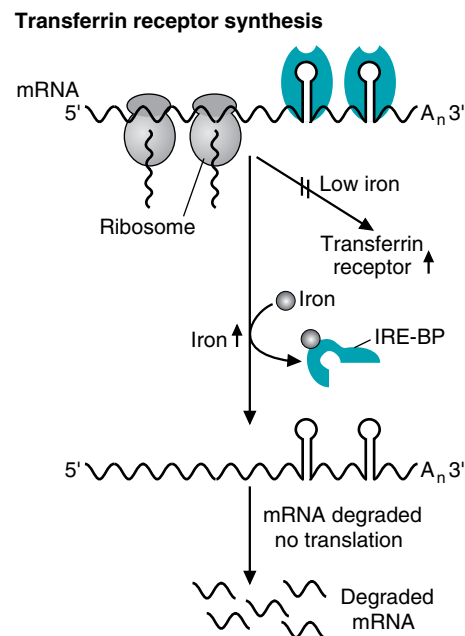


**Fig. 16.22.** Heme prevents inactivation of eIF2. When eIF2 is phosphorylated by heme kinase, it is inactive, and protein synthesis cannot be initiated. Heme inactivates heme kinase, thereby preventing phosphorylation of eIF2 and activating translation of the globin mRNA.



**Fig. 16.23.** Translational regulation of ferritin synthesis. The mRNA for ferritin has an iron response element (IRE). When the iron response element binding protein, IRE-BP does not contain bound iron, it binds to IRE, preventing translation. When IRE-BP binds iron, it dissociates, and the mRNA is translated.

protein located in cell membranes that permits cells to take up transferrin, the protein that transports iron in the blood. The rate of synthesis of the transferrin receptor increases when intracellular iron levels are low, enabling cells to take up more iron. Synthesis of the transferrin receptor, like that of the ferritin receptor, is regulated by the binding of the iron response element binding protein (IRE-BP) to the iron response elements (IRE). However, in the case of the transferrin receptor mRNA, the IREs are hairpin loops located at the 3'-end of the mRNA, and not at the 5'-end where translation is initiated. When the IRE-BP does not contain bound iron, it has a high affinity for the IRE hairpin loops. Consequently, IRE-BP prevents degradation of the mRNA when iron levels are low, thus permitting synthesis of more transferrin receptor so that the cell can take up more iron. Conversely, when iron levels are elevated, IRE-BP binds iron and has a low affinity for the IRE hairpin loops of the mRNA. Without bound IRE-BP at its 3'-end, the mRNA is rapidly degraded and the transferrin receptor is not synthesized. (Note: When IRE-BP is bound to IRE hairpin loops at the 3'-end of the mRNA, the IRE-BP prevents degradation of the mRNA)



**Fig. 16.24.** Regulation of degradation of the mRNA for the transferrin receptor. Degradation of the mRNA is prevented by binding of the iron response element binding protein (IRE-BP) to iron response elements (IRE), which are hairpin loops located at the 3'-end of the transferrin receptor mRNA. When iron levels are high, IRE-BP binds iron and is not bound to the mRNA. The mRNA is rapidly degraded, preventing synthesis of the transferrin receptor.

## CLINICAL COMMENTS



**Arlyn Foma.** Follicular lymphomas are the most common subset of non-Hodgkin's lymphomas (25–40% of cases). Patients with a more aggressive course, as seen in Arlyn Foma, die within 3 to 5 years after diagnosis if left untreated. In patients pretreated with multidrug chemotherapy (in this case AV/CM), a response rate of 50% has been reported when interferon- $\alpha$  is added to this regimen. In addition, a significantly longer event-free survival has been reported when using this approach.



Omnipotent stem cells in the bone marrow normally differentiate and mature in a highly selective and regulated manner, becoming red blood cells, white blood cells, or platelets. Cytokines stimulate differentiation of the stem cells into the lymphoid and myeloid lineages. The lymphoid lineage gives rise to B and T lymphocytes, which are white blood cells that generate antibodies for the immune response. The myeloid lineage gives rise to three types of progenitor cells: erythroid, granulocytic–monocytic, and megakaryocytic. The erythroid progenitor cells differentiate into red blood cells (erythrocytes), and the other myeloid progenitors give rise to nonlymphoid white blood cells and platelets. Various medical problems can affect this process. In **Mannie Weitzels**, who has chronic myelogenous leukemia (CML), a single line of primitive myeloid cells produces leukemic cells that proliferate abnormally, causing a large increase in the number of white blood cells in the circulation. In **Anne Niemick**, who has a deficiency of red blood cells caused by her  $\beta^+$  thalassemia (see Chapter 15), differentiation of precursor cells into mature red blood cells is stimulated to compensate for the anemia.



Ann O'Rexia has a hypochromic anemia, which means that her red blood cells are pale because they contain low levels of hemoglobin. Because of her iron deficiency, she is not producing adequate amounts of heme. Consequently, eIF2 is phosphorylated in her reticulocytes and cannot activate initiation of globin translation.



**Mannie Weitzels.** Mannie Weitzels has CML (chronic myelogenous leukemia), a hematologic disorder in which the proliferating leukemic cells are believed to originate from a single line of primitive myeloid cells. Although classified as one of the myeloproliferative disorders, CML is distinguished by the presence of a specific cytogenetic abnormality of the dividing marrow cells known as the Philadelphia chromosome, found in more than 90% of cases. In most instances, the cause of CML is unknown, but the disease occurs with an incidence of around 1.5 per 100,000 population in Western societies.



**Ann O'Rexia.** Ann O'Rexia's iron stores are depleted. Normally, about 16 to 18% of total body iron is contained in ferritin, which contains a spherical protein (apoferritin) that is capable of storing as many as 4,000 atoms of iron in its center. When an iron deficiency exists, serum and tissue ferritin levels fall. Conversely, the levels of transferrin (the blood protein that transports iron) and the levels of the transferrin receptor (the cell surface receptor for transferrin) increase.

## BIOCHEMICAL COMMENTS



**Regulation of transcription by iron.** A cell's ability to acquire and store iron is a carefully controlled process. Iron obtained from the diet is absorbed in the intestine and released into the circulation, where it is bound by transferrin, the iron transport protein in plasma. When a cell requires iron, the plasma iron–transferrin complex binds to the transferrin receptor in the cell membrane and is internalized into the cell. Once the iron is freed from transferrin, it then binds to ferritin, which is the cellular storage protein for iron. Ferritin has the capacity to store up to 4,000 molecules of iron per ferritin molecule. Both transcriptional and translational controls work to maintain intracellular levels of iron (see Figs. 16.23 and 16.24). When iron levels are low, the iron response element binding protein (IRE-BP) binds to specific looped structures on both the ferritin and transferrin receptor mRNAs. This binding event stabilizes the transferrin receptor mRNA so that it can be translated and the number of transferrin receptors in the cell membrane increased. Consequently, cells will take up more iron, even when plasma transferrin/iron levels are low. The binding of IRE-BP to the ferritin mRNA, however, blocks translation of the mRNA. With low levels of intracellular iron, there is little iron to store and less need for intracellular ferritin. Thus, the IRE-BP can stabilize one mRNA, and block translation from a different mRNA.

What happens when iron levels rise? Iron will bind to the IRE-BP, thereby decreasing its affinity for mRNA. When the IRE-BP dissociates from the transferrin receptor mRNA, the mRNA becomes destabilized and is degraded, leading to less receptor being synthesized. Conversely, dissociation of the IRE-BP from the ferritin mRNA allows that mRNA to be translated, thereby increasing intracellular levels of ferritin and increasing the cells capacity for iron storage.

Why does an anemia result from iron deficiency? When an individual is deficient in iron, the reticulocytes do not have sufficient iron to produce heme, the required prosthetic group of hemoglobin. When heme levels are low, the eukaryotic initiation factor eIF2 (see Fig. 16.22) is phosphorylated, and inactive. Thus, globin mRNA cannot be translated because of the lack of heme. This results in red blood cells with inadequate levels of hemoglobin for oxygen delivery, and an anemia.

## Suggested Readings

### Regulation of gene expression in prokaryotic and eukaryotic cells:

Lewin B. *Genes VII*. Oxford: Oxford University Press, 2000:273–318, 649–684.

### Thalassemias:

Watson J, Gilman M, Witkowski J, Zoller M. *Recombinant DNA*. Scientific American Books. New York: WH Freeman, 1992:540–544.

Weatherall DJ, Clegg JB, Higgs DR, Wood WG. The hemoglobinopathies. In Scriver CR, Beaudet AL, Sly WS, Valle D. *The Metabolic and Molecular Bases of Inherited Disease*. 8th Ed. New York: McGraw-Hill, 2001:4571–4636.

### Leukemias and Lymphomas:

Goldman L, Bennet JC. *Cecil Textbook of Medicine*. 21st Ed. Philadelphia: WB Saunders, 2000:945–977.



## REVIEW QUESTIONS—CHAPTER 16

- Which of the following explains why several different proteins can be synthesized from a typical prokaryotic mRNA?
  - Any of the three reading frames can be used.
  - There is redundancy in the choice of codon/tRNA interactions.
  - The gene contains several operator sequences from which to initiate translation.
  - Alternative splicing events are commonly found.
  - Many RNAs are organized in a series of consecutive translational cistrons.
- In *E. Coli*, under high lactose, high glucose conditions, which of the following could lead to maximal transcription activation of the *lac* operon?
  - A mutation in the *lac I* gene (which encodes the repressor)
  - A mutation in the CRP binding site leading to enhanced binding
  - A mutation in the operator sequence
  - A mutation leading to enhanced cAMP levels
  - A mutation leading to lower binding of repressor
- A mutation in the *I* (repressor) gene of a “non-inducible” strain of *E. coli* resulted in an inability to synthesize any of the proteins of the *lac* operon. Which of the following provides a rational explanation?
  - The repressor has lost its affinity for inducer.
  - The repressor has lost its affinity for operator.
  - A trans acting factor can no longer bind to the promoter.
  - The CAP protein is no longer made.
  - Lactose feedback inhibition becomes constitutive.
- Which of the following double-stranded DNA sequences shows perfect dyad symmetry (the same sequence of bases on both strands)?
  - GAACTGCTAGTCGC
  - GGCATCGCGATGCC
  - TAATCGGAACCAAT
  - GCAGATTTTAGACG
  - TGACCGGTGACCGG

5. Which of the following describes a common theme in the structure of DNA binding proteins?
- (A) The presence of a specific helix that lies across the major groove of DNA
  - (B) The ability to recognize RNA molecules with the same sequence
  - (C) The ability to form multiple hydrogen bonds between the protein peptide backbone and the DNA phosphodiester backbone
  - (D) The presence of zinc
  - (E) The ability to form dimers with disulfide linkages



## 17 Use of Recombinant DNA Techniques in Medicine

*The rapid development of techniques in the field of molecular biology is revolutionizing the practice of medicine. The potential uses of these techniques for the diagnosis and treatment of disease are vast.*

**Clinical applications.** **Polymorphisms**, inherited differences in DNA base sequences, are abundant in the human population, and many alterations in DNA sequences are associated with diseases. Tests for DNA sequence variations are more sensitive than many other techniques (such as enzyme assays) and permit recognition of diseases at earlier and therefore potentially more treatable stages. These tests can also identify carriers of inherited diseases so they can receive appropriate counseling. Because genetic variations are so distinctive, **DNA “fingerprinting”** (analysis of DNA sequence differences) can be used to determine family relationships or to help identify the perpetrators of a crime.

Techniques of molecular biology are used in the **prevention** and **treatment** of disease. For example, recombinant DNA techniques provide human insulin for the treatment of diabetes, Factor VIII for the treatment of hemophilia, and vaccines for the prevention of hepatitis. Although treatment of disease by gene therapy is in the experimental phase of development, the possibilities are limited only by the human imagination and, of course, by ethical considerations.

**Techniques.** To recognize normal or pathologic genetic variations, DNA must be isolated from the appropriate source, and adequate amounts must be available for study. Techniques for **isolating** and **amplifying** genes and studying and **manipulating** DNA sequences involve the use of **restriction enzymes**, **cloning vectors**, **polymerase chain reaction (PCR)**, **gel electrophoresis**, **blotting onto nitrocellulose paper**, and the preparation of **labeled probes that hybridize** to the appropriate target DNA sequences. **Gene therapy** involves isolating normal genes and inserting them into diseased cells so that the normal genes are expressed, permitting the diseased cells to return to a normal state. Students must have at least a general understanding of recombinant DNA techniques to appreciate their current use and the promise they hold for the future.



### THE WAITING ROOM



**Erna Nemdy**, a third-year medical student, has started working in the hospital blood bank two nights per week (see Chapter 15 for an introduction to Erna Nemdy and her daughter, Beverly). Because she will be handling human blood products, she must have a series of hepatitis B vaccinations. She has reservations about having these vaccinations and inquires about the efficacy and safety of the vaccines currently in use.



Cystic fibrosis is a disease caused by an inherited deficiency in the CFTR (cystic fibrosis transmembrane conductance regulator) protein, which is a chloride channel (see Chapter 10, Fig. 10.11). In the absence of chloride secretion, dried mucus blocks the pancreatic duct, resulting in decreased secretion of digestive enzymes into the intestinal lumen. The resulting malabsorption of fat and other foodstuffs decreases growth and may lead to varying degrees of small bowel obstruction. Liver and gallbladder secretions may be similarly affected. Eventually, atrophy of the secretory organs or ducts may occur. Dried mucus also blocks the airways, markedly diminishing air exchange and predisposing the patient to stasis of secretions, diminished immune defenses, and increased secondary infections. Defects in the CFTR chloride channel also affect sweat composition, increasing the sodium and chloride contents of the sweat, thereby providing a diagnostic tool.



**Sissy Fibrosa** is a 3-year-old Caucasian girl who is diagnosed with cystic fibrosis. Her growth rate has been in the lower 30th percentile over the last year. Since birth, she has had occasional episodes of spontaneously reversible and minor small bowel obstruction. These episodes are superimposed on gastrointestinal symptoms that suggest a degree of dietary fat malabsorption, such as bulky, glistening, foul-smelling stools two or three times per day. She has experienced recurrent flare-ups of bacterial bronchitis in the last 10 months, each time caused by *Pseudomonas aeruginosa*. A quantitative sweat test was unequivocally positive (Excessive sodium and chloride were found in her sweat on two occasions.). Based on these findings, the pediatrician informed Sissy's parents that Sissy probably has cystic fibrosis (CF). A sample of her blood was sent to a DNA testing laboratory to confirm the diagnosis and to determine specifically which one of the many potential genetic mutations known to cause CF was present in her cells.



**Carrie Sichel**, Will Sichel's 19-year-old sister, is considering marriage. Her growth and development have been normal, and she is free of symptoms of sickle cell anemia. Because a younger sister, Amanda, was tested and found to have sickle trait, and because of Will's repeated sickle crises, Carrie wants to know whether she also has sickle trait (see Chapters 6 and 7 for Will Sichel's history). A hemoglobin electrophoresis is performed that shows the composition of her hemoglobin to be 58% HbA, 39% HbS, 1% HbF, and 2% HbA<sub>2</sub>, a pattern consistent with the presence of sickle cell trait. The hematologist who saw her in the clinic on her first visit is studying the genetic mutations of sickle cell trait and asks Carrie for permission to draw additional blood for more sophisticated analysis of the genetic disturbance that causes her to produce HbS. Carrie informed her fiancé that she has sickle cell trait and that she wants to delay their marriage until he is tested.



**Victoria Tim** (Vicky Tim) was a 21-year-old woman who was the victim of a rape and murder. She left her home and drove to the local convenience store. When she had not returned home an hour later, her father drove to the store, looking for Vicky. He found her car still parked in front of the store and called the police. They searched the area around the store, and found Vicky Tim's body in a wooded area behind the building. She had been sexually assaulted and strangled. Medical technologists from the police laboratory collected a semen sample from vaginal fluid and took samples of dried blood from under the victim's fingernails. Witnesses identified three men who spoke to Vicky Tim while she was at the convenience store. DNA samples were obtained from these suspects to determine whether any of them was the perpetrator of the crime.



**Ivy Sharer's** cough is slightly improved on a multidrug regimen for pulmonary tuberculosis, but she continues to have night sweats. She is tolerating her current AIDS therapy well but complains of weakness and fatigue. The man with whom she has shared "dirty" needles to inject drugs accompanies Ivy to the clinic and requests that he be tested for the presence of HIV.

## I. RECOMBINANT DNA TECHNIQUES

Techniques for joining DNA sequences into new combinations (recombinant DNA) were originally developed as research tools to explore and manipulate genes but are now also being used to identify defective genes associated with disease and to correct genetic defects. Even a cursory survey of the current literature demonstrates that these techniques will soon replace many of the current clinical testing procedures. At least a basic appreciation of recombinant DNA techniques is required to understand the ways in which genetic variations among individuals are determined

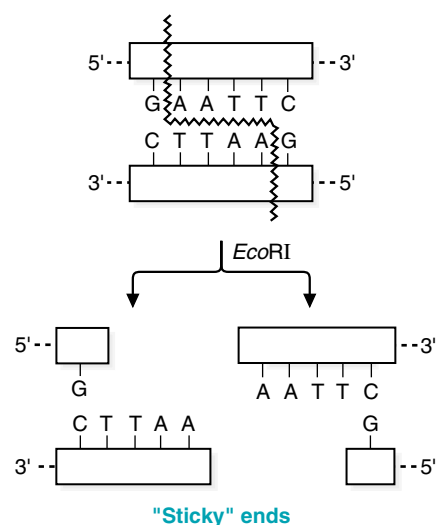
and how these differences can be used to diagnose disease. The first steps in determining individual variations in genes involve isolating the genes (or fragments of DNA) that contain variable sequences and obtaining adequate quantities for study.

## A. Strategies for Obtaining Fragments of DNA and Copies of Genes

### 1. RESTRICTION FRAGMENTS

Enzymes called restriction endonucleases enable molecular biologists to cleave segments of DNA from the genome of various types of cells or to fragment DNA obtained from other sources. A restriction enzyme is an endonuclease that specifically recognizes a short sequence of DNA, usually 4 to 6 base pairs (bp) in length, and cleaves a phosphodiester bond in both DNA strands within this sequence (Fig. 17.1). A key feature of restriction enzymes is their specificity. A restriction enzyme always cleaves at the same DNA sequence and only cleaves at that particular sequence. Most of the DNA sequences recognized by restriction enzymes are palindromes, that is, both strands of DNA have the same base sequence when read in a 5' to 3' direction. The cuts made by these enzymes are usually “sticky” (that is, the products are single-stranded at the ends, with one strand overhanging the other). However, sometimes they are blunt (the products are double-stranded at the ends, with no overhangs). Hundreds of restriction enzymes with different specificities have been isolated (Table 17.1).

Restriction fragments of DNA can be used to identify variations in base sequence in a gene. However, they also can be used to synthesize a recombinant DNA (also called chimeric DNA), which is composed of molecules of DNA from different



**Fig. 17.1.** Action of restriction enzymes. Note that the DNA sequence shown is a palindrome; each strand of the DNA, when read in a 5' to 3' direction, has the same sequence. Cleavage of this sequence by *EcoRI* produces single-stranded (or “sticky”) ends or tails. Not shown is an example of an enzyme that generates blunt ends (see Table 17.1).



In sickle cell anemia, the point mutation that converts a glutamate residue to a valine residue (GAG to GTG) occurs in a site that is cleaved by the restriction enzyme *MstII* (recognition sequence CCTNAGG, where N can be any base) within the normal  $\beta$ -globin gene. The sickle cell mutation causes the  $\beta$ -globin gene to lose this *MstII* restriction site. Therefore, because **Will Sichel** is homozygous for the sickle cell gene, neither of the two alleles of his  $\beta$ -globin gene will be cleaved at this site.

**Table 17.1. Sequences Cleaved by Selected Restriction Enzymes\***

| Restriction Enzyme | Source                                       | Cleavage Site                                        |
|--------------------|----------------------------------------------|------------------------------------------------------|
| <i>AluI</i>        | <i>Arthrobacter luteus</i>                   | 5' - A G C T - 3'<br>3' - T C G A - 5'               |
| <i>BamHI</i>       | <i>Bacillus amyloliquefaciens</i> H          | 5' - G G A T C C - 3'<br>3' - C C T A G G - 5'       |
| <i>EcoRI</i>       | <i>Escherichia coli</i> RY13                 | 5' - G A A T T C - 3'<br>3' - C T T A A G - 5'       |
| <i>HaeIII</i>      | <i>Haemophilus aegyptius</i>                 | 5' - G G C C - 3'<br>3' - C C G G - 5'               |
| <i>HindIII</i>     | <i>Haemophilus influenzae</i> R <sub>d</sub> | 5' - A A G C T T - 3'<br>3' - T T C G A A - 5'       |
| <i>MspI</i>        | <i>Moraxella</i> species                     | 5' - C C G G - 3'<br>3' - G G C C - 5'               |
| <i>MstII</i>       | <i>Microcoleus</i>                           | 5' - C C T N A G G - 3'<br>3' - G G A N T C C - 5'   |
| <i>NotI</i>        | <i>Nocardia otitidis</i>                     | 5' - G C G C C G C - 3'<br>3' - C G C C G G C G - 5' |
| <i>PstI</i>        | <i>Providencia stuartii</i> 164              | 5' - C T G C A G - 3'<br>3' - G A C G T C - 5'       |
| <i>SmaI</i>        | <i>Serratia marcescens</i> S <sub>b</sub>    | 5' - C C C G G G - 3'<br>3' - G G G C C C - 5'       |

\*Restriction enzymes are named for the bacterium from which they were isolated (e.g., *ECO* is from *Escherichia coli*).



Which **ONE** of the following sequences is most likely to be a restriction enzyme recognition sequence?

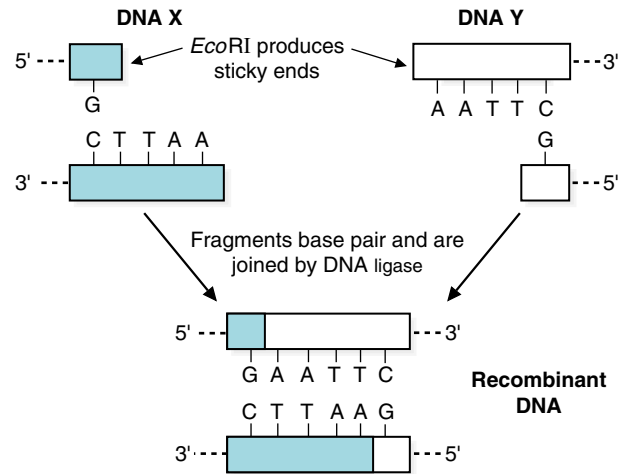
- (A) (5') G T C C T G (3')  
C A G G A C
- (B) (5') T A C G A T (3')  
A T G C T A
- (C) (5') C T G A G (3')  
G A C T C
- (D) (5') A T C C T A (3')  
T A G G A T



Restriction endonucleases were discovered in bacteria in the late 1960s and 1970s. These enzymes were named for the fact that bacteria use them to “restrict” the growth of viruses (bacteriophage) that infect the bacterial cells. They cleave the phage DNA into smaller pieces so the phage cannot reproduce in the bacterial cells. However, they do not cleave the bacterial DNA, because its bases are methylated at the restriction sites by DNA methylases. Restriction enzymes also restrict uptake of DNA from the environment, and they restrict mating with nonhomologous species.



The answer is C. C follows a palindromic sequence of CTNAG, where N can be any base. None of the other sequences is this close to a palindrome. Although most restriction enzymes recognize a “perfect” palindrome, where the sequence of bases in each strand are the same, others can have intervening bases between the regions of identity, as in this question. Note also the specificity of the enzyme *MstII* in Table 1.



**Fig. 17.2.** Production of recombinant DNA molecules with restriction enzymes and DNA ligase. The dashes at the 5' and 3'-ends indicate that this sequence is part of a longer DNA molecule.

sources that have been recombined in vitro (outside the organism, e.g., in a test tube). The sticky ends of two unrelated DNA fragments can be joined to each other if they have sticky ends that are complementary. Complementary ends are obtained by cleaving the unrelated DNAs with the same restriction enzyme (Fig. 17.2). After the sticky ends of the fragments base-pair with each other, the fragments can be covalently attached by the action of DNA ligase.

## 2. DNA PRODUCED BY REVERSE TRANSCRIPTASE

If mRNA transcribed from a gene is isolated, this mRNA can be used as a template by the enzyme reverse transcriptase, which produces a DNA copy (cDNA) of the RNA. In contrast to DNA fragments cleaved from the genome by restriction enzymes, DNA produced by reverse transcriptase does not contain introns because mRNA, which has no introns, is used as a template.

## 3. CHEMICAL SYNTHESIS OF DNA

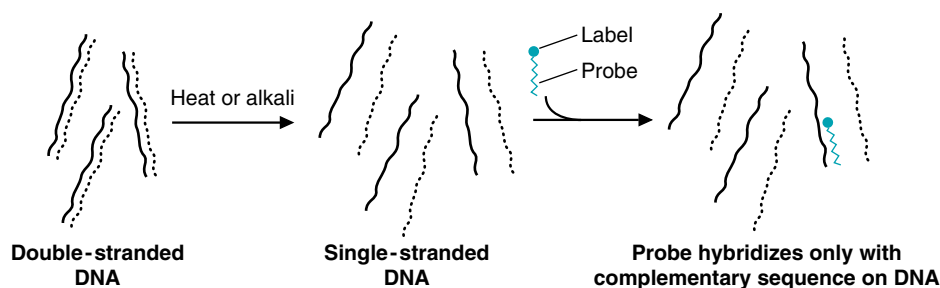
Automated machines can synthesize oligonucleotides (short molecules of single-stranded DNA) up to 100 nucleotides in length. These machines can be programmed to produce oligonucleotides with a specified base sequence. Although entire genes cannot yet be synthesized in one piece, oligonucleotides can be prepared that will base-pair with segments of genes. These oligonucleotides can be used in the process of identifying, isolating, and amplifying genes.

## B. Techniques for Identifying DNA Sequences

### 1. PROBES

A probe is a single-stranded polynucleotide of DNA or RNA that is used to identify a complementary sequence on a larger single-stranded DNA or RNA molecule (Fig. 17.3). Formation of base pairs with a complementary strand is called annealing or hybridization. Probes can be composed of cDNA (produced from mRNA by reverse transcriptase), fragments of genomic DNA (cleaved by restriction enzymes from the genome), chemically synthesized oligonucleotides, or, occasionally, RNA.

To identify the target sequence, the probe must carry a label (see Fig. 17.3). If the probe has a radioactive label such as  $^{32}\text{P}$ , it can be detected by autoradiography. An autoradiogram is produced by covering the material containing the probe with a sheet of x-ray film. Electrons ( $\beta$  particles) emitted by disintegration of the



**Fig. 17.3.** Use of probes to identify DNA sequences. The probe can be either DNA or RNA.

radioactive atoms expose the film in the region directly over the probe. A number of techniques can be used to introduce labels into these probes. Not all probes are radioactive. Some are chemical adducts (compounds that bind covalently to DNA) that can be identified, for example, by fluorescence.

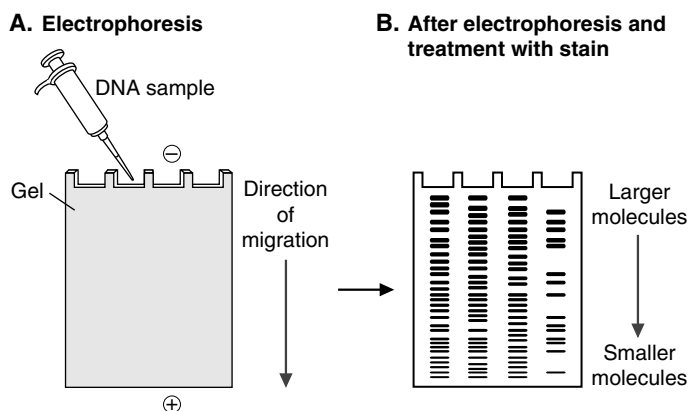
## 2. GEL ELECTROPHORESIS

Gel electrophoresis is a technique that uses an electrical field to separate molecules on the basis of size. Because DNA contains negatively charged phosphate groups, it will migrate in an electrical field toward the positive electrode (Fig. 17.4). Shorter molecules migrate more rapidly through the pores of a gel than do longer molecules, so separation is based on length. Gels composed of polyacrylamide, which can separate DNA molecules that differ in length by only one nucleotide, are used to determine the base sequence of DNA. Agarose gels are used to separate longer DNA fragments that have larger size differences.

The bands of DNA in the gel can be visualized by various techniques. Staining with dyes such as ethidium bromide allows direct visualization of DNA bands under ultra-violet light. Specific sequences are generally detected by means of a labeled probe.

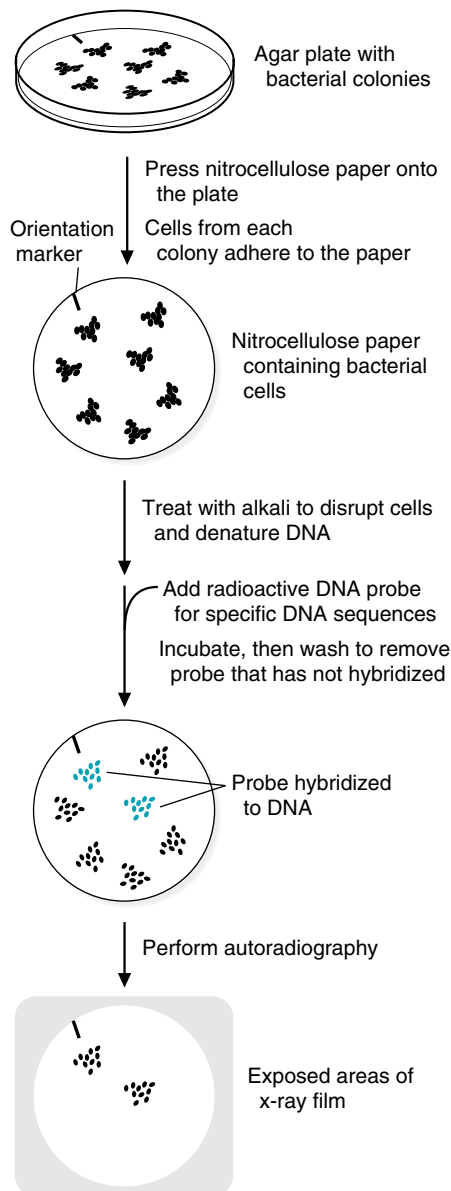
## 3. DETECTION OF SPECIFIC DNA SEQUENCES

To detect specific sequences, DNA is usually transferred to a solid support, such as a sheet of nitrocellulose paper. For example, if bacteria are growing on an agar plate, cells from each colony will adhere to a nitrocellulose sheet pressed against the



**Fig. 17.4.** Gel electrophoresis of DNA. **A.** DNA samples are placed into depressions (“wells”) at one end of a gel, and an electrical field is applied. The DNA migrates toward the positive electrode at a rate that depends on the size of the DNA molecules. Shorter molecules migrate more rapidly than longer molecules. **B.** The gel is removed from the apparatus. The bands are not visible until techniques are performed to visualize them (see Fig. 17.6).





**Fig. 17.5.** Identification of bacterial colonies containing specific DNA sequences. The autoradiogram can be used to identify bacterial colonies on the original agar plate that contain the desired DNA sequence. Note that an orientation marker is placed on the nitrocellulose and the agar plate so the results of the autoradiogram can be properly aligned with the original plate of bacteria.



Western blots are one of the tests for the AIDS virus. Viral proteins in the blood are detected by antibodies. Tests performed on **Ivy Sharer's** friend showed that he was HIV positive. Unlike Ivy, however, he has not yet developed the symptoms of AIDS.

agar, and an exact replica of the bacterial colonies can be transferred to the nitrocellulose paper (Fig. 17.5). A similar technique is used to transfer bands of DNA from electrophoretic gels to nitrocellulose sheets. After bacterial colonies or bands of DNA are transferred to nitrocellulose paper, the paper is treated with an alkaline solution. Alkaline solutions denature DNA (that is, separate the two strands of each double helix). The single-stranded DNA is then hybridized with a probe, and the regions on the nitrocellulose blot containing DNA that base-pairs with the probe are identified.

E. M. Southern developed the technique, which bears his name, for identifying DNA sequences on gels. Southern blots are produced when DNA on a nitrocellulose blot of an electrophoretic gel is hybridized with a DNA probe. Molecular biologists decided to continue with this geographic theme as they named two additional techniques. Northern blots are produced when RNA on a nitrocellulose blot is hybridized with a DNA probe. A slightly different but related technique, known as a Western blot, involves separating proteins by gel electrophoresis and probing with labeled antibodies for specific proteins (Fig. 17.6).

#### 4. DNA SEQUENCING

The most common procedure for determining the sequence of nucleotides in a DNA strand was developed by Frederick Sanger and involves the use of dideoxynucleotides. Dideoxynucleotides lack a 3'-hydroxyl group (in addition to lacking the 2' hydroxyl group normally absent from DNA deoxynucleotides). Thus, once they are incorporated into the growing chain, the next nucleotide cannot add, and polymerization is terminated. In this procedure, only one of the four dideoxynucleotides (ddATP, ddTTP, ddGTP, or ddCTP) is added to a tube containing all four normal deoxynucleotides, DNA polymerase, a primer, and the template strand for the DNA that is being sequenced (Fig. 17.7). As DNA polymerase catalyzes the sequential addition of complementary bases to the 3' end, the dideoxynucleotide competes with its corresponding normal nucleotide for insertion. Whenever the dideoxynucleotide is incorporated, further polymerization of the strand cannot occur, and synthesis is terminated. Some of the chains will terminate at each of the locations in the template strand that is complementary to the dideoxynucleotide. Consider, for example, a growing polynucleotide strand in which adenine (A) should add at positions 10, 15, and 17. Competition between ddATP and dATP for each position results in some chains terminating at position 10, some at 15, and some at 17. Thus, DNA strands of varying lengths are produced from a template. The shortest strands are closest to the 5'-end of the growing DNA strand because the strand grows in a 5' to 3' direction.

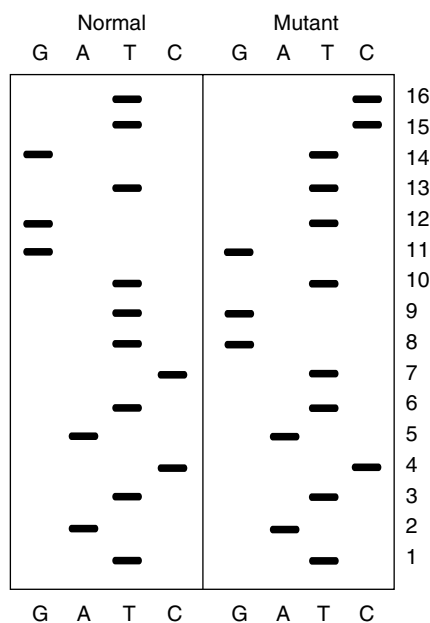
Four separate reactions are performed, each with only one of the dideoxynucleotides present (ddATP, ddTTP, ddGTP, ddCTP) plus a complete mixture of normal nucleotides (see Fig. 17.7B.). In each tube, some strands are terminated whenever the complementary base for that dideoxynucleotide is encountered. If these strands are subjected to gel electrophoresis, the sequence 5'→3' of the DNA strand complementary to the template can be determined by "reading" from the bottom to the top of the gel, that is, by noting the lanes (A, G, C, or T) in which bands appear, starting at the bottom of the gel and moving sequentially toward the top.



**Ivy Sharer** is being treated with didanosine. This drug is a purine nucleoside composed of the base hypoxanthine linked to dideoxyribose. In cells, didanosine is phosphorylated to form a nucleotide that adds to growing DNA strands. Because dideoxynucleotides lack both 2'- and 3'-hydroxyl groups, DNA synthesis is terminated. Reverse transcriptase has a higher affinity for the dideoxynucleotides than does the cellular DNA polymerase, so the use of this drug will affect reverse transcriptase to a greater extent than the cellular enzyme.



In the early studies on cystic fibrosis, DNA sequencing was used to determine the type of defect in patients. Buccal cells were obtained from washes of the mucous membranes of the mouth, DNA isolated from these cells was amplified by PCR, and DNA sequencing of the CF gene was performed. A sequencing gel for the region in which the normal gene differs from the mutant gene is shown below.



What is the difference between the normal and the mutant CF gene sequence shown on the gel, and what effect would this difference have on the protein produced from this gene?

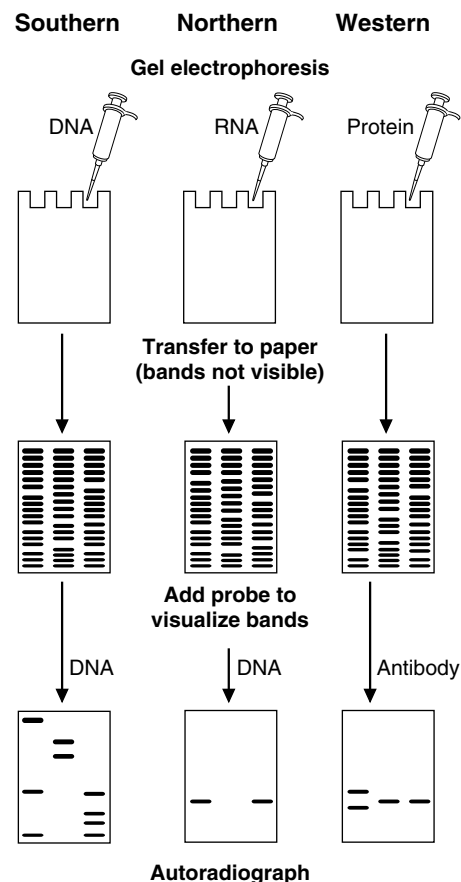
### C. Techniques for Amplifying DNA Sequences

To study genes or other DNA sequences, adequate quantities of material must be obtained. It is often difficult to isolate significant quantities of DNA from the original source. For example, an individual cannot usually afford to part with enough tissue to provide the amount of DNA required for clinical testing. Therefore, the available quantity of DNA has to be amplified.

#### 1. CLONING OF DNA

The first technique developed for amplifying the quantity of DNA is known as cloning (Fig. 17.8). The DNA that you want amplified (the “foreign” DNA) is attached to a vector (a carrier DNA), which is introduced into a host cell that makes multiple copies of the DNA. The foreign DNA and the vector DNA are usually cleaved with the same restriction enzyme, which produces complementary sticky ends in both DNAs. The foreign DNA is then added to the vector. Base pairs form between the complementary single-stranded regions, and DNA ligase joins the molecules to produce a chimera, or recombinant DNA. As the host cells divide, they replicate their own DNA, and they also replicate the DNA of the vector, which includes the foreign DNA.

If the host cells are bacteria, commonly used vectors are bacteriophage (viruses that infect bacteria), plasmids (extrachromosomal pieces of circular DNA that are taken up by bacteria), or cosmids (plasmids that contain DNA sequences from the lambda phage). When eukaryotic cells are used as the host, the vectors are often retroviruses, adenoviruses, free DNA, or DNA coated with a lipid layer (liposomes). The foreign DNA sometimes integrates into the host cell genome or it exists as eiposomes (extrachromosomal fragments of DNA) (See section III.D. of this chapter.)



**Fig. 17.6.** Southern, Northern, and Western blots. For Southern blots, DNA molecules are separated by electrophoresis, denatured, transferred to nitrocellulose paper (by “blotting”), and hybridized with a DNA probe. For Northern blots, RNA is electrophoresed and treated similarly except that alkali is not used. (First, alkali hydrolyzes RNA, and second, RNA is already single stranded.) For Western blots, proteins are electrophoresed, transferred to nitrocellulose, and probed with a specific antibody.



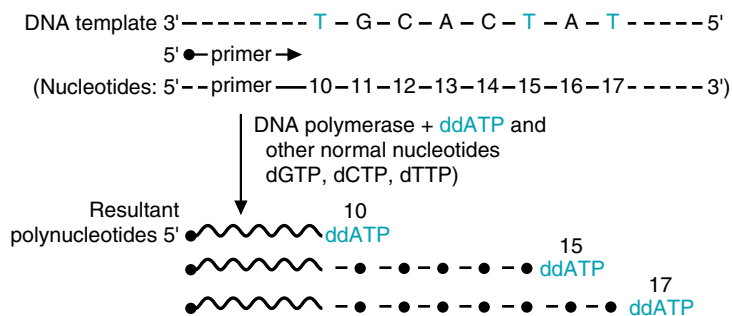
A genomic “library” in molecular biologists’ terms is a set of host cells that collectively contain all of the DNA sequences from the genome of another organism. A cDNA library is a set of host cells that collectively contain all the DNA sequences produced by reverse transcriptase from the mRNA obtained from cells of a particular type. Thus, a cDNA library contains all the genes expressed in that cell type, at the stage of differentiation when the mRNA was isolated.

**A:** In individuals of northern European descent, 70% of the cases of cystic fibrosis (CF) are caused by a deletion of three bases in the CF gene. In the region of the gene shown on the gels, the base sequence (read from the bottom to the top of the gel) is the same for the normal and mutant gene for the first 6 positions, and the bases in positions 10 through 16 of the normal gene are the same as the bases in positions 7 through 13 of the mutant gene. Therefore, a 3-base deletion in the mutant gene corresponds to bases 7 through 9 of the normal gene.

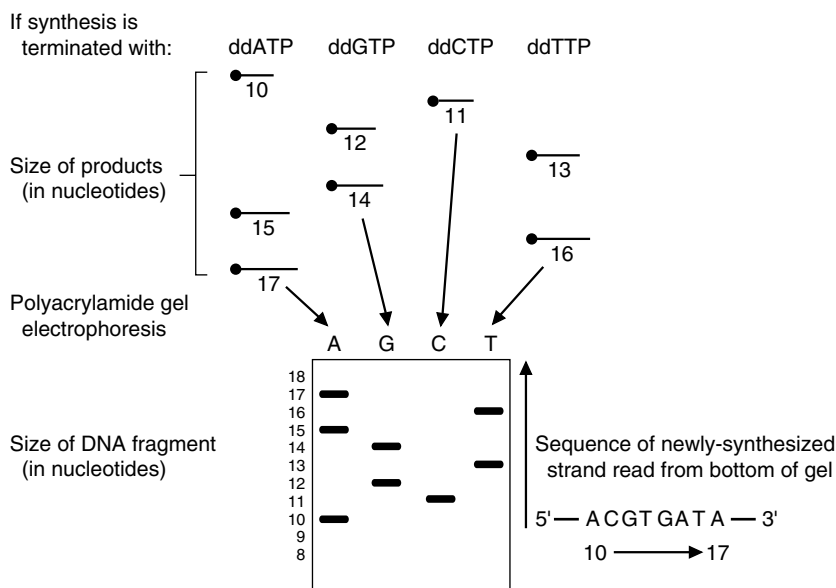
|                  |         |       |       |         |
|------------------|---------|-------|-------|---------|
|                  | Ile     | Ile   | Phe   | Gly     |
| Normal sequence: | T A T C | A T C | T T T | G G T   |
| CF sequence:     | T A T C | A T   | --    | T G G T |
|                  | Ile     | Ile   |       | Gly     |

Loss of 3 bp (indicated by the dashes) maintains the reading frame, so only the single amino acid phenylalanine (F) is lost. Phenylalanine would normally appear as residue 508 in the protein. Therefore, the deletion is referred to as  $\Delta F_{508}$ . The rest of the amino acid sequence of the normal and the mutant proteins is identical.

### A. Terminates with ddATP

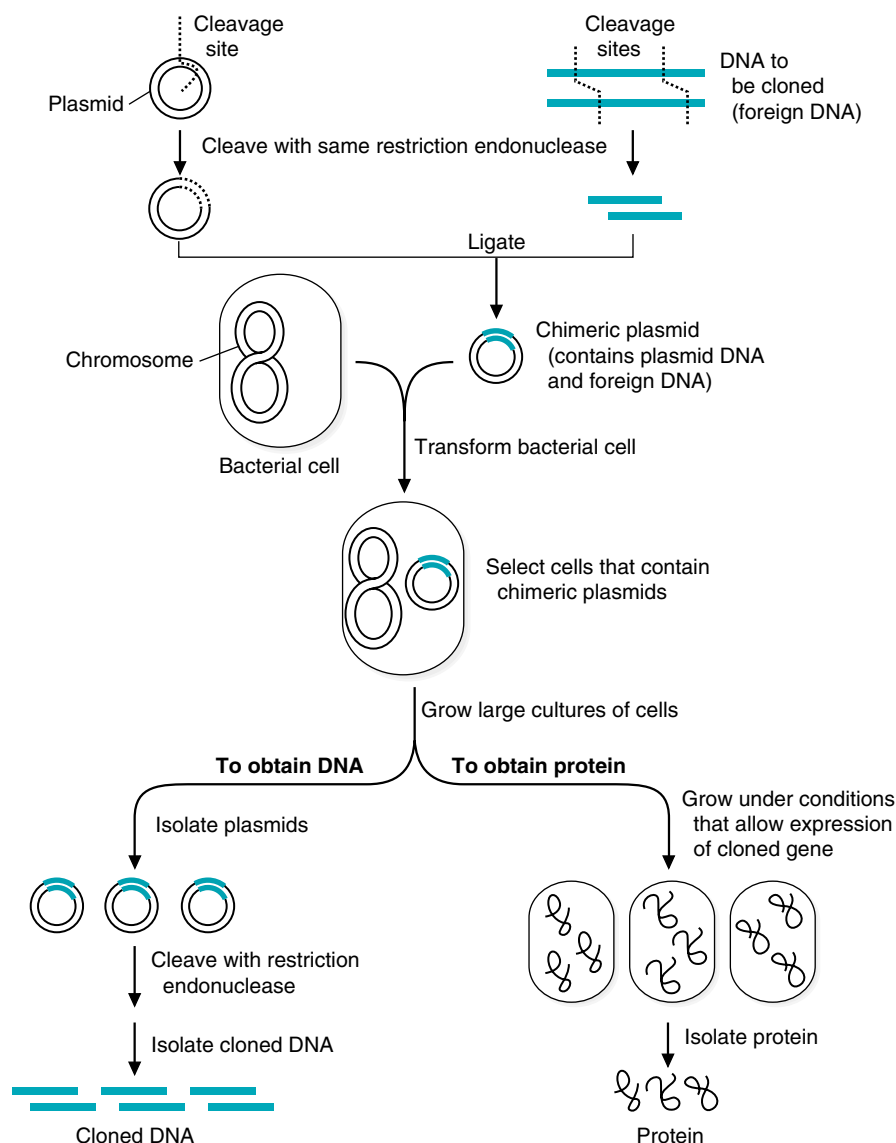


### B. If synthesis is terminated with:



**Fig. 17.7.** The Sanger method. (A). A reaction mixtures contain one of the dideoxynucleotides, such as ddATP, and some of the normal nucleotide, dATP, which compete for incorporation into the growing polypeptide chain. When a T is encountered on the template strand (position 10), some of the molecules will incorporate a ddATP, and the chain will be terminated. Those that incorporate a normal dATP will continue growing until position 15 is reached, where they will incorporate either a ddATP or the normal dATP. Only those that incorporate a dATP will continue growing to position 17. Thus, strands of different length from the 5' end are produced, corresponding to the position of a T in the template strand. (B). DNA sequencing by the dideoxynucleotide method. Four tubes are used. Each one contains DNA polymerase, a DNA template hybridized to a primer, plus dATP, dGTP, dCTP, and dTTP. Either the primer or the nucleotides must have a radioactive label, so bands can be visualized on the gel by autoradiography. Only one of the four dideoxyribonucleotides (ddNTPs) is added to each tube. Termination of synthesis occurs where the ddNTP is incorporated into the growing chain. The template is complementary to the sequence of the newly synthesized strand.

Host cells that contain recombinant DNA are called transformed cells if they are bacteria, or transfected cells if they are eukaryotes. Markers in the vector DNA are used to identify cells that have been transformed, and probes for the foreign DNA can be used to determine that the host cells actually contain the foreign DNA. If the host cells containing the foreign DNA are incubated under conditions in which they replicate rapidly, large quantities of the foreign DNA can be isolated from the cells. With the appropriate vector and growth conditions that



**Fig. 17.8.** Simplified scheme for cloning of DNA in bacteria. A plasmid is a specific type of vector, or carrier, which can contain inserts of foreign DNA of up to 2.0 kb in size. For clarity, the sizes of the pieces of DNA are not drawn to scale (for example, the bacterial chromosomal DNA should be much larger than the plasmid DNA).

permit expression of the foreign DNA, large quantities of the protein produced from this DNA can be isolated.

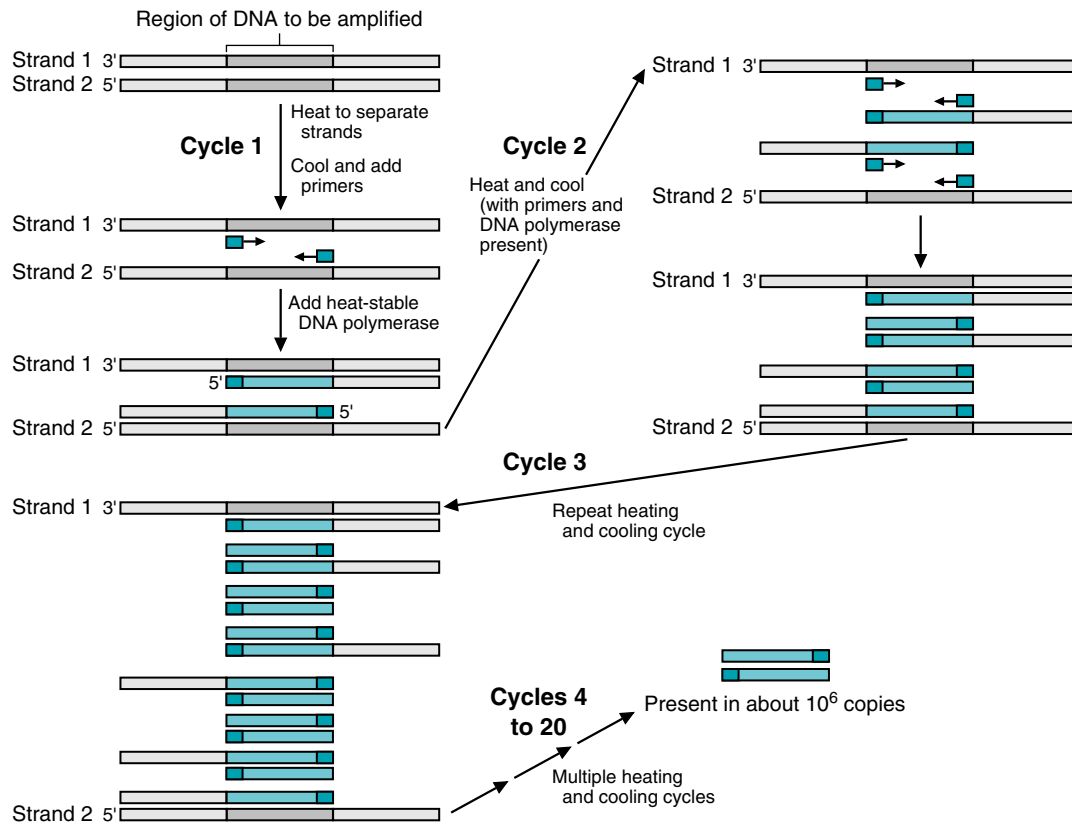
## 2. POLYMERASE CHAIN REACTION (PCR)

PCR is an *in vitro* method that can be used for rapid production of very large amounts of specific segments of DNA. It is particularly suited for amplifying regions of DNA for clinical or forensic testing procedures because only a very small sample of DNA is required as the starting material. Regions of DNA can be amplified by PCR from a single strand of hair or a single drop of blood or semen.

First, a sample of DNA containing the segment to be amplified must be isolated. Large quantities of primers, the four deoxyribonucleoside triphosphates, and a heat-stable DNA polymerase are added to a solution in which the DNA is heated to separate the strands (Fig. 17.9). The primers are two synthetic oligonu-



Although only small amounts of semen were obtained from **Vicky Tim's** body, the quantity of DNA in these specimens could be amplified by PCR. This technique provided sufficient amounts of DNA for comparison with DNA samples from the three suspects.



**Fig. 17.9.** Polymerase chain reaction (PCR). Strand 1 and strand 2 are the original DNA strands. The short dark blue fragments are the primers. After multiple heating and cooling cycles, the original strands remain, but most of the DNA consists of amplified copies of the segment (shown in lighter blue) synthesized by the heat-stable DNA polymerase.



The DNA polymerase used for PCR is isolated from *Thermus aquaticus*, a bacterium that grows in hot springs. This polymerase can withstand the heat required for separation of DNA strands.

cleotides; one oligonucleotide is complementary to a short sequence in one strand of the DNA to be amplified, and the other is complementary to a sequence in the other DNA strand. As the solution is cooled, the oligonucleotides form base pairs with the DNA and serve as primers for the synthesis of DNA strands by the heat-stable DNA polymerase. The process of heating, cooling, and new DNA synthesis is repeated many times until a large number of copies of the DNA are obtained. The process is automated, so that each round of replication takes only a few minutes and in 20 heating and cooling cycles, the DNA is amplified over a million-fold.

## II. USE OF RECOMBINANT DNA TECHNIQUES FOR DIAGNOSIS OF DISEASE

### A. DNA Polymorphisms

Polymorphisms are variations among individuals of a species in DNA sequences of the genome. They serve as the basis for using recombinant DNA techniques in the diagnosis of disease. The human genome probably contains millions of different polymorphisms. Some polymorphisms involve point mutations, the substitution of one base for another. Deletions and insertions are also responsible for variations in DNA sequences. Some polymorphisms occur within the coding region of genes. Others are found in noncoding regions closely linked to genes involved in the cause of inherited disease, in which case they can be used as a marker for the disease.



## B. Detection of Polymorphisms

### 1. RESTRICTION FRAGMENT LENGTH POLYMORPHISMS

Occasionally, a point mutation occurs in a recognition site for one of the restriction enzymes. The restriction enzyme therefore can cut at this restriction site in DNA from most individuals, but not in DNA from individuals with this mutation. Consequently, the restriction fragment that binds a probe for this region of the genome will be larger for a person with the mutation than for most members of the population. Mutations also can create restriction sites that are not commonly present. In this case, the restriction fragment from this region of the genome will be smaller for a person with the mutation than for most individuals. These variations in the length of restriction fragments are known as restriction fragment length polymorphisms (RFLPs).

In some cases, the mutation that causes a disease affects a restriction site within the coding region of a gene. However, in many cases, the mutation affects a restriction site that is outside the coding region but tightly linked (i.e., physically close on the DNA molecule) to the abnormal gene that causes the disease. This RFLP could still serve as a biologic marker for the disease. Both types of RFLPs can be used for genetic testing to determine whether an individual has the disease.

### 2. DETECTION OF MUTATIONS BY ALLELE-SPECIFIC OLIGONUCLEOTIDE PROBES

Other techniques have been developed to detect mutations, because many mutations associated with genetic diseases do not occur within restriction enzyme recognition sites or cause detectable restriction fragment length differences when digested with restriction enzymes. For example, oligonucleotide probes (containing 15–20 nucleotides) can be synthesized that are complementary to a DNA sequence that includes a mutation. Different probes are produced for alleles that contain mutations and for those that have a normal DNA sequence. The region of the genome that contains the abnormal gene is amplified by PCR, and the samples of DNA are placed in narrow bands on nitrocellulose paper (“slot blotting”). The paper is then treated with the radioactive probe for either the normal or the mutant sequence. Autoradiograms indicate whether the normal or mutant probe has preferentially base-paired (hybridized) with the DNA, that is, whether the alleles are normal or mutated. Carriers, of course, have two different alleles, one that binds to the normal probe and one that binds to the mutant probe.

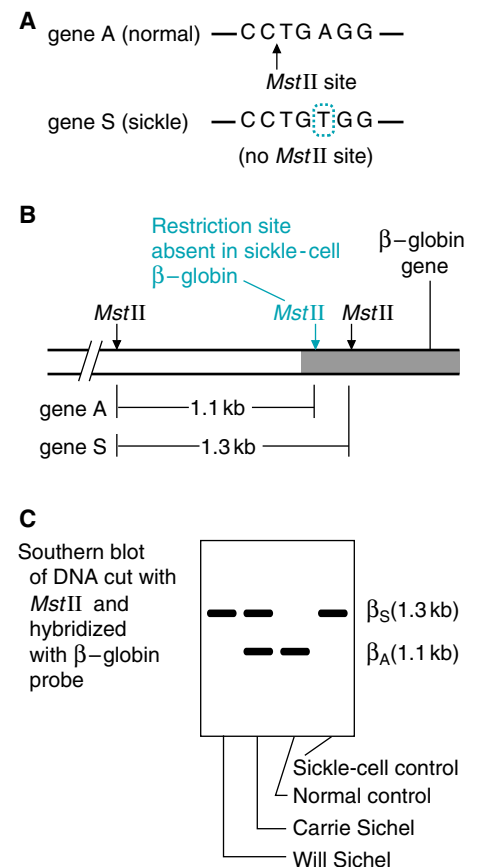


So how does one determine the DNA sequence of a gene that contains a mutation to develop specific probes to that mutation? Initially the gene causing the disease must be identified. This is done by a process known as positional cloning, which involves linking polymorphic markers to the disease. Individuals who express the disease should contain a specific variant of these polymorphic markers, whereas individuals who do not express the disease would not contain these markers. Once such polymorphic markers are identified, they are used as probes to screen a human genomic library. This will identify pieces of human DNA containing the polymorphic marker. These pieces of DNA are then used as probes to expand the region of the genome surrounding this marker. Potential genes within this region are identified (using data available from the sequencing of the human genome), and the sequence of bases within each gene compared with the sequence of bases in the genes of individuals with the disease. The one gene that shows an altered sequence in disease-carrying individuals as compared with normal individuals is the tentative disease gene. Through the sequencing of genes from many people afflicted with the disease, the types of mutations that lead to this disease can be characterized and specific tests developed to look for these specific mutations.



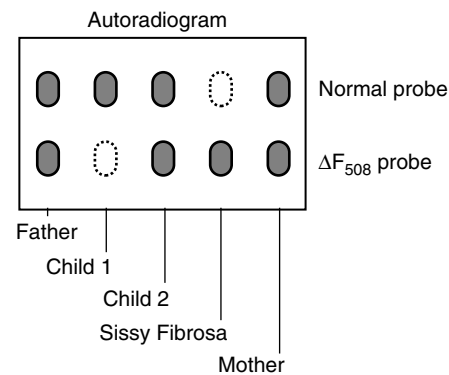
The mutation that causes sickle cell anemia abolishes a restriction site for the enzyme *MstII* in the  $\beta$ -globin gene. The consequence of this mutation is that the restriction fragment produced by *MstII* that includes the 5'-end of the  $\beta$ -globin gene is larger (1.3 kilobases [kb]) for individuals with sickle cell anemia than for normal individuals (1.1 kb). Analysis of restriction fragments provides a direct test for the mutation. In **Will Sichel's** case, both alleles for  $\beta$ -globin lack the *MstII* site and produce 1.3-kb restriction fragments; thus, only one band is seen in a Southern blot.

Carriers have both a normal and a mutant allele. Therefore, their DNA will produce both the larger and the smaller *MstII* restriction fragments. When **Will Sichel's** sister **Carrie Sichel** was tested, she was found to have both the small and the large restriction fragments, and her status as a carrier of sickle cell anemia, initially made on the basis of protein electrophoresis, was confirmed.



**Q:** Testing for cystic fibrosis by DNA sequencing is time-consuming and expensive. Therefore, another technique that uses allele-specific oligonucleotide probes has been developed. **Sissy Fibrosa** and her family were tested by this method. Oligonucleotide probes, complementary to the region where the 3-base deletion is located, have been synthesized. One probe binds to the mutant ( $\Delta F_{508}$ ) gene, and the other to the normal gene.

DNA was isolated from Sissy, her parents, and two siblings and amplified by PCR. Samples of the DNA were spotted on nitrocellulose paper, treated with the oligonucleotide probes, and the following results were obtained. (Dark spots indicate binding of the probe.)



Which members of Sissy's family have CF, which are normal, and which are carriers?

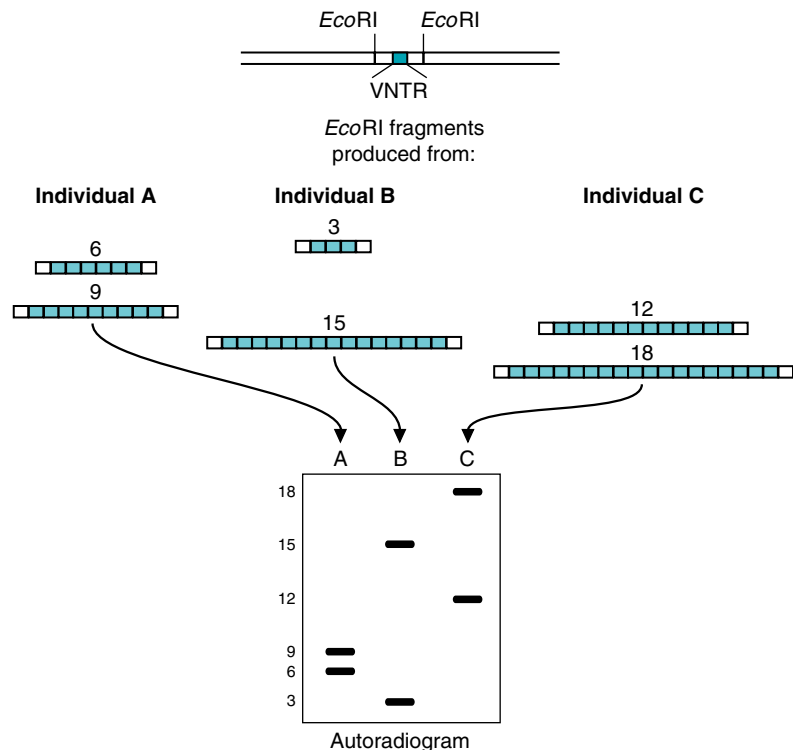
### 3. TESTING FOR MUTATIONS BY PCR

If an oligonucleotide complementary to a DNA sequence containing a mutation is used as a primer for PCR, a DNA sample used as the template will be amplified only if it contains the mutation. If the DNA is normal, the primer will not hybridize with it, and the DNA will not be amplified. This concept is extremely useful for clinical testing. In fact, a number of oligonucleotides, each specific for a different mutation and each containing a different label, can be used as primers in a single PCR reaction. This procedure results in rapid and relatively inexpensive testing for multiple mutations.

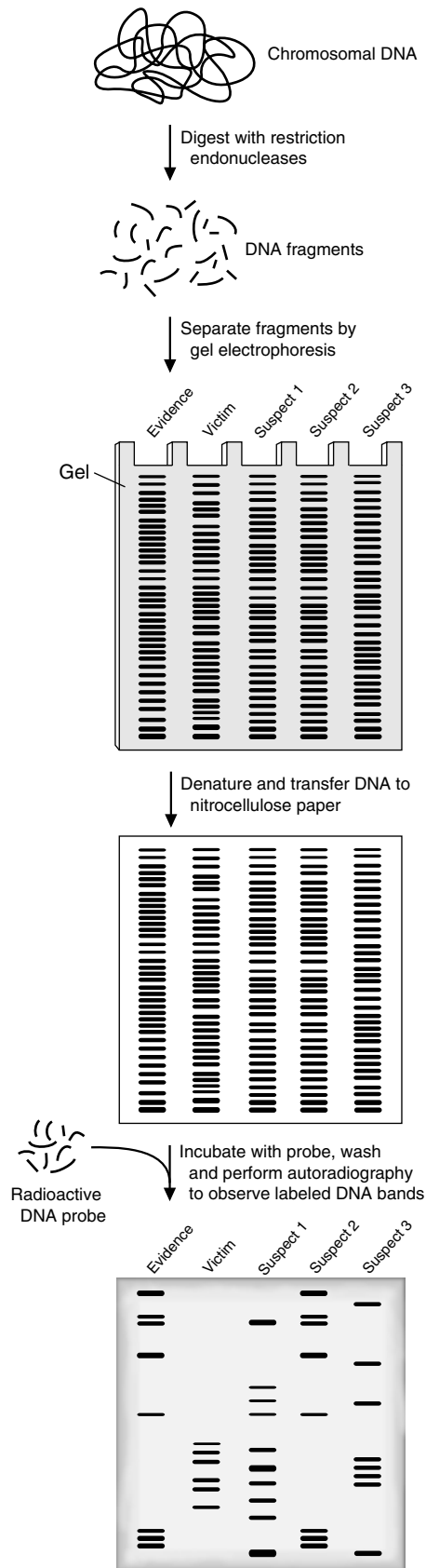
### 4. DETECTION OF POLYMORPHISMS CAUSED BY REPETITIVE DNA

Human DNA contains many sequences that are repeated in tandem a variable number of times at certain loci in the genome. These regions are called highly variable regions because they contain a variable number of tandem repeats (VNTR). Digestion with restriction enzymes that recognize sites that flank the VNTR region produces fragments containing these loci, which differ in size from one individual to another, depending on the number of repeats that are present. Probes used to identify these restriction fragments bind to or near the sequence that is repeated (Fig. 17.10).

The restriction fragment patterns produced from these loci can be used to identify individuals as accurately as the traditional fingerprint. In fact, this restriction fragment technique has been called "DNA fingerprinting" and is gaining widespread use in forensic analysis. Family relationships can be determined by this method, and it can be used to help acquit or convict suspects in criminal cases.



**Fig. 17.10.** Restriction fragments produced from a gene with a variable number of tandem repeats (VNTR). Each individual has two homologues of every somatic chromosome and thus two genes each containing this region with a VNTR. Cleavage of each individual's genomic DNA with a restriction enzyme produces two fragments containing this region. The length of the fragments depends on the number of repeats they contain. Electrophoresis separates the fragments, and a labeled probe that binds to the fragments allows them to be visualized. Each short blue block represents one repeat.



Obviously, the father and mother are both carriers of the defective allele, as is one of the two siblings (Child 2). Sissy has the disease, and the other sibling (Child 1) is normal.



DNA samples were obtained from each of the three suspects in **Vicky Tim's** rape and murder case, and these samples were compared with the victim's DNA by using DNA fingerprinting. Because **Vicky Tim's** sample size was small, PCR was used to amplify the regions containing the VNTRs. The results, using a probe for one of the repeated sequences in human DNA, are shown below to illustrate the process. For more positive identification, a number of different restriction enzymes and probes were used. The DNA from suspect 2 produced the same restriction pattern as the DNA from the semen obtained from the victim. If the other restriction enzymes and probes corroborate this finding, suspect 2 can be identified by DNA fingerprinting as the rapist/murderer.



The huge amount of information now available from the sequencing of the human genome, and the results available from gene chip experiments, has greatly expanded the field of Bioinformatics. Bioinformatics can be defined as the gathering, processing, data storage, data analysis, information extraction, and visualization of biologic data. Bioinformatics also provides the scientist with the capability of organizing vast amounts of data in a manageable form that allows easy access and retrieval of data. Powerful computers are required to perform these analyses. As an example of an experiment requiring these tools, suppose you want to compare the effects of two different immunosuppressant drugs on gene expression in lymphocytes. Lymphocytes would be treated with either nothing (the control) or with the drugs (experimental samples). RNA would be isolated from the cells during drug treatment, and the RNA converted to fluorescent cDNA using the enzyme reverse transcriptase and a fluorescent nucleotide analog. The cDNA produced from your three samples would be used as probes for a gene chip containing DNA fragments from more than 5,000 human genes. The samples would be allowed to hybridize to the chips, and you would then have 15,000 results to interpret (the extent of hybridization of each cDNA sample with each of the 5,000 genes on the chip). Computers are used to analyze the fluorescent spots on the chips and to compare the levels of fluorescent intensity from one chip to another. In this way, you can group genes showing similar levels of stimulation or inhibition in the presence of the drugs and compare the two drugs with respect to which genes have activated or inhibited expression.

Individuals who are closely related genetically will have restriction fragment patterns (DNA fingerprints) that are more similar than those who are more distantly related. Only monozygotic twins will have identical patterns.

## 5. DNA CHIPS

A recently introduced technique will permit the screening of many genes to determine which alleles of these genes are present in samples obtained from patients. The surface of a small chip is dotted with thousands of pieces of single-stranded DNA, each representing a different gene or segment of a gene. The chip is then incubated with a sample of a patient's DNA, and the pattern of hybridization is determined by computer analysis. The results of the hybridization analysis could be used, for example, to determine which one of the many known mutations for a particular genetic disease is the specific defect underlying a patient's problem. An individual's gene chip also may be used to determine which alleles of drug-metabolizing enzymes are present and, therefore, the likelihood of that individual having an adverse reaction to a particular drug.

Another use for a DNA chip is to determine which genes are being expressed. If the mRNA from a tissue specimen is used to produce a cDNA by reverse transcriptase, the cDNA will hybridize with only those genes being expressed in that tissue. In the case of a cancer patient, this technique could be used to determine the classification of the cancer much more rapidly and more accurately than the methods traditionally used by pathologists. The treatment then could be more specifically tailored to the individual patient. This technique also can be used to identify the genes required for tissue specificity (e.g., the difference between a muscle cell and a liver cell) and differentiation (the conversion of precursor cells into the different cell types). Experiments using gene chips are helping us to understand differentiation and may open the opportunity to artificially induce differentiation and tissue regeneration in the treatment of disease.

Although development of this DNA or biochip technology is in its infancy, the technique has astonishing potential for the future diagnosis and treatment of disease.

## III. USE OF RECOMBINANT DNA TECHNIQUES FOR THE PREVENTION AND TREATMENT OF DISEASE

### A. Vaccines

Before the advent of recombinant DNA technology, vaccines were made exclusively from infectious agents that had been either killed or attenuated (altered so that they can no longer multiply in an inoculated individual). Both types of vaccines were potentially dangerous because they could be contaminated with the live, infectious agent. In fact, in a small number of instances, disease has actually been caused by vaccination. The human immune system responds to antigenic proteins on the surface of an infectious agent. By recombinant DNA techniques, these antigenic proteins can be produced, completely free of the infectious agent, and used in a vaccine. Thus, any risk of infection is eliminated. The first successful recombinant DNA vaccine to be produced was for the hepatitis B virus.



When **Erna Nemdy** began working with patients, she received the hepatitis B vaccine. The hepatitis B virus (HBV) infects the liver, causing severe damage. The virus contains a surface antigen (HBsAg) or coat protein for which the gene has been isolated. However, because the protein is glycosylated, it could not be produced in *E. coli*. (Bacteria, because they lack subcellular organelles, cannot produce glycosylated proteins.) Therefore, a yeast (eukaryotic) expression system was used that produced a glycosylated form of the protein. The viral protein, separated from the small amount of contaminating yeast protein, is used as a vaccine for immunization against HBV infection.

## B. Production of Therapeutic Proteins

### 1. INSULIN AND GROWTH HORMONE

Recombinant DNA techniques are used to produce proteins that have therapeutic properties. One of the first such proteins to be produced was human insulin. Recombinant DNA corresponding to the A chain of human insulin was prepared and inserted into plasmids that were used to transform *E. coli* cells. The bacteria then synthesized the insulin chain, which was purified. A similar process was used to obtain B chains. The A and B chains were then mixed and allowed to fold and form disulfide bonds, producing active insulin molecules (Fig. 17.11). Insulin is not glycosylated, so there was no problem with differences in glycosyltransferase activity between *E. coli* and human cell types.

Human growth hormone has also been produced in *E. coli* and is used to treat children with growth hormone deficiencies. Before production of recombinant growth hormone, growth hormone isolated from cadaver pituitary tissue was used, which was in short supply.

### 2. COMPLEX HUMAN PROTEINS

More complex proteins have been produced in mammalian cell culture using recombinant DNA techniques. The gene for Factor VIII, a protein involved in blood clotting, is defective in individuals with hemophilia. Before genetically engineered Factor VIII became available, a number of hemophiliac patients died of AIDS or hepatitis that they contracted from transfusions of contaminated blood or from Factor VIII isolated from contaminated blood.

Tissue plasminogen activator (TPA) is a protease in blood that converts plasminogen to plasmin. Plasmin is a protease that cleaves fibrin (a major component of blood clots), and, thus, TPA administration dissolves blood clots. Recombinant TPA, produced in mammalian cell cultures, is frequently administered during or immediately after a heart attack to dissolve the thrombi that occlude coronary arteries and prevent oxygen from reaching the heart muscle.

Hematopoietic growth factors also have been produced in mammalian cell cultures by recombinant DNA techniques. Erythropoietin can be used in certain types of anemias to stimulate the production of red blood cells. Colony-stimulating factors (CSFs) and interleukins (ILs) can be used after bone marrow transplants and after chemotherapy to stimulate white blood cell production and decrease the risk of infection.

Recombinant  $\beta$ -interferon is the first drug known to decrease the frequency and severity of episodes resulting from the effects of demyelination in patients with multiple sclerosis.

A method for producing human proteins that is being tested involves transgenic animals. These animals (usually goat or sheep) have been genetically engineered to produce human proteins in the mammary gland and secrete them into milk. The gene of interest is engineered to contain a promoter that is only active in the mammary glands under lactating conditions. The vector containing the gene and promoter is inserted into the nucleus of a freshly fertilized egg, which is then implanted into a foster mother. The female animal progeny are tested for the presence of this transgene, and milk from the positive animals is collected. Large quantities of the protein of interest can then be isolated from the relatively small number of proteins present in milk.

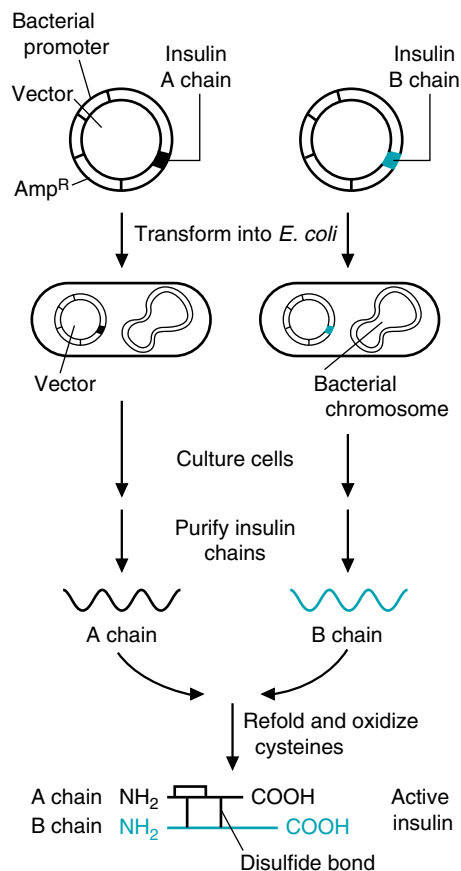
## C. Genetic Counseling

One means of preventing disease is to avoid passing defective genes to offspring. If individuals are tested for genetic diseases, particularly in families known to carry a



**Di Abietes** is using a recombinant human insulin called lispro (Humalog) (see Chapter 6, Fig. 6.13).

Lispro was genetically engineered so that lysine is at position 28 and proline is at position 29 of the B chain (the reverse of their positions in normal human insulin). Di injects a Humalog mixture that contains 75% lispro protamine suspension (intermediate-acting) and 25% lispro solution (rapid-acting). The switch of position of the two amino acids leads to a faster-acting insulin homolog. The lispro is absorbed from the site of injection much more quickly than other forms of insulin, and it acts to lower blood glucose levels much more rapidly than the other insulin forms.

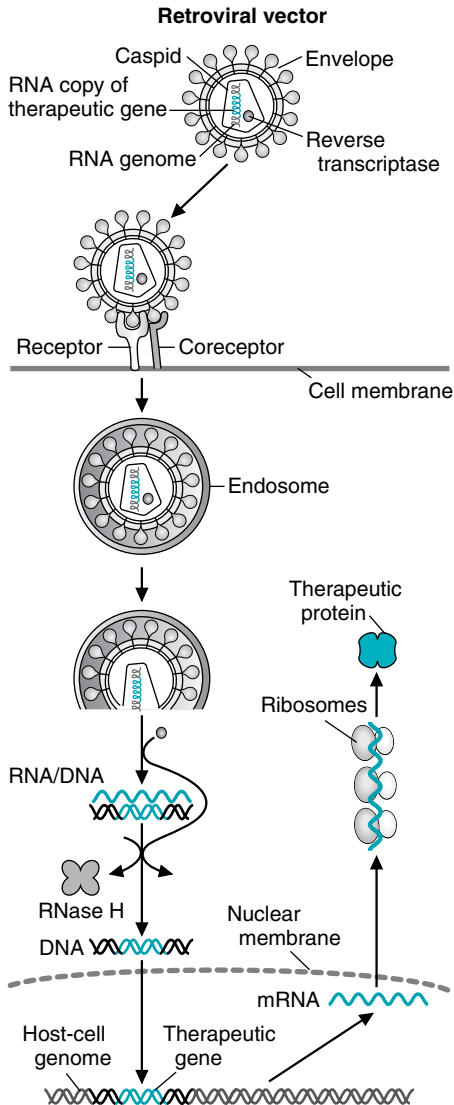


**Fig. 17.11.** Production of human insulin in *E. coli*. Amp<sup>R</sup> = the gene for ampicillin resistance. The presence of Amp<sup>R</sup> allows bacterial cells that contain the vector to grow in the presence of ampicillin. Cells lacking the Amp<sup>R</sup> gene die in the presence of ampicillin. Because *E. coli* cannot process preproinsulin, a synthetic scheme was developed whereby each individual chain of insulin was expressed, produced, and purified, and then the two chains were linked together in a test tube.





**Carrie Sichel's** fiancé decided to be tested for the sickle cell gene. He was found to have both the 1.3-kb and the 1.1-kb *Mst*II restriction fragments that include a portion of the  $\beta$ -globin gene. Therefore, like Carrie, he also is a carrier for the sickle cell gene.



**Fig. 17.12.** Use of retroviruses for gene therapy. The retrovirus carries an RNA copy of the therapeutic gene into the cell. The endosome that contains the virus dissolves, and the RNA and viral reverse transcriptase are released. This enzyme copies the RNA, making a double-stranded DNA that integrates into the host cell genome. Transcription and translation of this DNA (the therapeutic gene) produces the therapeutic protein. (The virus does not multiply because its genes were removed and replaced by the RNA copy of the therapeutic gene.)

defective gene, genetic counselors can inform them of their risks and options. With this information, individuals can decide in advance whether to have children.

Screening tests, based on the recombinant DNA techniques outlined in this chapter, have been developed for many inherited diseases. Although these tests are currently rather expensive, particularly if entire families have to be screened, the cost may be trivial compared with the burden of raising children with severe disabilities. Obviously, ethical considerations must be taken into account, but recombinant DNA technology has provided individuals with the opportunity to make choices.

Screening can be performed on the prospective parents before conception. If they decide to conceive, the fetus can be tested for the genetic defect. In some cases, if the fetus has the defect, treatment can be instituted at an early stage, even in utero. For certain diseases, early therapy leads to a more positive outcome.

## D. Gene Therapy

The ultimate cure for genetic diseases is to introduce normal genes into individuals who have defective genes. Currently, gene therapy is being attempted in animals, cell cultures, and human subjects. It is not possible at present to replace a defective gene with a normal gene at its usual location in the genome of the appropriate cells. However, as long as the gene is expressed at the appropriate time and produces adequate amounts of the protein to return the person to a normal state, the gene does not have to integrate into the precise place in the genome. Sometimes the gene does not even have to be in the cells that normally contain it.

Retroviruses were the first vectors used to introduce genes into human cells. Normally, retroviruses enter target cells, their RNA genome is copied by reverse transcriptase, and the double-stranded DNA copy is integrated into the host cell genome (see Fig. 14.22). If the retroviral genes (e.g., *gag*, *pol*, and *env*) are first removed and replaced with the therapeutic gene, the retroviral genes integrated into the host cell genome will produce the therapeutic protein rather than the viral proteins (Fig. 17.12). This process works only when the human host cells are undergoing division, so it has limited applicability. Other problems with this technique are that it can only be used with small genes ( $\approx 8$  kb), and it may disrupt other genes because the insertion point is random, thereby possibly resulting in cancer.



A defect in the adenosine deaminase (ADA) gene causes severe combined immunodeficiency syndrome (SCID). When ADA is defective, deoxyadenosine and dATP accumulate in rapidly dividing cells, such as lymphocytes, and prove toxic to these cells. Cells of the immune system cannot proliferate at a normal rate, and children with SCID usually die at an early age because they cannot combat infections. To survive, they must be confined to a sterile, environmental "bubble." When an appropriate donor is available, bone marrow transplantation can be performed with a reasonable degree of success.

In 1990, a 4-year-old girl, for whom no donor was available, was treated with infusions of her own lymphocytes that had been treated with a retrovirus containing a normal ADA gene. Although she had not responded to previous therapy, she improved significantly after this attempt at gene therapy. This disease is still being treated with gene therapy, in combination with replacement enzyme infusion.



In familial hypercholesterolemia, a condition associated with a high incidence of heart attacks, the low-density lipoprotein (LDL) receptor is deficient. Attempts to correct this defect with gene therapy involved removal of a segment of liver and preparation of hepatocytes that had been grown in tissue culture. After these dividing cells were infected with a retrovirus containing the gene for the LDL receptor, they were reinfused into the hepatic portal vein of the patient. The early efforts using this approach met with only limited success.

Adenoviruses, which are natural human pathogens, can also be used as vectors. As in retroviral gene therapy, the normal viral genes required for synthesis of viral particles are replaced with the therapeutic genes. The advantages to using an adenovirus are that the introduced gene can be quite large (~36 kb), and infection does not require division of host cells. The disadvantage is that genes carried by the adenovirus do not stably integrate into the host genome, resulting in only transient expression of the therapeutic proteins (but preventing disruption of host genes). Thus, the treatment must be repeated periodically. Another problem with adenoviral gene therapy is that the host can mount an immune response to the pathogenic adenovirus, causing complications.

To avoid the problems associated with viral vectors, researchers are employing treatment with DNA alone or with DNA coated with a layer of lipid (i.e., in liposomes). Adding a ligand for a receptor located on the target cells could aid delivery of the liposomes to the appropriate host cells. Many problems still plague the field of gene therapy. In many instances, the therapeutic genes must be targeted to the cells where they normally function—a difficult task at present. Deficiencies in dominant genes are more difficult to treat than those in recessive genes, and the expression of the therapeutic genes often needs to be carefully regulated. Although the field is moving forward, progress is slow.

Another approach to gene therapy involves the use of antisense oligonucleotides rather than vectors. These oligonucleotides are designed to hybridize either with the target gene to prevent transcription or with mRNA to prevent translation. Again technical problems have plagued the development of therapy based on this theoretically promising idea.

## E. Transgenic Animals

The introduction of normal genes into somatic cells with defective genes corrects the defect only in the treated individuals, not in their offspring. To eliminate the defect for future generations, the normal genes must be introduced into the germ cell line (the cells that produce sperm in males or eggs in females). Experiments with animals indicate that gene therapy in germ cells is feasible. Genes can be introduced into fertilized eggs from which transgenic animals develop, and these transgenic animals can produce apparently normal offspring.

In fact, if the nucleus isolated from the cell of one animal is injected into the enucleated egg from another animal of the same species and the egg is implanted in a foster mother, the resulting offspring is a “clone” of the animal from which the nucleus was derived (Fig. 17.13). Clones of sheep and pigs have been produced, and similar techniques could be used to clone humans. Obviously, these experiments raise many ethical questions that will be difficult to answer.



Adenoviral vectors have been used in a aerosol spray to deliver normal copies of the CFTR (cystic fibrosis transmembrane conductance regulator) gene to cells of the lung. Some cells took up this gene, and the patients experienced moderate improvement. However, stable integration of the gene into the genome did not occur, and cells affected by the disease other than those in the lung (e.g., pancreatic cells) did not benefit. Nevertheless, this approach marked a significant forward step in the development of gene therapy.



In an attempt to treat ornithine transcarbamoylase deficiency (a disorder of nitrogen metabolism) using adenoviral vectors, a volunteer died of a severe immune response to the vector. This unfortunate result has led to a reevaluation of the safety of viral vectors for gene therapy.



One approach to in vivo gene therapy involves the direct injection of DNA for certain HLA antigens into malignant melanomas (skin cancers). The HLA gene chosen for therapy should not be the one expressed by the patient. Thus, if the gene is incorporated into the cells and expressed, the body should recognize the tumor cells as foreign tissue and mount an immune attack. Preliminary results using this strategy were encouraging.

## CLINICAL COMMENTS



**Erna Nemdy.** In reading about development of the hepatitis B vaccine, **Erna Nemdy** learned that the first vaccine available for HBV, marketed in 1982, was a purified and “inactivated” vaccine containing HBV virus that had been chemically killed. The virus was derived from the blood of known HBV carriers. Later, “attenuated” vaccines were used in which the virus remained live but was altered so that it no longer multiplied in the inoculated host. Both the inactivated and the attenuated vaccines are potentially dangerous because they can be contaminated with live infectious HBV.

The modern “subunit” vaccines, first marketed in 1987, were made by recombinant DNA techniques described earlier in this chapter. Because this vaccine consists solely of the viral surface protein or antigen to which the immune system responds, there is no risk for infection with HBV.

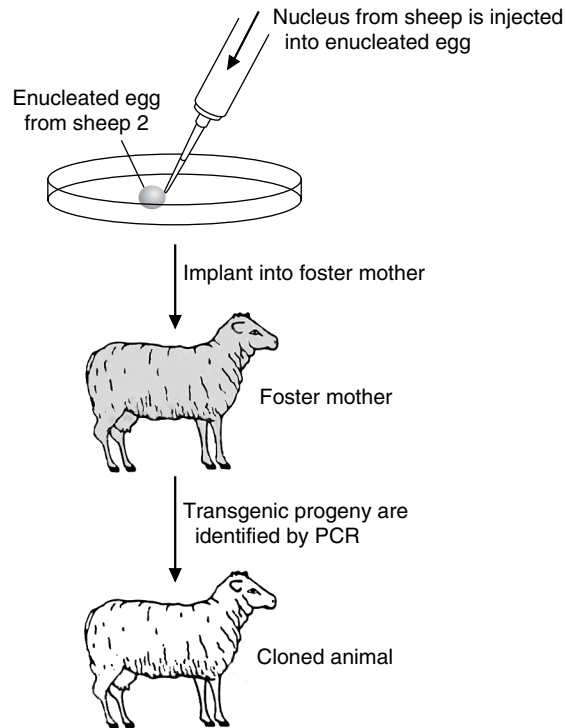


Fig. 17.13. Cloning of a mammalian organism.



The most common CF mutation is a 3-bp deletion that causes the loss of phenylalanine at position 508 (delta 508). This mutation is present in more than 70% of CFTR patients. The defective protein is synthesized in the endoplasmic reticulum, but is misfolded. It is therefore not transported to the Golgi, but is degraded by a proteolytic enzyme complex called the proteasome. Other mutations responsible for CF generate an incomplete mRNA because of premature stop signals, frame shifts, or abnormal splice sites or create a CFTR channel in the membrane that does not function properly.



**Sissy Fibrosa.** Cystic fibrosis (CF) is a genetically determined autosomal recessive disease that can be caused by a variety of mutations within the CF gene located on chromosome 7. **Sissy Fibrosa** was found to have a 3-bp deletion at residue 508 of the CF gene (the mutation present in approximately 70% of Caucasian patients with CF in the United States). This mutation is generally associated with a more severe clinical course than many other mutations causing the disease. However, other genes and environmental factors may modify the clinical course of the disease, so it is not currently possible to counsel patients accurately about prognosis based on their genotype.

CF is a relatively common genetic disorder in the United States, with a carrier rate of approximately 5% in Caucasians. The disease occurs in 1 per 1,600 to 2,000 Caucasian births in the country (1 per 17,000 in African Americans and 1 per 100,000 in Asians).



**Carrie Sichel.** After learning the results of their tests for the sickle cell gene, **Carrie Sichel** and her fiancé consulted a genetic counselor. The counselor informed them that, because they were both carriers of the sickle cell gene, their chance of having a child with sickle cell anemia was fairly high (approximately 1 in 4). She told them that prenatal testing was available with fetal DNA obtained from cells by amniocentesis or chorionic villus sampling. If these tests indicated that the fetus had sickle cell disease, abortion was a possibility. Carrie, because of her religious background, was not sure that abortion was an option for her. But having witnessed her brother's sickle cell crises for many years, she also was not sure that she wanted to risk having a child with the disease. Her fiancé also felt that, at 25 years of age, he was not ready to deal with such difficult problems. They mutually agreed to cancel their marriage plans.



**Vicky Tim.** DNA fingerprinting represents an important advance in forensic medicine. Before development of this technique, identification of criminals was far less scientific. The suspect in the rape and murder of **Vicky Tim** was arrested and convicted mainly on the basis of the results of DNA fingerprint analysis.

This technique has been challenged in some courts on the basis of technical problems in statistical interpretation of the data and sample collection. It is absolutely necessary for all of the appropriate controls to be run, including samples from the victim's DNA as well as the suspect's DNA. Another challenge to the fingerprinting procedure has been raised because PCR is such a powerful technique that it can amplify minute amounts of contaminating DNA from a source unrelated to the case.

## BIOCHEMICAL COMMENTS



**Mapping of the Human Genome.** The Human Genome Project began in 1990, and by the summer of 2000, the entire human genome had been mapped. This feat was accomplished in far less than the expected time as a result of both cooperative and competitive interactions of laboratories in the private as well as the public sector.

The human genome contains over  $3 \times 10^9$  (3 billion) bp. A large percentage of this genome (<95%) does not code for the amino acid sequences of proteins or for functional RNA (such as rRNA or tRNA) but is composed of repetitive sequences, introns, and other noncoding elements of unknown function. The human genome is estimated to contain only about 30,000 to 50,000 genes.

As the announcement of the identification of a wayward gene appears in the morning newspaper, the average citizen expects the cure for the genetic disease to be described in the evening edition. Although knowledge of the chromosomal location and the sequence of genes will result in the rapid development of tests to determine whether an individual carries a defective gene, the development of a treatment for the genetic disease caused by the defective gene will not be that easy or that rapid. As outlined in the section on gene therapy above, many technical problems need to be solved before gene therapy becomes commonplace. In addition to solving the molecular puzzles involved in gene therapy, we also will have to deal with many difficult questions.

Is it appropriate to replace defective genes in somatic cells to relieve human suffering? Many people may agree with this goal. But there is a related question: is it appropriate to replace defective genes **in the germ cell line** to relieve human suffering? Fewer people may agree with this goal. Genetic manipulation of somatic cells affects only one generation; these cells die with the individual. Germ cells, however, live on, producing each successive generation.

The techniques developed to explore the human genome could be used for many purposes. What are the limits for the application of the knowledge gained by advances in molecular biology? Who should decide what the limits are, and who should serve as the genetic police? If we permit experiments that involve genetic manipulation of the human germ cell line, however nobly conceived, could we, in our efforts to "improve" ourselves, genetically engineer the human race into extinction?



What are the statistical issues relating to DNA fingerprinting?

Through the analysis of 1,000 individuals of different ethnicities, one can determine the frequency of a particular DNA polymorphism within that distinct population. By matching four or five polymorphisms from DNA at the crime scene with DNA from a suspect, one can determine the odds of that match happening by chance. For example, let us assume that a suspect's DNA was compared with DNA found at the crime scene for four unique polymorphisms within the suspect's ethnic group. The frequency of polymorphism A in that population is 1 in 20; of polymorphism B, 1 in 30; of polymorphism C, 1 in 50; and of polymorphism D, 1 in 100. The odds of the suspect DNA matching the DNA found at the crime scene for all four polymorphisms would be the product of each individual probability, or  $(1/20) \times (1/30) \times (1/50) \times (1/100)$ . This comes out to a 1 in 3 million chance that an individual would have the same polymorphisms in their DNA as that found at the crime scene. The question left to the courts is whether the 1 in 3 million match is sufficient to convict the suspect of the crime. Given that there may be 30 million individuals in the United States within the same ethnic group as the suspect, there would then be 30 people within the country who would match the DNA polymorphisms found at the scene of the crime. Can the court be sure that the suspect is the correct individual? Clearly, the use of DNA fingerprinting is much clearer when a match is not made, for that immediately indicates that the suspect was not at the scene of the crime.

## Suggested References

- Housman D. DNA on trial—the molecular basis of DNA fingerprinting. *N Engl J Med* 1995;332:534–535.
- Trent RJ. *Molecular Medicine*. New York: Churchill Livingstone, 1993.

- Watson JD, Gilman M, Witkowski J, Zoller M. *Recombinant DNA*. 2nd Ed. New York: Scientific American Books, WH Freeman, 1992.
- Gelehrter TD, Collins FS, Ginsburg D. *Principles of Medical Genetics*. 2nd Ed. Baltimore: Williams & Wilkins, 1998.
- Welsh MJ. Cystic Fibrosis. Chapter 76. In: Goldman L, Bennett JC, eds. *Cecil Textbook of Medicine*. 21st Ed, Volume 1, pp 401–405. Philadelphia: WB Saunders, 2000.



## REVIEW QUESTIONS—CHAPTER 17

- Electrophoresis resolves double-stranded DNA fragments based on which of the following?
  - Sequence
  - Molecular weight
  - Isoelectric point
  - Frequency of CTG repeats
  - Secondary structure
- If a restriction enzyme recognizes a six-base sequence, how frequently, on average, will this enzyme cut a large piece of DNA?
  - Once every 16 bases
  - Once every 64 bases
  - Once every 256 bases
  - Once every 1,024 bases
  - Once every 4,096 bases
- Which of the following sets of reagents are required for dideoxy chain DNA synthesis in the Sanger technique for DNA sequencing?
  - Deoxyribonucleotides, Taq polymerase, DNA primer
  - Dideoxyribonucleotides, deoxyribonucleotides, template DNA
  - Dideoxyribonucleotides, DNA primer, reverse transcriptase
  - Two DNA primers, template DNA, Taq polymerase
  - mRNA, dideoxynucleotides, reverse transcriptase
- Which of the following statements correctly describe a feature of DNA electrophoresis?
  - Larger DNA fragments migrate farther in the gel.
  - DNA fragments migrate toward the negative charge (anode).
  - DNA can be visualized using UV light and the dye ethidium bromide.
  - Total human genomic DNA cut by a specific restriction endonuclease will generate three distinctly separable bands.
  - DNA must be denatured before it can be run in the gel.
- The best method to determine whether albumin is transcribed in the liver of a mouse model of hepatocarcinoma is which of the following?
  - Genomic library screening
  - Genomic Southern blot
  - Tissue Northern blot
  - Tissue Western blot
  - VNTR analysis



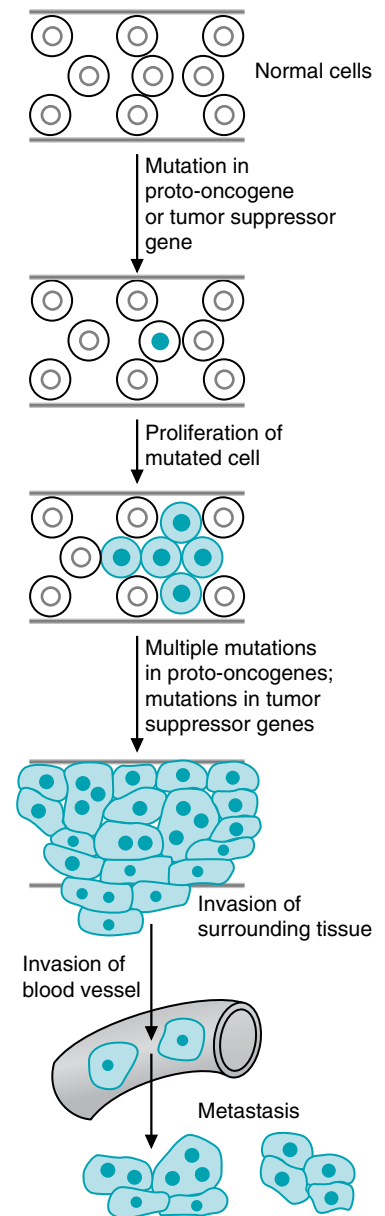
# 18 The Molecular Biology of Cancer

The term **cancer** applies to a group of diseases in which **cells grow abnormally** and form a **malignant tumor**. Malignant cells can invade nearby tissues and **metastasize** (i.e., travel to other sites in the body, where they establish secondary areas of growth). This aberrant growth pattern results from **mutations in genes that regulate proliferation, differentiation, and survival of cells** in a multicellular organism. Because of these genetic changes, cancer cells no longer respond to the signals that govern growth of normal cells (Fig. 18.1.)

**Oncogenes and Tumor Suppressor Genes.** The genes involved in the development of cancer are classified as **oncogenes** or **tumor suppressor genes**. **Oncogenes** are mutated derivatives of normal genes (**proto-oncogenes**) whose function is to promote proliferation or cell survival. These genes can code for **growth factors, growth factor receptors, signal transduction proteins, intracellular kinases, and transcription factors**. The process of transformation into a malignant cell may begin with a “**gain of function**” mutation in only one copy of a proto-oncogene. As the mutated cell proliferates, additional mutations can occur. **Tumor suppressor genes** (normal growth suppressor genes) encode proteins that **inhibit proliferation, promote cell death, or repair DNA**; both alleles need to be inactivated for transformation (**a loss of function**). Growth suppressor genes have been called the guardians of the cell.

**Cell Cycle Suppression and Apoptosis.** Normal cell growth depends on a balanced regulation of **cell cycle** progression and **apoptosis** (programmed cell death) by proto-oncogenes and growth suppressor genes. At **checkpoints** in the **cell cycle** products of **tumor suppressor genes** slow growth in response to signals from the cell's environment, including external growth inhibitory factors, or to allow time for repair of damaged DNA, or in response to other adverse circumstances in cells. Alternately, cells with damaged DNA are targeted for apoptosis so that they will not proliferate. Many growth-stimulatory pathways involving proto-oncogenes, and growth-inhibitory controls involving a variety of tumor suppressor genes, converge to regulate the activity of some key protein kinases, the **cyclin-dependent kinases**. These kinases act to control progression at specific points in the cell growth cycle. Apoptosis is initiated by either **death receptor activation** or intracellular signals leading to release of the **mitochondrial protein, cytochrome c**.

**Mutations.** Mutations in DNA that give rise to cancer may be inherited or caused by **chemical carcinogens, radiation, viruses**, and by **replication errors** that are not repaired. A cell population must accumulate **multiple mutations** for transformation to malignant cells.



**Fig. 18.1.** Development of cancer. Accumulation of mutations in a number of genes results in transformation. Cancer cells change morphologically, proliferate, invade other tissues, and metastasize.



## THE WAITING ROOM



Patients with leukemia experience a variety of hemorrhagic (bleeding) manifestations caused by a decreased number of platelets. Platelets are small cells that initiate clot formation at the site of endothelial injury. Because of the uncontrolled proliferation of white cells within the limited space of the marrow, the normal platelet precursor cells (the megakaryocytes) in the marrow are “squeezed” or crowded and fail to develop into mature platelets. Consequently, the number of mature platelets (thrombocytes) in the circulation falls, and a thrombocytopenia develops. Because there are fewer platelets to contribute to clot formation, bleeding problems are common.



Malignant neoplasms (new growth, a tumor) of epithelial cell origin (including the intestinal lining, cells of the skin, and cells lining the airways of the lungs) are called carcinomas. If the cancer grows in a gland-like pattern, it is an adenocarcinoma. Thus, **Nick O'Tyne** and **Colin Tuma** have adenocarcinomas. **Mel Anoma** had a carcinoma arising from melanocytes, which is technically a melanocarcinoma, but is usually referred to as a melanoma.



Moles (also called nevi) are tumors of the skin. They are formed by melanocytes that have been transformed from highly dendritic single cells interspersed among other skin cells to round oval cells that grow in aggregates or “nests.” (Melanocytes produce the dark pigment melanin that protects against sunlight by absorbing UV light.) Additional mutations may transform the mole into a malignant melanoma.



The study of cells in culture was a great impetus to the study of cancer, because the development of a tumor in animals could take months. Once cells could be removed from an animal and propagated in a tissue culture dish, the onset of transformation (the normal cell becoming a cancer cell) could be seen in days.

What are the criteria that distinguish transformed cells from normal cells in culture? The first is the requirement for serum in the cell culture medium to stimulate growth. Transformed cells have a reduced requirement for serum, approximately 10% that required for normal cells to grow. The second is the ability to grow without attachment to a supporting matrix (anchorage dependence). Normal cells (such as fibroblasts, smooth muscle cells) require adherence to a substratum (in this case, the bottom of the plastic dish) and will not grow if suspended in a soft agar mixture (the consistency of loose jello). Transformed cells, however, have lost this anchorage dependence for growth. An additional criterion used to demonstrate that cells are truly transformed is that they form tumors when injected into mice lacking an immune system.



**Mannie Weitzels** has chronic myelogenous leukemia, a disease in which a single line of myeloid cells in the bone marrow proliferate abnormally, causing a large increase in the number of nonlymphoid white blood cells (see Chapter 16). His myeloid cells contain the abnormal Philadelphia chromosome, which increases their proliferation. He has recently complained of pain and tenderness in various areas of his skeleton, possibly stemming from the expanding mass of myeloid cells within his bone marrow. He also reports a variety of hemorrhagic signs, including bruises (ecchymoses), bleeding gums, and the appearance of small red spots (petechiae caused by release of red cells into the skin).



**Nick O'Tyne** was diagnosed with a well-differentiated adenocarcinoma of the lungs (see Chapter 13). He underwent anatomic staging procedures to determine the location and severity of the tumor. As a result of these tests, he was considered a candidate for surgical resection of the primary tumor aimed at cure. He survived the surgery and was recovering uneventfully until 6 months later when he complained of an increasingly severe right temporal headache. A CT scan of his brain was performed. Results indicated that the cancer, which had originated in his lungs, had metastasized to his brain.



**Colin Tuma** has had an intestinal adenocarcinoma resected, but there were several metastatic nodules in his liver (see Chapters 12 and 13). He completed his second course of chemotherapy with 5-fluorouracil (5-FU) and had no serious side effects. He assured his physician at his most recent checkup that, this time, he intended to comply with any instructions his physicians gave him. He ruefully commented that he wished he had returned for regular examinations after his first round of surgery for benign intestinal polyps.



**Mel Anoma** returned to his physician after observing a brownish-black irregular mole on his forearm (see Chapter 13). His physician thought the mole looked suspiciously like a malignant melanoma, and performed an excision biopsy (surgical removal for the purposes of biopsy).

### I. CAUSES OF CANCER

*Cancer* is the term applied to a group of diseases in which cells no longer respond to normal restraints on growth. Normal cells in the body respond to signals, such as contact inhibition, that direct them to stop proliferating. Cancer cells do not require growth stimulatory signals and they are resistant to growth inhibitory signals. They are also resistant to apoptosis, the programmed death process whereby unwanted or

irreparably damaged cells self-destruct. They have an infinite proliferative capacity and do not become senescent (i.e., they are immortalized). Furthermore, they can grow independently of structural support, such as the extracellular matrix (loss of anchorage dependence).

A single cell that divides abnormally eventually forms a mass called a tumor. A tumor can be benign and harmless; the common wart is a benign tumor formed from a slowly expanding mass of cells. In contrast, a malignant neoplasm (malignant tumor) is a proliferation of rapidly growing cells that progressively infiltrate, invade, and destroy surrounding tissue. Tumors develop angiogenic potential, which is the capacity to form new blood vessels and capillaries. Thus, tumors can generate their own blood supply to bring in oxygen and nutrients. Cancer cells also can metastasize, separating from the growing mass of the tumor and traveling through the blood or lymph to unrelated organs, where they establish new growths of cancer cells.

The transformation of a normal cell to a cancer cell begins with damage to DNA (base changes or strand breaks) caused by chemical carcinogens, UV light, viruses, or replication errors (see Chapter 13). Mutations result from the damaged DNA if it is not repaired properly or if it is not repaired before replication occurs. A mutation that can lead to transformation also may be inherited. When a cell with one mutation proliferates, this clonal expansion (proliferation of cells arising from a single cell) results in a substantial population of cells containing this one mutation, from which one cell may acquire a second mutation relevant to control of cell growth or death. With each clonal expansion, the probability of another transforming mutation increases. As mutations accumulate in genes controlling proliferation, subsequent mutations occur even more rapidly until the cells acquire the multiple mutations (in the range of 4–7) necessary for full transformation.

The transforming mutations occur in genes that regulate cellular proliferation and differentiation (proto-oncogenes), suppress growth (tumor suppressor genes), target irreparably damaged cells for apoptosis, or repair damaged DNA. The genes regulating cellular growth are called proto-oncogenes, and their mutated forms are called oncogenes. The term *oncogene* is derived from the Greek word “onkos” meaning bulk or tumor. A transforming mutation in a proto-oncogene increases the activity or amount of the gene product (a gain-of-function mutation). Tumor suppressor genes (normal growth suppressor genes) and repair enzymes protect against uncontrolled cell proliferation. A transforming mutation in these protective genes results in a loss of activity or a decreased amount of the gene product. In summary, cancer is caused by the accumulation of mutations in the genes involved in normal cellular growth and differentiation. These mutations give rise to cancer cells capable of unregulated, autonomous, and infinite proliferation.

## II. DAMAGE TO DNA LEADING TO MUTATIONS

### A. Chemical and Physical Alterations in DNA

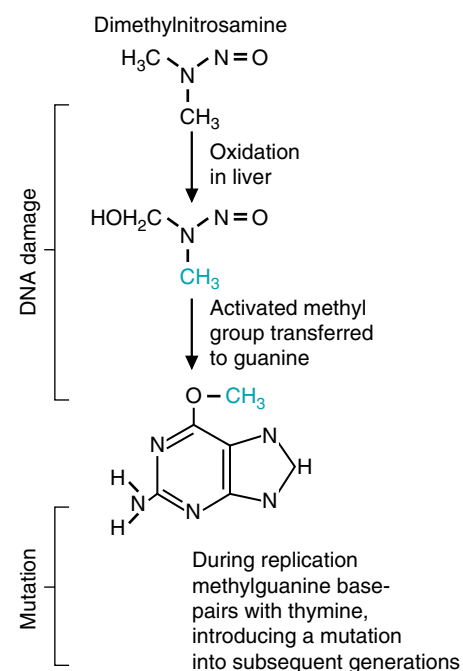
An alteration in the chemical structure of DNA, or of the sequence of bases in a gene, is an absolute requirement for the development of cancer. The function of DNA depends on the presence of various polar chemical groups in DNA bases, capable of forming hydrogen bonds between DNA strands or other chemical reactions. The oxygen and nitrogen atoms in DNA bases are targets for a variety of electrophiles (electron-seeking chemical groups). A typical sequence of events leading to a mutation is shown for dimethylnitrosamine in Figure 18.2. Chemical carcinogens (compounds that can cause transforming mutations) found in the environment and ingested in foods are generally stable lipophilic compounds that, like dimethylnitrosamine, must be activated by metabolism in the body to react with DNA (see also benz[*a*]pyrene, Action of Mutagens, Chapter 13, section III.A.). Many



Drs. Michael Bishop and Harold Varmus demonstrated that cancer is not caused by unusual and novel genes, but rather by mutation within existing cellular genes and that for every gene that causes cancer (an oncogene) there was a corresponding cellular gene, called the proto-oncogene. Although this concept seems straightforward today, it was a significant finding when it was first announced and, in 1989, Drs. Bishop and Varmus were awarded the Nobel Prize in Medicine.



The first experiments to show that oncogenes were mutant forms of proto-oncogenes in human tumors involved cells cultured from a human bladder carcinoma. The DNA sequence of the *ras* oncogene cloned from these cells differed from the normal *c-ras* proto-oncogene. Similar mutations were subsequently found in the *ras* gene of lung and colon tumors. **Colin Tuma's** malignant polyp had a mutation in the *ras* proto-oncogene.



**Fig. 18.2.** Mutations in DNA caused by nitrosamines. Nitrosamines are consumed in many natural products and are produced in the stomach from nitrites used as preservatives and secondary amines found in foods such as fish. They are believed to be responsible for the high incidence of gastric cancer found in Japan and Iceland, where salt-preserved fish was a major dietary item. Nitrosamine metabolites methylate guanine (the transferred methyl group is shown in blue).



UV rays derived from the skin induce an increased incidence of all skin cancers, including squamous cell carcinoma, basal cell carcinoma, and malignant melanoma of the skin. The wavelength of UV light most associated with skin cancer is UVB (280–320 nm), which forms pyrimidine dimers in DNA. This type of DNA damage is repaired by nucleotide excision repair pathways that require products of at least 20 genes. With excessive exposure to the sun, the nucleotide excision repair pathway is overwhelmed, and some damage remains unrepaired.



Burkitt's lymphoma, a general name for a number of types of B-cell malignancies, results from a translocation between chromosomes 8 and 14. The translocation of genetic material moves the proto-oncogene transcription factor *c-myc* (normally found on chromosome 8) to chromosome 14. The translocated gene is now under the control of the promoter region for the immunoglobulin heavy chain gene, which leads to inappropriate and overexpression of *c-myc*. The result may be uncontrolled cell proliferation and tumor development. All subtypes of Burkitt's lymphoma contain this translocation. Epstein-Barr virus infection of B cells is also associated with certain types of Burkitt's lymphoma.



**Mannie Weitzels'** bone marrow cells contain the Philadelphia chromosome, typical of chronic myelogenous leukemia (CML). The Philadelphia chromosome results from a reciprocal translocation between the long arms of chromosome 9 and 22. As a consequence, a fusion protein is produced containing the N-terminal region of the Bcr protein from chromosome 22 and the C-terminal region of the Abl protein from chromosome 9. *Abl* is a proto-oncogene, and the resulting fusion protein (Bcr-Abl) has lost its regulatory region and is constitutively active. When active, Abl stimulates the Ras pathway of signal transduction, leading to cell proliferation.



The oncogene *N-myc* (a cell proliferation transcription factor, related to *c-myc*) is amplified in some neuroblastomas, and amplification of the *erb-B2* oncogene (a growth factor receptor) is associated with several breast carcinomas.

chemotherapeutic agents, which are designed to kill proliferating cells by interacting with DNA, may also act as carcinogens and cause new mutations and tumors while eradicating the old. Structural alterations in DNA also occur through radiation and through UV light, which causes the formation of pyrimidine dimers. More than 90% of skin cancers occur in sunlight-exposed areas. Thus, each chemical carcinogen or reactant creates a characteristic modification in a DNA base. The DNA damage, if not repaired, introduces a mutation into the next generation when the cell proliferates.

## B. Gain-of-Function Mutations in Proto-oncogenes

Proto-oncogenes are converted to oncogenes by mutations in the DNA that cause a “gain-in-function,” that is, the protein can now function better in the absence of the normal activating events. Several mechanisms that lead to the conversion of proto-oncogenes to oncogenes are known:

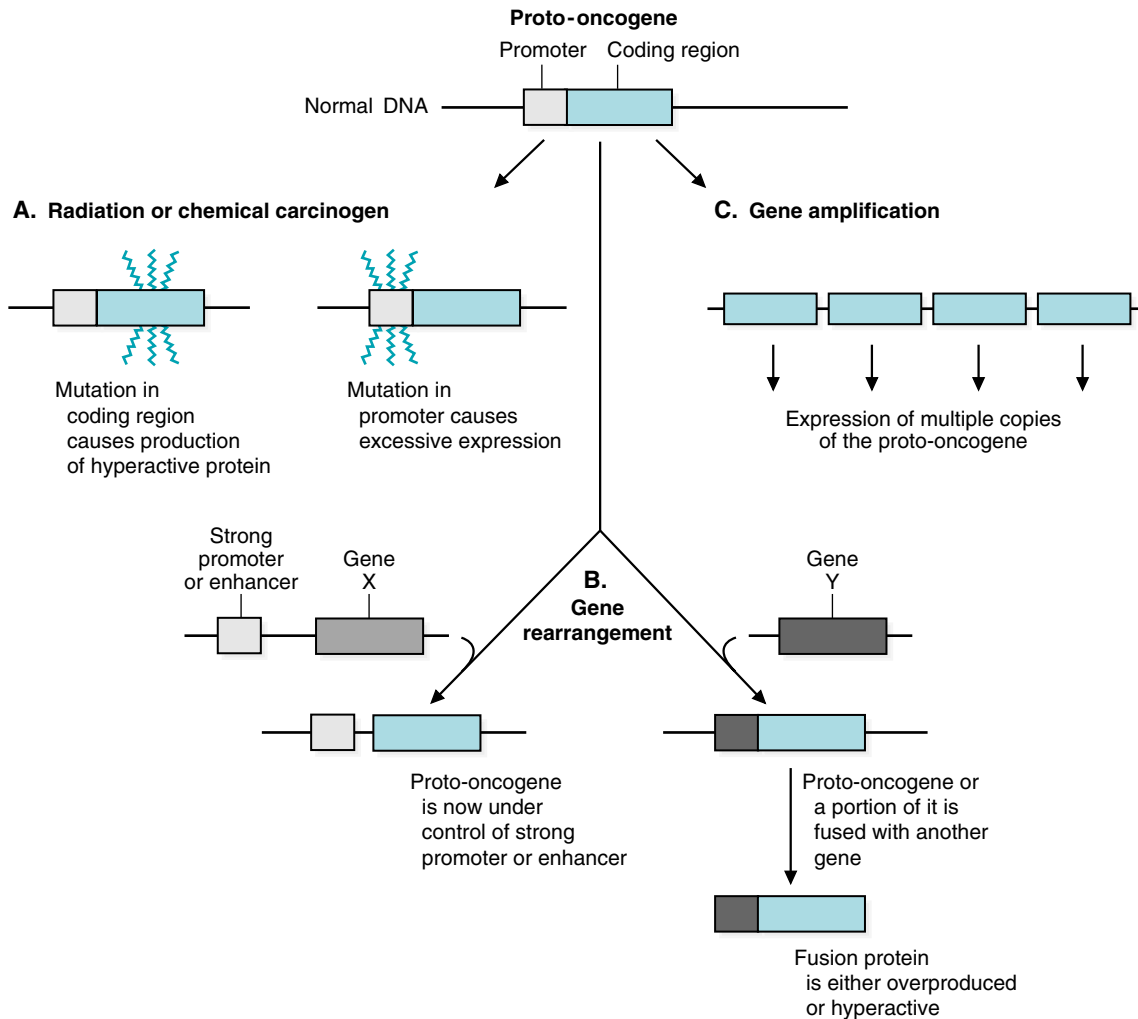
- Radiation and chemical carcinogens act (a) by causing a mutation in the regulatory region of a gene, increasing the rate of production of the proto-oncogene protein, or (b) by producing a mutation in the coding portion of the oncogene that results in the synthesis of a protein of slightly different amino acid composition capable of transforming the cell (Fig. 18.3A).
- The entire proto-oncogene or a portion of it may be transposed or translocated, that is, moved from one position in the genome to another (Fig. 18.3B). In its new location, the proto-oncogene may be controlled by a more active promoter and, therefore, overexpressed (increased amounts of the protein product may be produced). If only a portion of the proto-oncogene is translocated, it may be expressed as a truncated protein with altered properties, or it may fuse with another gene and produce a fusion protein containing portions of what normally were two separate proteins. The truncated or fusion protein would be hyperactive and cause inappropriate cell growth.
- The proto-oncogene may be amplified (Fig. 18.3C), so that multiple copies of the gene are produced in a single cell. If more genes are active, more proto-oncogene protein will be produced, increasing the growth rate of the cells.
- If an oncogenic virus infects a cell, its oncogene may integrate into the host cell genome, permitting production of the abnormal oncogene protein. The cell may be transformed and exhibit an abnormal pattern of growth. Rather than inserting an oncogene, a virus may simply insert a strong promoter into the host cell genome. This promoter may cause an increased or untimely expression of a normal proto-oncogene.

The important point to remember is that transformation results from abnormalities in the normal growth regulatory program caused by gain-of-function mutations in proto-oncogenes. However, loss-of-function mutations also must occur in the tumor suppressor genes, repair enzymes, or activators of apoptosis for full transformation to a cancer cell.

## C. Mutations in Repair Enzymes

Repair enzymes are the first line of defense preventing conversion of chemical damage in DNA to a mutation (see Chapter 13, section III.B). DNA repair enzymes are tumor suppressor genes in the sense that errors repaired before replication do not become mutagenic. DNA damage is constantly occurring from exposure to sunlight, background radiation, toxins, and replication error. If DNA repair enzymes are absent, mutations accumulate much more rapidly, and once a mutation develops in a growth regulatory gene, a cancer may arise. As an example, inherited mutations in the tumor suppressor genes *brca1* and *brca2* predispose women to the development of breast cancer.





**Fig. 18.3.** Transforming mutations in proto-oncogenes. (A) Effect of radiation or chemical carcinogens on proto-oncogenes or their promoters. The mutations may be point mutations, deletions, or insertions. (B) Gene rearrangements as caused by transposition or translocation of a proto-oncogene or proto-oncogene fragment. (C) Amplification of a proto-oncogene allows more protein to be produced. The proto-oncogene and the resulting oncogene are shown in blue.

The protein products of these genes play roles in DNA repair, recombination, and regulation of transcription. A second example, HNPCC (hereditary non-polyposis colorectal cancer), was previously introduced in Chapter 13. It results from inherited mutations in enzymes involved in the DNA mismatch repair system.

### III. ONCOGENES

Proto-oncogenes control normal cell growth and division. These genes encode proteins that are growth factors, growth factor receptors, signal transduction proteins, transcription factors, cell cycle regulators, and regulators of apoptosis (Table 18.1). (The name representing the gene of an oncogene is referred to in lowercase letters and italics [e.g., *myc*], but the name of the protein product is capitalized and italics are not used [e.g., *Myc*]). The mutations in oncogenes giving rise to transformation are usually gain-of-function mutations; either a more active protein is produced or an increased amount of the normal protein is synthesized.



Although mutations in both the *brca1* and *brca2* genes are linked to breast cancer development in women, there are some fundamental differences in how these genes interact with other proteins at the molecular level. As a result, there are some differences in the diseases expressed by mutations within these genes. For example, *brca1* mutations are also linked to ovarian cancer and *brca2* mutations are not. *Brca2* mutations have been linked to pancreatic cancer, whereas *brca1* mutations have not. Men with inherited *brca2* mutations develop breast cancer; but men who carry *brca1* mutations do not.



**Table 18.1 Classes of Oncogenes, Mechanism of Activation, and Associated Human Tumors**

| Class                                         | Proto-oncogene                 | Activation Mechanism            | Location                       | Disease                                                                      |
|-----------------------------------------------|--------------------------------|---------------------------------|--------------------------------|------------------------------------------------------------------------------|
| <b>Growth Factors</b>                         |                                |                                 |                                |                                                                              |
| Platelet-Derived Growth Factor- $\beta$ Chain | <i>sis</i>                     | Overexpression                  | Secreted                       | Astrocytoma<br>Osteosarcoma                                                  |
| Fibroblast growth factors                     | <i>int-2</i>                   | Amplification                   | Secreted                       | Breast cancer<br>Bladder cancer<br>Melanoma                                  |
| <b>Growth Factor Receptors</b>                |                                |                                 |                                |                                                                              |
| Epidermal Growth Factor Receptor Family       | <i>erb-B1</i><br><i>erb-B2</i> | Overexpression<br>Amplification | Cell membrane<br>Cell membrane | Squamous cell carcinoma of the lung<br>Breast, ovarian, lung, stomach cancer |
| Platelet-Derived Growth Factor-Receptor       | <i>PDGFR</i>                   | Translocation                   | Cell membrane                  | Chronic myelomonocytic leukemia                                              |
| Hedgehog Receptor                             | <i>SMO</i>                     | Point mutation                  | Cell membrane                  | Basal cell carcinoma                                                         |
| <b>Signal Transduction Proteins</b>           |                                |                                 |                                |                                                                              |
| G proteins                                    | <i>ras</i>                     | Point mutation                  | Cytoplasm                      | Multiple cancers including lung, colon, thyroid, pancreas, many leukemias    |
| Serine/threonine kinase                       | <i>akt2</i>                    | Amplification                   | Cytoplasm                      | Ovarian carcinoma                                                            |
|                                               | <i>raf</i>                     | Overexpression                  | Cytoplasm                      | Myeloid leukemia                                                             |
| Tyrosine kinase                               | <i>abl</i>                     | Translocation                   | Cytoplasm                      | Chronic myeloid leukemia<br>Acute lymphoblastic leukemia                     |
|                                               | <i>src</i>                     | Overexpression                  | Cytoplasm                      | Breast                                                                       |
| <b>Hormone Receptors</b>                      |                                |                                 |                                |                                                                              |
| Retinoid receptor                             | <i>RAR<math>\alpha</math></i>  | Translocation                   | Nucleus                        | Acute promyelocytic leukemia                                                 |
| <b>Transcription Factors</b>                  |                                |                                 |                                |                                                                              |
|                                               | <i>Hox11</i>                   | Translocation                   | Nucleus                        | Acute T cell leukemia                                                        |
|                                               | <i>Myc</i>                     | Translocation<br>Amplification  | Nucleus<br>nucleus             | Burkitt's lymphoma<br>Neuroblastoma, small cell carcinoma of the lung        |
|                                               | <i>fos, jun</i>                | Phosphorylation                 | Nucleus                        | Breast cancer                                                                |
| <b>Apoptosis Regulators</b>                   |                                |                                 |                                |                                                                              |
|                                               | <i>Bcl-2</i>                   | Translocation                   | Mitochondria                   | Follicular B-cell lymphoma                                                   |
| <b>Cell Cycle Regulators</b>                  |                                |                                 |                                |                                                                              |
| Cyclins                                       | Cyclin D                       | Translocation<br>Amplification  | Nucleus<br>Nucleus             | Lymphoma<br>Breast, liver, esophageal cancer                                 |
| Cyclin-dependent kinase                       | CDK4                           | Amplification<br>Point mutation | Nucleus<br>Nucleus             | Gglioblastoma, sarcoma<br>Melanoma                                           |



The gene for the human epidermal growth factor receptor (*HER2*, *c-erbB-2*) is over-expressed in 10 to 20% of breast cancer cases. When this gene is over-expressed, the prognosis for recovery is poor, as the patients display shorter disease-free intervals, increased risks for metastasis, and a resistance to therapy. Clinical trials are underway using a monoclonal antibody directed against this receptor (herceptin) to inactivate it. Preliminary results are encouraging in that use of this drug, either alone or in combination with others, appears to control the growth of some tumors overexpressing the *HER2* gene. However, not all tumors overexpressing *HER2* are responsive to herceptin. Thus, it appears that a complete genotyping of breast cancer cells may be necessary (using the microarray techniques described in Chapter 17) to specifically develop an effective therapy for each patient with the disease.

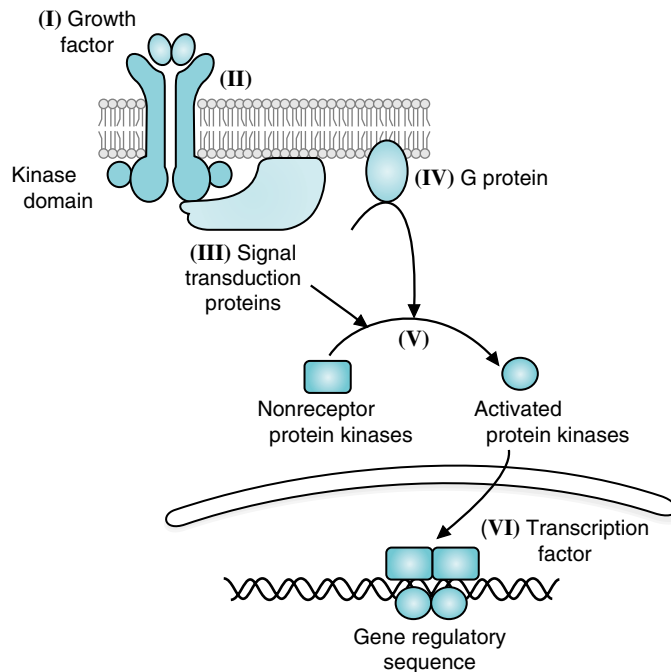
## A. Oncogenes and Signal Transduction Cascades

All of the proteins in growth factor signal transduction cascades are proto-oncogenes (Fig18.4).

### 1. GROWTH FACTORS AND GROWTH FACTOR RECEPTORS

The genes for both growth factors and growth factor receptors are oncogenes.

Growth factors generally regulate growth by serving as ligands that bind to cellular receptors located on the plasma membrane (cell-surface receptors) (see Chapter 11). Binding of ligands to these receptors stimulates a signal transduction pathway in the cell activating the transcription of certain genes. If too much of a growth factor or a growth factor receptor is produced, the target cells may respond by proliferating inappropriately. Growth factors receptors may also become oncogenic through translocation or point mutations in domains affecting binding of the growth factor, dimerization, kinase activity, or some other aspect of their signal transmission. In such cases, the receptor transmits a proliferative signal even though the growth factor normally required to activate the receptor is absent. In other words, the receptor is stuck in the “on” position.



**Fig. 18.4.** Proto-oncogene sites for transforming mutations in growth factor signaling pathways. I. The amount of growth factor. II. The receptor, which normally must bind the growth factor to dimerize and activate a kinase domain. III. Signal transduction proteins. Some, such as PI 3-kinase, form second messengers. IV. G proteins, and their regulators, which are also signal transduction proteins. V. Nonreceptor protein kinase cascades, which lead to phosphorylation of transcription factors. VI. Nuclear transcription factors normally activated through phosphorylation or binding of a ligand.

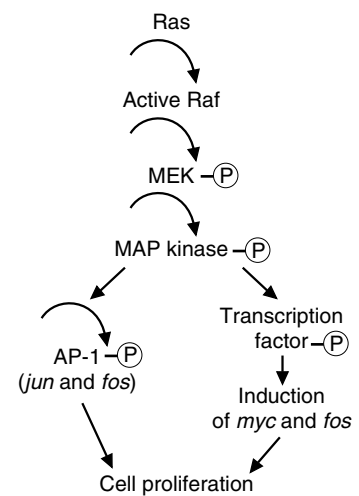
## 2. SIGNAL TRANSDUCTION PROTEINS

The genes encoding proteins involved in growth factor signal transduction cascades may also be proto-oncogenes. Consider, for example, the monomeric G protein Ras. Binding of growth factor leads to the activation of Ras (see Fig 11.11). When Ras binds GTP, it is active, but Ras slowly inactivates itself by hydrolyzing its bound GTP to GDP and Pi. This controls the length of time that Ras is active. Ras is converted to an oncogenic form by point mutations that decrease the activity of the GTPase domain of Ras, thereby increasing the length of time it remains in the active form.

Ras, when active, activates the serine/threonine kinase Raf (a MAP kinase kinase kinase), which activates MEK (a MAP kinase kinase), which activates MAP kinase (Fig. 18.5). Activation of MAP kinase results in the phosphorylation of cytoplasmic and nuclear proteins, followed by increased transcription of the transcription factor proto-oncogenes *myc* and *fos* (see below). Note that mutations in the genes for any of the proteins that regulate MAP kinase activity, as well as those proteins induced by MAP kinase activation, can lead to uncontrolled cell proliferation.

## 3. TRANSCRIPTION FACTORS

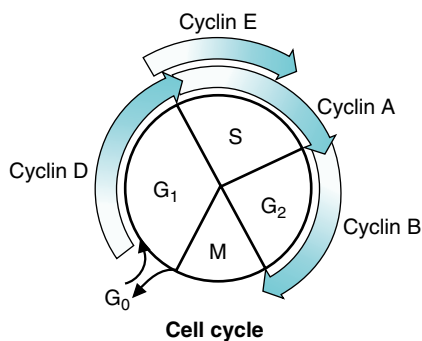
Many transcription factors, such as Myc and Fos, are proto-oncoproteins (the products of proto-oncogenes). MAP kinase, in addition to inducing *myc* and *fos*, also directly activates the AP-1 transcription factor through phosphorylation (see Fig.18.5). AP-1 is a heterodimer formed by the protein products of the *fos* and *jun* families of proto-oncogenes. The targets of AP-1 activation are genes involved in



**Fig. 18.5.** Phosphorylation cascade leading to activation of proto-oncogene transcription factors *myc*, *fos*, and *jun*.



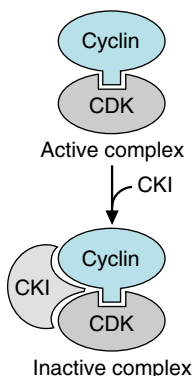
The synthesis of the transcription factor C-myc is tightly regulated in normal cells, and it is expressed only during the S phase of the cell cycle. In a large number of tumor types, this regulated expression is lost, and c-myc becomes inappropriately expressed or overexpressed throughout the cell cycle, driving cells continuously to proliferation.



**Fig. 18.6.** Cyclin synthesis during different phases of the cell cycle.



The proteins induced by E2F include cyclin E, cyclin A, cdc25A (an activating protein phosphatase), and proteins required to bind at origins of replication to initiate DNA synthesis. The synthesis of cyclin E allows it to complex with cdk2, forming another active cyclin complex that retains activity into S phase (see Fig. 18.6). Cyclin A activates Cdk2. Thus, each phase of the cell cycle activates the next through cyclin synthesis. The cyclins are removed by regulated proteolysis.



**Fig. 18.7.** CKI inhibition of cyclin/CDK activity.

cellular proliferation and progression through the cell cycle, as are the targets of the *myc* transcription factor. The net result is increased production of the proteins that carry out the processes required for proliferation.

## B. Oncogenes and the Cell Cycle

The growth of human cells, involving DNA replication and cell division in the cell cycle, is activated by growth factors, hormones, and other messengers. These activators work through cyclins and cyclin-dependent kinases (CDKs) that control progression from one phase of the cycle to another (Fig 18.6). For quiescent cells to proliferate, they must leave  $G_0$  and enter the  $G_1$  phase of the cell cycle (see Chapter 13, Fig. 13.7). If the proper sequence of events occurs during  $G_1$ , the cells enter the S phase and are committed to DNA replication and cell division. Similarly, during  $G_2$  cells make a commitment to mitotic division. CDKs are made constantly throughout the cell cycle but require binding of a specific cyclin to be active. Different cyclins made at different times in the cell cycle control each of the transitions ( $G_1/S$ ,  $S/G_2$ ,  $G_2/M$ ).

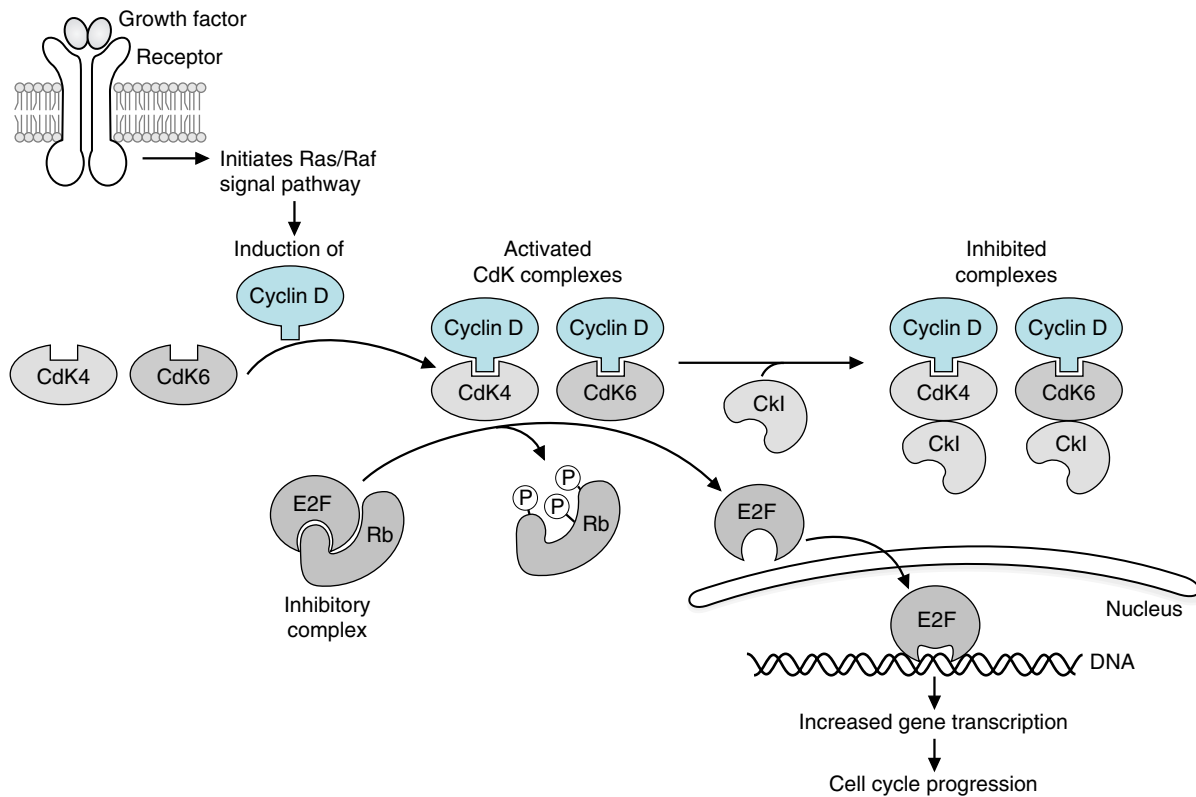
The activity of the cyclin-CDK complex is further regulated through phosphorylation and through inhibitory proteins called cyclin-dependent kinase inhibitors (CKIs) (Fig. 18.7). CKIs slow cell cycle progression by binding and inhibiting the CDK-cyclin complexes. CDKs are also controlled through activating phosphorylation by CAK (cyclin-activating kinases) and inhibitory hyperphosphorylation kinases.

To illustrate the role of these proteins, consider some of the events occurring at the  $G_1/S$  checkpoint (Fig. 18.8). Because the cell is committed to DNA replication and division once it enters the S phase, multiple regulatory proteins are involved in determining whether the cell is ready to pass this checkpoint. These regulatory proteins include cdk4 and cdk6 (which are constitutively produced throughout the cell cycle), cyclin D (whose synthesis is only induced after growth factor stimulation of a quiescent cell), the retinoblastoma gene product (Rb), and a class of transcription factors known collectively as E2F. In quiescent cells, Rb is complexed with E2F, resulting in inhibition of these transcription factors. On growth factor stimulation, the cyclin Ds are induced (there are three types of cyclin D; D1, D2, and D3). They bind to cdk4 and cdk6, converting them to active protein kinases. One of the targets of cyclin/cdk phosphorylation is the Rb protein. Phosphorylation of Rb releases it from E2F, and E2F is then free to activate the transcription of genes required for entry into S. The Rb protein is a tumor suppressor gene (more below).

Progression through the cell cycle is opposed by the CKIs (see Fig. 18.8). The CKIs regulating cyclin/cdk expression in the  $G_1$  phase of the cell cycle fall into two categories: the Cip/Kip family and the INK4 family. The Cip/Kip family members (p21, p27, and p57) have a broad specificity and inhibit all cyclin-CDK complexes. The INK4 family, which consists of p15, p16, p18, and p19, are specific for the cyclin D-cdk4/6 family of complexes (inhibitors of cyclin-dependent kinase-4). The regulation of synthesis of different CKIs is complex, but some are induced by DNA damage to the cell and halt cell cycle progression until the damage can be repaired. For example, the CKI p21 (a protein of 21,000 Daltons) is a key member of this group that responds to specific signals to block cell proliferation. If the damage cannot be repaired, an apoptotic pathway is selected, and the cell dies.



In addition to sunlight and a preexisting nevus, hereditary factors also play a role in the development of malignant melanoma. Ten percent of melanomas tend to run in families. Some of the suspected melanoma-associated genes include the tumor suppressor gene p16 (an inhibitor of cdk 4) and CDK4. **Mel Anoma** was the single child of parents who had died of a car accident in their 50s, and thus, a familial tendency could not be assessed.



**Fig. 18.8.** Control of the G1/S transition in the cell cycle. The genes encoding cyclins and CDKs are oncogenes, and the gene encoding the Retinoblastoma protein (Rb) is a tumor suppressor gene. Abbreviations: CDK, cyclin- dependent kinase; CDKI, cyclin-dependent kinase inhibitor.

#### IV. TUMOR SUPPRESSOR GENES

Like the oncogenes, the tumor suppressor genes encode molecules involved in the regulation of cell proliferation. Table 18.2 provides several examples. The normal function of tumor suppressor proteins is generally to inhibit proliferation in response to certain signals such as DNA damage. The signal is removed when the cell is fully equipped to proliferate; the effect of their elimination of tumor suppressor genes is to remove the brakes on cell growth. They affect cell cycle regulation, signal transduction, transcription, and cell adhesion. The products of tumor suppressor genes frequently modulate pathways that are activated by the products of proto-oncogenes.

**Table 18.2 Tumor Suppressors**

| Class                                        | Protein               | Location            | Associated Diseases               |
|----------------------------------------------|-----------------------|---------------------|-----------------------------------|
| Adhesion protein<br>Receptor                 | E-cadherin            | Cell membrane       | Stomach cancer                    |
|                                              | PATCHED               | Cell membrane       | Basal cell carcinoma              |
|                                              | TGF- $\beta$ receptor | Cell membrane       | Colon cancer                      |
| Signal transduction                          | NF-1                  | Under cell membrane | Neurofibrosarcoma                 |
|                                              | SMAD4/DPC             | Cytoplasm/nucleus   | Pancreatic and colorectal cancer  |
| Transcription factor<br>Cell cycle regulator | WT-1                  | Nucleus             | Wilms tumor                       |
|                                              | p16(INK4)             | Nucleus             | Melanoma, lung, pancreatic cancer |
|                                              | Retinoblastoma        | Nucleus             | Retinoblastoma, sarcomas          |
| Cell cycle/apoptosis<br>DNA repair           | p53                   | Nucleus             | Most cancers                      |
|                                              | BRACA 1               | Nucleus             | Breast cancer                     |

Tumor suppressor genes contribute to the development of cancer when both copies of the gene are inactivated. This is different from the case of proto-oncogene mutations because only one allele of a proto-oncogene needs to be converted to an oncogene to initiate transformation.

### A. Tumor Suppressor Genes That Directly Regulate the Cell Cycle

The two best understood cell cycle regulators that are also tumor suppressors are the retinoblastoma (*rb*) and *p53* genes.

#### 1. THE RETINOBLASTOMA (*rb*) GENE

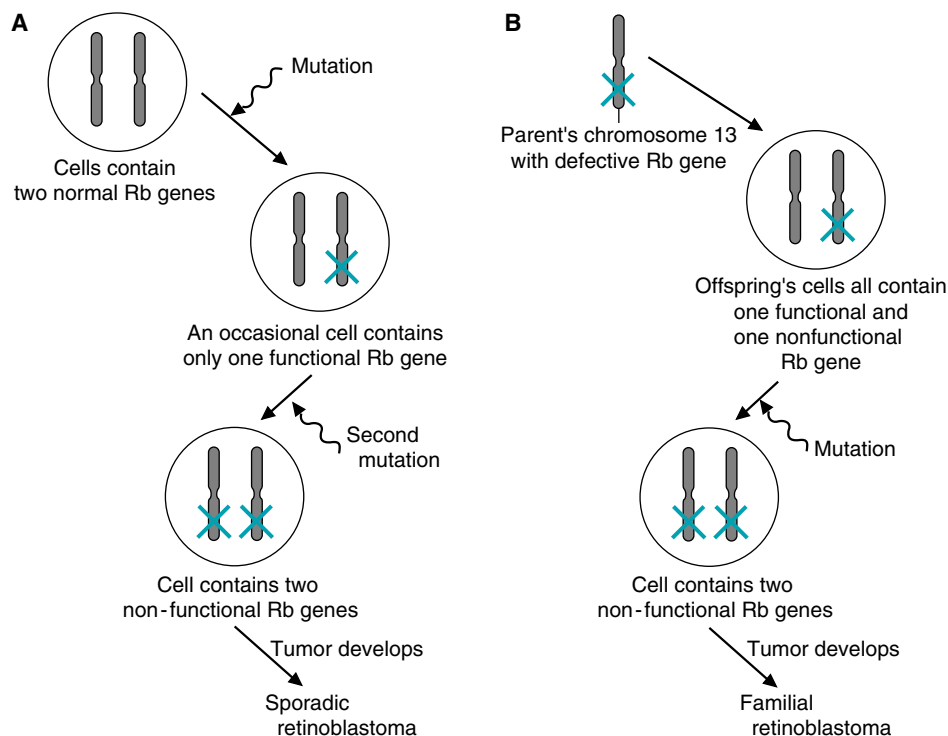
As previously discussed, the retinoblastoma gene product, Rb, functions in the transition from G1 to S phase and regulates the activation of members of the E2F family of transcription factors (see Fig. 18.8). If an individual inherits a mutated copy of the *rb* allele, there is a 100% chance of that individual developing retinoblastoma, because of high probability that the second allele of *rb* will gain a mutation (Fig. 18.9). This is considered familial retinoblastoma. Individuals who do not inherit mutations in *rb*, but who develop retinoblastoma, are said to have sporadic retinoblastoma, and acquire two specific mutations, one in each *rb* allele of the retinoblast, during their lifetime.

#### 2. *p53*, THE GUARDIAN OF THE GENOME

The *p53* protein is a transcription factor regulating the cell cycle and apoptosis, programmed cell death. Loss of both *p53* alleles is found in over more than 50% of human tumors. *p53* acts as the “guardian of the genome” by halting replication in cells that have suffered DNA damage and targeting unrepaired cells to apoptosis.

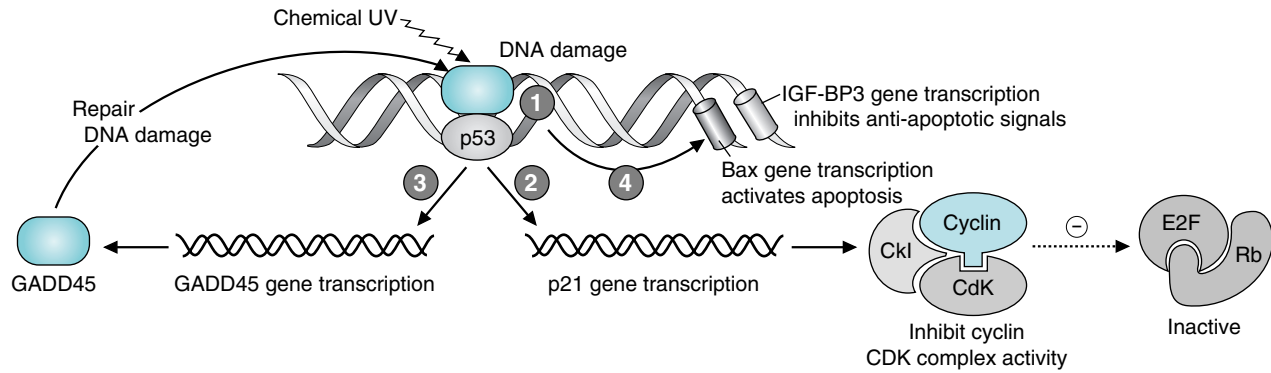


Inheritance of a mutation in *p53* leads to Li-Fraumeni syndrome, which is characterized by multiple types of tumors. Mutations in *p53* are present in more than 50% of human tumors. These are secondary mutations within the cell, and if *p53* is mutated the overall rate of cellular mutation will increase because there is no *p53* to check for DNA damage, to initiate the repair of the damaged DNA, or to initiate apoptosis if the damage is not repaired. Thus, damaged DNA is replicated, and the frequency of additional mutations within the same cell increases remarkably.



**Fig. 18.9.** Mutations in the retinoblastoma (*Rb*) gene. A. Sporadic retinoblastoma. B. Familial retinoblastoma.





**Fig. 18.10.** p53 and cell cycle arrest. Mechanisms that recognize DNA damage stop p53 degradation and modify the p53 protein (circle 1). p53 stimulates the transcription of p21 (circle 2) and GADD45 (circle 3). p21 blocks the cyclin/CDK phosphorylation of Rb, which continues to inhibit the E2F family of transcription factors, thereby blocking cell progression through the cell cycle. GADD45 allows the DNA damage to be repaired. If the damage is not repaired, apoptotic genes are activated (circle 4).

In response to DNA-damaging mutagens, ionizing radiation, or ultraviolet light, the level of p53 rises (Fig. 18.10, circle 1). p53, acting as a transcription factor, stimulates transcription of *p21* (a member of the Cip/Kip family of CKIs), as shown in Figure 18.10, circle 2. The p21 gene product inhibits the cyclin/CDK complexes, which prevents the phosphorylation of Rb and release of E2F proteins. The cell is thus prevented from entering S phase. P53 also stimulates the transcription of GADD45 (Growth Arrest and DNA Damage), a DNA repair enzyme (Fig. 18.10, circle 3). If the DNA is successfully repaired, p53 induces its own downregulation. If the DNA repair was not successful, p53 activates two apoptosis genes, *bax* (discussed below) and *IGF-BP3* (Fig. 18.10, circle 4). The IGF-BP3 protein product binds the receptor for insulin-like growth factor, which presumably induces apoptosis by blocking the anti-apoptotic signaling by growth factors, and the cell enters a growth factor deprivation mode.

## B. Tumor Suppressor Genes That Affect Receptors and Signal Transduction

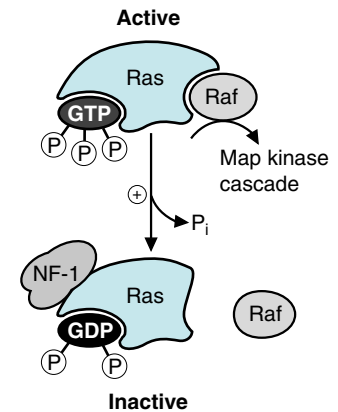
Tumor suppressor genes may encode receptors, components of the signaling transduction pathway, or transcription factors.

### 1. REGULATORS OF RAS

The Ras family of proteins is involved in signal transduction for many hormones and growth factors (see above), and is, therefore, oncogenic. The activity of these pathways is interrupted by GAPs (GTPase-activating proteins; see Chapter 9), which vary among cell types. Neurofibromin, the product of the tumor suppressor gene *NF-1*, is a nervous system-specific GAP that regulates the activity of Ras in neuronal tissues (Fig. 18.11). The growth signal is transmitted so long as the Ras protein binds GTP. Binding of NF-1 to Ras activates the GTPase domain of Ras, which hydrolyzes GTP to GDP, thereby inactivating Ras. Without a functional neurofibromin molecule, Ras is perpetually active.

### 2. PATCHED AND SMOOTHENED

A good example of tumor suppressors and oncogenes working together is provided by the co-receptor genes *patched* and *smoothened*, which encode the receptor for the hedgehog class of signaling peptides. These co-receptors normally function to control growth during embryogenesis and illustrate the importance of maintaining



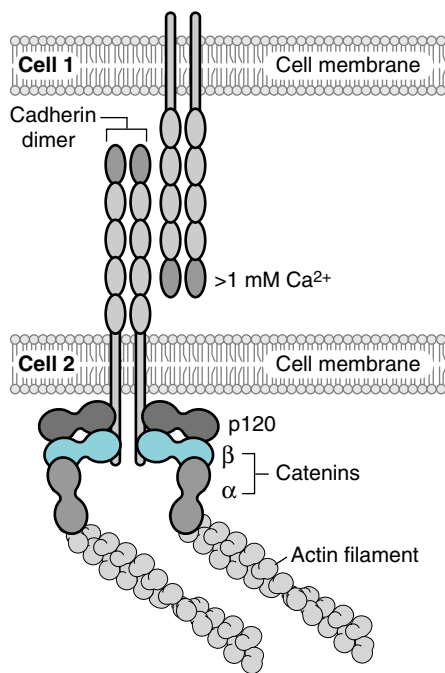
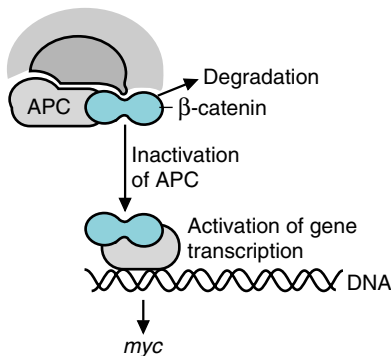
**Fig. 18.11.** NF-1 (neurofibromin) binds to Ras and activates its GTPase activity, thereby decreasing cell proliferation. A mutation in neurofibromin leads to prolonged GTP binding and activation of Ras.



An inherited mutation in *NF-1* can lead to neurofibromatosis, a disease primarily of numerous benign, but painful, tumors of the nervous system. The movie *Elephant Man* was based on an individual who was believed to have this disease. Recent analysis of the patient's remains, however, indicates that he may have suffered from the rare Proteus syndrome, and not neurofibromatosis.



The strange names of some of the tumor suppressor genes arose because they were first discovered in *Drosophila* (fruit fly), and the names of *Drosophila* mutations are often based on the appearance of the fly expressing the mutation. Once the human homolog is found, it is given the same name as the *Drosophila* gene.

**A. Catenins and cadherins in cell attachment****B. β-catenin and APC in gene transcription**

**Fig. 18.12.** A. Catenins and cadherins. E-cadherin molecules form intercellular, calcium-dependent homodimers with cadherins from another cell, resulting in cell–cell adhesion. The cytoplasmic portion of E-cadherin is complexed to a variety of catenins, which anchor the cadherin to the actin cytoskeleton. B. β-Catenin and APC in transcription. The APC complex activates β-catenin for proteolytic degradation. If APC is inactivated, β-catenin levels increase. It acts as a transcription factor that increases synthesis of *myc* and other genes regulating cell cycle progression.

a balance between oncogenes and tumor suppressor genes. The Patched receptor protein inhibits Smoothed, its co-receptor protein. Binding of a hedgehog ligand to Patched releases the inhibition of Smoothed, which then transmits an activating signal to the nucleus, stimulating new gene transcription. *Smoothed* is a proto-oncogene, and *patched* is a tumor suppressor gene. If *patched* loses its function (definition of a tumor suppressor), then Smoothed can signal the cell to proliferate, even in the absence of a hedgehog signal. Conversely, if *smoothed* undergoes a gain of function mutation (definition of an oncogene), it can signal in the absence of the hedgehog signal, even in the presence of Patched. Inherited mutations in either *smoothed* or *patched* will lead to an increased incidence of basal cell carcinoma.

**C. Tumor Suppressor Genes That Affect Cell Adhesion**

The cadherin family of glycoproteins mediates calcium-dependent cell–cell adhesion. Cadherins form intercellular complexes binding cells together (Fig. 18.12A). They are anchored intracellularly by catenins, which bind to actin filaments. Loss of E-cadherin expression may contribute to the ability of cancer cells to detach and migrate in metastasis. Individuals who inherit a mutation in E cadherin (this mutation is designated as *CDH1*) are sharply predisposed to developing diffuse type gastric cancer.

The catenin proteins have two functions; in addition to anchoring cadherins to the cytoskeleton, they act as transcription factors (Fig. 18.12B). β-Catenin also binds to a complex containing the regulatory protein APC (adenomatous polyposis coli), which activates it for degradation. When the appropriate signal activates APC, β-catenin levels increase, and it travels to the nucleus, where it activates *myc* and *cyclin D1* transcription, leading to cell proliferation. APC is a tumor suppressor gene. If it is inactivated, it cannot bind β-catenin and inhibit cell proliferation. Mutations in APC or proteins interacting with it are found in the vast majority of sporadic human colon cancer. Inherited mutations in APC lead to the most common form of hereditary colon cancer, familial adenomatous polyposis.

**V. CANCER AND APOPTOSIS**

In the body, superfluous or unwanted cells are destroyed by a pathway called apoptosis, or programmed cell death. Apoptosis is a regulated energy-dependent sequence of events by which a cell self-destructs. In this suicidal process, the cell shrinks, the chromatin condenses, and the nucleus fragments. The cell membrane forms blebs (outpouches), and the cell breaks up into membrane-enclosed apoptotic vesicles (apoptotic bodies) containing varying amounts of cytoplasm, organelles, and DNA fragments. Phosphatidylserine, a lipid on the inner leaflet of the cell membrane, is exposed on the external surface of these apoptotic vesicles. It is one of the phagocytic markers recognized by macrophages and other nearby phagocytic cells that engulf the apoptotic bodies.

Apoptosis is a normal part of multiple processes in complex organisms: embryogenesis, the maintenance of proper cell number in tissues, the removal of infected or otherwise injured cells, the maintenance of the immune system, and aging. It can be initiated by injury, radiation, free radicals or other toxins, withdrawal of growth



A form of apoptosis is a normal part of embryonic development. For example, the development of the nervous system uses apoptosis to destroy neurons that have not made the proper connections with target cells. Neurons are produced in excess, and more than 50% of developing neurons are eliminated by programmed cell death. Those neurons that have made the correct connections survive by secreting growth factors that block apoptosis.

factors or hormones, binding of pro-apoptotic cytokines, or interactions with cytotoxic T cells in the immune system. Apoptosis can protect organisms from the negative effect of mutations by destroying cells with irreparably damaged DNA before they proliferate. Just as an excess of a growth signal can produce an excess of unwanted cells, the failure of apoptosis to remove excess or damaged cells can contribute to the development of cancer.

## A. Normal Pathways to Apoptosis

Apoptosis can be divided into three general phases: an initiation phase, a signal integration phase, and an execution phase. Apoptosis can be initiated by external signals that work through death receptors, such as tumor necrosis factor (TNF), or deprivation of growth hormones (Fig. 18.13). It can also be initiated by intracellular events that affect mitochondrial integrity (e.g., oxygen deprivation, radiation), and irreparably damaged DNA. In the signal integration phase, these pro-apoptotic signals are balanced against anti-apoptotic cell survival signals by several pathways, including members of the Bcl-2 family of proteins. The execution phase is carried out by proteolytic enzymes called caspases.

### 1. CASPASES

Caspases are cysteine proteases that cleave peptide bonds next to an aspartate residue. They are present in the cell as procaspases, zymogen-type enzyme precursors activated by proteolytic cleavage of the inhibitory portion of their polypeptide chain. The different caspases are generally divided into two groups according to their function: initiator caspases, which specifically cleave other procaspases, and execution caspases, which cleave other cellular proteins involved in maintaining cellular integrity (see Fig. 18.13). The initiator caspases are activated through two major signaling pathways; the death receptor pathway and the mitochondrial integrity pathway. They activate the execution caspases, which cleave protein kinases involved in cell adhesion, lamins that form the inner lining of the nuclear envelope, actin and other proteins required for cell structure, and DNA repair enzymes. They also cleave an inhibitor protein of the endonuclease CAD (caspase-activated DNase). With destruction of the nuclear envelope, additional endonucleases ( $\text{Ca}^{2+}$ - and  $\text{Mg}^{2+}$ -dependent) also become activated.

### 2. THE DEATH RECEPTOR PATHWAY TO APOPTOSIS

The death receptors are a subset of TNF-1 receptors, which includes Fas/CD95, TNF-Receptor 1 (TNF-R1) and Death Receptor 3 (DR3). These receptors form a trimer that binds TNF-1 or another death ligand on its external domain and adaptor proteins to its intracellular domain (Fig. 18.14). The activated TNF-receptor complex forms the scaffold for binding two molecules of procaspase 8, which autocatalytically cleave each other to form active caspase 8. Caspase 8 is an initiator caspase that activates execution caspases 3, 6, and 7. Caspase 8 also cleaves a Bcl-2 protein, Bid, to a form that activates the mitochondrial integrity pathway to apoptosis.

### 3. THE MITOCHONDRIAL INTEGRITY PATHWAY TO APOPTOSIS

Apoptosis is also induced by intracellular signals indicating that cell death should occur. Examples of these signals include growth factor withdrawal, cell injury, the release of certain steroids, and an inability to maintain low levels of intracellular calcium. All of these treatments, or changes, lead to release of cytochrome c from the mitochondria (Fig. 18.15). Cytochrome c is a necessary protein component of the mitochondrial electron transport chain that is loosely bound to the outside of the inner mitochondrial membrane. Its release initiates apoptosis.

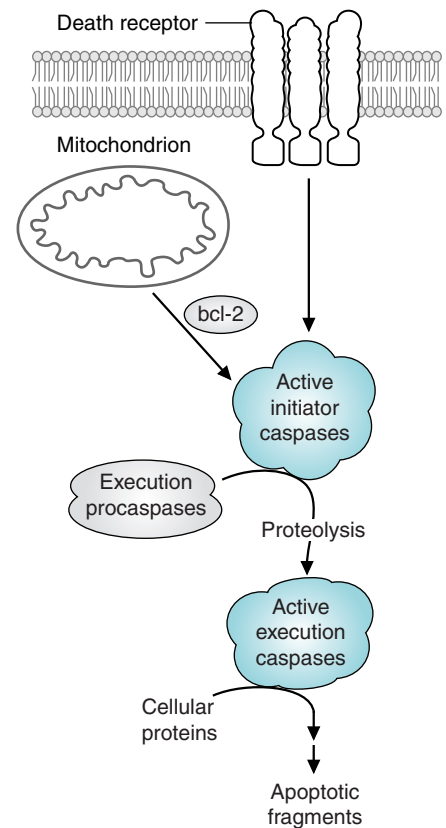
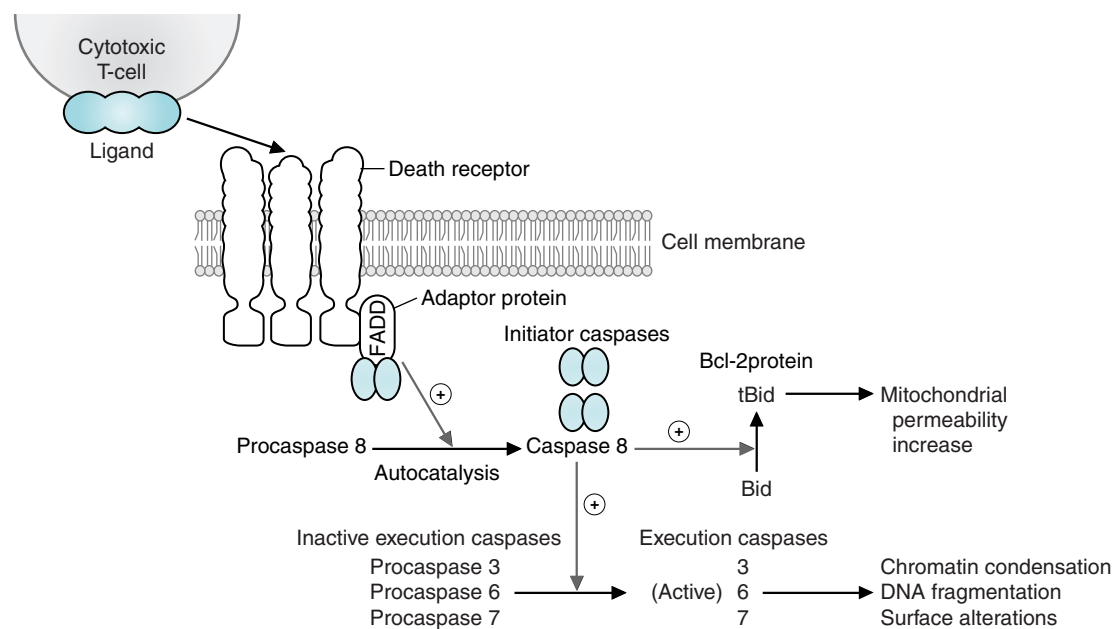


Fig. 18.13. Major components in apoptosis.



**Fig. 18.14.** The death receptor pathway to apoptosis. The ligand (usually a cell surface protein on another cell) binds to the death receptor, which makes a scaffold for autocatalytic activation of caspase 8. Active caspase 8 directly cleaves apoptotic execution caspases. However, the pathway also activates Bid, which acts on mitochondrial membrane integrity.

In the cytosol, cytochrome c binds Apaf (pro-apoptotic protease activating factor). The Apaf/cytochrome c complex binds caspase 9, an initiator caspase, to form an active complex called the apoptosome. The apoptosome in turn activates execution caspases by zymogen cleavage.

4. INTEGRATION OF PRO- AND ANTI-APOPTOTIC SIGNALS BY THE BCL-2 FAMILY OF PROTEINS

**Table 18.3 Bcl-2 Family Members**

|                        |
|------------------------|
| <i>Anti-apoptotic</i>  |
| Bcl-2                  |
| Bcl-x                  |
| Bcl-w                  |
| <i>Proapoptotic</i>    |
| <i>Channel Forming</i> |
| Bax                    |
| Bak                    |
| Bok                    |
| <i>Pro-apoptotic</i>   |
| <i>BH3-Only</i>        |
| Bad                    |
| Bid                    |
| Bod/Bim                |

Roughly 30 Bcl-2 family members are currently known. These proteins play tissue-specific as well as signal pathway-specific roles in regulating apoptosis. The tissue-specificity is overlapping. For example, Bcl-2 is expressed in hair follicles, kidney, small intestines, neurons, and the lymphoid system, whereas Bcl-x is expressed in the nervous system and hematopoietic cells.

The Bcl-2 family members are decision-makers that integrate prodeath and anti-death signals to determine whether the cell should commit suicide. Both pro-apoptotic and anti-apoptotic members of the *Bcl-2* family exist (Table 18.3).

The antiapoptotic *Bcl-2* –type proteins (including Bcl-2, Bcl-xL, Bcl-wL) have at least two ways of antagonizing death signals. They insert into the outer mitochondrial membrane to antagonize channel-forming pro-apoptotic factors, thereby decreasing cytochrome c release. They may also bind cytoplasmic Apaf so that it cannot form the apoptosome complex (Fig. 18.16).

These anti-apoptotic Bcl-2 proteins are opposed by pro-apoptotic family members that fall into two categories: ion-channel forming members and the BH3-only members. The pro-death ion channel forming members, such as Bax, are very similar to the anti-apoptotic family members, except that they do not contain the binding domain for Apaf. They have the other structural domains, however, and when they dimerize with



When *Bcl-2* is mutated, and oncogenic, it is usually overexpressed, for example, in follicular lymphoma and CML (chronic myelogenous leukemia). Overexpression of *Bcl-2* disrupts the normal regulation of pro and anti-apoptotic factors and tips the balance to an anti-apoptotic stand. This leads to an inability to destroy cells with damaged DNA, such that mutations can accumulate within the cell. Bcl-2 is also a multi-drug resistance protein and if over-expressed will block the induction of apoptosis by antitumor agents by rapidly removing them from the cell. Thus, strategies are being developed to reduce Bcl-2 levels in tumors over-expressing it before initiating drug or radiation treatment.

pro-apoptotic BH3-only members in the outer mitochondrial membrane, they form an ion channel that promotes cytochrome c release rather than inhibiting it (see Fig. 18.16). The pro-death BH3-only proteins (e.g., Bim and Bid) contain only the structural domain that allows them to bind to other bcl-2 family members (the BH3 domain), and not the domains for binding to the membrane or forming ion channels. Their binding activates the pro-death family members and inactivates the anti-apoptotic members. When the cell receives a signal from a pro-death agonist, a BH3 protein like Bid is activated (see Fig. 18.16). The BH3 protein activates Bax (an ion-channel forming pro-apoptotic channel member), which stimulates release of cytochrome c. Normally Bcl-2 acts as a death antagonist by binding Apaf and keeping it in an inactive state. However, at the same time that Bid is activating Bax, Bid also binds to Bcl-2, thereby disrupting the Bcl-2/Apaf complex and freeing Apaf to bind to released cytochrome c to form the apoptosome.

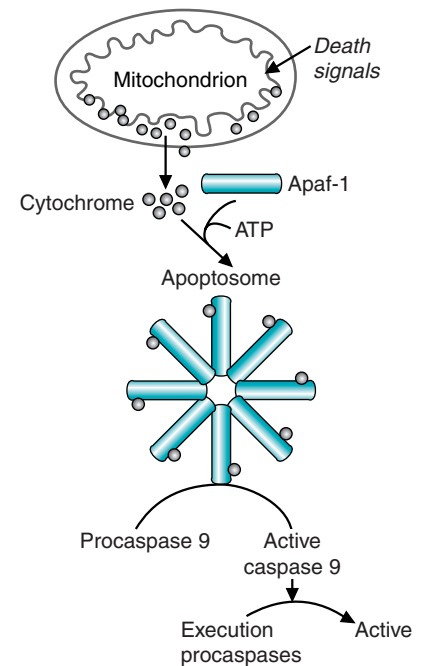
## B. Cancer Cells Bypass Apoptosis

Apoptosis should be triggered by a number of stimuli, such as withdrawal of growth factors, elevation of p53 in response to DNA damage, monitoring of DNA damage by repair enzymes, or by release of TNF or other immune factors. However, mutations in oncogenes can create apoptosis-resistant cells.

One of the ways this occurs is through activation of growth factor-dependent signaling pathways that inhibit apoptosis, such as the PDGF/Akt/BAD pathway. Nonphosphorylated BAD acts like Bid in promoting apoptosis (see Fig. 18.16). Binding of the platelet-derived growth factor to its receptor activates PI-3 kinase, which phosphorylates and activates the serine/threonine kinase Akt (protein kinase B, see Chapter 11, section III.B.3). Activation of Akt results in the phosphorylation of the pro-apoptotic BH3-only protein BAD, which inactivates it. The PDGF/Akt/BAD pathway illustrates the requirement of normal cells for growth factor stimulation to prevent cell death. One of the features of neoplastic transformation is the loss of growth factor dependence for survival.



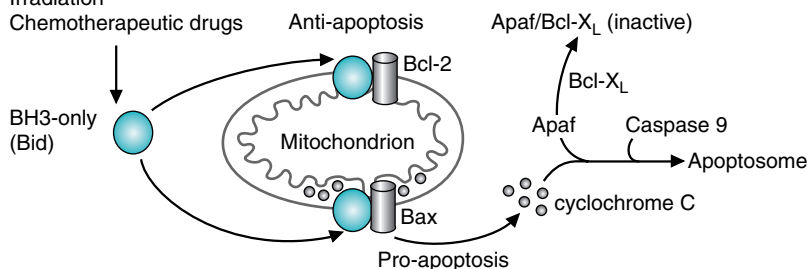
The MAP kinase pathway is also involved in regulating apoptosis and sends cell survival signals. MAP kinase kinase phosphorylates and activates another protein kinase known as RSK. Like Akt, RSK phosphorylates BAD and inhibits its activity. Thus, BAD acts as a site of convergence for the PI-3 kinase/Akt and MAP kinase pathways in signaling cell survival. Gain-of-function mutations in the genes controlling these pathways, such as *ras*, creates apoptosis-resistant cells.



**Fig. 18.15.** The mitochondrial integrity pathway releases cytochrome c, which binds to Apaf and forms a multimeric complex called the apoptosome. The apoptosome converts procaspase 9 to active caspase, which it releases into the cytosol.

### Stimulus

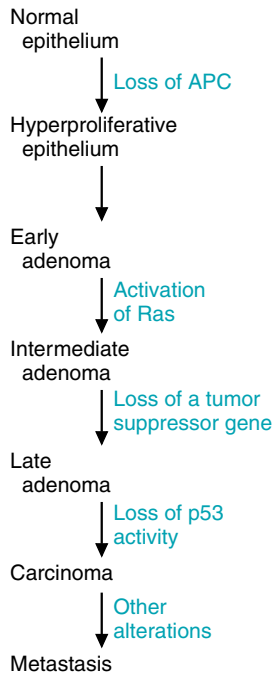
Growth factor deprivation  
Steroids  
Irradiation  
Chemotherapeutic drugs



**Fig. 18.16.** Roles of the Bcl-2 family members in regulating apoptosis. Bcl-2, which is anti-apoptotic, binds Bid (or tBid) and blocks formation of channels that allow cytochrome c release from the mitochondria. Death signals result in activation of a BH3-only protein such as Bid, which can lead to mitochondrial pore formation, swelling, and release of cytochrome c. Bid binds to and activates the membrane ion-channel protein Bax, activating cytochrome c release, which binds to Apaf and leads to formation of the apoptosome.



| Cell type | Gene alteration |
|-----------|-----------------|
|-----------|-----------------|



**Fig. 18.17.** Possible steps in the development of colon cancer. The changes do not always occur in this order, but the most benign tumors have the lowest frequency of mutations, and the most malignant have the highest frequency.

## VI. CANCER REQUIRES MULTIPLE MUTATIONS

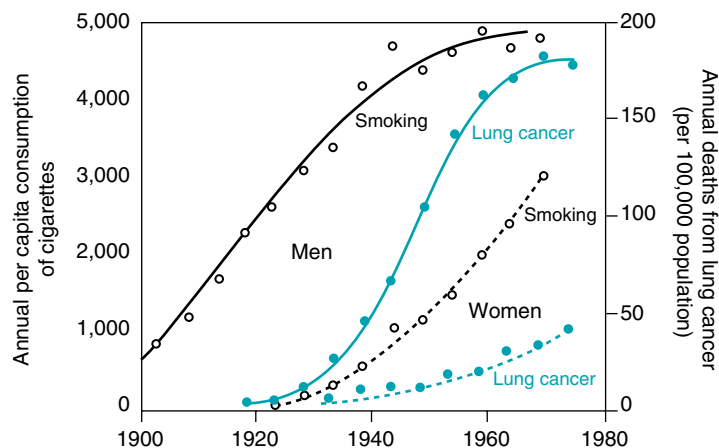
Cancer takes a long time to develop in humans because multiple genetic alterations are required to transform normal cells into malignant cells (see Fig. 18.1). A single change in one oncogene or tumor suppressor gene in an individual cell is not adequate for transformation. For example, if cells derived from biopsy specimens of normal cells are not already “immortalized,” that is, able to grow in culture indefinitely, addition of the *ras* oncogene to the cells is not sufficient for transformation. However, additional mutations in a combination of oncogenes, for example *ras* and *myc*, can result in transformation (Fig. 18.17). Epidemiologists have estimated that four to seven mutations are required for normal cells to be transformed.

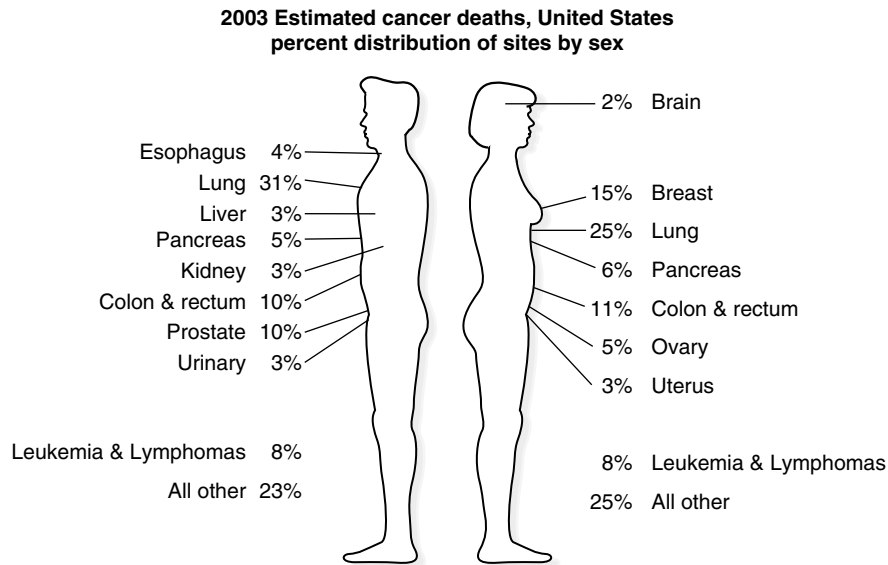
Cells accumulate multiple mutations through clonal expansion. When DNA damage occurs in a normally proliferative cell, a population of cells with that mutation is produced. Expansion of the mutated population enormously increases the probability of a second mutation in a cell containing the first mutation. After one or more mutations in proto-oncogenes or tumor suppressor genes, a cell may proliferate more rapidly in the presence of growth stimuli and with further mutations grow autonomously, that is, independent of normal growth controls. Enhanced growth increases the probability of further mutations. Some families have a strong predisposition to cancer. Individuals in these families have inherited a mutation or deletion of one allele of a tumor suppressor gene, and as progeny of that cell proliferate, mutations can occur in the second allele, leading to a loss of control of cellular proliferation. These familial cancers include familial retinoblastoma, familial adenomatous polyps of the colon, and multiple endocrine neoplasia, one form of which involves tumors of the thyroid, parathyroid, and adrenal medulla (MEN type II).

Studies of benign and malignant polyps of the colon show that these tumors have a number of different genetic abnormalities. The incidence of these mutations increases with the level of malignancy. In the early stages, normal cells of the intestinal epithelium proliferate, develop mutations in the APC gene, and polyps develop. This change is associated with a mutation in the *ras* proto-oncogene that converts it to an active oncogene. Progression to the next stage is associated with a deletion or alteration of a tumor suppressor gene on chromosome 5. Subsequently, mutations occur in chromosome 18, inactivating a gene that may be involved in cell adhesion, and in chromosome 17, inactivating the *p53* tumor suppressor gene. The



**Nick O'Tyne** had been smoking for 40 years before he developed lung cancer. The fact that cancer takes so long to develop has made it difficult to prove that the carcinogens in cigarette smoke cause lung cancer. Studies in England and Wales show that cigarette consumption by men began to increase in the early 1900s. Followed by a 20-year lag, the incidence in lung cancer in men also began to rise. Women began smoking later, in the 1920s. Again the incidence of lung cancer began to increase after a 20-year lag.





**Fig. 18.18.** Estimated cancer deaths by site and sex. Data from The American Cancer Society, Inc: Cancer Facts and Figures, 2003.

cells become malignant, and further mutations result in growth that is more aggressive and metastatic. This sequence of mutations is not always followed precisely, but an accumulation of mutations in these genes is found in a large percentage of colon carcinomas.

## VII. AT THE MOLECULAR LEVEL, CANCER IS MANY DIFFERENT DISEASES

More than 20% of the deaths in the United States each year are caused by cancer, with tumors of the lung, large intestine, and the breast being the most common (Fig. 18.18). Different cell types typically use different mechanisms through which they lose the ability to control their own growth. An examination of the genes involved in the development of cancer shows that a particular type of cancer can arise in multiple ways. For example, *Patched* and *Smoothened* are the receptor and co-receptor for the signaling peptide, sonic hedgehog. Either mutation of *smoothened*, an oncogene, or inactivation of *patched*, a tumor suppressor gene, can give rise to basal cell carcinoma. Similarly, transforming growth factor  $\beta$  and its signal transduction proteins SMAD4/DPC are part of the same growth-inhibiting pathway, and either may be absent in colon cancer. Thus, treatments which are successful for one patient with colon cancer may not be successful in a second patient with colon cancer because of the differences in the molecular basis of each individual's disease (this now also appears to be the case with breast cancer as well). Medical practice in the future will require identifying the molecular lesions involved in a particular disease and developing appropriate treatments accordingly. The use of gene chip technology (see Chapter 17) to genotype tumor tissues will aid greatly in allowing patient specific treatments to be developed.



A new treatment for CML based on rational drug design was recently introduced. The fusion protein Bcr-Abl is found only in the transformed cells expressing the Philadelphia chromosome and not in normal cells. Once the structure of Bcr-Abl was determined, the drug Gleevec was designed to specifically bind to and inhibit only the active site of the fusion protein and not the normal protein. Gleevec was successful in blocking Bcr-Abl function, thereby stopping cell proliferation, and in some cells would induce apoptosis, so the cells would die. Because normal cells do not express the hybrid protein, they were not affected by the drug. The problem with this treatment is that some patients suffered relapses, and when their Bcr-Abl proteins were studied it was found in some patients that the fusion protein had a single amino acid substitution near the active site that prevented Gleevec from binding to the protein. Other patients had an amplification of the Bcr-Abl gene product. Thus, Gleevec is a promising first step in designing drugs specifically targeted to tumor cells and is leading the way for rational drug design in the treatment of cancer.

## CLINICAL COMMENTS



**Mannie Weitzels.** The treatment of a symptomatic patient with CML whose white blood cell count is in excess of 50,000 cells/mL is usually initiated with busulfan. Alkylating agents such as cyclophos-



The TNM system standardizes the classification of tumors. The T stands for the stage of tumor (the higher the number, the worse the prognosis), the N stands for the number of lymph nodes that are affected by the tumor (again, the higher the number, the worse the prognosis), and M stands for the presence of metastasis (0 for none, 1 for the presence of metastatic cells).



Mutations associated with malignant melanomas include *ras* (gain-of-function in growth signal transduction oncogene), *p53* (loss of function of tumor suppressor gene), *p16* (loss of function in Cdk inhibitor tumor suppressor gene), *Cdk4* (gain of function in a cell cycle progression oncogene) and *cadherin/β-catenin* regulation (loss of regulation that requires attachment).

phamide have been used alone or in combination with busulfan. Purine and pyrimidine antagonists and hydroxyurea (an inhibitor of the enzyme ribonucleotide reductase, which converts ribonucleotides to deoxyribonucleotides for DNA synthesis) are sometimes effective in CML as well. In addition, trials with both  $\gamma$ - and  $\beta$ -interferon have shown promise in increasing survival in these patients. Interestingly, the latter agents have been associated with the disappearance of the Philadelphia chromosome in dividing marrow cells of some patients treated in this way.



**Nick O' Tyne.** Surgical resection of the primary lung cancer with an attempt at cure was justified in Nick O'Tyne, who had a good prognosis with a  $T_1, N_1, M_0$  staging classification preoperatively. Without some evidence of spread to the central nervous system at that time, a preoperative CT scan of the brain would not have been justified. This conservative approach would require scanning of all of the potential sites for metastatic disease from a non-small cell cancer of the lung in all patients who present in this way. In an era of runaway costs of health care delivery, such an approach could not be considered cost-effective.

Unfortunately, Mr. O'Tyne developed a metastatic lesion in the right temporal cortex of his brain. Because metastases were almost certainly present in other organs, Mr. O'Tyne's brain tumor was not treated surgically. In spite of palliative radiation therapy to the brain, Mr. O'Tyne succumbed to his disease just 9 months after its discovery, an unusually virulent course for this malignancy. On postmortem examination, it was found that his body was riddled with metastatic disease.



**Colin Tuma.** Colin Tuma requires yearly colonoscopies to check for new polyps in his intestinal tract. Because the development of a metastatic adenoma requires a number of years (because of the large numbers of mutations that must occur), yearly checks will enable new polyps to be identified and removed before malignant tumors develop.



**Mel Anoma.** The biopsy of Mel Anoma's excised mole showed that it was not malignant. The most important clinical sign of a malignant melanoma is a change in color in a pigmented lesion. Unlike benign (nondysplastic) nevi, melanomas exhibit striking variations in pigmentation, appearing in shades of black, brown, red, dark blue, and gray. Additional clinical warning signs of a melanoma are: enlargement of a preexisting mole, itching or pain in a preexisting mole, development of a new pigmented lesion during adult life, and irregularity of the borders of a pigmented lesion. **Mel Anoma** was advised to conduct a monthly self-examination, to have a clinical skin examination once or twice yearly, to avoid sunlight, and to use appropriate sunscreens.

## BIOCHEMICAL COMMENTS



**Viruses and human cancer.** Three RNA retroviruses are associated with the development of cancer in humans: HTLV-1, HIV, and hepatitis C. There are also DNA viruses associated with cancer.



**HTLV-1.** HTLV-1 causes adult T-cell leukemia. The HTLV-1 genome encodes a protein Tax, which is a transcriptional coactivator. The cellular proto-oncogenes *c-sis* and *c-fos* are activated by Tax, thereby altering the normal controls on cellular proliferation and leading to malignancy. Thus, *tax* is a viral oncogene without a counterpart in the host cell genome.



**HIV.** Infection with HIV, the virus causing acquired immunodeficiency disease (AIDS), leads to the development of neoplastic disease through several mechanisms. HIV infection leads to immunosuppression and, consequently, loss of immune-mediated tumor surveillance. HIV-infected individuals are predisposed to non-Hodgkins lymphoma, which results from an overproduction of T cell lymphocytes. The HIV genome encodes a protein, Tat, a transcription factor that activates transcription of the *interleukin-6* and *interleukin-10* genes in infected T cells. IL-6 and IL-10 are growth factors that promote proliferation of T cells and, thus, their increased production may contribute to the development of non-Hodgkins lymphoma. Tat can also be released from infected cells and act as an angiogenic (blood vessel forming) growth factor. This property is thought to contribute to the development of Kaposi's sarcoma.



**DNA viruses.** Some DNA viruses also cause human cancer, but by different mechanisms. Three DNA tumor virus families, SV40, papillomavirus, and adenovirus, encode proteins that inactivate pRb and p53. By interfering with the G1/S checkpoint, these oncoproteins increase the probability that mutations in oncogenes and tumor suppressor genes will be incorporated into the genome of infected cells, thereby increasing the probability of transformation. The Epstein-Barr virus encodes a Bcl-2 protein that restricts apoptosis of the infected cell.

### Suggested References

- Culotta E, Koshland D Jr. Molecule of the year: p53 sweeps through cancer research. *Science* 1993;262:1958–1959.
- Tamm I, Schriever F, and Dorhen B. Apoptosis: implications of basic research for clinical oncology. *The Lancet Oncology* 2001;2:33–42.
- Vogelstein B, Fearon ER, Hamilton SR, et al. Genetic alterations during colorectal tumor development. *N Engl J Med* 1988;319:525.
- Bertram JS. The molecular biology of cancer. *Molecular Aspects Med* 2000;21:167–223.
- Alberts B, Johnson A, Lewis J, Raff M, et al. Cancer. Chapter 23. In *Molecular Biology of the Cell*, 4th Ed. New York: Garland Scientific, 2002:1313–1362.
- Weinberg RA. How cancer arises. *Sci Am* 1996;275:62–70.



### REVIEW QUESTIONS—CHAPTER 18

- The *ras* oncogene in Colin Tuma's malignant polyp differs from the *c-ras* proto-oncogene only in the region that encodes the N-terminus of the protein. This portion of the normal and mutant sequences is shown below:

|        |                                         |    |    |
|--------|-----------------------------------------|----|----|
|        | 10                                      | 20 | 30 |
| Normal | ATGACGGAATATAAGCTGGTGGTGGTGGGCGCCGGCGGT |    |    |
| Mutant | ATGACGGAATATAAGCTGGTGGTGGTGGGCGCCGTCGGT |    |    |

This mutation is similar to the mutation found in the *ras* oncogene in various tumors. What type of mutation converts the *ras* proto-oncogene to an oncogene?

- An insertion that disrupts the reading frame of the protein
- A deletion that disrupts the reading frame of the protein
- A missense mutation that changes one amino acid within the protein
- A silent mutation that has no change in amino acid sequence of the protein
- An early termination that creates a stop codon in the reading frame of the protein

2. The mechanism through which Ras becomes an oncogenic protein is which of the following?
  - (A) Ras remains bound to GAP.
  - (B) Ras can no longer bind cAMP.
  - (C) Ras has lost its GTPase activity.
  - (D) Ras can no longer bind GTP.
  - (E) Ras can no longer be phosphorylated by MAP kinase.
3. Which of the following statements best describes a characteristic of oncogenes?
  - (A) All retroviruses contain at least one oncogene.
  - (B) Retroviral oncogenes were originally obtained from a cellular host chromosome.
  - (C) Proto-oncogenes are genes, found in retroviruses, that have the potential to transform normal cells when inappropriately expressed.
  - (D) The oncogenes that lead to human disease are different from those that lead to tumors in animals.
  - (E) Oncogenes are mutated versions of normal viral gene products.
4. When p53 increases in response to DNA damage, which of the following events occur?
  - (A) p53 induces transcription of cdk4.
  - (B) p53 induces transcription of cyclin D.
  - (C) p53 binds E2F to activate transcription.
  - (D) p53 induces transcription of p21.
  - (E) p53 directly phosphorylates the transcription factor jun.
5. A tumor suppressor gene is best described by which of the following?
  - (A) A gain-of-function mutation leads to uncontrolled proliferation.
  - (B) A loss-of-function mutation leads to uncontrolled proliferation.
  - (C) When expressed, the gene suppresses viral genes from being expressed.
  - (D) When expressed, the gene specifically blocks the G1/S checkpoint.
  - (E) When expressed, the gene induces tumor formation.



## Fuel Oxidation and The Generation of ATP

All physiologic processes in living cells require energy transformation. Cells convert the chemical bond energy in foods into other forms, such as an electrochemical gradient across the plasma membrane, or the movement of muscle fibers in an arm, or assembly of complex molecules such as DNA (Fig. 1). These energy transformations can be divided into three principal phases: (1) oxidation of fuels (fat, carbohydrate, and protein), (2) conversion of energy from fuel oxidation into the high-energy phosphate bonds of ATP, and (3) utilization of ATP phosphate bond energy to drive energy-requiring processes.

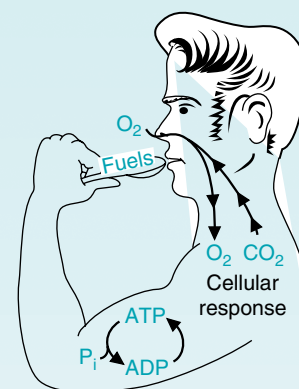
The first two phases of energy transformation are part of cellular respiration, the overall process of using  $O_2$  and energy derived from oxidizing fuels to generate ATP. We need to breathe principally because our cells require  $O_2$  to generate adequate amounts of ATP from the oxidation of fuels to  $CO_2$ . Cellular respiration uses over 90% of the  $O_2$  we inhale.

In phase 1 of respiration, energy is conserved from fuel oxidation by enzymes that transfer electrons from the fuels to the electron-accepting coenzymes  $NAD^+$  and FAD, which are reduced to NADH and FAD(2H), respectively (Fig. 2). The pathways for the oxidation of most fuels (glucose, fatty acids, ketone bodies, and many amino acids) converge in the generation of the activated 2-carbon acetyl group in acetyl CoA. The complete oxidation of the acetyl group to  $CO_2$  occurs in the tricarboxylic acid (TCA) cycle, which collects the energy mostly as NADH and FAD(2H).

In phase 2 of cellular respiration, the energy derived from fuel oxidation is converted to the high-energy phosphate bonds of ATP by the process of oxidative phosphorylation (see Fig. 2). Electrons are transferred from NADH and FAD(2H) to  $O_2$  by the electron transport chain, a series of electron transfer proteins that are located in the inner mitochondrial membrane. Oxidation of NADH and FAD(2H) by  $O_2$  generates an electrochemical potential across the inner mitochondrial membrane in the form of a transmembrane proton gradient ( $\Delta p$ ). This electrochemical potential drives the synthesis of ATP from ADP and  $P_i$  by a transmembrane enzyme called ATP synthase (or  $F_0F_1$ ATPase).

In phase 3 of cellular respiration, the high-energy phosphate bonds of ATP are used for processes such as muscle contraction (mechanical work), maintaining low intracellular  $Na^+$  concentrations (transport work), synthesis of larger molecules such as DNA in anabolic pathways (biosynthetic work), or detoxification (biochemical work). As a consequence of these processes, ATP is either directly or indirectly hydrolyzed to ADP and inorganic phosphate ( $P_i$ ), or to AMP and pyrophosphate (PPi).

Cellular respiration occurs in mitochondria (Fig. 3). The mitochondrial matrix, which is the compartment enclosed by the inner mitochondrial membrane, contains almost all of the enzymes for the TCA cycle and oxidation of fatty acids,



**Fig. 1.** Energy transformations in fuel metabolism. When ATP energy is transformed into cellular responses, such as muscle contraction, ATP is cleaved to ADP and  $P_i$  (inorganic phosphate). In cellular respiration,  $O_2$  is used for regenerating ATP from oxidation of fuels to  $CO_2$ .

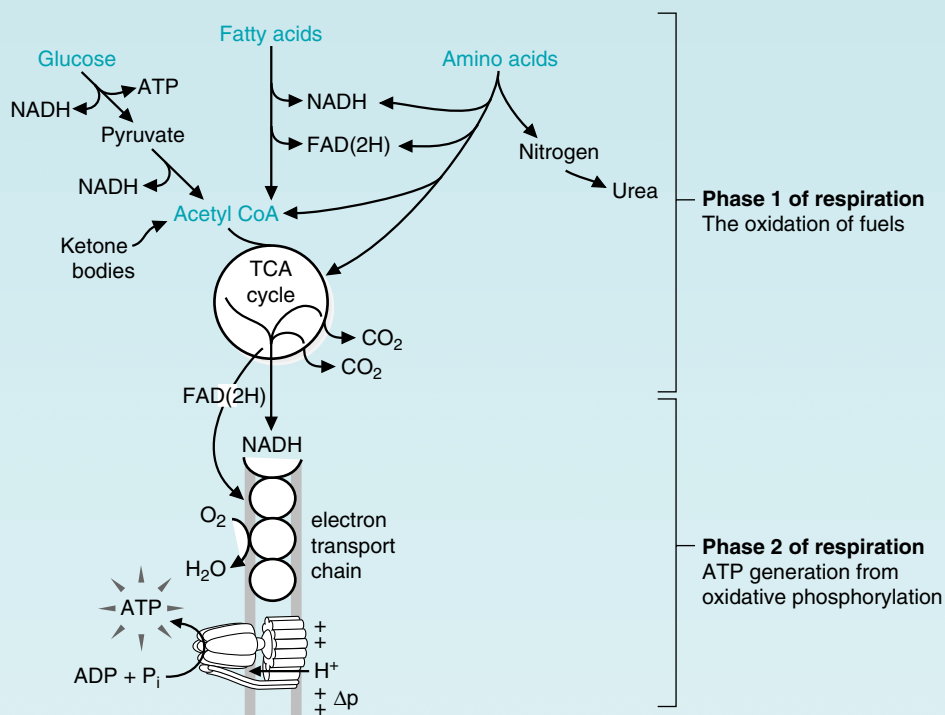


Fig. 2. Cellular respiration.  $\Delta p$  = the proton gradient.



Definitions of prefixes and suffixes used in describing clinical conditions:

|                |                            |
|----------------|----------------------------|
| <b>an-:</b>    | Without                    |
| <b>-emia:</b>  | Blood                      |
| <b>hyper-:</b> | Excessive, above normal    |
| <b>hypo-:</b>  | Deficient, below normal    |
| <b>-osis:</b>  | Abnormal or diseased state |
| <b>-uria:</b>  | Urine                      |

ketone bodies, and most amino acids. The inner mitochondrial membrane contains the protein complexes of the electron transport chain and ATP synthase, the enzyme complex that generates ATP from ADP and  $\text{P}_i$ . Some of the subunits of these complexes are encoded by mitochondrial DNA, which resides in the matrix. ATP is generated in the matrix, but most of the energy-using processes in the cell occur outside of the mitochondrion. As a consequence, newly generated ATP must be continuously transported to the cytosol by protein transporters in the impermeable inner mitochondrial membrane and by diffusion through pores in the more permeable outer mitochondrial membrane.

The rates of fuel oxidation and ATP utilization are tightly coordinated through feedback regulation of the electron transport chain and the pathways of fuel oxidation. Thus, if less energy is required for work, more fuel is stored as glycogen or fat in adipose tissue. The basal metabolic rate (BMR), caloric balance, and  $\Delta G$  (the change in Gibbs free energy, which is the amount of energy available to do useful work) are quantitative ways of describing energy requirements and the energy that can be derived from fuel oxidation. The various types of enzyme regulation described in Chapter 9 are all used to regulate the rate of oxidation of different fuels to meet energy requirements.

Fatty acids are a major fuel in the body. After eating, we store excess fatty acids and carbohydrates that are not oxidized as fat (triacylglycerols) in adipose tissue. Between meals, these fatty acids are released and circulate in blood bound to albumin. In muscle, liver, and other tissues, fatty acids are oxidized to acetyl CoA in the pathway of  $\beta$ -oxidation. NADH and FAD(2H) generated from  $\beta$ -oxidation are reoxidized by  $\text{O}_2$  in the electron transport chain, thereby generating ATP (see Fig. 2). Small amounts of certain fatty acids are oxidized through other pathways that convert them to either oxidizable fuels or urinary excretion products (e.g., peroxisomal  $\beta$ -oxidation).

Not all acetyl CoA generated from  $\beta$ -oxidation enters the TCA cycle. In the liver, acetyl CoA generated from  $\beta$ -oxidation of fatty acids can also be converted to the

ketone bodies acetoacetate and  $\beta$ -hydroxybutyrate. Ketone bodies are taken up by muscle and other tissues, which convert them back to acetyl CoA for oxidation in the TCA cycle. They become a major fuel for the brain during prolonged fasting.

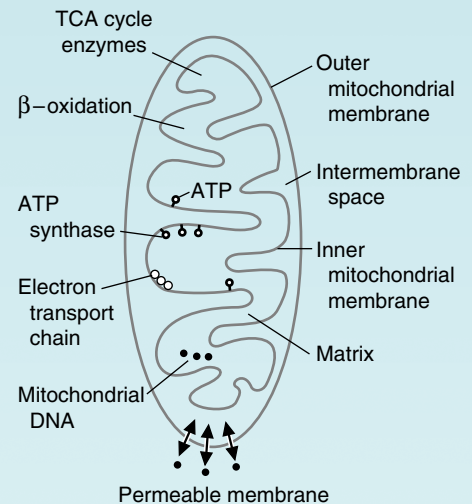
Amino acids derived from dietary or body proteins are also potential fuels that can be oxidized to acetyl CoA, or converted to glucose and then oxidized (see Fig. 2). These oxidation pathways, like those of fatty acids, generate NADH or FAD(2H). Ammonia, which can be formed during amino acid oxidation, is toxic. It is therefore converted to urea in the liver and excreted in the urine. There are more than 20 different amino acids, each with a somewhat different pathway for oxidation of the carbon skeleton and conversion of its nitrogen to urea. Because of the complexity of amino acid metabolism, use of amino acids as fuels is considered separately in Section Seven, Nitrogen Metabolism.

Glucose is a universal fuel used to generate ATP in every cell type in the body (Fig. 4). In glycolysis, 1 mole of glucose is converted to 2 moles of pyruvate and 2 moles of NADH by cytosolic enzymes. Small amounts of ATP are generated when high-energy pathway intermediates transfer phosphate to ADP in a process termed substrate level phosphorylation. In aerobic glycolysis, the NADH produced from glycolysis is reoxidized by  $O_2$  via the electron transport chain, and pyruvate enters the TCA cycle. In anaerobic glycolysis, the NADH is reoxidized by conversion of pyruvate to lactate, which enters the blood. Although anaerobic glycolysis has a low ATP yield, it is important for tissues with a low oxygen supply and few mitochondria (e.g., the kidney medulla), or tissues experiencing diminished blood flow (ischemia).

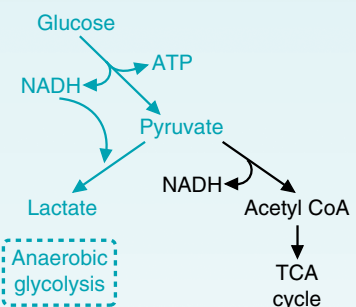
All cells continuously use ATP and require a constant supply of fuels to provide energy for the generation of ATP. Chapters 1 through 3 of this text outline the basic patterns of fuel utilization in the human and provide information about dietary components.

The pathologic consequences of metabolic problems in fuel oxidation can be grouped into 2 categories: (1) lack of a required product, or (2) excess of a substrate or pathway intermediate. The product of fuel oxidation is ATP, and an inadequate rate of ATP production occurs under a wide variety of medical conditions. Extreme conditions that interfere with ATP generation from oxidative phosphorylation, such as complete oxygen deprivation (anoxia), or cyanide poisoning, are fatal. A myocardial infarction is caused by a lack of adequate blood flow to regions of the heart (ischemia), thereby depriving cardiomyocytes of oxygen and fuel. Hyperthyroidism is associated with excessive heat generation from fuel oxidation, and in hypothyroidism, ATP generation can decrease to a fatal level. Conditions such as malnutrition, anorexia nervosa, or excessive alcohol consumption may decrease availability of thiamine,  $Fe^{2+}$ , and other vitamins and minerals required by the enzymes of fuel oxidation. Mutations in mitochondrial DNA or nuclear DNA result in deficient ATP generation from oxidative metabolism.

In contrast, problems arising from an excess of substrate or fuel are seen in diabetes mellitus, which may result in a potentially fatal ketoacidosis. Lactic acidosis occurs with problems of oxidative metabolism.



**Fig. 3.** Oxidative metabolism in mitochondria. The inner mitochondrial membrane forms infoldings, called cristae, which enclose the mitochondrial matrix. Most of the enzymes for the TCA cycle, the  $\beta$ -oxidation of fatty acids, and for mitochondrial DNA synthesis are found in the matrix. ATP synthase and the protein complexes of the electron transport chain are embedded in the inner mitochondrial membrane. The outer mitochondrial membrane is permeable to small ions, but the inner mitochondrial membrane is impermeable.



**Fig. 4.** Glycolysis. In glycolysis, glucose is converted to pyruvate. If the pyruvate is reduced to lactate, the pathway does not require  $O_2$  and is called anaerobic glycolysis. If this pyruvate is converted instead to acetyl CoA and oxidized in the TCA cycle, glycolysis requires  $O_2$  and is aerobic.



# 19 Cellular Bioenergetics:

## ATP And O<sub>2</sub>

**Bioenergetics** refers to **cellular energy transformations**.

**The ATP-ADP cycle.** In cells, the **chemical bond energy** of fuels is transformed into the physiologic responses necessary for life. The central role of the **high-energy phosphate bonds of ATP** in these processes is summarized in the **ATP-ADP cycle** (Fig. 19.1). To generate ATP through cellular respiration, fuels are degraded by oxidative reactions that transfer most of their **chemical bond energy** to NAD<sup>+</sup> and FAD to generate the reduced form of these coenzymes, **NADH** and **FAD(2H)**. When NADH and FAD(2H) are oxidized by O<sub>2</sub> in the electron transport chain, the energy is used to regenerate ATP in the process of **oxidative phosphorylation**. Energy available from cleavage of the high-energy phosphate bonds of ATP can be used directly for **mechanical work** (e.g., muscle contraction) or for **transport work** (e.g., a Na<sup>+</sup> gradient generated by Na<sup>+</sup>, K<sup>+</sup>-ATPase). It can also be used for **biochemical work** (energy-requiring chemical reactions), such as **anabolic pathways** (biosynthesis of large molecules like proteins) or detoxification reactions. **Phosphoryl transfer** reactions, **protein conformational changes**, and the formation of **activated intermediates** containing **high energy bonds** (e.g., UDP-sugars) facilitate these energy transformations. Energy released from foods that is not used for work against the environment is transformed into **heat**.

**ATP homeostasis.** Fuel oxidation is regulated to maintain **ATP homeostasis** (homeo, same; stasis, state). Regardless of whether the level of cellular fuel utilization is high (with increased ATP consumption), or low (with decreased ATP consumption), the available ATP within the cell is maintained at a constant level by appropriate increases or decreases in the rate of fuel oxidation. Problems in ATP homeostasis and energy balance occur in obesity, hyperthyroidism, and myocardial infarction.

**Energy from Fuel Oxidation.** Fuel oxidation is **exergonic**; it releases energy. The maximum quantity of energy released that is available for useful work (e.g., ATP synthesis) is called  $\Delta G^{0'}$ , the **change in Gibbs free energy** at pH 7.0 under standard conditions. Fuel oxidation has a **negative  $\Delta G^{0'}$** , that is, the products have a lower chemical bond energy than the reactants and their formation is energetically favored. ATP synthesis from ADP and inorganic phosphate is **endergonic**; it requires energy and has a positive  $\Delta G^{0'}$ . To proceed in our cells, all pathways must have a negative  $\Delta G^{0'}$ . How is this accomplished for anabolic pathways such as glycogen synthesis? These metabolic pathways incorporate reactions that expend high-energy bonds to compensate for the energy-requiring steps. Because the  $\Delta G^{0'}$ s for a sequence of reactions **are additive**, the overall pathway becomes energetically favorable.

Fuels are oxidized principally by donating electrons to NAD<sup>+</sup> and FAD, which then donate electrons to O<sub>2</sub> in the electron transport chain. The **caloric value** of a fuel is related to its  $\Delta G^{0'}$  for transfer of electrons to O<sub>2</sub>, and its **reduction potential, E<sup>0'</sup>** (a measure of its willingness to donate, or accept,

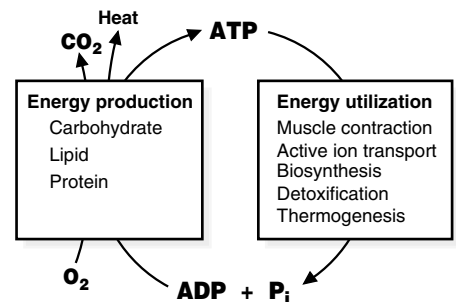
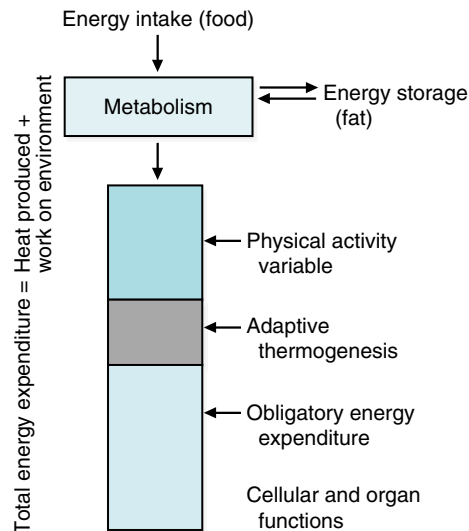


Fig. 19.1. ATP-ADP cycle.





In the thermodynamic perspective of energy expenditure, where energy intake to the body exceeds energy expended, the difference is effectively stored as fat.



The portion of food that is metabolized is regulated to match the total energy expenditure of the body. A certain amount of the energy is obligatory (the amount of energy expended to do the work of the cells, the BMR). Some energy is also expended for adaptive thermogenesis, heat generated in response to cold or diet. An additional amount of energy is used for physical exercise (work against the environment). To voluntarily store less energy as fat, we can vary our caloric intake through dietary changes or our energy expenditure through changes in our physical exercise.



**Cora Nari** suffered a heart attack 8 months ago and had a significant loss of functional heart muscle. The pain she is experiencing is called angina pectoris, which is a crushing or constricting pain located in the center of the chest, often radiating to the neck or arms (see Ann Jeina, Chapters 6 and 7). The most common cause of angina pectoris is partial blockage of coronary arteries from atherosclerosis. The heart muscle cells beyond the block receive an inadequate blood flow and oxygen, and die when ATP production falls too low.

electrons). Because fatty acids are more reduced than carbohydrates, they have a higher caloric value. The high affinity of oxygen for electrons (a high positive reduction potential) drives fuel oxidation forward, with release of energy that can be used for ATP synthesis in oxidative phosphorylation. However, smaller amounts of ATP can be generated without the use of  $O_2$  in **anaerobic glycolysis**.

Fuel oxidation can also generate **NADPH**, which usually donates electrons to biosynthetic pathways and detoxification reactions. For example, in some reactions catalyzed by **oxygenases**, NADPH is the electron donor and  $O_2$  the electron acceptor.



## THE WAITING ROOM



**Otto Shape** is a 26-year old medical student who has completed his first year of medical school. He is 70 inches tall and began medical school weighing 154 lb, within his ideal weight range (see Chapter 1). By the time he finished his last examination in his first year, he weighed 187 lb. He had calculated his BMR at approximately 1,680 kcal, and his energy expenditure for physical exercise equal to 30% of his BMR. He planned on returning to his premedical school weight in 6 weeks over the summer by eating 576 kcal less each day and playing 7 hours of tennis every day. However, he did a summer internship instead of playing tennis. When Otto started his second year of medical school, he weighed 210 lb.



**X.S. Teefore** (excess  $T_4$ ) is a 26-year-old man who noted heat intolerance with heavy sweating, heart palpitations, and tremulousness. Over the past 4 months, he has lost weight in spite of a good appetite. He is sleeping poorly and describes himself as feeling “jittery inside.”

On physical examination, his heart rate is rapid (116 beats/min) and he appears restless and fidgety. His skin feels warm, and he is perspiring profusely. A fine hand tremor is observed as he extends his arms in front of his chest. His thyroid gland appears to be diffusely enlarged and, on palpation, is approximately 3 times normal size. Thyroid function tests confirm that Mr. Teefore’s thyroid gland is secreting excessive amounts of the thyroid hormones  $T_4$  (tetraiodothyronine) and  $T_3$  (triiodothyronine), the major thyroid hormones present in the blood.



**Cora Nari** is a 64-year-old woman who had a myocardial infarction 8 months ago. Although she managed to lose 6 lb since that time, she remains overweight and has not reduced the fat content of her diet adequately. The graded aerobic exercise program she started 5 weeks after her infarction is now followed irregularly, falling far short of the cardiac conditioning intensity prescribed by her cardiologist. She is readmitted to the hospital cardiac care unit (CCU) after experiencing a severe “viselike pressure” in the mid-chest area while cleaning ice from the windshield of her car. The electrocardiogram (ECG) shows evidence of a new posterior wall myocardial infarction. Signs and symptoms of left ventricular failure are present.

## I. ENERGY AVAILABLE TO DO WORK

The basic principle of the ATP-ADP cycle is that fuel oxidation generates ATP, and hydrolysis of ATP to ADP provides the energy to perform most of the work required in the cell. ATP has therefore been called the energy currency of our cells. Like the

one dollar bill, it has a defined value, is required to obtain goods and services, and disappears before we know it. To keep up with the demand, we must constantly replenish our ATP supply through the use of O<sub>2</sub> for fuel oxidation.

The amount of energy from ATP cleavage available to do useful work is related to the difference in energy levels between the products and substrates of the reaction and is called the change in Gibbs free energy,  $\Delta G$  ( $\Delta$ , difference; G, Gibbs free energy). In cells, the  $\Delta G$  for energy production from fuel oxidation must be greater than the  $\Delta G$  of energy-requiring processes, such as protein synthesis and muscle contraction, for life to continue.

### A. The High-Energy Phosphate Bonds of ATP

The amount of energy released or required by bond cleavage or formation is determined by the chemical properties of the substrates and products. The bonds between the phosphate groups in ATP are called phosphoanhydride bonds (Fig. 19.2). When these bonds are hydrolyzed, energy is released because the products of the reaction (ADP and phosphate) are more stable, with lower bond energies, than the reactants (ATP and H<sub>2</sub>O). The instability of the phosphoanhydride bonds arises from their negatively charged phosphate groups, which repel each other and strain the bonds between them. It takes energy to make the phosphate groups stay together. In contrast, there are fewer negative charges in ADP to repel each other. The phosphate group as a free anion is more stable than it is in ATP because of an increase in resonance structures (i.e., the electrons of the oxygen double bond are shared by all the oxygen atoms). As a consequence, ATP hydrolysis is energetically favorable and proceeds with release of energy as heat.

In the cell, ATP is not directly hydrolyzed. Energy released as heat from ATP hydrolysis cannot be transferred efficiently into energy-requiring processes, such as biosynthetic reactions or maintaining an ion gradient. Instead, cellular enzymes directly transfer the phosphate group to a metabolic intermediate or protein that is part of the energy-requiring process (a phosphoryl transfer reaction).

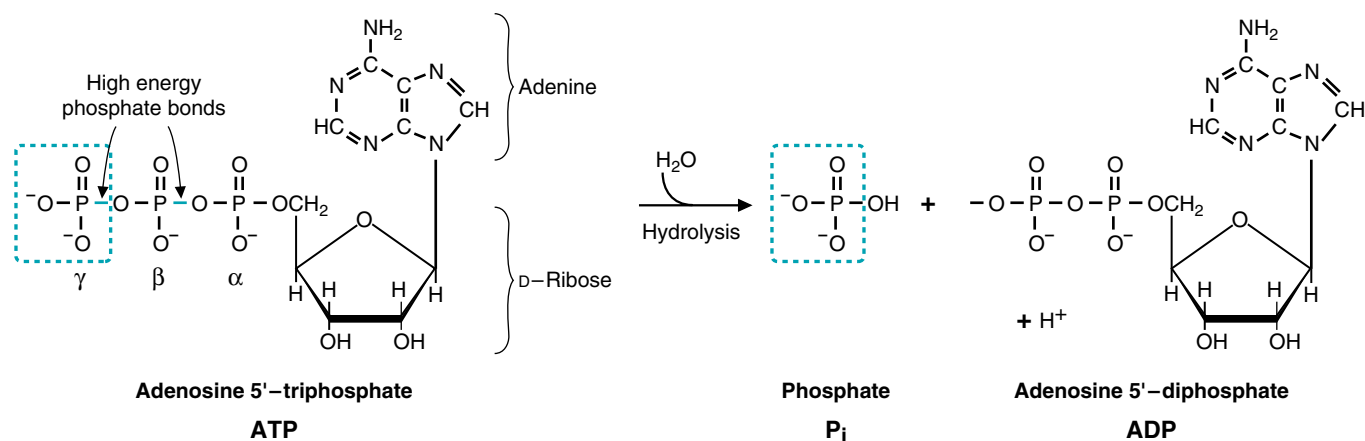
### B. Change in Free Energy ( $\Delta G$ ) During a Reaction

How much energy can be obtained from ATP hydrolysis to do the work required in the cell? The maximum amount of useful energy that can be obtained from a



The heart is a specialist in the transformation of ATP chemical bond energy into mechanical work.

Each single heartbeat uses approximately 2% of the ATP in the heart. If the heart were not able to regenerate ATP, all its ATP would be hydrolyzed in less than 1 minute. Because the amount of ATP required by the heart is so high, it must rely on the pathway of oxidative phosphorylation for generation of this ATP. In **Cora Nari's** heart, hypoxia is affecting her ability to generate ATP.



**Fig. 19.2.** Hydrolysis of ATP to ADP and inorganic phosphate (P<sub>i</sub>). Cleavage of the phosphoanhydride bonds between either the  $\beta$  and  $\gamma$  phosphates or the  $\alpha$  and  $\beta$  phosphates releases the same amount of energy, approximately  $-7.3$  kcal/mole. However, hydrolysis of the phosphate-adenosine bond (a phosphoester bond) releases less energy ( $-3.4$  kcal/mole), and consequently, this bond is not considered a high-energy phosphate bond. During ATP hydrolysis, the change in disorder during the reaction is small and so  $\Delta G$  values at physiologic temperature ( $37^\circ\text{C}$ ) are similar to those at standard temperature ( $25^\circ\text{C}$ ).  $\Delta G$  is affected by pH, which alters the ionization state of the phosphate groups of ATP and by the intracellular concentration of Mg<sup>2+</sup> ions, which bind to the  $\beta$  and  $\gamma$  phosphate groups of ATP.

reaction is called  $\Delta G$ , the change in Gibbs free energy. The value of  $\Delta G$  for a reaction can be influenced by the initial concentration of substrates and products, by temperature, pH, and pressure. The  $\Delta G^0$  for a reaction refers to the energy change for a reaction starting at 1 M substrate and product concentrations and proceeding to equilibrium (equilibrium, by definition, occurs when there is no change in substrate and product concentrations with time).  $\Delta G^{0'}$  is the value for  $\Delta G^0$  under standard conditions (pH = 7.0,  $[\text{H}_2\text{O}] = 55 \text{ M}$ , and  $25^\circ\text{C}$ ), as well as standard concentrations (Table 19.1).

$\Delta G^{0'}$  is equivalent to the chemical bond energy of the products minus that of the reactants, corrected for energy that has gone into entropy (an increase in amount of molecular disorder). This correction for change in entropy is very small for most reactions occurring in cells, and, thus, the  $\Delta G^{0'}$  for hydrolysis of various chemical bonds reflects the amount of energy available from that bond.

The value of  $-7.3 \text{ kcal/mole}$  ( $-30.5 \text{ kJ/mole}$ ) that is generally used for the  $\Delta G^{0'}$  of ATP hydrolysis is, thus, the amount of energy available from hydrolysis of ATP under standard conditions that can be spent on energy-requiring processes; it defines the “monetary value” of our “ATP currency.” Although the difference between cellular conditions (pH 7.3,  $37^\circ\text{C}$ ) and standard conditions is very small, the difference between cellular concentrations of ATP, ADP, and Pi and the standard 1-M concentrations is huge and greatly affects the availability of energy in the cell.

### C. Exothermic and Endothermic Reactions

The value of  $\Delta G^{0'}$  tells you whether the reaction requires or releases energy, the amount of energy involved, and the ratio of products to substrates at equilibrium. The negative value for the  $\Delta G^{0'}$  of ATP hydrolysis indicates that, if you begin with equimolar (1 M) concentrations of substrates and products, the reaction proceeds in

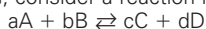
**Q:** The reaction catalyzed by phosphoglucomutase (PGM) is reversible and functions in the synthesis of glycogen from glucose as well as the degradation of glycogen back to glucose. If the  $\Delta G^{0'}$  for conversion of glucose 6-P to glucose 1-P is  $+1.65 \text{ kcal/mole}$ , what is the  $\Delta G^{0'}$  of the reverse reaction?

**Table 19.1. Thermodynamic Expressions, Laws, and Constants**

| Definitions                                                                                                                                                              |                                                                                                                                                                   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\Delta G$                                                                                                                                                               | Change in free energy, or Gibbs free energy                                                                                                                       |
| $\Delta G^0$                                                                                                                                                             | Standard free energy change, $\Delta G$ at 1 M concentrations of substrates and products                                                                          |
| $\Delta G^{0'}$                                                                                                                                                          | Standard free energy change at $25^\circ\text{C}$ , pH 7.0                                                                                                        |
| $\Delta H$                                                                                                                                                               | Change in enthalpy, or heat content                                                                                                                               |
| $\Delta S$                                                                                                                                                               | Change in entropy, or increase in disorder                                                                                                                        |
| $K'_{\text{eq}}$                                                                                                                                                         | Equilibrium constant at $25^\circ\text{C}$ , pH 7.0, incorporating $[\text{H}_2\text{O}] = 55.5 \text{ M}$ and $[\text{H}^+] = 10^{-7} \text{ M}$ in the constant |
| $\Delta E^{0'}$                                                                                                                                                          | Change in reduction potential                                                                                                                                     |
| $\sim\text{P}$                                                                                                                                                           | Biochemical symbol for a high-energy phosphate bond, i.e., a bond which is hydrolyzed with the release of more than about 7 kcal/mole of heat                     |
| Laws of thermodynamics                                                                                                                                                   |                                                                                                                                                                   |
| First law of thermodynamics, the conservation of energy: In any physical or chemical change, the total energy of a system, including its surroundings, remains constant. |                                                                                                                                                                   |
| Second law of thermodynamics: The universe tends toward disorder. In all natural processes, the total entropy of a system always increases.                              |                                                                                                                                                                   |
| Constants                                                                                                                                                                |                                                                                                                                                                   |
| Units of $\Delta G$ and $\Delta H = \text{cal/mole}$ or $\text{J/mole}$ : $1 \text{ cal} = 4.18 \text{ J}$                                                               |                                                                                                                                                                   |
| $T$ , Absolute temperature: K, Kelvin = $273 + ^\circ\text{C}$ ( $25^\circ\text{C} = 298^\circ\text{K}$ )                                                                |                                                                                                                                                                   |
| $R$ , Universal gas constant: $1.99 \text{ cal/mole}\cdot\text{K}$ or $8.31 \text{ J/mole}\cdot\text{K}$                                                                 |                                                                                                                                                                   |
| $F$ , Faraday constant: $F = 23 \text{ kcal/mole-volt}$ or $96,500 \text{ J/V}\cdot\text{mole}$                                                                          |                                                                                                                                                                   |
| Units of $E_0'$ , volts                                                                                                                                                  |                                                                                                                                                                   |
| Formulas                                                                                                                                                                 |                                                                                                                                                                   |
| $\Delta G = \Delta H - T\Delta S$                                                                                                                                        |                                                                                                                                                                   |
| $\Delta G^{0'} = -RT \ln K'_{\text{eq}}$                                                                                                                                 |                                                                                                                                                                   |
| $\Delta G^{0'} = -nF\Delta E^{0'}$                                                                                                                                       |                                                                                                                                                                   |
| $\ln = 2.303 \log_{10}$                                                                                                                                                  |                                                                                                                                                                   |

**Table 19.2. A General Expression for  $\Delta G$** 

To generalize the expression for  $\Delta G$ , consider a reaction in which



The small letters denote a moles of A will combine with b moles of B to produce c moles of C and d moles of D.

$$\Delta G^{0'} = -RT \ln K'_{eq} = -RT \ln \frac{[C]_{eq}^c [D]_{eq}^d}{[A]_{eq}^a [B]_{eq}^b}$$

and

$$\Delta G = \Delta G^{0'} + RT \ln \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

the forward direction with the release of energy. From initial concentrations of 1 M, the ATP concentration will decrease, and ADP and Pi will increase until equilibrium is reached.

For a reaction in which a substrate S is converted to a product P, the ratio of the product concentration to the substrate concentration at equilibrium is given by:

**Equation 1:**

$$\Delta G^{0'} = -RT \ln [P]/[S]$$

(see Table 19.2. for a more general form of this equation; R is equal to the gas constant (1.98 calories/mole-degree Kelvin), and T is equal to the temperature in degrees Kelvin).

Thus, the difference in chemical bond energies of the substrate and product ( $\Delta G^{0'}$ ) determines the concentration of each at equilibrium.

Reactions such as ATP hydrolysis are exergonic (release energy) or exothermic (release heat). They have a negative  $\Delta G^{0'}$  and release energy while proceeding in the forward direction to equilibrium. Endergonic, or endothermic, reactions have a positive  $\Delta G^{0'}$  for the forward direction (the direction shown), and the backward direction is favored. For example, in the pathway of glycogen synthesis, phosphoglucomutase converts glucose-6-P to glucose-1-P. Glucose-1-P has a higher phosphate bond energy than glucose-6-P because the phosphate is on the aldehyde carbon (Fig 19.3). The  $\Delta G^{0'}$  for the forward direction (glucose-1-P  $\rightarrow$  glucose-6-P) is, therefore, positive. Beginning at equimolar concentrations of both compounds, there is a net conversion of glucose-1-P back to glucose-6-P and, at equilibrium, the concentration of glucose-6-P is higher than glucose-1-P. The exact ratio is determined by  $\Delta G^{0'}$  for the reaction.

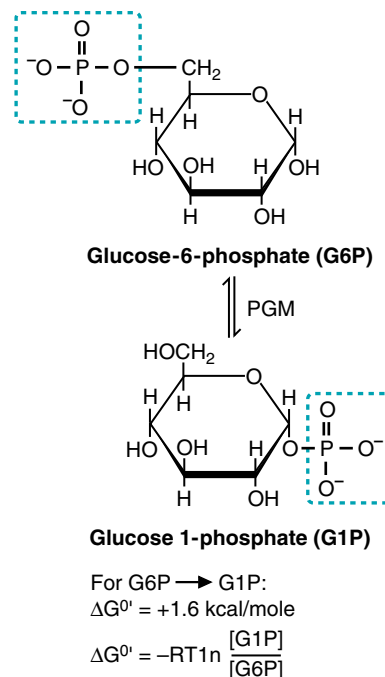
It is often said that a reaction with a negative  $\Delta G'$  proceeds spontaneously in the forward direction, meaning that products accumulate at the expense of reactants. However,  $\Delta G'$  is not an indicator of the velocity of the reaction, or the rate at which equilibrium can be reached. In the cell, the velocity of the reaction depends on the efficiency and amount of enzyme available to catalyze the reaction (see Chapter 9), and, therefore, “spontaneously” in this context can be misleading.

## II. ENERGY TRANSFORMATIONS TO DO MECHANICAL AND TRANSPORT WORK

To do work in the cell, a mechanism must be available for converting the chemical bond energy of ATP into another form, such as an Na<sup>+</sup> gradient across a membrane. These energy transformations usually involve intermediate steps in which ATP is bound to a protein, and cleavage of the bound ATP results in a conformational change of the protein.



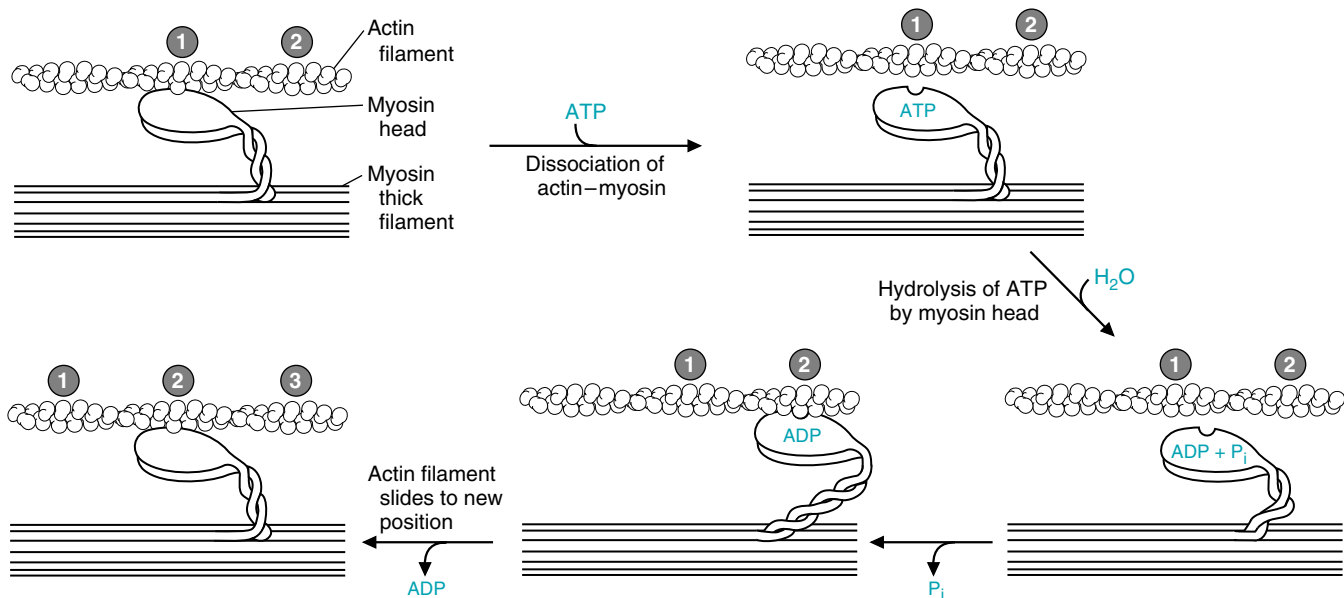
The  $\Delta G^{0'}$  for the reverse reaction is  $-1.65$  kcal. The change in free energy is the same for the forward and reverse directions, but has an opposite sign. Because negative  $\Delta G^{0'}$  values indicate favorable reactions, this reaction under standard conditions favors the conversion of glucose 1-P to glucose 6-P.



**Fig. 19.3.** The phosphoglucomutase reaction. The forward direction is involved in converting glucose to glycogen, and the reverse direction in converting glycogen to glucose 6-P.



The  $\Delta G^{0'}$  for the conversion of glucose 6-P to glucose 1-P is  $+1.65$  kcal/mole. What is the ratio of [glucose-1-P] to [glucose-6-P] at equilibrium?



**Fig. 19.4.** A simplified diagram of myosin ATPase. Muscle fiber is made of thick filaments composed of bundles of the protein myosin, and thin filaments composed of the protein actin (which is activated by  $\text{Ca}^{2+}$  binding). At many positions along the actin filament, a terminal domain of a myosin molecule, referred to as the “head,” binds to a specific site on the actin. The myosin head has an ATP binding site and is an ATPase; it can hydrolyze ATP to ADP and  $\text{P}_i$ . (1) As ATP binds to myosin, the conformation of myosin changes, and it dissociates from the actin. (2) Myosin hydrolyzes the ATP, again changing conformation. (3) When  $\text{P}_i$  dissociates, the myosin head reassociates with activated actin at a new position. (4) As ADP dissociates, the myosin again changes conformation, or tightens. This change of conformation at multiple association points between actin and myosin slides the actin filament forward (5).

**A:** In equation 1 of Table 19.2,  $\Delta G^{0'}$  =  $RT \ln K_{\text{eq}}$ . For this reaction,  $K_{\text{eq}}$  =  $[\text{glucose-1-phosphate}]/[\text{glucose-6-phosphate}]$ . The constant  $R$  is  $1.99 \times 10^{-3}$  kcal/mole-°K, and  $T$  is  $(273 + 25)$  °K, so  $RT$  equals 0.593 kcal/mole. Substituting in equation 1 then gives  $1.65 = -0.593 \ln [\text{glucose-1-P}]/[\text{glucose-6-P}]$ . Thus,  $\ln[\text{glucose-1-P}]/[\text{glucose-6-P}] = -2.78$ , and  $[\text{glucose-1-P}]/[\text{glucose-6-phosphate}] = e^{-2.78}$ , or 0.062. So the ratio of [glucose-1-P] to [glucose-6-P] at equilibrium is 0.062.

**Otto Shape** has not followed his proposed diet and exercise regimen and has been gaining weight. He has a positive caloric balance, because his daily energy expenditure is less than his daily energy intake (see Chapter 2). Although the energy expenditure for physical exercise is only approximately 30% of the BMR (basal metabolic rate) in a sedentary individual, it can be 100% or more of the BMR in a person who exercises strenuously for several hours or more. The large increase in ATP utilization for muscle contraction during exercise accounts for its contribution to the daily energy expenditure.

## A. Mechanical Work

In mechanical work, the high-energy phosphate bond of ATP is converted into movement by changing the conformation of a protein (Fig.19.4.). For example, in contracting muscle fibers, the hydrolysis of ATP while it is bound to myosin ATPase changes the conformation of myosin so that it is in a “cocked” position ready to associate with the sliding actin filament. Thus, exercising muscle fibers have almost a hundred-fold higher rate of ATP utilization and caloric requirements than resting muscle fibers. Motor proteins, such as kinesins that transport chemicals along fibers, provide another example of mechanical work in a cell.

## B. Transport Work

In transport work, called active transport, the high-energy phosphate bond of ATP is used to transport compounds against a concentration gradient (see Chapter10,

The equations for calculating  $\Delta G$  are based on the first law of thermodynamics (see Table 19.1). The change in chemical bond energy that occurs during a reaction is  $\Delta H$ , the change in enthalpy of the reaction. At constant temperature and pressure,  $\Delta H$  is equivalent to the chemical bond energy of the products minus that of the reactants.  $\Delta G$ , the maximum amount of useful work available from a reaction, is equal to  $\Delta H$  minus  $T\Delta S$ .  $T\Delta S$  is a correction for the amount of energy that has gone into an increase in the entropy (disorder in arrangement of molecules) of the system.

$$\Delta G = \Delta H - T\Delta S$$

where  $\Delta H$  = the change in enthalpy,  $T$  is the temperature of the system in Kelvin, and  $\Delta S$  is the change in entropy, or increased disorder of the system.  $\Delta S$  is often negligible in reactions such as ATP hydrolysis in which the number of substrates ( $\text{H}_2\text{O}$ , ATP) and products (ADP,  $\text{P}_i$ ) are equal and no gas is formed. Under these conditions, the values for  $\Delta G$  at physiologic temperature (37°C) are similar to those at standard temperature (25°C)



Fig.10.12). In P-ATPases (plasma membrane ATPases) and V-ATPases (vesicular ATPases), the chemical bond energy of ATP is used to reversibly phosphorylate the transport protein and change its conformation. For example, as Na<sup>+</sup>, K<sup>+</sup>-ATPase binds and cleaves ATP, it becomes phosphorylated and changes its conformation to release 3 Na<sup>+</sup> ions to the outside of the cell, thereby building up a higher extracellular than intracellular concentration of Na<sup>+</sup>. Na<sup>+</sup> re-enters the cell on cotransport proteins that drive the uptake of amino acids and many other compounds into the cell. Thus, Na<sup>+</sup> must be continuously transported back out. The expenditure of ATP for Na<sup>+</sup> transport occurs even while we sleep and is estimated to account for 10 to 30% of our BMR.

A large number of other active transporters also convert ATP chemical bond energy into an ion gradient (membrane potential). Vesicular ATPases pump protons into lysosomes. Ca<sup>2+</sup> ATPases in the plasma membrane move Ca<sup>2+</sup> out of the cell against a concentration gradient. Similar Ca<sup>2+</sup> ATPases pump Ca<sup>2+</sup> into the lumen of the endoplasmic reticulum and the sarcoplasmic reticulum (in muscle). Thus, a considerable amount of energy is expended in maintaining a low cytoplasmic Ca<sup>2+</sup> level.

### III. BIOCHEMICAL WORK

The high-energy phosphate bonds of ATP are also used for biochemical work. Biochemical work occurs in anabolic pathways, which are pathways that synthesize large molecules (e.g., DNA, glycogen, triacylglycerols, and proteins) from smaller compounds. Biochemical work also occurs when toxic compounds are converted to nontoxic compounds that can be excreted (e.g., the liver converts NH<sub>4</sub><sup>+</sup> ions to urea in the urea cycle). In general, formation of chemical bonds between two organic molecules (e.g., C-C bonds in fatty acid synthesis or C-N bonds in protein synthesis) requires energy and is therefore biochemical work. How do our cells get these necessary energy-requiring reactions to occur?

To answer this question, the next sections consider how energy is used to synthesize glycogen from glucose (Fig 19.5). Glycogen is a storage polysaccharide consisting of glucosyl units linked together through glycosidic bonds. If an anabolic pathway, such as glycogen synthesis, were to have an overall positive  $\Delta G^{0'}$ , the cell would be full of glucose and intermediates of the pathway, but very little glycogen would be formed. To avoid this, cells do biochemical work and spend enough of their ATP currency to give anabolic pathways an overall negative  $\Delta G^{0'}$ .



Approximately 70% of our resting daily energy requirement arises from work carried out by our largest organs: the heart, brain, kidneys, and liver. Using their rate of oxygen consumption and a P/O ratio of 2.5, it can be estimated that each of these organs is using and producing several times its own weight in ATP each day. The heart, which rhythmically contracts, is using this ATP for mechanical work. In contrast, skeletal muscles in a resting individual use far less ATP per gram of tissue. The kidney has an ATP consumption per gram of tissue similar to that of the heart and is using this ATP largely for transport work to recover usable nutrients and maintain pH and electrolyte balance. The brain, likewise, uses most of its ATP for transport work, maintaining the ion gradients necessary for conduction of the nerve impulse. The liver, in contrast, has a high rate of ATP consumption and utilization to carry out metabolic work (biosynthesis and detoxification).

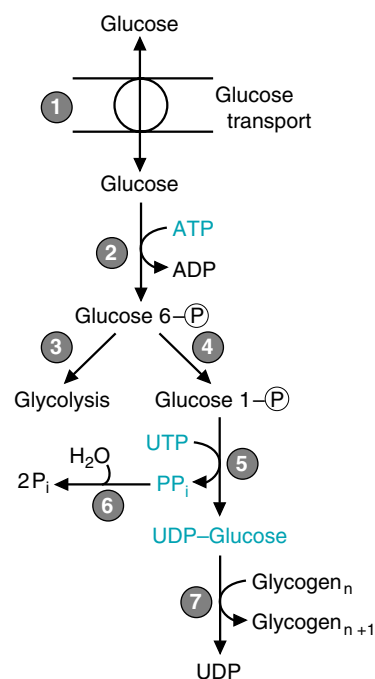
Estimated Daily Use of ATP

(g ATP/g tissue)

|                           |      |
|---------------------------|------|
| Heart                     | 16   |
| Brain                     | 6    |
| Kidneys                   | 24   |
| Liver                     | 6    |
| Skeletal Muscle (rest)    | 0.3  |
| Skeletal Muscle (running) | 23.6 |



**X.S. Teefore** has increased blood levels of thyroid hormones, which accelerate basal metabolic processes that use ATP in our organs (e.g., Na<sup>+</sup>,K<sup>+</sup>-ATPase), thereby increasing the BMR. An increased BMR was used for a presumptive diagnosis of hyperthyroidism before development of the tests to measure T3 and T4. Because **X.S. Teefore** did not fully compensate for his increased ATP requirements with an increased caloric intake, he was in negative caloric balance and lost weight.



**Fig. 19.5.** Energetics of glycogen synthesis. Compounds containing high-energy bonds are shown in blue. (1) Glucose is transported into the cell. (2) Glucose phosphorylation uses the high-energy phosphate bond (~P) of ATP in a phosphoryl transfer step. (4) Conversion of glucose 6-phosphate to glucose 1-phosphate by phosphoglucomutase. (5) UDP-Glucose pyrophosphorylase cleaves a ~P bond in UTP, releasing pyrophosphate and forming UDP-glucose, an activated intermediate. (6) The pyrophosphate is hydrolyzed, releasing additional energy. (7) The phosphoester bond of UDP-glucose is cleaved during the addition of a glucosyl unit to the end of a glycogen polysaccharide chain. The UDP-glucose acts as the leaving group in this reaction. Glucose-6-phosphate also can be metabolized via glycolysis (3) when energy is required.

## A. $\Delta G^0$ Values Are Additive

Reactions in which chemical bonds are formed between two organic molecules are usually catalyzed by enzymes that transfer energy from cleavage of ATP in a phosphoryl transfer reaction or by enzymes that cleave a high-energy bond in an activated intermediate of the pathway. Because the  $\Delta G^{0'}$  values in a reaction sequence are additive, the pathway acquires an overall negative  $\Delta G^{0'}$ , and the reactions in the pathway will occur to move toward an equilibrium state where the concentration of final products is greater than that of the initial reactants.

### 1. PHOSPHORYL TRANSFER REACTIONS

One of the characteristics of Gibbs free energy is that  $\Delta G^0$  values for consecutive steps or reactions in a sequence can be added together to obtain a single value for the overall process. Thus, the high-energy phosphate bonds of ATP can be used to drive a reaction forward that would otherwise be highly unfavorable energetically. Consider, for example, synthesis of glucose 6-P from glucose, the first step in glycolysis and glycogen synthesis (see Fig. 19.5, circle 2). If the reaction were to proceed by addition of inorganic phosphate to glucose, glucose-6-P synthesis would have a positive  $\Delta G^{0'}$  value of 3.3 kcal/mole (Table 19.3). However, when this reaction is coupled to cleavage of the high-energy ATP bond through a phosphoryl transfer reaction, the  $\Delta G^{0'}$  for glucose-6-P synthesis acquires a net negative value of minus 4.0 kcal/mole, which can be calculated from the sum of the two reactions. Glucose 6-P cannot be transported back out of the cell, and therefore the net negative  $\Delta G^{0'}$  for glucose 6-P synthesis helps the cell to trap glucose for its own metabolic needs.

The net value for synthesis of glucose 6-P from glucose and ATP would be the same whether or not the two reactions were catalyzed by the same enzyme, were catalyzed by two separate enzymes, or were not catalyzed by an enzyme at all, because it is dictated by the amount of energy in the chemical bonds being broken and formed.

### 2. ACTIVATED INTERMEDIATES IN GLYCOGEN SYNTHESIS

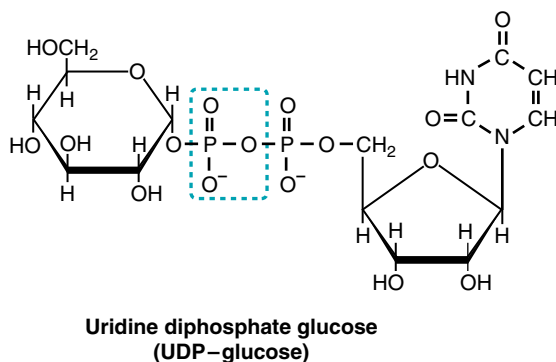
To synthesize glycogen from glucose, energy is provided by the cleavage of 3 high-energy phosphate bonds in ATP, UTP, and pyrophosphate (PPi) (see Fig. 19.5, Steps 2, 5, and 6). Energy transfer is facilitated by phosphoryl group transfer and by formation of an activated intermediate (UDP-glucose). Step 4, the conversion of glucose 6-phosphate to glucose 1-P, has a positive  $\Delta G^{0'}$ . This step is pulled and pushed in the desired direction by the accumulation of substrate and removal of product in reactions that have a negative  $\Delta G^{0'}$  from cleavage of high-energy bonds. In Step 5, the UTP high-energy phosphate bond is cleaved to form the activated sugar, UDP-glucose (Fig 19.6). This reaction is further facilitated by cleavage of the high-energy bond in the pyrophosphate (Step 6) that is released in Step 5 (approximately  $-7.7$  kcal). In Step 7, cleavage of the bond between UDP and glucose in the activated intermediate provides the energy for attaching the glucose moiety to the end of the glycogen molecule (approximately  $-3.3$  kcal). In general, the amount of ATP phosphate bond energy used in an anabolic pathway, or detoxification pathway, must provide the pathway with an overall negative  $\Delta G^{0'}$ , so that the concentration of products is favored over that of reactants.



**Q:** Given a  $\Delta G^{0'}$  of  $+1.65$  kcal/mole for the conversion of glucose-6-P to glucose-1-P, and a  $\Delta G^{0'}$  of  $-4.0$  kcal/mole for the conversion of glucose + ATP to glucose-6-P + ADP, what is the value of  $\Delta G^{0'}$  for the conversion of glucose to glucose-1-P?

**Table 19.3.  $\Delta G^{0'}$  for the Transfer of a Phosphate from ATP to Glucose**

|                                                           |                                  |
|-----------------------------------------------------------|----------------------------------|
| Glucose + Pi $\rightarrow$ glucose-6-P + H <sub>2</sub> O | $\Delta G^{0'} = +3.3$ kcal/mole |
| ATP + H <sub>2</sub> O $\rightarrow$ ADP + Pi             | $\Delta G^{0'} = -7.3$ kcal/mole |
| Sum: glucose + ATP $\rightarrow$ glucose-6-P + ADP        | $\Delta G^{0'} = -4.0$ kcal/mole |



**Fig. 19.6.** UDP-glucose contains a high-energy pyrophosphate bond, shown in blue.

## B. $\Delta G$ Depends on Substrate and Product Concentration

$\Delta G^{0'}$  reflects the energy difference between reactants and products at specific concentrations (each at 1 M) and standard conditions (pH 7.0, 25°C). However, these are not the conditions prevailing in cells, where variations from “standard conditions” are relevant to determining actual free energy changes and hence the direction in which reactions are likely to occur. One aspect of free energy changes contributing to the forward direction of anabolic pathways is the dependence of  $\Delta G$ , the free energy change of a reaction, on the initial substrate and product concentrations. Reactions in the cell with a positive  $\Delta G^{0'}$  can proceed in the forward direction if the concentration of substrate is raised to high enough levels, or if the concentration of product is decreased to very low levels. Product concentrations can be very low if, for example, the product is rapidly used in a subsequent energetically favorable reaction, or if the product diffuses or is transported away.

### 1. THE DIFFERENCE BETWEEN $\Delta G$ AND $\Delta G^{0'}$

The driving force toward equilibrium starting at any concentration of substrate and product is expressed by  $\Delta G$ , and not by  $\Delta G^{0'}$ , which is the free energy change to reach equilibrium starting with 1 M concentrations of substrate and product. For a reaction in which the substrate S is converted to the product P:

#### Equation 2

$$\Delta G = \Delta G^{0'} + RT \ln [P]/[S]$$

(see Table 19.2, for the general form of this equation).

The expression for  $\Delta G$  has two terms:  $\Delta G^{0'}$ , the energy change to reach equilibrium starting at equal and 1 M concentrations of substrates and products, and the second term, the energy change to reach equal concentrations of substrate and product starting from any initial concentration. (When  $[P] = [S]$  and  $[P]/[S] = 1$ , the  $\ln$  of  $[P]/[S]$  is 0, and  $\Delta G = \Delta G^{0'}$ ). The second term will be negative for all concentrations of substrate greater than product, and the greater the substrate concentration, the more negative this term will be. Thus, if the substrate concentration is suddenly raised high enough or the product concentration decreased low enough,  $\Delta G$  (the sum of the first and second terms) will also be negative, and conversion of substrate to product becomes thermodynamically favorable.

### 2. THE REVERSIBILITY OF THE PHOSPHOGLUCOMUTASE REACTION IN THE CELL

The effect of substrate and product concentration on  $\Delta G$  and the direction of a reaction in the cell can be illustrated with conversion of glucose-6-P to glucose-1-P, the



$\Delta G^{0'}$  for the overall reaction is the sum of the individual reactions, and is  $-2.35$  kcal. The individual reactions are:

Glucose + ATP  $\rightarrow$  glucose 6-P + ADP

$\Delta G^{0'} = -4.0$  kcal/mole

Glucose 6-P  $\rightarrow$  glucose 1-P

$\Delta G^{0'} = +1.65$  kcal/mole

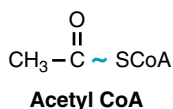
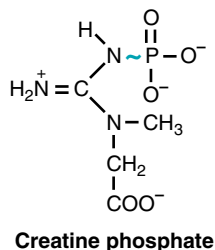
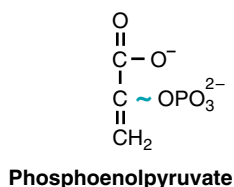
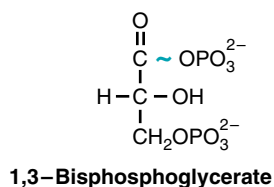
Sum: Glucose + ATP  $\rightarrow$  glucose 1-P + ADP

$\Delta G^{0'} = -2.35$  kcal/mole

Thus, the cleavage of ATP has made the synthesis of glucose-1-P from glucose energetically favorable.



The  $\Delta G$  of ATP hydrolysis in cells can be very different than  $\Delta G^{\circ}$ . The synthesis of ATP rather than its hydrolysis becomes the energetically favorable direction if the [ATP] drops just a little or the [ADP] or [Pi] increase just a little over equilibrium values. Thus, as ATP is hydrolyzed in energy-requiring reactions, the change in the sign of  $\Delta G$  promotes ATP synthesis.



**Fig. 19.7.** Some compounds with high-energy bonds. 1,3-bisphosphoglycerate and phosphoenolpyruvate are intermediates of glycolysis. Creatine phosphate is a high-energy phosphate reservoir and shuttle in brain, muscle, and spermatozoa. Acetyl CoA is a precursor of the TCA cycle. The high-energy bonds are shown in blue.

reaction catalyzed by phosphoglucomutase in the pathway of glycogen synthesis (see Fig. 19.3). The reaction has a small positive  $\Delta G^{\circ}$  for glucose 1-P synthesis (+1.65 kcal/mole) and at equilibrium, the ratio of [glucose 1-P]/[glucose 6-P] is approximately 6 to 94 (which you calculated in Question 2). However, if another reaction uses glucose 1-P such that this ratio suddenly becomes 3 to 94, there is now a driving force for converting more glucose 6-P to glucose 1-P and restoring the equilibrium ratio. Substitution in equation 2 gives  $\Delta G$ , the driving force to equilibrium, as  $+1.65 + RT \ln [\text{G1P}]/[\text{G6P}] = 1.65 + (-2.06) = -0.41$ , which is a negative value. Thus, a decrease in the ratio of product to substrate has converted the synthesis of glucose 1-P from a thermodynamically unfavorable to a thermodynamically favorable reaction that will proceed in the forward direction until equilibrium is reached.

### C. Activated Intermediates with High Energy Bonds

Many biochemical pathways form activated intermediates containing high-energy bonds to facilitate biochemical work. The term “high-energy bond” is a biologic term defined by the  $\Delta G^{\circ}$  for ATP hydrolysis; any bond that can be hydrolyzed with the release of approximately as much, or more, energy than ATP is called a high-energy bond. The high-energy bond in activated intermediates, such as UDP-glucose in glycogen synthesis, facilitate energy transfer.

#### 1. ATP, UTP, GTP, AND CTP

Cells use GTP and CTP, as well as UTP and ATP, to form activated intermediates. Different anabolic pathways generally use different nucleotides as their direct source of high phosphate bond energy: UTP is used for combining sugars, CTP in lipid synthesis, and GTP in protein synthesis.

The high-energy phosphate bonds of UTP, GTP, and CTP are energetically equivalent to ATP and are synthesized from ATP by nucleoside diphosphokinases and nucleoside monophosphokinases. For example, UTP is formed from UDP by a nucleoside diphosphokinase in the reaction:



ADP is converted back to ATP by the process of oxidative phosphorylation, using energy supplied by fuel oxidation.

Energy-requiring reactions often generate the nucleoside diphosphate ADP. Adenylate kinase, an important enzyme in cellular energy balance, is a nucleoside monophosphate kinase that transfers a phosphate from one ADP to another ADP to form ATP and AMP:



This enzyme, thus, can regenerate ATP under conditions in which ATP utilization is required.

#### 2. OTHER COMPOUNDS WITH HIGH-ENERGY BONDS

In addition to the nucleoside triphosphates, other compounds containing high-energy bonds are formed to facilitate energy transfer in anabolic and catabolic pathways (e.g., 1,3-bisphosphoglycerate in glycolysis and acetyl CoA in the TCA cycle) (Fig. 19.7). Creatine phosphate contains a high-energy phosphate bond that allows it to serve as an energy reservoir for ATP synthesis and transport in muscle cells, neurons, and spermatozoa. All of these high-energy bonds are “unstable,” and their hydrolysis yields substantial free energy because the products are much more stable, as a result of electron resonance within their structures.

## IV. THERMOGENESIS

According to the first law of thermodynamics, energy cannot be destroyed. Thus, energy from oxidation of a fuel (its caloric content) must be equal to the amount of heat released, the work performed against the environment, and the increase in order of molecules in our bodies. Some of the energy from fuel oxidation is converted into heat as the fuel is oxidized and some heat is generated as ATP is used to do work. If we become less efficient in converting energy from fuel oxidation into ATP, or if we use an additional amount of ATP for muscular contraction, we will oxidize an additional amount of fuel to maintain ATP homeostasis (constant cellular ATP levels). With the oxidation of additional fuel, we release additional heat. Thus, heat production is a natural consequence of “burning fuel.”

*Thermogenesis* refers to energy expended for the purpose of generating heat in addition to that expended for ATP production. To maintain our body at 37°C, despite changes in environmental temperature, it is necessary to regulate fuel oxidation and its efficiency (as well as heat dissipation). In shivering thermogenesis, we respond to sudden cold with asynchronous muscle contractions (shivers) that increase ATP utilization and, therefore, fuel oxidation and the release of energy as heat. In non-shivering thermogenesis (adaptive thermogenesis), the efficiency of converting energy from fuel oxidation into ATP is decreased. More fuel needs to be oxidized to maintain constant ATP levels and, thus, more heat is generated.

## V. ENERGY FROM FUEL OXIDATION

Fuel oxidation provides energy for bodily processes principally through generation of the reduced coenzymes, NADH and FAD(2H). They are used principally to generate ATP in oxidative phosphorylation. However, fuel oxidation also generates NADPH, which is most often used directly in energy-requiring processes. Carbohydrates also may be used to generate ATP through a nonoxidative pathway, called anaerobic glycolysis.

### A. Energy Transfer from Fuels through Oxidative Phosphorylation

Fuel oxidation is our major source of ATP and our major means of transferring energy from the chemical bonds of the fuels to cellular energy-requiring processes. The amount of energy available from a fuel is equivalent to the amount of heat that is generated when a fuel is burned. To conserve this energy for the generation of ATP, the process of cellular respiration transforms the energy from the chemical bonds of fuels into the reduction state of electron-accepting coenzymes, NAD<sup>+</sup> and FAD (circle 1, Fig. 19.8). As these compounds transfer electrons to O<sub>2</sub> in the electron transport chain, most of this energy is transformed into an electrochemical gradient across the inner mitochondrial membrane (circle 2, Fig. 19.8). Much of the energy in the electrochemical gradient is used to regenerate ATP from ADP in oxidative phosphorylation (phosphorylation that requires O<sub>2</sub>).

### 1. OXIDATION-REDUCTION REACTIONS

Oxidation-reduction reactions always involve a pair of chemicals: an electron donor, which is oxidized in the reactions, and an electron acceptor, which is reduced in the reaction. In fuel metabolism, the fuel donates electrons, and is oxidized, and NAD<sup>+</sup> and FAD accept electrons, and are reduced.

When is NAD<sup>+</sup>, rather than FAD, used in a particular oxidation-reduction reaction? It depends on the chemical properties of the electron donor and the enzyme catalyzing the reaction. In oxidation reactions, NAD<sup>+</sup> accepts two electrons as a hydride ion to form NADH, and a proton (H<sup>+</sup>) is released into the medium (Fig 19.9). It is generally used for metabolic reactions involving oxidation of alcohols and aldehydes. In contrast,



**X.S. Teefore** has increased thyroid hormone levels that increase his rate of ATP utilization and fuel oxidation. An excess of thyroid hormones also may affect the efficiency of ATP production, resulting in fewer ATP produced for a given O<sub>2</sub> consumption. The increased rate of ATP utilization and diminished efficiency stimulates oxidative metabolism, resulting in a much greater rate of heat production. The hyperthyroid patient, therefore, complains of constantly feeling hot (heat intolerance) and sweaty. (Perspiration allows dissipation of excess heat through evaporation from the skin surface.)



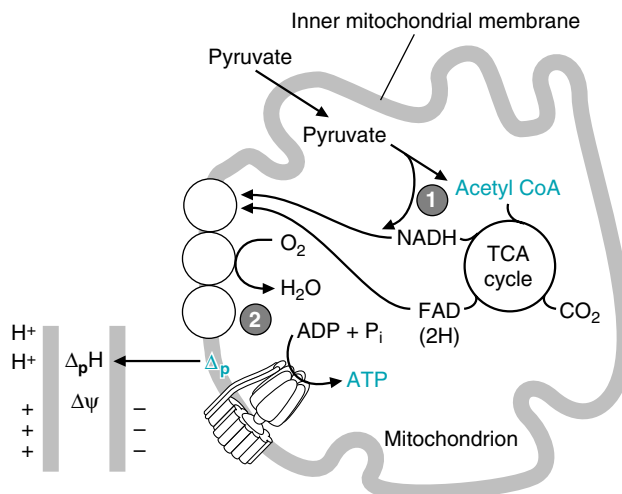
Oxidation is the loss of electrons, and reduction is the gain of electrons. Remember LEO GER:

Loss of Electrons = Oxidation;

Gain of Electrons = Reduction.

Compounds are oxidized in the body in essentially three ways: (1) the transfer of electrons from the compound as a hydrogen atom or a hydride ion, (2) the direct addition of oxygen from O<sub>2</sub>, and (3) the direct donation of electrons (e.g., Fe<sup>2+</sup> → Fe<sup>3+</sup>) (see Chapter 5). Fuel oxidation involves the transfer of electrons as a hydrogen atom or a hydride ion and, thus, reduced compounds have more hydrogen relative to oxygen than the oxidized compounds. Consequently, aldehydes are more reduced than acids, and alcohols are more reduced than aldehydes.

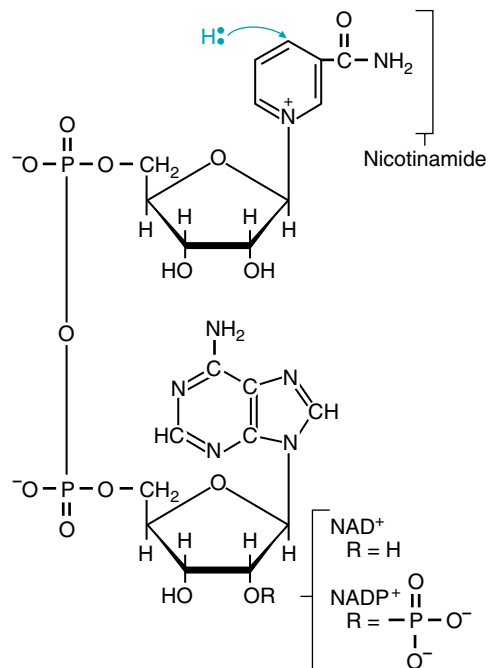




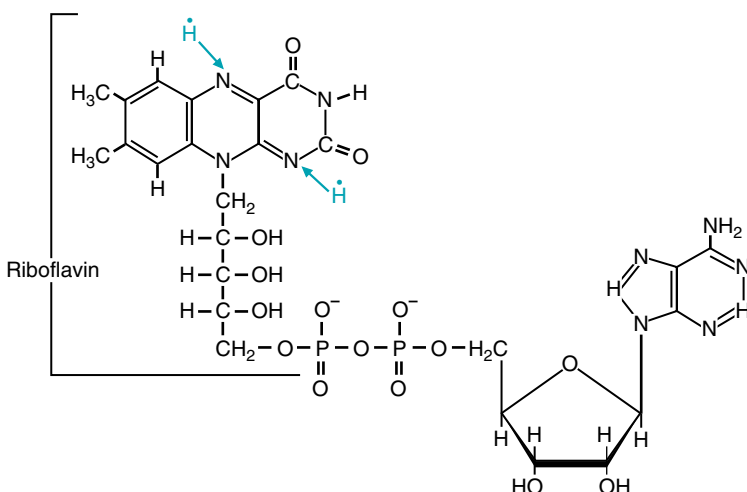
**Fig. 19.8.** Overview of energy transformations in oxidative phosphorylation. The electrochemical potential gradient across the mitochondrial membrane is represented by  $\Delta p\text{H}$ , the proton gradient, and  $\Delta\psi$ , the membrane potential. The role of the electrochemical potential in oxidative phosphorylation is discussed in more depth in Chapter 21.

FAD accepts two electrons as hydrogen atoms, which are donated singly from separate atoms (e.g., formation of a double bond or a disulfide)(Fig. 19.10).

As the reduced coenzymes donate these electrons to  $\text{O}_2$  through the electron transport chain, they are reoxidized. The energy derived from reoxidation of NADH and FAD(2H) is available for the generation of ATP by oxidative phosphorylation. In our analogy of ATP as currency, the reduced coenzymes are our “paychecks” for oxidizing fuels. Because our cells spend ATP so fast, we must immediately convert our paychecks into ATP cash.



**Fig. 19.9.** Reduction of NAD<sup>+</sup> and NADP<sup>+</sup>. These structurally related coenzymes are reduced by accepting two electrons as H<sup>-</sup>, the hydride ion.



**Fig. 19.10.** Reduction of FAD. FAD accepts two electrons as two hydrogen atoms and is reduced. The reduced coenzyme is denoted in this text as FAD(2H) because it often accepts a total of two electrons one at a time, never going to the fully reduced form, FADH<sub>2</sub>. FMN (flavin mononucleotide) consists of riboflavin with one phosphate group attached.

## 2. REDUCTION POTENTIAL

Each oxidation/reduction reaction makes or takes a fixed amount of energy, ( $\Delta G^{0'}$ ), which is directly proportional to the  $\Delta E^{0'}$  (the difference in reduction potentials of the oxidation-reduction pair). The reduction potential of a compound,  $E^{0'}$ , is a measure in volts of the energy change when that compound accepts electrons (becomes reduced); minus  $E^{0'}$  is the energy change when the compound donates electrons (becomes oxidized).  $E^{0'}$  can be considered an expression of the willingness of the compound to accept electrons. Some examples of reduction potentials are shown in Table 19.4. Oxygen, which is the best electron acceptor, has the largest positive reduction potential (i.e., is the most willing to accept electrons and be reduced). As a consequence, the transfer of electrons from all compounds to O<sub>2</sub> is energetically favorable and occurs with energy release.

The more negative the reduction potential of a compound, the greater is the energy available for ATP generation when that compound passes its electrons to oxygen. The  $\Delta G^{0'}$  for transfer of electrons from NADH to O<sub>2</sub> is greater than the transfer from FAD(2H) to O<sub>2</sub> (see the reduction potential values for NADH and FAD(2H) in Table 19.4). Thus, the energy available for ATP synthesis from NADH is approximately  $-53$  kcal, and approximately  $-41$  kcal from the FAD-containing flavoproteins in the electron transport chain.



To calculate the free energy change of an oxidation-reduction reaction, the reduction potential of the electron donor (NADH) is added to that of the acceptor (O<sub>2</sub>). The  $\Delta E^{0'}$  for the net reaction is calculated from the sum of the half reactions. For NADH donation of electrons, it is  $+0.320$  volts, opposite of that shown in Table 4 (remember, Table 4 shows the  $E^{0'}$  for accepting electrons), and for O<sub>2</sub> acceptance, it is  $+0.816$ . The number of electrons being transferred is 2 (so,  $n = 2$ ). The direct relationship between the energy changes in oxidation-reduction reactions and  $\Delta G^{0'}$  is expressed by the equation

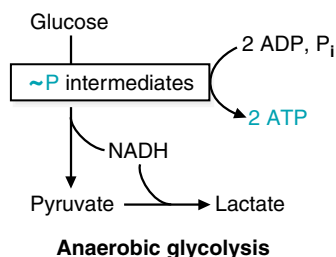
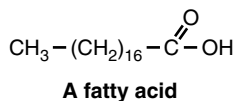
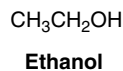
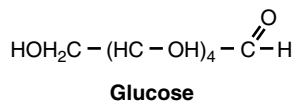
$$\Delta G^{0'} = -n F \Delta E^{0'}$$

where  $n$  is the number of electrons transferred and  $F$  is Faraday's constant (23 kcal/mole  $\cdot$  volt). Thus, a value of approximately  $-53$  kcal/mole is obtained for the energy available for ATP synthesis by transferring two electrons from NADH to oxygen.

**Table 19.4. Reduction Potentials of Some Oxidation-Reduction Half-Reactions**

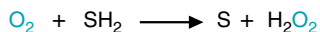
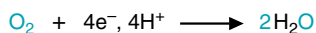
| Reduction Half-Reactions                                                              | $E^{0'}$ at pH 7.0 |
|---------------------------------------------------------------------------------------|--------------------|
| $1/2 \text{ O}_2 + 2\text{H}^+ + 2 \text{ e}^- \rightarrow \text{H}_2\text{O}$        | 0.816              |
| $\text{Cytochrome a-Fe}^{3+} + 1 \text{ e}^- \rightarrow \text{cytochrome a-Fe}^{2+}$ | 0.290              |
| $\text{CoQ} + 2\text{H}^+ + 2 \text{ e}^- \rightarrow \text{CoQH}_2$                  | 0.060              |
| $\text{Fumarate} + 2\text{H}^+ + 2 \text{ e}^- \rightarrow \text{succinate}$          | 0.030              |
| $\text{Oxalacetate} + 2\text{H}^+ + 2 \text{ e}^- \rightarrow \text{malate}$          | $-0.102$           |
| $\text{Acetaldehyde} + 2\text{H}^+ + 2 \text{ e}^- \rightarrow \text{ethanol}$        | $-0.163$           |
| $\text{Pyruvate} + 2\text{H}^+ + 2 \text{ e}^- \rightarrow \text{lactate}$            | $-0.190$           |
| $\text{Riboflavin} + 2\text{H}^+ + 2 \text{ e}^- \rightarrow \text{riboflavin-H}_2$   | $-0.200$           |
| $\text{NAD}^+ + 2\text{H}^+ + 2 \text{ e}^- \rightarrow \text{NADH} + \text{H}^+$     | $-0.320$           |
| $\text{Acetate} + 2\text{H}^+ + 2 \text{ e}^- \rightarrow \text{acetaldehyde}$        | $-0.468$           |

**Q:** **Otto Shape** decided to lose weight by decreasing his intake of fat and alcohol (ethanol), and increasing his content of carbohydrates. Compare the structure of ethanol with that of glucose and fatty acids (below). On the basis of their oxidation state, which compound provides the most energy (calories) per gram?

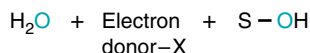
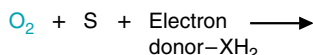


**Fig. 19.11.** Anaerobic glycolysis. Phosphate is transferred from high-energy intermediates of the pathway to ADP. Because NADH from the pathway is reoxidized by reduction of pyruvate to lactate, no oxygen is required.

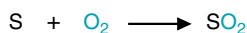
#### Oxidases



#### Monooxygenases



#### Dioxygenases



**Fig. 19.12.** Oxidases and oxygenases. The fate of  $\text{O}_2$  is shown in blue. S represents an organic substrate.

### 3. CALORIC VALUES OF FUELS

The caloric value of a food is directly related to its oxidation state, which is a measure of  $\Delta G^{0'}$  for transfer of electrons from that fuel to  $\text{O}_2$ . The electrons donated by the fuel are from its C-H and C-C bonds. Fatty acids such as palmitate ( $\text{CH}_3(\text{CH}_2)_{14}\text{COOH}$ ) have a caloric value of roughly 9 kcal/g. Glucose is already partially oxidized and has a caloric value of only about 4 kcal/g. The carbons, on an average, contain fewer C-H bonds from which to donate electrons.

The caloric value of a food is applicable in humans only if our cells have enzymes that can oxidize that fuel by transferring electrons from the fuel to  $\text{NAD}^+$ ,  $\text{NADP}^+$ , or  $\text{FAD}$ . When we burn wood in a fireplace, electrons are transferred from cellulose and other carbohydrates to  $\text{O}_2$ , releasing energy as heat. However, wood has no caloric content for humans; we cannot digest it and convert cellulose to a form that can be oxidized by our enzymes. Cholesterol, although a lipid, also has no caloric value for us because we cannot oxidize the carbons in its complex ring structure in reactions that generate  $\text{NADH}$ ,  $\text{FAD(2H)}$ , or  $\text{NADPH}$ .

### B. NADPH in Oxidation-Reduction Reactions

$\text{NADP}^+$  is similar to  $\text{NAD}^+$  and has the same reduction potential. However,  $\text{NADP}^+$  has an extra phosphate group on the ribose, which affects its enzyme binding (see Fig. 19.9). Consequently, most enzymes use either  $\text{NAD}^+$  or  $\text{NADP}^+$ , but seldom both. In certain reactions, fuels are oxidized by transfer of electrons to  $\text{NADP}^+$  to form  $\text{NADPH}$ . For example, glucose 6-P dehydrogenase, in the pentose phosphate pathway, transfers electrons from glucose 6-P to  $\text{NADP}^+$  instead of  $\text{NAD}^+$ .  $\text{NADPH}$  usually donates electrons to biosynthetic reactions such as fatty acid synthesis, and to detoxification reactions that use oxygen directly. Consequently, the energy in its reduction potential is usually used in energy-requiring reactions without first being converted to ATP currency.

### C. Anaerobic Glycolysis

Not all ATP is generated by fuel oxidation. In anaerobic glycolysis, glucose is degraded in reactions that form high-energy phosphorylated intermediates of the pathway (Fig. 19.11). These activated high-energy intermediates provide the energy for the generation of ATP from ADP without involving electron transfer to  $\text{O}_2$ . Therefore, this pathway is called anaerobic glycolysis, and ATP is generated from substrate level phosphorylation rather than oxidative phosphorylation (see Chapter 22). Anaerobic glycolysis is a critical source of ATP for cells that have a decreased  $\text{O}_2$  supply, either because they are physiologically designed that way (e.g., cells in the kidney medulla), or because their supply of  $\text{O}_2$  has been pathologically decreased (e.g., coronary artery disease).

## VI. OXYGENASES AND OXIDASES NOT INVOLVED IN ATP GENERATION

Approximately 90 to 95% of the oxygen we consume is used by the terminal oxidase in the electron transport chain for ATP generation via oxidative phosphorylation. The remainder of the  $\text{O}_2$  is used directly by oxygenases and other oxidases, enzymes that oxidize a compound in the body by transferring electrons directly to  $\text{O}_2$  (Fig. 19.12). The large positive reduction potential of  $\text{O}_2$  makes all of these reactions extremely favorable thermodynamically, but the electronic structure of  $\text{O}_2$  slows the speed of electron transfer. These enzymes, therefore, contain a metal ion that facilitates reduction of  $\text{O}_2$ .

## A. Oxidases

Oxidases transfer electrons from the substrate to O<sub>2</sub>, which is reduced to water (H<sub>2</sub>O) or to hydrogen peroxide (H<sub>2</sub>O<sub>2</sub>). The terminal protein complex in the electron transport chain, called cytochrome oxidase, is an oxidase because it accepts electrons donated to the chain by NADH and FAD(2H) and uses these to reduce O<sub>2</sub> to water. Most of the other oxidases in the cell form hydrogen peroxide (H<sub>2</sub>O<sub>2</sub>), instead of H<sub>2</sub>O, and are called peroxidases. Peroxidases are generally confined to peroxisomes to protect DNA and other cellular components from toxic free radicals (compounds containing single electrons in an outer orbital) generated by hydrogen peroxide.

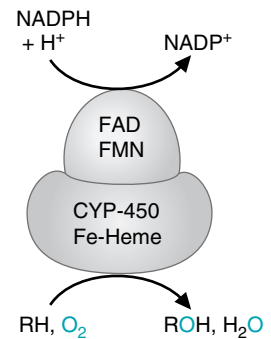
## B. Oxygenases

Oxygenases, in contrast to oxidases, incorporate one or both of the atoms of oxygen into the organic substrate (see Fig 19.12). Monooxygenases, enzymes that incorporate one atom of oxygen into the substrate and the other into H<sub>2</sub>O, are often named hydroxylases (e.g., phenylalanine hydroxylase, which adds a hydroxyl group to phenylalanine to form tyrosine) or mixed function oxidases. Monooxygenases require an electron donor-substrate, such as NADPH, a coenzyme such as FAD, which can transfer single electrons, and a metal or similar compound that can form a reactive oxygen complex (Fig.19.13). They are usually found in the endoplasmic reticulum, and occasionally in mitochondria. Dioxygenases, enzymes that incorporate both atoms of oxygen into the substrate, are used in the pathways for converting arachidonate into prostaglandins, thromboxanes, and leukotrienes.

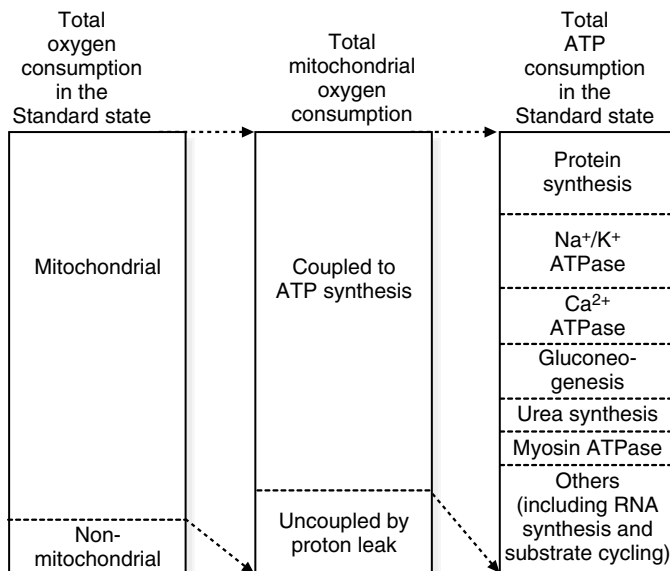
## VII. ENERGY BALANCE

Our total energy expenditure is equivalent to our oxygen consumption (Fig. 19.14). The resting metabolic rate (energy expenditure of a person at rest, at 25°C, after an overnight fast) accounts for approximately 60 to 70% of our total energy expenditure and O<sub>2</sub> consumption, and physical exercise accounts for the

**A:** In palmitate and other fatty acids, most carbons are more reduced than those in glucose or ethanol (more of the carbons have electrons in carbon-hydrogen bonds). Therefore, fatty acids have the greatest caloric content/gram, 9 kcal. In glucose, the carbons have already formed bonds with oxygen, and fewer electrons in C-H bonds are available to generate energy. Thus, the complete oxidation of glucose gives roughly 4 kcal/g. In ethanol, one carbon is a methyl group with C-H bonds, and one has an OH group. Therefore the oxidation state is intermediate between glucose and fatty acids, and ethanol thus has 7 kcal/g.



**Fig. 19.13.** Cytochrome P450 mono-oxygenases. Electrons are donated by NADPH to O<sub>2</sub> and the substrate. The flavin coenzymes FAD and FMN in one subunit transfer single electrons to cytochrome P450, which is an Fe-heme containing protein that absorbs light at a wavelength of 450 nm. The enzyme is embedded in a membrane, usually the endoplasmic reticulum.



**Fig. 19.14.** Estimated contribution of processes to energy utilization in standard state. Copied, with permission, from Rolfe DFS, Brown GC. Cellular energy utilization and molecular origin of standard metabolic rate in mammals. *Physiol Rev* 1997;77:731-758.

remainder. Of the resting metabolic rate, approximately 90 to 95% of  $O_2$  consumption is used by the mitochondrial electron transport chain, and only 5 to 10% is required for nonmitochondrial oxidases and oxygenases and is not related to ATP synthesis. Approximately 20 to 30% of the energy from this mitochondrial  $O_2$  consumption is lost by proton leak back across the mitochondrial membrane, which dissipates the electrochemical gradient without ATP synthesis. The remainder of our  $O_2$  consumption is used for ATPases that maintain ion gradients and for biosynthetic pathways.

ATP homeostasis refers to the ability of our cells to maintain constant levels of ATP despite fluctuations in the rate of utilization. Thus, increased utilization of ATP for exercise or biosynthetic reactions increases the rate of fuel oxidation. The major mechanism employed is feedback regulation; all of the pathways of fuel oxidation leading to generation of ATP are feedback-regulated by ATP levels, or by compounds related to the concentration of ATP. In general, the less ATP used, the less fuel will be oxidized to generate ATP.

According to the first law of thermodynamics, the energy (cal) in our consumed fuel can never be lost. Consumed fuel is either oxidized to meet the energy demands of the basal metabolic rate + exercise, or it is stored as fat. Thus, an intake of calories in excess of those expended results in weight gain. The simple statement, “If you eat too much and don’t exercise, you will get fat,” is really a summary of the bioenergetics of the ATP-ADP cycle.

## CLINICAL COMMENTS



**Otto Shape.** Otto Shape visited his physician, who noted the increased weight. He recommended several diet modifications to Otto that would decrease the caloric content of his diet and pointed out the importance of exercise for weight reduction. He reminded Otto that the American Heart Association and the American Cancer Society recommended 45 to 60 minutes of moderate-to-vigorous exercise 5 to 7 days per week. He also reminded Otto that he would want to be a role model for his patients. Otto decided to begin an exercise regimen that includes an hour of running each day.



**X.S. Teefore.** Mr. Teefore exhibited the classical signs and symptoms of hyperthyroidism (increased secretion of the thyroid hormones,  $T_3$  and  $T_4$ ) including a goiter (enlarged thyroid gland). Thyroid function tests confirmed this diagnosis.

Thyroid hormones (principally  $T_3$ ) modulate cellular energy production and utilization through their ability to increase the gene transcription of many proteins involved in intermediary metabolism, including enzymes in the TCA cycle and oxidative phosphorylation. They increase the rate of ATP utilization by  $Na^+$ ,  $K^+$ -ATPase, and other enzymes. They also affect the efficiency of energy transformations, so that either more fuel must be oxidized to maintain a given level of ATP, or more ATP must be expended to achieve the desired physiological response. The loss of weight experienced by **X.S. Teefore**, in spite of a very good appetite, reflects his increased caloric requirements and a less efficient utilization of fuels. The result is an enhanced oxidation of adipose tissue stores as well as a catabolic effect on muscle and other protein-containing tissues. Through mechanisms that are not well understood, increased levels of thyroid hormone in the blood also increase the activity or “tone” of the sympathetic (adrenergic) nervous system. An activated sympathetic nervous system leads to a more rapid and forceful heartbeat (tachycardia and palpitations), increased nervousness (anxiety and insomnia), tremulousness (a sense of shakiness or jitteriness), and other symptoms.



The thyroid gland secretes the thyroid hormones tetraiodothyronine ( $T_4$ ) and triiodothyronine ( $T_3$ ) (see Fig. 11.8 for the structure of  $T_3$ ).  $T_3$  is the most active form of the hormone.  $T_4$  is synthesized and secreted in approximately 10 times greater amounts than  $T_3$ . Hepatocytes (liver cells) and other cells contain a deiodinase that removes one of the iodines from  $T_4$ , converting it to  $T_3$ .  $T_3$  exerts its effects on tissues by regulating the transcription of specific genes involved in energy metabolism (see Chapter 16, section III.C.2., Fig. 16.14).





**Cora Nari.** Cora Nari was in left ventricular heart failure (LVF) when she presented to the hospital with her second heart attack in 8 months. The diagnosis of LVF was based, in part, on her rapid heart rate (104 beats/min) and respiratory rate. On examining her lungs, her physician heard respiratory rales, caused by inspired air bubbling in fluid that had filled her lung air spaces secondary to LVF. This condition is referred to as congestive heart failure.

**Cora Nari's** rapid heart rate (tachycardia) resulted from a reduced capacity of her ischemic, failing left ventricular muscle to eject a normal amount of blood into the arteries leading away from the heart with each contraction. The resultant drop in intraarterial pressure signaled a reflex response in the central nervous system that, in turn, caused an increase in heart rate in an attempt to bring the total amount of blood leaving the left ventricle each minute (the cardiac output) back toward a more appropriate level to maintain systemic blood pressure.

Treatment of Cora's congestive heart failure will include efforts to reduce the workload of the heart with diuretics and other "load reducers," attempts to improve the force of left ventricular contraction with digitalis and other "inotropes," and the administration of oxygen by nasal cannula to reduce the injury caused by lack of blood flow (ischemia) to the viable heart tissue in the vicinity of the infarction.

## BIOCHEMICAL COMMENTS



**Active Transport and Cell Death.** Most of us cannot remember when we first learned that we would die if we stopped breathing. But exactly how cells die from a lack of oxygen is an intriguing question. Pathologists generally describe two histologically distinct types of cell death: necrosis and apoptosis (programmed cell death). Cell death from a lack of O<sub>2</sub>, such as occurs during a myocardial infarction, can be very rapid, and is considered necrosis. The lack of ATP for the active transport of Na<sup>+</sup> and Ca<sup>2+</sup> triggers some of the death cascades leading to necrosis (Fig. 19.15).

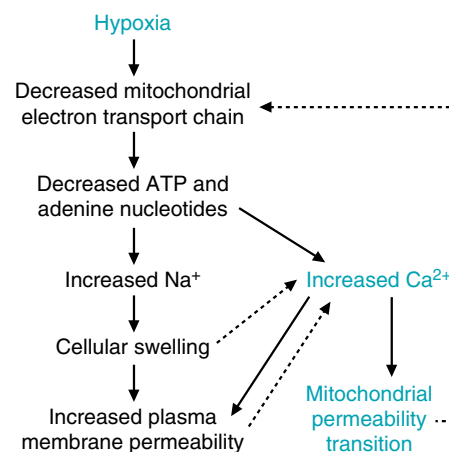
The influx of Na<sup>+</sup> and loss of the Na<sup>+</sup> gradient across the plasma membrane is an early event accompanying ATP depletion during interruption of the O<sub>2</sub> supply. One consequence of the increased intracellular Na<sup>+</sup> concentration is that other transport processes driven by the Na<sup>+</sup> gradient are impaired. For example, the Na<sup>+</sup> / H<sup>+</sup> exchanger, which normally pumps out H<sup>+</sup> generated from metabolism in exchange for extracellular Na<sup>+</sup>, can no longer function, and intracellular pH may drop. The increased intracellular H<sup>+</sup> may impair ATP generation from anaerobic glycolysis. As a consequence of increased intracellular ion concentrations, water enters the cells and hydropic swelling occurs. Swelling is accompanied by the release of creatine kinase MB subunits, troponin I, and troponin C into the blood. These enzymes are measured in the blood as indicators of a myocardial infarction (see Chapters 6 and 7). Swelling is an early event and is considered a reversible stage of cell injury.

Normally, intracellular Ca<sup>2+</sup> concentration is carefully regulated to fluctuate at low levels (intracellular Ca<sup>2+</sup> concentration is less than 10<sup>-7</sup> M, compared with approximately 10<sup>-3</sup> M in extracellular fluid). Fluctuations of Ca<sup>2+</sup> concentration at these low levels regulate myofibrillar contraction, energy metabolism, and other cellular processes. However, when Ca<sup>2+</sup> concentration is increased above this normal range, it triggers cell death (necrosis). High Ca<sup>2+</sup> concentrations activate a phospholipase that increases membrane permeability, resulting in further loss of ion gradients across the cell membrane. They also trigger opening of the mitochondrial permeability transition pore, which results in loss of mitochondrial function and further impairs oxidative phosphorylation.

Intracellular Ca<sup>2+</sup> levels may increase as a result of cell swelling, the lack of ATP for ATP-dependent Ca<sup>2+</sup> pumps, or the loss of the Na<sup>+</sup> gradient. Normally,



Congestive heart failure occurs when the weakened pumping action of the ischemic left ventricular heart muscle causes back pressure to increase in the vessels which bring oxygenated blood from the lungs to the left side of the heart. The pressure inside these pulmonary vessels eventually reaches a critical level above which water from the blood moves down a "pressure gradient" from the capillary lumen into alveolar air spaces of the lung (transudation). The patient experiences shortness of breath as the fluid in the air spaces interferes with oxygen exchange from the inspired air into arterial blood, causing hypoxia. The hypoxia then stimulates the respiratory center in the central nervous system, leading to a more rapid respiratory rate in an effort to increase the oxygen content of the blood. As the patient inhales deeply, the physician hears gurgling sounds (known as inspiratory rales) with a stethoscope placed over the posterior lung bases. These sounds represent the bubbling of inspired air as it enters the fluid-filled pulmonary alveolar air spaces.



**Fig. 19.15.** Hypoxia, Ca<sup>2+</sup>, Na<sup>+</sup>, and cell death. Without an adequate O<sub>2</sub> supply, decreased ATP synthesis from oxidative phosphorylation results in an increase of cytoplasmic Na<sup>+</sup> and Ca<sup>2+</sup> ions. Increased ions levels can trigger death cascades that involve increased permeability of the plasma membrane, loss of ion gradients, decreased cytosolic pH, mitochondrial Ca<sup>2+</sup> overload, and a change in mitochondrial permeability called the mitochondrial permeability transition. The solid lines show the first sequence of events; the dashed lines show how these events feed-back to accelerate the mitochondrial deterioration, making recovery of oxidative phosphorylation impossible.

$\text{Ca}^{2+}$ -ATPases located in the plasma membrane pump  $\text{Ca}^{2+}$  out of the cell.  $\text{Ca}^{2+}$ -ATPases in the endoplasmic reticulum, and in the sarcoplasmic reticulum of heart and other muscles, sequester  $\text{Ca}^{2+}$  within the membranes, where it is bound by a low-affinity binding protein.  $\text{Ca}^{2+}$  is released from the sarcoplasmic reticulum in response to a nerve impulse, which signals contraction, and the increase of  $\text{Ca}^{2+}$  stimulates both muscle contraction and the oxidation of fuels. Within the heart, another  $\text{Ca}^{2+}$  transporter protein, the  $\text{Na}^+/\text{Ca}^{2+}$  exchange transporter, coordinates the efflux of  $\text{Ca}^{2+}$  in exchange for  $\text{Na}^+$ , so that  $\text{Ca}^{2+}$  is extruded with each contraction.

### Suggested References

Nelson DL, Lehninger AL, Cox MM. Lehninger Principles of biochemistry. Chapter 14. In: Principles of Bioenergetics. New York: Worth, 2000:490–526.  
 Hanson RW. The role of ATP in metabolism. *Biochem Educ* 1989;17:86–92.  
 Rolfe DES, Brown GC. Cellular energy utilization and molecular origin of standard metabolic rate in mammals. *Physiol Rev* 1997;77:731–758.



## REVIEW QUESTIONS—CHAPTER 19

- The highest-energy phosphate bond in ATP is located between which of the following groups?
  - Adenosine and phosphate
  - Ribose and phosphate
  - Ribose and adenine
  - Two hydroxyl groups in the ribose ring
  - Two phosphate groups
- Which of the following bioenergetic terms or phrases is correctly defined?
  - The first law of thermodynamics states that the universe tends towards a state of increased order.
  - The second law of thermodynamics states that the total energy of a system remains constant.
  - The change in enthalpy of a reaction is a measure of the total amount of heat that can be released from changes in the chemical bonds.
  - $\Delta G^{\circ'}$  of a reaction is the standard free energy change measured at 37°C and a pH of 7.4.
  - A high-energy bond is a bond that releases more than 3 kcal/mole of heat when it is hydrolyzed.
- Which statement best describes the direction a chemical reaction will follow?
  - A reaction with a positive free energy will proceed in the forward direction if the substrate concentration is raised high enough.
  - Under standard conditions, a reaction will proceed in the forward direction if the free energy  $\Delta G^{\circ'}$  is positive.
  - The direction of a reaction is independent of the initial substrate and product concentrations because the direction is determined by the change in free energy.
  - The concentration of all of the substrates must be higher than all of the products to proceed in the forward direction.
  - The enzyme for the reaction must be working at better than 50% of its maximum efficiency for the reaction to proceed in the forward direction.
- A patient, Mr. Perkins, has just suffered a heart attack. As a consequence, his heart would display which of the following changes?
  - An increased intracellular  $\text{O}_2$  concentration
  - An increased intracellular ATP concentration
  - An increased intracellular  $\text{H}^+$  concentration
  - A decreased intracellular  $\text{Ca}^{2+}$  concentration
  - A decreased intracellular  $\text{Na}^+$  concentration

5. Which of the following statements correctly describes reduction of one of the electron carriers, NAD<sup>+</sup> or FAD?
- (A) NAD<sup>+</sup> accepts two electrons as hydrogen atoms to form NADH<sub>2</sub>.
  - (B) NAD<sup>+</sup> accepts two electrons that are each donated from a separate atom of the substrate.
  - (C) NAD<sup>+</sup> accepts two electrons as a hydride ion to form NADH.
  - (D) FAD releases a proton as it accepts two electrons.
  - (E) FAD must accept two electrons at a time.

## 20 Tricarboxylic Acid Cycle



The TCA cycle is frequently called the Krebs cycle because Sir Hans Krebs first formulated its reactions into a cycle. It is also called the “citric acid cycle” because citrate was one of the first compounds known to participate. The most common name for this pathway, the tricarboxylic acid or TCA cycle, denotes the involvement of the tricarboxylates citrate and isocitrate.



The major pathways of fuel oxidation generate acetyl CoA, which is the substrate for the TCA cycle. In the first step of the TCA cycle, the acetyl portion of acetyl CoA combines with the 4-carbon intermediate oxaloacetate to form citrate (6 carbons), which is rearranged to form isocitrate. In the next two oxidative decarboxylation reactions, electrons are transferred to  $\text{NAD}^+$  to form NADH, and 2 molecules of electron-depleted  $\text{CO}_2$  are released. Subsequently, a high-energy phosphate bond in GTP is generated from substrate level phosphorylation. In the remaining portion of the TCA cycle, succinate is oxidized to oxaloacetate with the generation of one FAD(2H) and one NADH. The net reaction of the TCA cycle, which is the sum of the equations for individual steps, shows that the two carbons of the acetyl group have been oxidized to two molecules of  $\text{CO}_2$ , with conservation of energy as three molecules of NADH, one of FAD(2H), and one of GTP.

*The **TCA cycle** (tricarboxylic acid cycle) accounts for over two thirds of the ATP generated from fuel oxidation. The pathways for oxidation of fatty acids, glucose, amino acids, acetate, and ketone bodies all generate **acetyl CoA**, which is the substrate for the TCA cycle. As the activated 2-carbon acetyl group is oxidized to two molecules of  $\text{CO}_2$ , energy is conserved as **NADH**, **FAD(2H)**, and **GTP** (Fig. 20.1). NADH and FAD(2H) subsequently donate electrons to  $\text{O}_2$  via the electron transport chain, with the generation of ATP from oxidative phosphorylation. Thus, the TCA cycle is central to energy generation from **cellular respiration**.*

*Within the TCA cycle, the oxidative decarboxylation of  $\alpha$ -ketoglutarate is catalyzed by the multisubunit  **$\alpha$ -ketoglutarate dehydrogenase complex**, which contains the coenzymes **thiamine-pyrophosphate**, **lipoate**, and **FAD**. A similar complex, the **pyruvate dehydrogenase complex (PDC)**, catalyzes the oxidation of pyruvate to acetyl CoA, thereby providing a link between the pathways of glycolysis and the TCA cycle (see Fig. 20.1)*

*The two-carbon acetyl group is the ultimate source of the electrons that are transferred to  $\text{NAD}^+$  and FAD and also the carbon in the two  $\text{CO}_2$  molecules that are produced. Oxaloacetate is used and regenerated in each turn of the cycle (see Fig. 20.1). However, when cells use intermediates of the TCA cycle for*

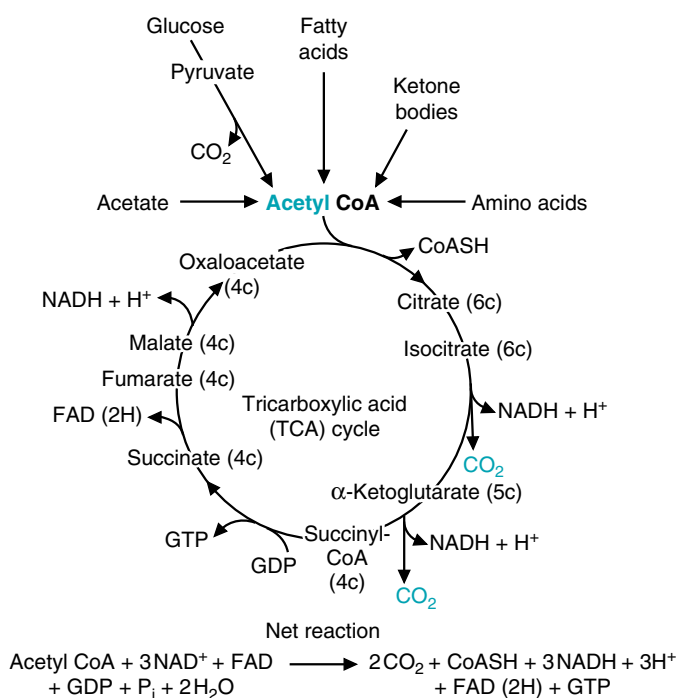


Fig. 20.1. Summary of the TCA cycle.

biosynthetic reactions, the carbons of oxaloacetate must be replaced by **anaplerotic** (filling up) reactions, such as the **pyruvate carboxylase reaction**.

The TCA cycle occurs in the mitochondrion, where its flux is tightly coordinated with the rate of the electron transport chain and oxidative phosphorylation through **feedback regulation that reflects the demand for ATP**. The rate of the TCA cycle is increased when ATP utilization in the cell is increased through the response of several enzymes to **ADP levels**, the **NADH/ NAD<sup>+</sup> ratio**, the rate of **FAD(2H) oxidation** or the **Ca<sup>2+</sup> concentration**. For example, **isocitrate dehydrogenase is allosterically activated by ADP**.

There are two general consequences to impaired functioning of the TCA cycle: (1) an inability to generate ATP from fuel oxidation, and (2) an accumulation of TCA cycle precursors. For example, inhibition of pyruvate oxidation in the TCA cycle results in its reduction to lactate, which can cause a **lactic acidosis**. The most common situation leading to an impaired function of the TCA cycle is a relative lack of oxygen to accept electrons in the electron transport chain.



## THE WAITING ROOM



**Otto Shape**, a 26-year-old medical student, has faithfully followed his diet and aerobic exercise program of daily tennis and jogging (see Chapter 19). He has lost a total of 33 lb and is just 23 lb from his college weight of 154 lb. His exercise capacity has markedly improved; he can run for a longer time at a faster pace before noting shortness of breath or palpitations of his heart. Even his test scores in his medical school classes have improved.



**Ann O' Rexia** suffers from anorexia nervosa (see Chapters 1, 3, and 9). In addition to a low body weight, decreased muscle mass, glycogen, and fat stores, she has iron-deficiency anemia (see Chapter 16). She has started to gain weight, and is trying a daily exercise program. However, she constantly feels weak and tired. When she walks, she feels pain in her calf muscles. On this visit to her nutritionist, they discuss the vitamin content of her diet, and its role in energy metabolism.

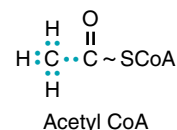


**Al Martini** has been hospitalized for congestive heart failure (see Chapter 8) and for head injuries sustained while driving under the influence of alcohol (Chapters 9 and 10). He completed an alcohol detoxification program, enrolled in a local Alcoholics Anonymous (AA) group, and began seeing a psychologist. During this time, his alcohol-related neurologic and cardiac manifestations of thiamine deficiency partially cleared. However, in spite of the support he was receiving, he began drinking excessive amounts of alcohol again while eating poorly. Three weeks later, he was readmitted with symptoms of "high output" heart failure.



Vitamins and minerals required for the TCA cycle and anaplerotic reactions

Niacin (NAD<sup>+</sup>)  
Riboflavin (FAD)  
Pantothenate (CoA)  
Thiamine  
Biotin  
Mg<sup>2+</sup>  
Ca<sup>2+</sup>  
Fe<sup>2+</sup>  
Phosphate



**Fig. 20.2.** The acetyl group of acetyl CoA. Acetyl CoA donates eight electrons to the TCA cycle, which are shown in blue, and two carbons. The high-energy bond is shown by a ~. The acetyl group is the ultimate source of the carbons in the two molecules of CO<sub>2</sub> that are produced, and the source of electrons in the one molecule of FAD(2H) and 3 molecules of NADH, which have each accepted two electrons. However, the same carbon atoms and electrons that enter from one molecule of acetyl CoA do not leave as CO<sub>2</sub>, NADH, or FAD(2H) within the same turn of the cycle.

## I. REACTIONS OF THE TCA CYCLE

In the TCA cycle, the 2-carbon acetyl group of acetyl CoA is oxidized to 2 CO<sub>2</sub> molecules (see Fig. 20.1). The function of the cycle is to conserve the energy from this oxidation, which it accomplishes principally by transferring electrons from intermediates of the cycle to NAD<sup>+</sup> and FAD. The eight electrons donated by the acetyl group eventually end up in three molecules of NADH and one of FAD(2H) (Fig. 20.2). As a consequence, ATP can be generated from oxidative phosphorylation when NADH and FAD(2H) donate these electrons to O<sub>2</sub> via the electron transport chain.



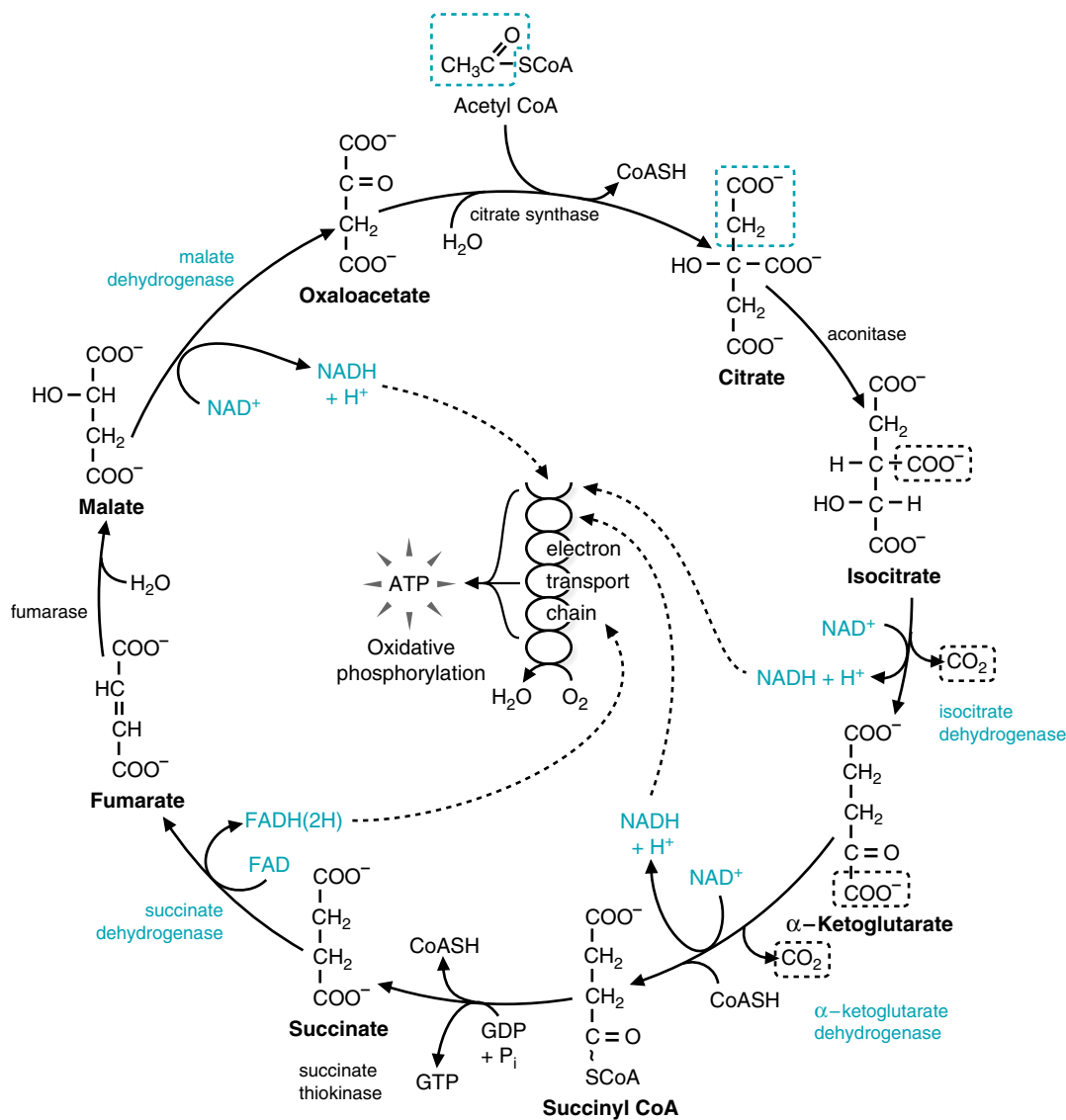


Synthases, such as citrate synthase, catalyze condensation of two organic molecules to form a carbon-carbon bond. Dehydrogenases, such as isocitrate dehydrogenase, are enzymes that remove electron-containing hydrogen or hydride atoms from a substrate and transfer them to electron-accepting coenzymes, such as  $\text{NAD}^+$  or  $\text{FAD}$ . Aconitase is an isomerase, an enzyme that catalyzes an internal rearrangement of atoms or electrons. In aconitase, a hydroxyl group is being transferred from one carbon to another. An iron cofactor in the enzyme facilitates this transfer.

Initially, the acetyl group is incorporated into citrate, an intermediate of the TCA cycle (Fig. 20.3). As citrate progresses through the cycle to oxaloacetate, it is oxidized by four dehydrogenases (isocitrate dehydrogenase,  $\alpha$ -ketoglutarate dehydrogenase, succinate dehydrogenase, and malate dehydrogenase), which transfer electrons to  $\text{NAD}^+$  or  $\text{FAD}$ . The isomerase aconitase rearranges electrons in citrate, thereby forming isocitrate, to facilitate an electron transfer to  $\text{NAD}^+$ .

Although no  $\text{O}_2$  is introduced into the TCA cycle, the two molecules of  $\text{CO}_2$  produced have more oxygen than the acetyl group. These oxygen atoms are ultimately derived from the carbonyl group of acetyl CoA, two molecules of water added by fumarase and citrate synthase, and the  $\text{PO}_4^{2-}$  added to GDP.

The overall yield of energy-containing compounds from the TCA cycle is 3  $\text{NADH}$ , 1  $\text{FAD(2H)}$ , and 1  $\text{GTP}$ . The high-energy phosphate bond of  $\text{GTP}$  is generated from substrate level phosphorylation catalyzed by succinate thiokinase (succinyl CoA synthetase). As the  $\text{NADH}$  and  $\text{FAD(2H)}$  are reoxidized in the electron transport chain, approximately 2.5  $\text{ATP}$  are generated for each  $\text{NADH}$ , and 1.5  $\text{ATP}$



**Fig. 20.3.** Reactions of the TCA cycle. The oxidation-reduction enzymes and coenzymes are shown in blue. Entry of the two carbons of acetyl CoA into the TCA cycle are indicated with blue dashed boxes. The carbons released as  $\text{CO}_2$  are shown with black dashed boxes.

for the FAD(2H). Consequently, the net energy yield from the TCA cycle and oxidative phosphorylation is about 10 high-energy phosphate bonds for each acetyl group oxidized.

### A. Formation and Oxidation of Isocitrate

The TCA cycle begins with condensation of the activated acetyl group and oxaloacetate to form the 6-carbon intermediate citrate, a reaction catalyzed by the enzyme citrate synthase (see Fig. 20.3). Because oxaloacetate is regenerated with each turn of the cycle, it is not really considered a substrate of the cycle, or a source of electrons or carbon.

In the next step of the TCA cycle, the hydroxyl (alcohol) group of citrate is moved to an adjacent carbon so that it can be oxidized to form a keto group. The isomerization of citrate to isocitrate is catalyzed by the enzyme aconitase, which is named for an intermediate of the reaction. The enzyme isocitrate dehydrogenase catalyzes the oxidation of the alcohol group and the subsequent cleavage of the carboxyl group to release CO<sub>2</sub> (an oxidative decarboxylation).

### B. $\alpha$ -Ketoglutarate to Succinyl CoA

The next step of the TCA cycle is the oxidative decarboxylation of  $\alpha$ -ketoglutarate to succinyl CoA, catalyzed by the  $\alpha$ -ketoglutarate dehydrogenase complex (see Fig. 20.3). The dehydrogenase complex contains the coenzymes thiamine pyrophosphate, lipoic acid, and FAD.

In this reaction, one of the carboxyl groups of  $\alpha$ -ketoglutarate is released as CO<sub>2</sub>, and the adjacent keto group is oxidized to the level of an acid, which then combines with CoASH to form succinyl CoA (see Fig. 20.3). Energy from the reaction is conserved principally in the reduction state of NADH, with a smaller amount present in the high-energy thioester bond of succinyl CoA.

### C. Generation of GTP

Energy from the succinyl CoA thioester bond is used to generate GTP from GDP and Pi in the reaction catalyzed by succinate thiokinase (see Fig. 20.3). This reaction is an example of substrate level phosphorylation. By definition, substrate level phosphorylation is the formation of a high-energy phosphate bond where none previously existed without the use of molecular O<sub>2</sub> (in other words, NOT oxidative phosphorylation). The high-energy phosphate bond of GTP is energetically equivalent to that of ATP, and can be used directly for energy-requiring reactions like protein synthesis.

### D. Oxidation of Succinate to Oxaloacetate

Up to this stage of the TCA cycle, two carbons have been stripped of their available electrons and released as CO<sub>2</sub>. Two pairs of these electrons have been transferred to 2 NAD<sup>+</sup>, and one GTP has been generated. However, two additional pairs of electrons arising from acetyl CoA still remain in the TCA cycle as part of succinate. The remaining steps of the TCA cycle transfer these two pairs of electrons to FAD and NAD<sup>+</sup> and add H<sub>2</sub>O, thereby regenerating oxaloacetate.

The sequence of reactions converting succinate to oxaloacetate begins with the oxidation of succinate to fumarate (see Fig. 20.3). Single electrons are transferred from the two adjacent -CH<sub>2</sub>- methylene groups of succinate to an FAD bound to succinate dehydrogenase, thereby forming the double bond of fumarate. From the reduced enzyme-bound FAD, the electrons are passed into the electron transport chain. An OH<sup>-</sup> group and a proton from water add to the double bond of fumarate, converting it to malate. In the last reaction of the TCA cycle, the alcohol group of malate is oxidized to a keto group through the donation of electrons to NAD<sup>+</sup>.



**Otto Shape's** exercise program increases his rate of ATP utilization and his rate of fuel oxidation in the TCA cycle. The TCA cycle generates NADH and FAD(2H), and the electron transport chain transfers electrons from NADH and FAD(2H) to O<sub>2</sub>, thereby creating the electrochemical potential that drives ATP synthesis from ADP. As ATP is used in the cell, the rate of the electron transport chain increases. The TCA cycle and other fuel oxidative pathways respond by increasing their rates of NADH and FAD(2H) production.



Succinate thiokinase is also known as succinyl CoA synthetase. Both names refer to the reverse direction of the reaction, i.e., the conversion of succinate to the thioester succinyl CoA, utilizing energy from GTP. Synthetases, such as citrate synthase, differ from synthetases in that synthetases cleave a high-energy phosphate bond in ATP, UTP, CTP, or GTP, and synthetases do not.



**Q:** From Figure 20.3, which enzymes in the TCA cycle release CO<sub>2</sub>? How many moles of oxaloacetate are consumed in the TCA cycle for each mole of CO<sub>2</sub> produced?



The succinate to oxaloacetate sequence of reactions—oxidation through formation of a double bond, addition of water to the double bond, and oxidation of the resultant alcohol to a ketone is found in many oxidative pathways in the cell, such as the pathways for the oxidation of fatty acids, and oxidation of the branched chain amino acids.



**Ann O'Rexia** has been malnourished for some time, and has developed subclinical deficiencies of many vitamins, including riboflavin. The coenzymes FAD (flavin adenine dinucleotide) and FMN (flavin mononucleotide) are synthesized from the vitamin riboflavin. Riboflavin is actively transported into cells, where the enzyme flavokinase adds a phosphate to form FMN. FAD synthetase then adds AMP to form FAD. FAD is the major coenzyme in tissues and is generally found tightly bound to proteins, with about 10% being covalently bound. Its turnover in the body is very slow, and people can live for long periods on low intakes without displaying any signs of a riboflavin deficiency.

**A:**



With regeneration of oxaloacetate, the TCA cycle is complete; the chemical bond energy, carbon, and electrons donated by the acetyl group have been converted to  $\text{CO}_2$ , NADH, FAD(2H), GTP, and heat.

## II. COENZYMES OF THE TCA CYCLE

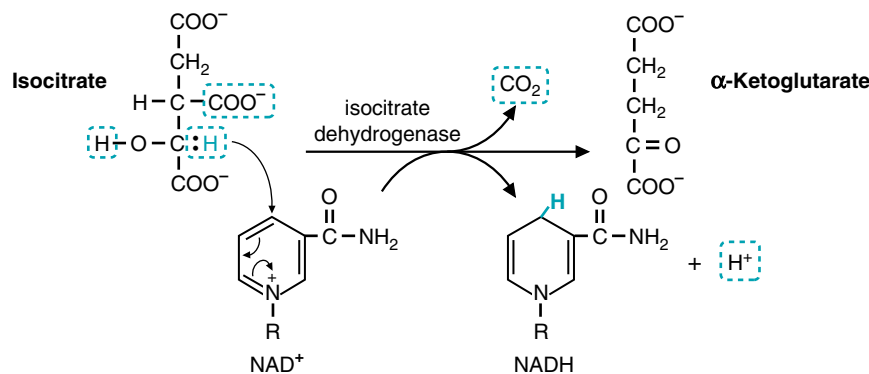
The enzymes of the TCA cycle rely heavily on coenzymes for their catalytic function. Isocitrate dehydrogenase and malate dehydrogenase use  $\text{NAD}^+$  as a coenzyme, and succinate dehydrogenase uses FAD. Citrate synthase catalyzes a reaction that uses a CoA derivative, acetyl CoA. The  $\alpha$ -ketoglutarate dehydrogenase complex uses thiamine pyrophosphate, lipoate and FAD as bound coenzymes, and  $\text{NAD}^+$  and CoASH as substrates. Each of these coenzymes has unique structural features that enable it to fulfill its role in the TCA cycle.

### A. FAD and NAD<sup>+</sup>

Both FAD and  $\text{NAD}^+$  are electron-accepting coenzymes. Why is FAD used in some reactions and  $\text{NAD}^+$  in others? Their unique structural features enable FAD and  $\text{NAD}^+$  to act as electron acceptors in different types of reactions, and play different physiological roles in the cell. FAD is able to accept single electrons ( $\text{H}\bullet$ ), and forms a half-reduced single electron intermediate (Fig. 20.4). It thus participates in reactions in which single electrons are transferred independently from two different atoms, which occurs in double bond formation (e.g., succinate to fumarate) and disulfide bond formation (e.g., lipoate to lipoate disulfide in the  $\alpha$ -ketoglutarate



**Fig. 20.4.** One-electron steps in the reduction of FAD. When FAD and FMN accept single electrons, they are converted to the half-reduced semiquinone, a semistable free radical form. They can also accept two electrons to form the fully reduced form, FADH<sub>2</sub>. However, in most dehydrogenases, FADH<sub>2</sub> is never formed. Instead, the first electron is shared with a group on the protein as the next electron is transferred. Therefore, in this text, overall acceptance of two electrons by FAD has been denoted by the more general abbreviation, FAD(2H).



**Fig. 20.5.** Oxidation and decarboxylation of isocitrate. The alcohol group (C—OH) is oxidized to a ketone, with the C—H electrons donated to  $\text{NAD}^+$  as the hydride ion. Subsequent electron shifts in the pyridine ring remove the positive charge. The H of the OH group dissociates into water as a proton,  $\text{H}^+$ .  $\text{NAD}^+$ , the electron acceptor, is reduced.

dehydrogenase reaction). In contrast,  $\text{NAD}^+$  accepts a pair of electrons as the hydride ion ( $\text{H}^-$ ), which is attracted to the carbon opposite the positively-charged pyridine ring (Fig. 20.5). This occurs, for example, in the oxidation of alcohols to ketones by malate dehydrogenase and isocitrate dehydrogenase. The nicotinamide ring accepts a hydride ion from the C—H bond, and the alcoholic hydrogen is released into the medium as a positively charged proton,  $\text{H}^+$ .

The free radical, single-electron forms of FAD are very reactive, and FADH can lose its electron through exposure to water or the initiation of chain reactions. As a consequence, FAD must remain very tightly, sometimes covalently, attached to its enzyme while it accepts and transfers electrons to another group bound on the enzyme (Fig 20.6). Because FAD interacts with many functional groups on amino acid side chains in the active site, the  $E^{0'}$  for enzyme-bound FAD varies greatly and can be greater or much less than that of  $\text{NAD}^+$ . In contrast,  $\text{NAD}^+$  and NADH are more like substrate and product than coenzymes.

NADH plays a regulatory role in balancing energy metabolism that FAD(2H) cannot because FAD(2H) remains attached to its enzyme. Free  $\text{NAD}^+$  binds to a dehydrogenase and is reduced to NADH, which is then released into the medium where it can bind and inhibit a different dehydrogenase. Consequently, oxidative enzymes are controlled by the  $\text{NADH}/\text{NAD}^+$  ratio, and do not generate NADH faster than it can be reoxidized in the electron transport chain. The regulation of the TCA cycle and other pathways of fuel oxidation by the  $\text{NADH}/\text{NAD}^+$  ratio is part of the mechanism for coordinating the rate of fuel oxidation to the rate of ATP utilization.

## B. Role of CoA in the TCA Cycle

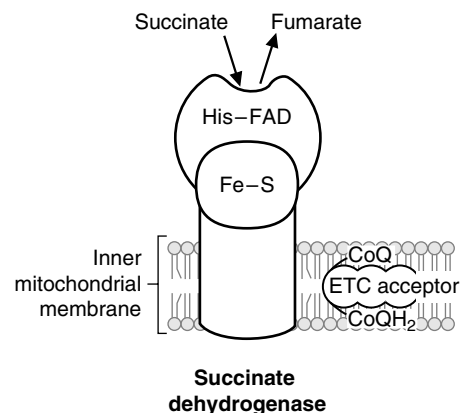
CoASH, the acylation coenzyme, participates in reactions through the formation of a thioester bond between the sulfur (S) of CoASH and an acyl group (e.g., acetyl



FAD has been referred to as a married coenzyme, and  $\text{NAD}^+$  is its promiscuous cousin. FAD faithfully accepts only electrons from a substrate that is bound to the same enzyme (or enzyme complex), and donates these without leaving that enzyme. It does this repeatedly while still attached to its enzyme.  $\text{NAD}^+$ , conversely, may accept electrons when bound to any dehydrogenase, and leaves the enzyme immediately afterward. It donates these electrons while bound to a different dehydrogenase, such as NADH dehydrogenase in the electron transport chain. It really gets around!



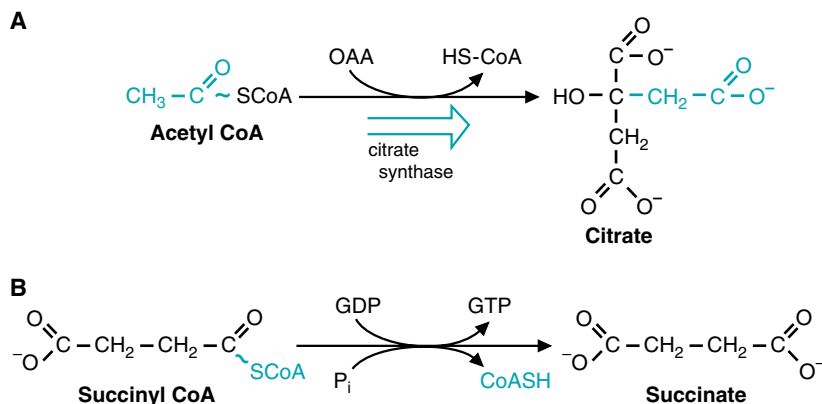
The claim that succinate oxidation could produce energy without oxygen is wrong. It was probably based on the fact that succinate is oxidized to fumarate by the donation of electrons to FAD. However, ATP can only be generated from this process when these electrons are donated to oxygen in the electron transport chain. The energy generated by the electron transport chain is used for ATP synthesis in the process of oxidative phosphorylation. After the covalently bound FAD(2H) is oxidized back to FAD by the electron transport chain, succinate dehydrogenase can oxidize another succinate molecule.



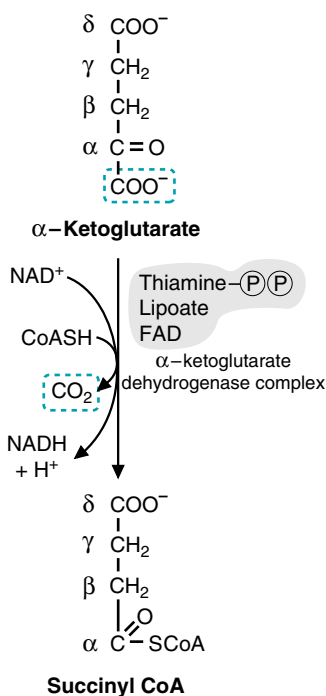
**Fig. 20.6.** Succinate dehydrogenase contains covalently bound FAD. As a consequence, succinate dehydrogenase and similar flavoproteins reside in the inner mitochondrial membrane where they can directly transfer electrons into the electron transport chain. The electrons are transferred from the covalently bound FAD to an Fe-S complex on the enzyme, and then to coenzyme Q in the electron transport chain (see Chapter 21). Thus, FAD does not have to dissociate from the enzyme to transfer its electrons. All the other enzymes of the TCA cycle are found in the mitochondrial matrix.



CoASH is synthesized from the vitamin pantothenate in a sequence of reactions which phosphorylate pantothenate, add the sulfhydryl portion of CoA from cysteine, and then add AMP and an additional phosphate group from ATP (see Fig. 8.12). Pantothenate is widely distributed in foods (pantos means everywhere), so it is unlikely that **Ann O'Rexia** has developed a pantothenate deficiency. Although CoA is required in approximately 100 different reactions in mammalian cells, no Recommended Daily Allowance (RDA) has been established for pantothenate, in part because indicators have not yet been found which specifically and sensitively reflect a deficiency of this vitamin in the human. The reported symptoms of pantothenate deficiency (fatigue, nausea, and loss of appetite) are characteristic of vitamin deficiencies in general.



**Fig. 20.7.** Utilization of the high-energy thioester bond of acyl CoAs. Energy transformations are shown in blue. **A.** The energy released by hydrolysis of the thioester bond of acetyl CoA in the citrate synthase reaction contributes a large negative  $\Delta G^{0'}$  to the forward direction of the TCA cycle. **B.** The energy of the succinyl CoA thioester bond is used for the synthesis of the high-energy phosphate bond of GTP.



**Fig. 20.8.** Oxidative decarboxylation of  $\alpha$ -ketoglutarate. The  $\alpha$ -ketoglutarate dehydrogenase complex oxidizes  $\alpha$ -ketoglutarate to succinyl CoA. The carboxyl group is released as  $\text{CO}_2$ . The keto group on the  $\alpha$ -carbon is oxidized, and then forms the acyl CoA thioester, succinyl CoA. The  $\alpha$ ,  $\beta$ ,  $\gamma$ , and  $\delta$  on succinyl CoA refer to the sequence of atoms in  $\alpha$ -ketoglutarate.

CoA, succinyl CoA) (Fig. 20.7). The complete structure of CoASH and its vitamin precursor, pantothenate, is shown in Figure 8.12. A thioester bond differs from a typical oxygen ester bond because S, unlike O, does not share its electrons and participate in resonance formations. One of the consequences of this feature of sulfur chemistry is that the carbonyl carbon, the  $\alpha$ -carbon and the  $\beta$ -carbon of the acyl group in a CoA thioester can be activated for participation in different types of reactions (e.g., in the citrate synthase reaction, the  $\alpha$ -carbon methyl group is activated for condensation with oxaloacetate, see Figs. 20.3 and 20.7A). Another consequence is that the thioester bond is a high-energy bond that has a large negative  $\Delta G^{0'}$  of hydrolysis (approximately  $-13$  kcal/mole).

The energy from cleavage of the high-energy thioester bonds of succinyl CoA and acetyl CoA is used in two different ways in the TCA cycle. When the succinyl CoA thioester bond is cleaved by succinate thiokinase, the energy is used directly for activating an enzyme-bound phosphate that is transferred to GDP (see Fig. 20.7B). In contrast, when the thioester bond of acetyl CoA is cleaved in the citrate synthase reaction, the energy is released, giving the reaction a large negative  $\Delta G^{0'}$  of  $-7.7$  kcal/mole. The large negative  $\Delta G^{0'}$  for citrate formation helps to keep the TCA cycle going in the forward direction.

### C. The $\alpha$ -Ketoacid Dehydrogenase Complexes

The  $\alpha$ -ketoglutarate dehydrogenase complex is one of a three-member family of similar  $\alpha$ -keto acid dehydrogenase complexes. The other members of this family are the pyruvate dehydrogenase complex, and the branched chain amino acid  $\alpha$ -keto acid dehydrogenase complex. Each of these complexes is specific for a different  $\alpha$ -keto acid structure. In the sequence of reactions catalyzed by the complexes, the  $\alpha$ -ketoacid is decarboxylated (i.e., releases the carboxyl group as  $\text{CO}_2$ ) (Fig. 20.8). The keto group is oxidized to the level of a carboxylic acid, and then combined with CoASH to form an acyl CoA thioester (e.g., succinyl CoA).

All of the  $\alpha$ -ketoacid dehydrogenase complexes are huge enzyme complexes composed of multiple subunits of three different enzymes,  $E_1$ ,  $E_2$ , and  $E_3$  (Fig. 20.9).  $E_1$  is an  $\alpha$ -ketoacid decarboxylase which contains thiamine pyrophosphate (TPP); it cleaves off the carboxyl group of the  $\alpha$ -keto acid.  $E_2$  is a transacylase containing lipoate; it transfers the acyl portion of the  $\alpha$ -keto acid from thiamine to CoASH.  $E_3$  is dihydrolipoyl dehydrogenase, which contains